Thesis Project by Group CLS

THESIS 1-3

Thesis Project

Group Members:

- Celles, Aaron Kent M.
- Leviste, Lee Ryan A.
- Santeco, Enrique S.

Section 1. Introduction

1.1 Problem Overview

Large-scale global optimization (LSGO) presents a critical challenge in computational intelligence, where the "curse of dimensionality" leads to an explosive increase in local optima and a high risk of premature convergence. While foundational algorithms like Differential Evolution (DE) are robust, their dependence on crossover and random mutation may restrict guided exploration — reducing particle diversity over time and leading to premature convergence.

This notebook implements and evaluates the proposed **Diversity-Preserving Level-Based Learning Differential Evolution (DP-LLDE)** algorithm, tested against the **CEC 2017 Large-Scale Global Optimization Benchmark** across 29 benchmark functions (F1–F30, excluding deprecated F2) at **10 dimensions**.

1.2 Research Objectives

---

1. Design and implement DP-LLDE, integrating sequential relative particle mutation and dynamic level-based learning into the Differential Evolution framework.
2. Evaluate DP-LLDE on the CEC 2017 benchmark suite (F1–F30) at 10 dimensions.
3. Examine convergence behavior, solution quality, and the effect of diversity-preserving strategies across unimodal, multimodal, hybrid,

and composition functions.

1.3 Algorithm Overview - DP-LLDE

DP-LLDE extends LBLDE (Qiao et al., 2022) with three core contributions:

1. **Sequential Level Assignment** — each individual is assigned a learning source level drawn randomly from {2, …, NL} on the first generation, decaying toward the elite level (level 1) in subsequent generations.
2. **Diversity-Aware Exemplar Selection** — if two exemplars e1, e2 fall below a diversity threshold, one is replaced with the most distant candidate in that level, preventing exemplar collapse.
3. **Normalized Mutation Pressure (Eq. 10):**

$$V_i = e_1 + F_i \times (e_2 - X_i) \times \frac{NL - k}{NL}$$

The factor $(NL-k)/NL$ decays to zero as $k \to NL$, naturally reducing mutation pressure as learning quality improves.

Parameter	Value
Population Size (NP)	100
Number of Levels (NL)	4
Initial μ_CR	0.35
Max FES	10,000 × D
Dimensions (D)	10
Seed	42

Section 2. Benchmark Function Description

2.1 Overview

The **CEC 2017 Large-Scale Global Optimization Benchmark** (Awad et al., 2017) is a standardized suite of 30 test functions designed to evaluate optimization algorithms on complex, high-dimensional search spaces. The benchmark tests an

algorithm's ability to avoid false local optima, convergence speed, and solution accuracy.

All functions are defined over the search space $[-100, 100]^D$ with their respective global optima offset by a function-specific bias value.

> **Library used:** `cec2017` by Duncan Tilley — provides the official shift vectors, rotation matrices, and shuffle permutations from the CEC 2017 specification.

2.2 Unimodal Function Overview

Unimodal functions contain a **single global optimum and no local optima**. They are used to test the convergence speed and accuracy of an algorithm in a straightforward landscape.

No.	Function Name	Global Optimum
F1	Shifted and Rotated Bent Cigar	100
F2	Shifted and Rotated Sum of Different Power *(Deprecated)*	200
F3	Shifted and Rotated Zakharov	300

> **Note:** F2 is officially deprecated from the CEC 2017 benchmark but is included here for completeness as it remains implemented in the `cec2017` library.

2.3 Multimodal Function Overview

Multimodal functions contain **multiple local optima**, making them significantly harder to solve. They are used to evaluate an algorithm's capacity to avoid premature convergence and escape false local optima.

No.	Function Name	Global Optimum
F4	Shifted and Rotated Rosenbrock's	400
F5	Shifted and Rotated Rastrigin's	500
F6	Shifted and Rotated Expanded Schaffer's F6	600
F7	Shifted and Rotated Lunacek Bi-Rastrigin	700
F8	Shifted and Rotated Non-Continuous Rastrigin's	800

No.	Function Name	Global Optimum
F9	Shifted and Rotated Levy	900
F10	Shifted and Rotated Schwefel's	1000

2.4 Hybrid Function Overview

Hybrid functions divide the search space into several regions, each governed by a **different basic function**. They assess an algorithm's adaptability to heterogeneous landscapes and mixed search behaviors.

No.	Component Functions	Global Optimum
F11	Zakharov; Rosenbrock; Rastrigin	1100
F12	Elliptic; Schwefel; Bent Cigar	1200
F13	Bent Cigar; Rosenbrock; Bi-Rastrigin	1300
F14	Elliptic; Ackley; Schaffer F7; Rastrigin	1400
F15	Bent Cigar; HGBat; Rastrigin; Rosenbrock	1500
F16	Schaffer F6; HGBat; Rosenbrock; Schwefel	1600
F17	Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin	1700
F18	Elliptic; Ackley; Rastrigin; HGBat; Discus	1800
F19	Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstrass; Schaffer F6	1900
F20	HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer F7	2000

2.5 Composite Function Overview

Composition functions **combine multiple shifted and rotated basic functions** into a single search space, creating highly complex and deceptive optimization landscapes. They are the most challenging category in the CEC 2017 suite and significantly test an algorithm's overall robustness and global search capability.

No.	Component Functions	Global Optimum
F21	Rosenbrock; Elliptic; Rastrigin	2100
F22	Rastrigin; Griewank; Schwefel	2200
F23	Rosenbrock; Ackley; Schwefel; Rastrigin	2300

No.	Component Functions	Global Optimum
F24	Ackley; Elliptic; Griewank; Rastrigin	2400
F25	Rastrigin; HappyCat; Ackley; Discus; Rosenbrock	2500
F26	Schaffer F6; Schwefel; Griewank; Rosenbrock; Rastrigin	2600
F27	HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic; Schaffer F6	2700
F28	Ackley; Griewank; Discus; Rosenbrock; HappyCat; Schaffer F6	2800
F29	F15; F16; F17	2900
F30	F15; F18; F19	3000

Section 3. List of Required Libraries

3.1 List of Libraries and Versions

The following libraries and modules were used for optimization, benchmarking, and visualization in this project:

Library / Module	Purpose	Version
`python`	Core programming language	3.10+
`numpy`	Numerical computing and array operations	2.0.2
`matplotlib`	Plotting and saving convergence graphs	3.10.0
`cec2017`	CEC 2017 benchmark functions (simple, hybrid, composition)	Latest (pip)
`cec2017.simple`	Simple benchmark functions F1–F10	—
`cec2017.hybrid`	Hybrid benchmark functions F11–F20	—
`cec2017.composition`	Composition benchmark functions F21–F30	—
`dpllde`	DP-LLDE optimization algorithm (uploaded locally)	Custom
`google.colab.files`	File upload utility for loading `dpllde.py` into the Colab environment	Built-in
`sys`	Clears cached module before re-importing `dpllde.py`	Built-in

Section 4. Environment Setup and Validation

4.1 Installation

Install `cec2017`

In [1]:
```
!pip install cec2017 --quiet
```

```
ERROR: Could not find a version that satisfies the requirement cec2017 (from ve
rsions: none)
ERROR: No matching distribution found for cec2017
```

4.2 File Validation

Import `simple.py`, `hybrid.py`, and `composition.py` python files for benchmark functions

In [2]:
```
from cec2017 import simple, hybrid, composition
print('cec2017 loaded!')
```

```
cec2017 loaded!
```

Import `sys` control over warning messages for verification that `dpllde.py` exists

In [3]:
```
from google.colab import files
import sys

uploaded = files.upload()

if 'dpllde.py' in uploaded:
    if 'dpllde' in sys.modules:
        del sys.modules['dpllde']
    import dpllde
    from dpllde import DPLLDE
    print('dpllde.py uploaded and imported successfully!')
else:
    print('ERROR: Please upload a file named dpllde.py')
```

Browse...

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving dpllde.py to dpllde.py
dpllde.py uploaded and imported successfully!
```

Section 5. Configuration and Helper Functions

5.1 Library Imports

Import libraries `numpy`, `matplotlib`, and `cec2017` for numerical computation, visualization, and CEC 2017 benchmark functions.

In [4]:
```
import numpy as np
import matplotlib.pyplot as plt
from cec2017 import simple, hybrid, composition
```

5.2 Global Parameters

Define global parameters

In [5]:
```
D = 10 # Change this when testing different dimensions
```

In [6]:
```
bounds = np.array([[-100.0, 100.0]] * D)
MAX_FES = 10000 * D
NP, NL, MU_CR = 100, 4, 0.35
SEED = 42
```

Store all results for summary table under `all_results` dict

In [7]:
```
all_results = {}
```

5.3 Function Definitions

Define `make_wrapper` to adapt CEC 2017 functions to accept a 1D input array from DP-LLDE, returning a large penalty value if the output is invalid (nan/inf).

In [8]:
```
def make_wrapper(fn):
    """Wrap a cec2017 function (expects (1,D) input) to accept (D,) input."""
    def wrapper(x):
        val = fn(x.reshape(1, -1))
        if hasattr(val, '__len__'):
            val = float(val[0])
        else:
            val = float(val)
        return val if np.isfinite(val) else 1e18
    return wrapper
```

Define `run_and_plot` to execute a single DP-LLDE optimization run on a given benchmark function, print the results, store them for the summary table, and save the convergence graph.

In [9]:
```
def run_and_plot(fn_name, fn_eval, f_global, fn_label):
    """Run DP-LLDE on a function and plot convergence graph."""
    print('=' * 80)
    print(f'DP-LLDE on CEC 2017 {fn_name} — Single Run')
    print('=' * 80)
    print(f'Function       : {fn_label} ({D}D)')
    print(f'Bounds         : [-100, 100]')
    print(f'Global optimum : {f_global}')

    opt = DPLLDE(
        objective_func=fn_eval,
        bounds=bounds,
        NP=NP, NL=NL, mu_CR_ini=MU_CR,
        max_fes=MAX_FES,
        seed=SEED
    )
    best_sol, best_fit, history = opt.optimize(verbose=True)

    error = abs(best_fit - f_global)
    print(f'\n{"="*80}')
    print('Results:')
    print(f'  Best Fitness           : {best_fit:.6e}')
    print(f'  Global Optimum         : {f_global:.6e}')
    print(f'  Error (|best-optimum|) : {error:.6e}')
    print(f'  Best Solution (5 dim)  : {best_sol[:5]}')
    print(f'  D^(0) (init. diversity): {opt.D_0:.4e}')
    print(f'{"="*80}')

    # Store for summary
    all_results[fn_name] = {
        'label': fn_label,
        'best_fit': best_fit,
        'f_global': f_global,
        'error': error
    }

    # Plot
    fes_axis = np.linspace(0, MAX_FES, len(history))
    plt.figure(figsize=(10, 6))
    plt.plot(fes_axis, history, color='steelblue', linewidth=1.5)
    plt.xlabel('Function Evaluations (FEs)', fontsize=12)
    plt.ylabel('Best Fitness', fontsize=12)
    plt.title(f'DP-LLDE Convergence — CEC 2017 {fn_name}: {fn_label} ({D}D)',
    plt.grid(True, which='both', linestyle='--', alpha=0.6)
    plt.tight_layout()
    fname = f'convergence_{fn_name}.png'
    plt.savefig(fname, dpi=150)
    plt.show()
    print(f'Graph saved as {fname}\n')
```

```
print('All wrappers and helpers defined. Ready to run functions.')
```

```
All wrappers and helpers defined. Ready to run functions.
```

Section 6. CEC2017 Benchmark Functions

6.1 Unimodal Functions (F1-F3)

In [10]:
```
# ============================================================
# F1 — Shifted and Rotated Bent Cigar
# ============================================================
run_and_plot('F1', make_wrapper(simple.f1), 100.0, 'Shifted and Rotated Bent C
```

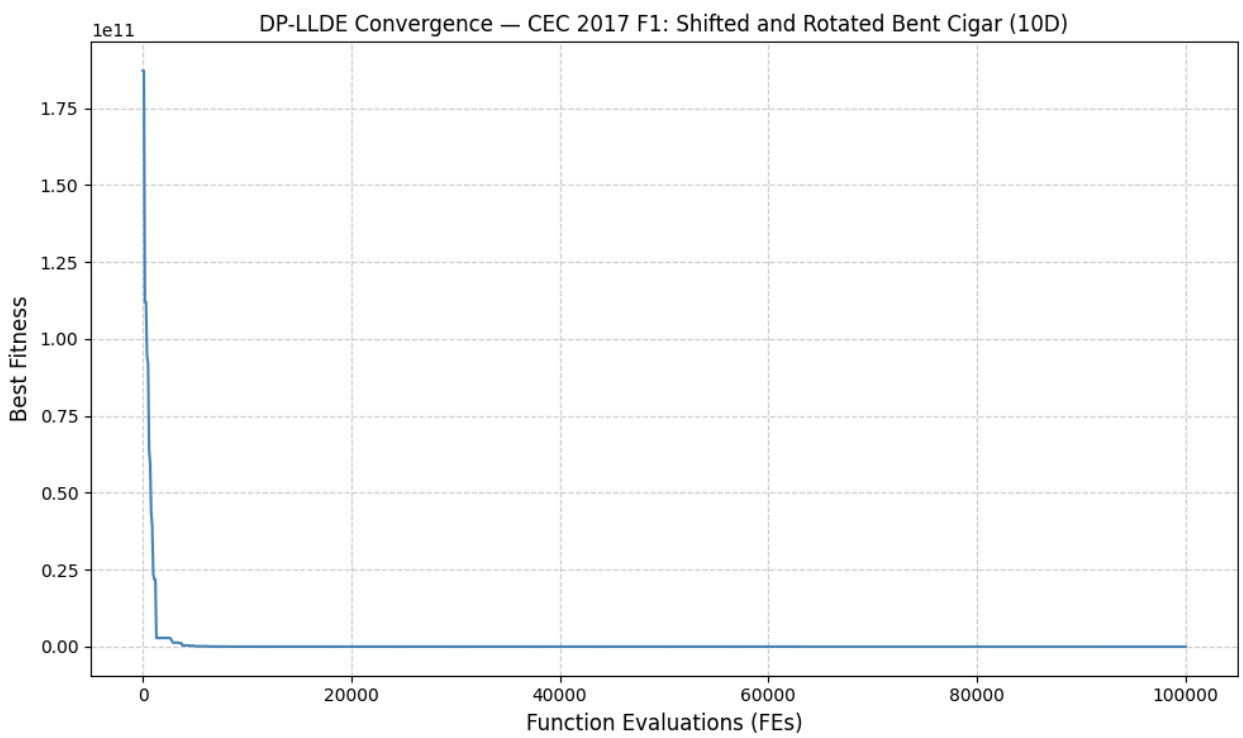
```
================================================================================
=
DP-LLDE on CEC 2017 F1 — Single Run
================================================================================
=
Function       : Shifted and Rotated Bent Cigar (10D)
Bounds         : [-100, 100]
Global optimum : 100.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 1.871674e+11
Generation   10: Best = 2.358231e+10  FES =   1100  μ_F = 0.595  μ_CR = 0.353
Generation   20: Best = 2.836411e+09  FES =   2100  μ_F = 0.648  μ_CR = 0.360
Generation   30: Best = 1.320003e+09  FES =   3100  μ_F = 0.682  μ_CR = 0.371
Generation   40: Best = 4.067197e+08  FES =   4100  μ_F = 0.694  μ_CR = 0.367
Generation   50: Best = 1.425156e+08  FES =   5100  μ_F = 0.719  μ_CR = 0.366
Generation   60: Best = 1.425156e+08  FES =   6100  μ_F = 0.707  μ_CR = 0.348
Generation   70: Best = 6.727673e+07  FES =   7100  μ_F = 0.732  μ_CR = 0.340
Generation   80: Best = 1.587221e+07  FES =   8100  μ_F = 0.743  μ_CR = 0.329
Generation   90: Best = 1.587221e+07  FES =   9100  μ_F = 0.751  μ_CR = 0.328
Generation  100: Best = 4.114547e+06  FES =  10100  μ_F = 0.780  μ_CR = 0.308
Generation  110: Best = 4.114547e+06  FES =  11100  μ_F = 0.789  μ_CR = 0.292
Generation  120: Best = 4.114547e+06  FES =  12100  μ_F = 0.795  μ_CR = 0.289
Generation  130: Best = 2.432890e+06  FES =  13100  μ_F = 0.804  μ_CR = 0.264
Generation  140: Best = 1.662757e+06  FES =  14100  μ_F = 0.822  μ_CR = 0.247
Generation  150: Best = 8.981751e+05  FES =  15100  μ_F = 0.820  μ_CR = 0.231
Generation  160: Best = 6.853996e+05  FES =  16100  μ_F = 0.825  μ_CR = 0.214
Generation  170: Best = 5.134766e+05  FES =  17100  μ_F = 0.816  μ_CR = 0.201
Generation  180: Best = 3.677691e+05  FES =  18100  μ_F = 0.833  μ_CR = 0.179
Generation  190: Best = 2.422047e+05  FES =  19100  μ_F = 0.830  μ_CR = 0.162
Generation  200: Best = 2.281525e+05  FES =  20100  μ_F = 0.822  μ_CR = 0.160
Generation  210: Best = 1.443699e+05  FES =  21100  μ_F = 0.838  μ_CR = 0.147
Generation  220: Best = 7.551887e+04  FES =  22100  μ_F = 0.833  μ_CR = 0.158
Generation  230: Best = 5.279316e+04  FES =  23100  μ_F = 0.821  μ_CR = 0.153
Generation  240: Best = 3.965924e+04  FES =  24100  μ_F = 0.825  μ_CR = 0.150
Generation  250: Best = 2.310615e+04  FES =  25100  μ_F = 0.829  μ_CR = 0.158
Generation  260: Best = 1.587307e+04  FES =  26100  μ_F = 0.831  μ_CR = 0.153
Generation  270: Best = 1.286715e+04  FES =  27100  μ_F = 0.832  μ_CR = 0.152
Generation  280: Best = 1.264177e+04  FES =  28100  μ_F = 0.841  μ_CR = 0.145
Generation  290: Best = 1.248883e+04  FES =  29100  μ_F = 0.844  μ_CR = 0.135
Generation  300: Best = 1.248883e+04  FES =  30100  μ_F = 0.842  μ_CR = 0.136
Generation  310: Best = 1.075090e+04  FES =  31100  μ_F = 0.845  μ_CR = 0.142
Generation  320: Best = 1.011591e+04  FES =  32100  μ_F = 0.839  μ_CR = 0.138
Generation  330: Best = 8.869260e+03  FES =  33100  μ_F = 0.852  μ_CR = 0.139
Generation  340: Best = 8.822684e+03  FES =  34100  μ_F = 0.860  μ_CR = 0.132
Generation  350: Best = 8.632403e+03  FES =  35100  μ_F = 0.856  μ_CR = 0.134
Generation  360: Best = 8.419257e+03  FES =  36100  μ_F = 0.847  μ_CR = 0.141
Generation  370: Best = 8.383438e+03  FES =  37100  μ_F = 0.855  μ_CR = 0.141
Generation  380: Best = 7.211943e+03  FES =  38100  μ_F = 0.838  μ_CR = 0.140
Generation  390: Best = 7.085575e+03  FES =  39100  μ_F = 0.827  μ_CR = 0.139
Generation  400: Best = 7.085575e+03  FES =  40100  μ_F = 0.836  μ_CR = 0.140
Generation  410: Best = 6.523104e+03  FES =  41100  μ_F = 0.850  μ_CR = 0.143
Generation  420: Best = 6.523104e+03  FES =  42100  μ_F = 0.847  μ_CR = 0.155
Generation  430: Best = 6.378162e+03  FES =  43100  μ_F = 0.849  μ_CR = 0.139
Generation  440: Best = 6.095120e+03  FES =  44100  μ_F = 0.849  μ_CR = 0.127
```

```
Generation  450: Best = 6.088545e+03  FES =  45100  μ_F = 0.836  μ_CR = 0.130
Generation  460: Best = 5.985528e+03  FES =  46100  μ_F = 0.850  μ_CR = 0.129
Generation  470: Best = 5.950637e+03  FES =  47100  μ_F = 0.844  μ_CR = 0.129
Generation  480: Best = 5.947824e+03  FES =  48100  μ_F = 0.849  μ_CR = 0.135
Generation  490: Best = 5.947824e+03  FES =  49100  μ_F = 0.849  μ_CR = 0.139
Generation  500: Best = 5.922445e+03  FES =  50100  μ_F = 0.846  μ_CR = 0.153
Generation  510: Best = 5.918047e+03  FES =  51100  μ_F = 0.860  μ_CR = 0.146
Generation  520: Best = 5.912230e+03  FES =  52100  μ_F = 0.849  μ_CR = 0.135
Generation  530: Best = 5.911949e+03  FES =  53100  μ_F = 0.827  μ_CR = 0.125
Generation  540: Best = 5.907537e+03  FES =  54100  μ_F = 0.844  μ_CR = 0.122
Generation  550: Best = 5.906846e+03  FES =  55100  μ_F = 0.852  μ_CR = 0.118
Generation  560: Best = 5.903454e+03  FES =  56100  μ_F = 0.852  μ_CR = 0.109
Generation  570: Best = 5.903442e+03  FES =  57100  μ_F = 0.840  μ_CR = 0.108
Generation  580: Best = 5.623351e+03  FES =  58100  μ_F = 0.799  μ_CR = 0.114
Generation  590: Best = 5.515573e+03  FES =  59100  μ_F = 0.806  μ_CR = 0.117
Generation  600: Best = 4.913624e+03  FES =  60100  μ_F = 0.810  μ_CR = 0.115
Generation  610: Best = 4.913306e+03  FES =  61100  μ_F = 0.825  μ_CR = 0.125
Generation  620: Best = 4.913306e+03  FES =  62100  μ_F = 0.848  μ_CR = 0.124
Generation  630: Best = 4.899038e+03  FES =  63100  μ_F = 0.842  μ_CR = 0.126
Generation  640: Best = 4.084883e+03  FES =  64100  μ_F = 0.846  μ_CR = 0.130
Generation  650: Best = 4.015533e+03  FES =  65100  μ_F = 0.850  μ_CR = 0.141
Generation  660: Best = 3.962495e+03  FES =  66100  μ_F = 0.873  μ_CR = 0.146
Generation  670: Best = 3.510592e+03  FES =  67100  μ_F = 0.856  μ_CR = 0.139
Generation  680: Best = 2.596341e+03  FES =  68100  μ_F = 0.862  μ_CR = 0.135
Generation  690: Best = 2.394881e+03  FES =  69100  μ_F = 0.849  μ_CR = 0.132
Generation  700: Best = 1.551455e+03  FES =  70100  μ_F = 0.854  μ_CR = 0.138
Generation  710: Best = 1.085011e+03  FES =  71100  μ_F = 0.865  μ_CR = 0.134
Generation  720: Best = 3.282809e+02  FES =  72100  μ_F = 0.858  μ_CR = 0.135
Generation  730: Best = 2.394696e+02  FES =  73100  μ_F = 0.852  μ_CR = 0.133
Generation  740: Best = 2.394687e+02  FES =  74100  μ_F = 0.860  μ_CR = 0.128
Generation  750: Best = 2.394686e+02  FES =  75100  μ_F = 0.840  μ_CR = 0.129
Generation  760: Best = 2.394686e+02  FES =  76100  μ_F = 0.854  μ_CR = 0.137
Generation  770: Best = 2.394686e+02  FES =  77100  μ_F = 0.856  μ_CR = 0.121
Generation  780: Best = 2.394644e+02  FES =  78100  μ_F = 0.859  μ_CR = 0.116
Generation  790: Best = 2.394644e+02  FES =  79100  μ_F = 0.852  μ_CR = 0.115
Generation  800: Best = 2.394644e+02  FES =  80100  μ_F = 0.874  μ_CR = 0.125
Generation  810: Best = 2.394644e+02  FES =  81100  μ_F = 0.863  μ_CR = 0.119
Generation  820: Best = 2.394643e+02  FES =  82100  μ_F = 0.844  μ_CR = 0.118
Generation  830: Best = 1.855644e+02  FES =  83100  μ_F = 0.837  μ_CR = 0.113
Generation  840: Best = 1.855634e+02  FES =  84100  μ_F = 0.833  μ_CR = 0.124
Generation  850: Best = 1.202298e+02  FES =  85100  μ_F = 0.846  μ_CR = 0.127
Generation  860: Best = 1.202298e+02  FES =  86100  μ_F = 0.856  μ_CR = 0.126
Generation  870: Best = 1.202297e+02  FES =  87100  μ_F = 0.857  μ_CR = 0.122
Generation  880: Best = 1.202255e+02  FES =  88100  μ_F = 0.854  μ_CR = 0.137
Generation  890: Best = 1.202255e+02  FES =  89100  μ_F = 0.854  μ_CR = 0.134
Generation  900: Best = 1.202254e+02  FES =  90100  μ_F = 0.859  μ_CR = 0.145
Generation  910: Best = 1.202254e+02  FES =  91100  μ_F = 0.861  μ_CR = 0.139
Generation  920: Best = 1.202230e+02  FES =  92100  μ_F = 0.846  μ_CR = 0.136
Generation  930: Best = 1.202226e+02  FES =  93100  μ_F = 0.823  μ_CR = 0.124
Generation  940: Best = 1.202226e+02  FES =  94100  μ_F = 0.840  μ_CR = 0.116
Generation  950: Best = 1.137058e+02  FES =  95100  μ_F = 0.850  μ_CR = 0.118
Generation  960: Best = 1.137057e+02  FES =  96100  μ_F = 0.860  μ_CR = 0.125
Generation  970: Best = 1.137057e+02  FES =  97100  μ_F = 0.866  μ_CR = 0.123
Generation  980: Best = 1.137057e+02  FES =  98100  μ_F = 0.861  μ_CR = 0.127
```

```
Generation  990: Best = 1.137057e+02  FES =  99100  μ_F = 0.861  μ_CR = 0.129

Optimization complete.
Best Fitness             = 1.137057e+02
Total Function Evals     = 100000
Total Successful Updates = 16003

================================================================================
=
Results:
  Best Fitness           : 1.137057e+02
  Global Optimum         : 1.000000e+02
  Error (|best-optimum|) : 1.370568e+01
  Best Solution (5 dim)  : [-55.26773493 -70.42956005 -29.61018237 -58.32676383
22.08960164]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F1.png
```

In [11]:
```
# ============================================================
# F2 — Shifted and Rotated Sum of Different Power
# ============================================================
run_and_plot('F2', make_wrapper(simple.f2), 200.0, 'Shifted and Rotated Sum of
```

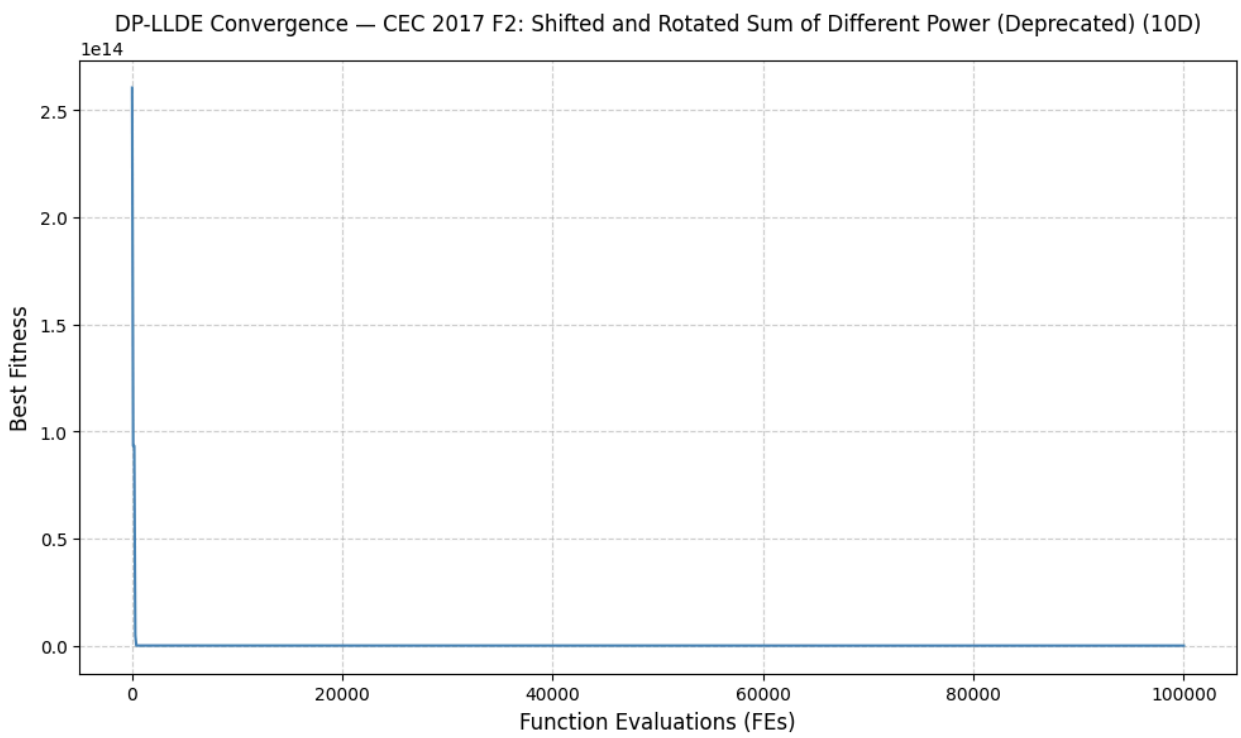
```
================================================================================
=
DP-LLDE on CEC 2017 F2 — Single Run
================================================================================
=
Function       : Shifted and Rotated Sum of Different Power (Deprecated) (10D)
Bounds         : [-100, 100]
Global optimum : 200.0
WARNING: f2 has been deprecated from the CEC 2017 benchmark suite
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 2.603565e+14
Generation   10: Best = 8.009984e+09  FES =   1100  μ_F = 0.575  μ_CR = 0.349
Generation   20: Best = 1.835752e+08  FES =   2100  μ_F = 0.636  μ_CR = 0.366
Generation   30: Best = 1.305099e+07  FES =   3100  μ_F = 0.656  μ_CR = 0.363
Generation   40: Best = 6.032078e+06  FES =   4100  μ_F = 0.706  μ_CR = 0.351
Generation   50: Best = 1.392279e+06  FES =   5100  μ_F = 0.725  μ_CR = 0.359
Generation   60: Best = 6.604761e+05  FES =   6100  μ_F = 0.744  μ_CR = 0.345
Generation   70: Best = 1.323279e+05  FES =   7100  μ_F = 0.753  μ_CR = 0.349
Generation   80: Best = 8.908620e+04  FES =   8100  μ_F = 0.779  μ_CR = 0.353
Generation   90: Best = 8.908620e+04  FES =   9100  μ_F = 0.792  μ_CR = 0.338
Generation  100: Best = 3.800582e+04  FES =  10100  μ_F = 0.830  μ_CR = 0.309
Generation  110: Best = 3.800582e+04  FES =  11100  μ_F = 0.825  μ_CR = 0.270
Generation  120: Best = 1.912656e+04  FES =  12100  μ_F = 0.830  μ_CR = 0.262
Generation  130: Best = 1.691174e+04  FES =  13100  μ_F = 0.827  μ_CR = 0.236
Generation  140: Best = 1.691174e+04  FES =  14100  μ_F = 0.798  μ_CR = 0.222
Generation  150: Best = 1.272211e+04  FES =  15100  μ_F = 0.737  μ_CR = 0.208
Generation  160: Best = 5.668771e+03  FES =  16100  μ_F = 0.701  μ_CR = 0.200
Generation  170: Best = 3.600888e+03  FES =  17100  μ_F = 0.698  μ_CR = 0.181
Generation  180: Best = 3.600888e+03  FES =  18100  μ_F = 0.748  μ_CR = 0.150
Generation  190: Best = 1.855716e+03  FES =  19100  μ_F = 0.795  μ_CR = 0.145
Generation  200: Best = 1.855716e+03  FES =  20100  μ_F = 0.749  μ_CR = 0.138
Generation  210: Best = 1.855716e+03  FES =  21100  μ_F = 0.801  μ_CR = 0.126
Generation  220: Best = 1.854917e+03  FES =  22100  μ_F = 0.759  μ_CR = 0.135
Generation  230: Best = 1.854790e+03  FES =  23100  μ_F = 0.791  μ_CR = 0.144
Generation  240: Best = 1.854457e+03  FES =  24100  μ_F = 0.791  μ_CR = 0.159
Generation  250: Best = 1.474120e+03  FES =  25100  μ_F = 0.790  μ_CR = 0.155
Generation  260: Best = 1.474120e+03  FES =  26100  μ_F = 0.771  μ_CR = 0.145
Generation  270: Best = 1.474120e+03  FES =  27100  μ_F = 0.781  μ_CR = 0.137
Generation  280: Best = 4.296051e+02  FES =  28100  μ_F = 0.804  μ_CR = 0.142
Generation  290: Best = 4.296051e+02  FES =  29100  μ_F = 0.836  μ_CR = 0.138
Generation  300: Best = 4.219487e+02  FES =  30100  μ_F = 0.782  μ_CR = 0.132
Generation  310: Best = 4.209684e+02  FES =  31100  μ_F = 0.772  μ_CR = 0.127
Generation  320: Best = 3.485209e+02  FES =  32100  μ_F = 0.781  μ_CR = 0.135
Generation  330: Best = 3.345932e+02  FES =  33100  μ_F = 0.791  μ_CR = 0.136
Generation  340: Best = 2.503644e+02  FES =  34100  μ_F = 0.772  μ_CR = 0.124
Generation  350: Best = 2.503644e+02  FES =  35100  μ_F = 0.766  μ_CR = 0.126
Generation  360: Best = 2.332108e+02  FES =  36100  μ_F = 0.759  μ_CR = 0.125
Generation  370: Best = 2.107630e+02  FES =  37100  μ_F = 0.761  μ_CR = 0.128
Generation  380: Best = 2.107614e+02  FES =  38100  μ_F = 0.791  μ_CR = 0.120
Generation  390: Best = 2.107614e+02  FES =  39100  μ_F = 0.843  μ_CR = 0.119
Generation  400: Best = 2.107614e+02  FES =  40100  μ_F = 0.835  μ_CR = 0.109
Generation  410: Best = 2.103034e+02  FES =  41100  μ_F = 0.836  μ_CR = 0.111
Generation  420: Best = 2.103034e+02  FES =  42100  μ_F = 0.814  μ_CR = 0.132
Generation  430: Best = 2.085952e+02  FES =  43100  μ_F = 0.792  μ_CR = 0.114
```

```
Generation  440: Best = 2.020615e+02  FES =  44100  μ_F = 0.785  μ_CR = 0.108
Generation  450: Best = 2.020615e+02  FES =  45100  μ_F = 0.788  μ_CR = 0.112
Generation  460: Best = 2.020615e+02  FES =  46100  μ_F = 0.806  μ_CR = 0.115
Generation  470: Best = 2.020615e+02  FES =  47100  μ_F = 0.823  μ_CR = 0.107
Generation  480: Best = 2.019675e+02  FES =  48100  μ_F = 0.832  μ_CR = 0.109
Generation  490: Best = 2.019675e+02  FES =  49100  μ_F = 0.822  μ_CR = 0.109
Generation  500: Best = 2.019675e+02  FES =  50100  μ_F = 0.815  μ_CR = 0.118
Generation  510: Best = 2.016886e+02  FES =  51100  μ_F = 0.807  μ_CR = 0.122
Generation  520: Best = 2.016764e+02  FES =  52100  μ_F = 0.808  μ_CR = 0.105
Generation  530: Best = 2.016764e+02  FES =  53100  μ_F = 0.786  μ_CR = 0.098
Generation  540: Best = 2.016764e+02  FES =  54100  μ_F = 0.789  μ_CR = 0.097
Generation  550: Best = 2.016163e+02  FES =  55100  μ_F = 0.783  μ_CR = 0.103
Generation  560: Best = 2.016162e+02  FES =  56100  μ_F = 0.795  μ_CR = 0.107
Generation  570: Best = 2.016127e+02  FES =  57100  μ_F = 0.790  μ_CR = 0.108
Generation  580: Best = 2.015994e+02  FES =  58100  μ_F = 0.802  μ_CR = 0.133
Generation  590: Best = 2.015703e+02  FES =  59100  μ_F = 0.775  μ_CR = 0.128
Generation  600: Best = 2.015703e+02  FES =  60100  μ_F = 0.785  μ_CR = 0.128
Generation  610: Best = 2.003509e+02  FES =  61100  μ_F = 0.810  μ_CR = 0.119
Generation  620: Best = 2.001377e+02  FES =  62100  μ_F = 0.819  μ_CR = 0.114
Generation  630: Best = 2.001377e+02  FES =  63100  μ_F = 0.803  μ_CR = 0.114
Generation  640: Best = 2.001077e+02  FES =  64100  μ_F = 0.804  μ_CR = 0.115
Generation  650: Best = 2.001058e+02  FES =  65100  μ_F = 0.817  μ_CR = 0.132
Generation  660: Best = 2.001036e+02  FES =  66100  μ_F = 0.808  μ_CR = 0.138
Generation  670: Best = 2.000510e+02  FES =  67100  μ_F = 0.808  μ_CR = 0.131
Generation  680: Best = 2.000070e+02  FES =  68100  μ_F = 0.826  μ_CR = 0.137
Generation  690: Best = 2.000070e+02  FES =  69100  μ_F = 0.835  μ_CR = 0.146
Generation  700: Best = 2.000033e+02  FES =  70100  μ_F = 0.841  μ_CR = 0.132
Generation  710: Best = 2.000016e+02  FES =  71100  μ_F = 0.853  μ_CR = 0.146
Generation  720: Best = 2.000004e+02  FES =  72100  μ_F = 0.856  μ_CR = 0.147
Generation  730: Best = 2.000004e+02  FES =  73100  μ_F = 0.840  μ_CR = 0.142
Generation  740: Best = 2.000004e+02  FES =  74100  μ_F = 0.859  μ_CR = 0.141
Generation  750: Best = 2.000004e+02  FES =  75100  μ_F = 0.844  μ_CR = 0.133
Generation  760: Best = 2.000004e+02  FES =  76100  μ_F = 0.857  μ_CR = 0.131
Generation  770: Best = 2.000004e+02  FES =  77100  μ_F = 0.865  μ_CR = 0.131
Generation  780: Best = 2.000004e+02  FES =  78100  μ_F = 0.859  μ_CR = 0.132
Generation  790: Best = 2.000004e+02  FES =  79100  μ_F = 0.839  μ_CR = 0.145
Generation  800: Best = 2.000002e+02  FES =  80100  μ_F = 0.837  μ_CR = 0.139
Generation  810: Best = 2.000002e+02  FES =  81100  μ_F = 0.831  μ_CR = 0.147
Generation  820: Best = 2.000001e+02  FES =  82100  μ_F = 0.815  μ_CR = 0.138
Generation  830: Best = 2.000001e+02  FES =  83100  μ_F = 0.812  μ_CR = 0.143
Generation  840: Best = 2.000001e+02  FES =  84100  μ_F = 0.812  μ_CR = 0.134
Generation  850: Best = 2.000001e+02  FES =  85100  μ_F = 0.818  μ_CR = 0.129
Generation  860: Best = 2.000001e+02  FES =  86100  μ_F = 0.813  μ_CR = 0.135
Generation  870: Best = 2.000001e+02  FES =  87100  μ_F = 0.819  μ_CR = 0.135
Generation  880: Best = 2.000001e+02  FES =  88100  μ_F = 0.839  μ_CR = 0.136
Generation  890: Best = 2.000001e+02  FES =  89100  μ_F = 0.847  μ_CR = 0.161
Generation  900: Best = 2.000001e+02  FES =  90100  μ_F = 0.838  μ_CR = 0.172
Generation  910: Best = 2.000001e+02  FES =  91100  μ_F = 0.837  μ_CR = 0.174
Generation  920: Best = 2.000001e+02  FES =  92100  μ_F = 0.859  μ_CR = 0.179
Generation  930: Best = 2.000001e+02  FES =  93100  μ_F = 0.809  μ_CR = 0.176
Generation  940: Best = 2.000000e+02  FES =  94100  μ_F = 0.812  μ_CR = 0.163
Generation  950: Best = 2.000000e+02  FES =  95100  μ_F = 0.831  μ_CR = 0.155
Generation  960: Best = 2.000000e+02  FES =  96100  μ_F = 0.833  μ_CR = 0.159
Generation  970: Best = 2.000000e+02  FES =  97100  μ_F = 0.858  μ_CR = 0.156
```

```
Generation  980: Best = 2.000000e+02  FES =  98100  μ_F = 0.872  μ_CR = 0.145
Generation  990: Best = 2.000000e+02  FES =  99100  μ_F = 0.875  μ_CR = 0.149

Optimization complete.
Best Fitness             = 2.000000e+02
Total Function Evals     = 100000
Total Successful Updates = 8429

================================================================================
=
Results:
  Best Fitness           : 2.000000e+02
  Global Optimum         : 2.000000e+02
  Error (|best-optimum|) : 3.642452e-05
  Best Solution (5 dim)  : [-29.34308373 -17.05070158 -75.4906634   -0.69524913
4.26862114]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F2.png
```

In [12]:
```
# ============================================================
# F3 — Shifted and Rotated Zakharov
# ============================================================
run_and_plot('F3', make_wrapper(simple.f3), 300.0, 'Shifted and Rotated Zakhar
```

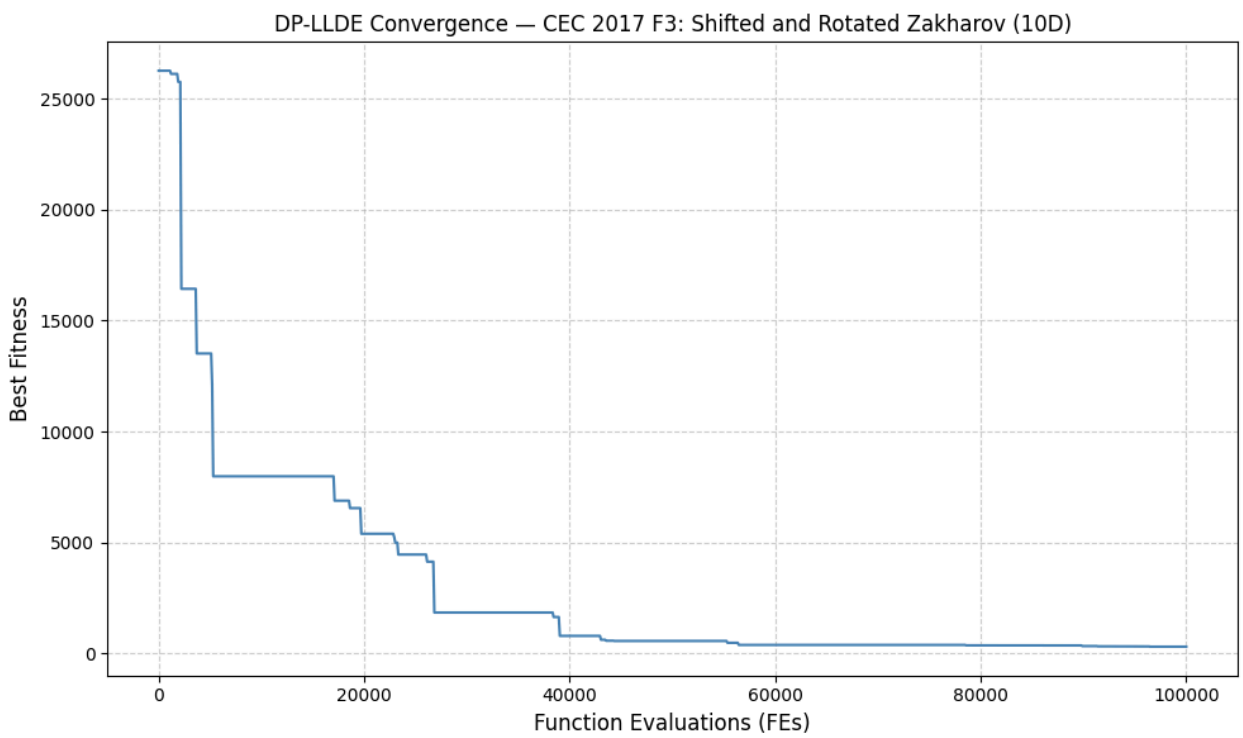
```
================================================================================
=
DP-LLDE on CEC 2017 F3 — Single Run
================================================================================
=
Function       : Shifted and Rotated Zakharov (10D)
Bounds         : [-100, 100]
Global optimum : 300.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 2.625937e+04
Generation   10: Best = 2.625937e+04  FES =   1100  μ_F = 0.565  μ_CR = 0.348
Generation   20: Best = 2.575657e+04  FES =   2100  μ_F = 0.662  μ_CR = 0.358
Generation   30: Best = 1.643475e+04  FES =   3100  μ_F = 0.740  μ_CR = 0.329
Generation   40: Best = 1.352076e+04  FES =   4100  μ_F = 0.770  μ_CR = 0.316
Generation   50: Best = 1.352076e+04  FES =   5100  μ_F = 0.828  μ_CR = 0.326
Generation   60: Best = 7.987234e+03  FES =   6100  μ_F = 0.824  μ_CR = 0.282
Generation   70: Best = 7.987234e+03  FES =   7100  μ_F = 0.812  μ_CR = 0.288
Generation   80: Best = 7.987234e+03  FES =   8100  μ_F = 0.813  μ_CR = 0.303
Generation   90: Best = 7.987234e+03  FES =   9100  μ_F = 0.818  μ_CR = 0.319
Generation  100: Best = 7.987234e+03  FES =  10100  μ_F = 0.815  μ_CR = 0.341
Generation  110: Best = 7.987234e+03  FES =  11100  μ_F = 0.830  μ_CR = 0.364
Generation  120: Best = 7.987234e+03  FES =  12100  μ_F = 0.827  μ_CR = 0.358
Generation  130: Best = 7.987234e+03  FES =  13100  μ_F = 0.760  μ_CR = 0.360
Generation  140: Best = 7.987234e+03  FES =  14100  μ_F = 0.734  μ_CR = 0.329
Generation  150: Best = 7.987234e+03  FES =  15100  μ_F = 0.712  μ_CR = 0.307
Generation  160: Best = 7.987234e+03  FES =  16100  μ_F = 0.660  μ_CR = 0.325
Generation  170: Best = 7.987234e+03  FES =  17100  μ_F = 0.590  μ_CR = 0.313
Generation  180: Best = 6.888868e+03  FES =  18100  μ_F = 0.508  μ_CR = 0.335
Generation  190: Best = 6.554982e+03  FES =  19100  μ_F = 0.507  μ_CR = 0.329
Generation  200: Best = 5.398807e+03  FES =  20100  μ_F = 0.485  μ_CR = 0.343
Generation  210: Best = 5.398807e+03  FES =  21100  μ_F = 0.481  μ_CR = 0.391
Generation  220: Best = 5.398807e+03  FES =  22100  μ_F = 0.512  μ_CR = 0.396
Generation  230: Best = 5.000794e+03  FES =  23100  μ_F = 0.483  μ_CR = 0.405
Generation  240: Best = 4.462200e+03  FES =  24100  μ_F = 0.616  μ_CR = 0.358
Generation  250: Best = 4.462200e+03  FES =  25100  μ_F = 0.610  μ_CR = 0.353
Generation  260: Best = 4.462200e+03  FES =  26100  μ_F = 0.650  μ_CR = 0.338
Generation  270: Best = 1.850178e+03  FES =  27100  μ_F = 0.638  μ_CR = 0.307
Generation  280: Best = 1.850178e+03  FES =  28100  μ_F = 0.632  μ_CR = 0.258
Generation  290: Best = 1.850178e+03  FES =  29100  μ_F = 0.720  μ_CR = 0.281
Generation  300: Best = 1.850178e+03  FES =  30100  μ_F = 0.687  μ_CR = 0.239
Generation  310: Best = 1.850178e+03  FES =  31100  μ_F = 0.694  μ_CR = 0.258
Generation  320: Best = 1.850178e+03  FES =  32100  μ_F = 0.684  μ_CR = 0.249
Generation  330: Best = 1.850178e+03  FES =  33100  μ_F = 0.645  μ_CR = 0.257
Generation  340: Best = 1.850178e+03  FES =  34100  μ_F = 0.666  μ_CR = 0.210
Generation  350: Best = 1.850178e+03  FES =  35100  μ_F = 0.664  μ_CR = 0.202
Generation  360: Best = 1.850178e+03  FES =  36100  μ_F = 0.711  μ_CR = 0.190
Generation  370: Best = 1.850178e+03  FES =  37100  μ_F = 0.694  μ_CR = 0.192
Generation  380: Best = 1.850178e+03  FES =  38100  μ_F = 0.678  μ_CR = 0.188
Generation  390: Best = 7.985681e+02  FES =  39100  μ_F = 0.663  μ_CR = 0.159
Generation  400: Best = 7.985681e+02  FES =  40100  μ_F = 0.700  μ_CR = 0.164
Generation  410: Best = 7.985681e+02  FES =  41100  μ_F = 0.677  μ_CR = 0.159
Generation  420: Best = 7.985681e+02  FES =  42100  μ_F = 0.646  μ_CR = 0.148
Generation  430: Best = 6.243696e+02  FES =  43100  μ_F = 0.615  μ_CR = 0.147
Generation  440: Best = 5.806641e+02  FES =  44100  μ_F = 0.650  μ_CR = 0.159
```

```
Generation  450: Best = 5.697207e+02  FES =  45100  μ_F = 0.678  μ_CR = 0.134
Generation  460: Best = 5.697207e+02  FES =  46100  μ_F = 0.638  μ_CR = 0.131
Generation  470: Best = 5.697207e+02  FES =  47100  μ_F = 0.629  μ_CR = 0.115
Generation  480: Best = 5.697207e+02  FES =  48100  μ_F = 0.605  μ_CR = 0.123
Generation  490: Best = 5.697207e+02  FES =  49100  μ_F = 0.644  μ_CR = 0.127
Generation  500: Best = 5.697207e+02  FES =  50100  μ_F = 0.684  μ_CR = 0.139
Generation  510: Best = 5.697207e+02  FES =  51100  μ_F = 0.736  μ_CR = 0.160
Generation  520: Best = 5.697207e+02  FES =  52100  μ_F = 0.785  μ_CR = 0.147
Generation  530: Best = 5.697207e+02  FES =  53100  μ_F = 0.790  μ_CR = 0.151
Generation  540: Best = 5.697207e+02  FES =  54100  μ_F = 0.816  μ_CR = 0.140
Generation  550: Best = 5.697207e+02  FES =  55100  μ_F = 0.795  μ_CR = 0.147
Generation  560: Best = 4.862010e+02  FES =  56100  μ_F = 0.817  μ_CR = 0.177
Generation  570: Best = 3.907888e+02  FES =  57100  μ_F = 0.704  μ_CR = 0.178
Generation  580: Best = 3.907888e+02  FES =  58100  μ_F = 0.746  μ_CR = 0.175
Generation  590: Best = 3.907888e+02  FES =  59100  μ_F = 0.782  μ_CR = 0.148
Generation  600: Best = 3.907888e+02  FES =  60100  μ_F = 0.739  μ_CR = 0.120
Generation  610: Best = 3.907888e+02  FES =  61100  μ_F = 0.733  μ_CR = 0.101
Generation  620: Best = 3.907888e+02  FES =  62100  μ_F = 0.780  μ_CR = 0.121
Generation  630: Best = 3.907888e+02  FES =  63100  μ_F = 0.807  μ_CR = 0.146
Generation  640: Best = 3.907888e+02  FES =  64100  μ_F = 0.789  μ_CR = 0.123
Generation  650: Best = 3.907888e+02  FES =  65100  μ_F = 0.795  μ_CR = 0.136
Generation  660: Best = 3.907888e+02  FES =  66100  μ_F = 0.752  μ_CR = 0.156
Generation  670: Best = 3.907888e+02  FES =  67100  μ_F = 0.766  μ_CR = 0.138
Generation  680: Best = 3.907888e+02  FES =  68100  μ_F = 0.758  μ_CR = 0.151
Generation  690: Best = 3.907888e+02  FES =  69100  μ_F = 0.688  μ_CR = 0.123
Generation  700: Best = 3.907888e+02  FES =  70100  μ_F = 0.716  μ_CR = 0.128
Generation  710: Best = 3.907888e+02  FES =  71100  μ_F = 0.757  μ_CR = 0.120
Generation  720: Best = 3.907888e+02  FES =  72100  μ_F = 0.779  μ_CR = 0.133
Generation  730: Best = 3.907888e+02  FES =  73100  μ_F = 0.745  μ_CR = 0.145
Generation  740: Best = 3.907888e+02  FES =  74100  μ_F = 0.727  μ_CR = 0.149
Generation  750: Best = 3.907888e+02  FES =  75100  μ_F = 0.767  μ_CR = 0.144
Generation  760: Best = 3.907888e+02  FES =  76100  μ_F = 0.841  μ_CR = 0.113
Generation  770: Best = 3.907888e+02  FES =  77100  μ_F = 0.845  μ_CR = 0.116
Generation  780: Best = 3.907888e+02  FES =  78100  μ_F = 0.857  μ_CR = 0.103
Generation  790: Best = 3.724688e+02  FES =  79100  μ_F = 0.861  μ_CR = 0.133
Generation  800: Best = 3.724688e+02  FES =  80100  μ_F = 0.767  μ_CR = 0.128
Generation  810: Best = 3.698802e+02  FES =  81100  μ_F = 0.815  μ_CR = 0.130
Generation  820: Best = 3.698802e+02  FES =  82100  μ_F = 0.811  μ_CR = 0.114
Generation  830: Best = 3.698802e+02  FES =  83100  μ_F = 0.765  μ_CR = 0.100
Generation  840: Best = 3.695352e+02  FES =  84100  μ_F = 0.698  μ_CR = 0.112
Generation  850: Best = 3.695352e+02  FES =  85100  μ_F = 0.701  μ_CR = 0.113
Generation  860: Best = 3.695352e+02  FES =  86100  μ_F = 0.695  μ_CR = 0.115
Generation  870: Best = 3.695352e+02  FES =  87100  μ_F = 0.749  μ_CR = 0.115
Generation  880: Best = 3.695352e+02  FES =  88100  μ_F = 0.717  μ_CR = 0.114
Generation  890: Best = 3.695352e+02  FES =  89100  μ_F = 0.736  μ_CR = 0.116
Generation  900: Best = 3.398887e+02  FES =  90100  μ_F = 0.771  μ_CR = 0.123
Generation  910: Best = 3.398887e+02  FES =  91100  μ_F = 0.817  μ_CR = 0.138
Generation  920: Best = 3.284945e+02  FES =  92100  μ_F = 0.775  μ_CR = 0.141
Generation  930: Best = 3.284945e+02  FES =  93100  μ_F = 0.728  μ_CR = 0.146
Generation  940: Best = 3.284945e+02  FES =  94100  μ_F = 0.781  μ_CR = 0.111
Generation  950: Best = 3.272788e+02  FES =  95100  μ_F = 0.780  μ_CR = 0.131
Generation  960: Best = 3.263567e+02  FES =  96100  μ_F = 0.786  μ_CR = 0.113
Generation  970: Best = 3.118899e+02  FES =  97100  μ_F = 0.773  μ_CR = 0.104
Generation  980: Best = 3.118899e+02  FES =  98100  μ_F = 0.748  μ_CR = 0.104
```

```
Generation  990: Best = 3.118899e+02  FES =  99100  μ_F = 0.758  μ_CR = 0.108

Optimization complete.
Best Fitness             = 3.118899e+02
Total Function Evals     = 100000
Total Successful Updates = 2581

================================================================================
=
Results:
  Best Fitness           : 3.118899e+02
  Global Optimum         : 3.000000e+02
  Error (|best-optimum|) : 1.188993e+01
  Best Solution (5 dim)  : [-56.91480489   5.94569397  35.48248189   8.34810005
-46.20553692]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F3.png
```

6.2 Multimodal Functions (F4-F10)

In [13]:
```
# ============================================================
# F4 — Shifted and Rotated Rosenbrock
# ============================================================
run_and_plot('F4', make_wrapper(simple.f4), 400.0, "Shifted and Rotated Rosenb
```

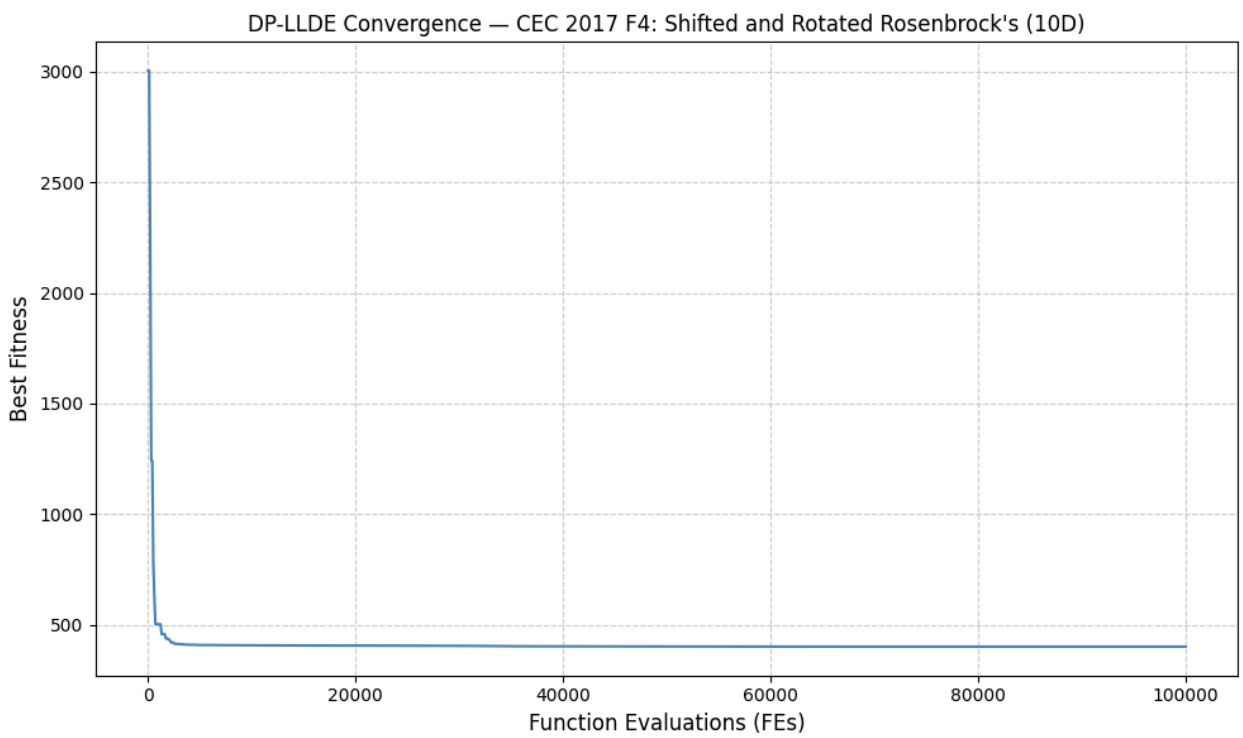
```
================================================================================
=
DP-LLDE on CEC 2017 F4 — Single Run
================================================================================
=
Function       : Shifted and Rotated Rosenbrock's (10D)
Bounds         : [-100, 100]
Global optimum : 400.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 3.004346e+03
Generation   10: Best = 5.016087e+02  FES =   1100  μ_F = 0.591  μ_CR = 0.352
Generation   20: Best = 4.334252e+02  FES =   2100  μ_F = 0.632  μ_CR = 0.358
Generation   30: Best = 4.122972e+02  FES =   3100  μ_F = 0.680  μ_CR = 0.366
Generation   40: Best = 4.090799e+02  FES =   4100  μ_F = 0.707  μ_CR = 0.362
Generation   50: Best = 4.077873e+02  FES =   5100  μ_F = 0.737  μ_CR = 0.361
Generation   60: Best = 4.073258e+02  FES =   6100  μ_F = 0.754  μ_CR = 0.366
Generation   70: Best = 4.070575e+02  FES =   7100  μ_F = 0.766  μ_CR = 0.387
Generation   80: Best = 4.067181e+02  FES =   8100  μ_F = 0.786  μ_CR = 0.385
Generation   90: Best = 4.063728e+02  FES =   9100  μ_F = 0.806  μ_CR = 0.410
Generation  100: Best = 4.062912e+02  FES =  10100  μ_F = 0.762  μ_CR = 0.423
Generation  110: Best = 4.060093e+02  FES =  11100  μ_F = 0.762  μ_CR = 0.465
Generation  120: Best = 4.059361e+02  FES =  12100  μ_F = 0.783  μ_CR = 0.477
Generation  130: Best = 4.058132e+02  FES =  13100  μ_F = 0.773  μ_CR = 0.440
Generation  140: Best = 4.057079e+02  FES =  14100  μ_F = 0.788  μ_CR = 0.436
Generation  150: Best = 4.054588e+02  FES =  15100  μ_F = 0.797  μ_CR = 0.417
Generation  160: Best = 4.052492e+02  FES =  16100  μ_F = 0.801  μ_CR = 0.448
Generation  170: Best = 4.051524e+02  FES =  17100  μ_F = 0.815  μ_CR = 0.484
Generation  180: Best = 4.050387e+02  FES =  18100  μ_F = 0.792  μ_CR = 0.492
Generation  190: Best = 4.049628e+02  FES =  19100  μ_F = 0.787  μ_CR = 0.487
Generation  200: Best = 4.048900e+02  FES =  20100  μ_F = 0.795  μ_CR = 0.489
Generation  210: Best = 4.047585e+02  FES =  21100  μ_F = 0.741  μ_CR = 0.490
Generation  220: Best = 4.046715e+02  FES =  22100  μ_F = 0.741  μ_CR = 0.494
Generation  230: Best = 4.045887e+02  FES =  23100  μ_F = 0.713  μ_CR = 0.514
Generation  240: Best = 4.044813e+02  FES =  24100  μ_F = 0.740  μ_CR = 0.533
Generation  250: Best = 4.043758e+02  FES =  25100  μ_F = 0.749  μ_CR = 0.549
Generation  260: Best = 4.043139e+02  FES =  26100  μ_F = 0.735  μ_CR = 0.563
Generation  270: Best = 4.042284e+02  FES =  27100  μ_F = 0.769  μ_CR = 0.574
Generation  280: Best = 4.040515e+02  FES =  28100  μ_F = 0.790  μ_CR = 0.608
Generation  290: Best = 4.039686e+02  FES =  29100  μ_F = 0.786  μ_CR = 0.641
Generation  300: Best = 4.038250e+02  FES =  30100  μ_F = 0.767  μ_CR = 0.673
Generation  310: Best = 4.035230e+02  FES =  31100  μ_F = 0.765  μ_CR = 0.697
Generation  320: Best = 4.031741e+02  FES =  32100  μ_F = 0.749  μ_CR = 0.749
Generation  330: Best = 4.030303e+02  FES =  33100  μ_F = 0.755  μ_CR = 0.762
Generation  340: Best = 4.027106e+02  FES =  34100  μ_F = 0.762  μ_CR = 0.785
Generation  350: Best = 4.021063e+02  FES =  35100  μ_F = 0.767  μ_CR = 0.825
Generation  360: Best = 4.017102e+02  FES =  36100  μ_F = 0.770  μ_CR = 0.851
Generation  370: Best = 4.014595e+02  FES =  37100  μ_F = 0.779  μ_CR = 0.870
Generation  380: Best = 4.013564e+02  FES =  38100  μ_F = 0.783  μ_CR = 0.893
Generation  390: Best = 4.012985e+02  FES =  39100  μ_F = 0.797  μ_CR = 0.891
Generation  400: Best = 4.012640e+02  FES =  40100  μ_F = 0.806  μ_CR = 0.881
Generation  410: Best = 4.012366e+02  FES =  41100  μ_F = 0.828  μ_CR = 0.885
Generation  420: Best = 4.012123e+02  FES =  42100  μ_F = 0.846  μ_CR = 0.884
Generation  430: Best = 4.011776e+02  FES =  43100  μ_F = 0.852  μ_CR = 0.887
Generation  440: Best = 4.011457e+02  FES =  44100  μ_F = 0.857  μ_CR = 0.882
```

```
Generation  450: Best = 4.011160e+02  FES =  45100  μ_F = 0.862  μ_CR = 0.875
Generation  460: Best = 4.010882e+02  FES =  46100  μ_F = 0.868  μ_CR = 0.874
Generation  470: Best = 4.010708e+02  FES =  47100  μ_F = 0.864  μ_CR = 0.875
Generation  480: Best = 4.010377e+02  FES =  48100  μ_F = 0.867  μ_CR = 0.876
Generation  490: Best = 4.010043e+02  FES =  49100  μ_F = 0.864  μ_CR = 0.879
Generation  500: Best = 4.009709e+02  FES =  50100  μ_F = 0.867  μ_CR = 0.885
Generation  510: Best = 4.009427e+02  FES =  51100  μ_F = 0.870  μ_CR = 0.882
Generation  520: Best = 4.009108e+02  FES =  52100  μ_F = 0.863  μ_CR = 0.879
Generation  530: Best = 4.008905e+02  FES =  53100  μ_F = 0.866  μ_CR = 0.880
Generation  540: Best = 4.008693e+02  FES =  54100  μ_F = 0.866  μ_CR = 0.878
Generation  550: Best = 4.008549e+02  FES =  55100  μ_F = 0.867  μ_CR = 0.874
Generation  560: Best = 4.008348e+02  FES =  56100  μ_F = 0.867  μ_CR = 0.873
Generation  570: Best = 4.007961e+02  FES =  57100  μ_F = 0.868  μ_CR = 0.871
Generation  580: Best = 4.007535e+02  FES =  58100  μ_F = 0.861  μ_CR = 0.876
Generation  590: Best = 4.007269e+02  FES =  59100  μ_F = 0.869  μ_CR = 0.873
Generation  600: Best = 4.006812e+02  FES =  60100  μ_F = 0.868  μ_CR = 0.878
Generation  610: Best = 4.005952e+02  FES =  61100  μ_F = 0.870  μ_CR = 0.883
Generation  620: Best = 4.003912e+02  FES =  62100  μ_F = 0.867  μ_CR = 0.905
Generation  630: Best = 4.002576e+02  FES =  63100  μ_F = 0.847  μ_CR = 0.919
Generation  640: Best = 4.000903e+02  FES =  64100  μ_F = 0.844  μ_CR = 0.932
Generation  650: Best = 4.000272e+02  FES =  65100  μ_F = 0.829  μ_CR = 0.944
Generation  660: Best = 4.000128e+02  FES =  66100  μ_F = 0.834  μ_CR = 0.950
Generation  670: Best = 4.000057e+02  FES =  67100  μ_F = 0.833  μ_CR = 0.954
Generation  680: Best = 4.000037e+02  FES =  68100  μ_F = 0.819  μ_CR = 0.952
Generation  690: Best = 4.000013e+02  FES =  69100  μ_F = 0.827  μ_CR = 0.953
Generation  700: Best = 4.000006e+02  FES =  70100  μ_F = 0.811  μ_CR = 0.954
Generation  710: Best = 4.000003e+02  FES =  71100  μ_F = 0.811  μ_CR = 0.955
Generation  720: Best = 4.000002e+02  FES =  72100  μ_F = 0.813  μ_CR = 0.955
Generation  730: Best = 4.000001e+02  FES =  73100  μ_F = 0.821  μ_CR = 0.961
Generation  740: Best = 4.000000e+02  FES =  74100  μ_F = 0.824  μ_CR = 0.959
Generation  750: Best = 4.000000e+02  FES =  75100  μ_F = 0.828  μ_CR = 0.957
Generation  760: Best = 4.000000e+02  FES =  76100  μ_F = 0.827  μ_CR = 0.958
Generation  770: Best = 4.000000e+02  FES =  77100  μ_F = 0.829  μ_CR = 0.959
Generation  780: Best = 4.000000e+02  FES =  78100  μ_F = 0.830  μ_CR = 0.961
Generation  790: Best = 4.000000e+02  FES =  79100  μ_F = 0.831  μ_CR = 0.961
Generation  800: Best = 4.000000e+02  FES =  80100  μ_F = 0.827  μ_CR = 0.964
Generation  810: Best = 4.000000e+02  FES =  81100  μ_F = 0.818  μ_CR = 0.968
Generation  820: Best = 4.000000e+02  FES =  82100  μ_F = 0.815  μ_CR = 0.966
Generation  830: Best = 4.000000e+02  FES =  83100  μ_F = 0.799  μ_CR = 0.962
Generation  840: Best = 4.000000e+02  FES =  84100  μ_F = 0.806  μ_CR = 0.963
Generation  850: Best = 4.000000e+02  FES =  85100  μ_F = 0.806  μ_CR = 0.964
Generation  860: Best = 4.000000e+02  FES =  86100  μ_F = 0.805  μ_CR = 0.960
Generation  870: Best = 4.000000e+02  FES =  87100  μ_F = 0.809  μ_CR = 0.962
Generation  880: Best = 4.000000e+02  FES =  88100  μ_F = 0.811  μ_CR = 0.961
Generation  890: Best = 4.000000e+02  FES =  89100  μ_F = 0.810  μ_CR = 0.959
Generation  900: Best = 4.000000e+02  FES =  90100  μ_F = 0.811  μ_CR = 0.953
Generation  910: Best = 4.000000e+02  FES =  91100  μ_F = 0.815  μ_CR = 0.940
Generation  920: Best = 4.000000e+02  FES =  92100  μ_F = 0.825  μ_CR = 0.923
Generation  930: Best = 4.000000e+02  FES =  93100  μ_F = 0.836  μ_CR = 0.907
Generation  940: Best = 4.000000e+02  FES =  94100  μ_F = 0.843  μ_CR = 0.892
Generation  950: Best = 4.000000e+02  FES =  95100  μ_F = 0.841  μ_CR = 0.890
Generation  960: Best = 4.000000e+02  FES =  96100  μ_F = 0.842  μ_CR = 0.885
Generation  970: Best = 4.000000e+02  FES =  97100  μ_F = 0.837  μ_CR = 0.877
Generation  980: Best = 4.000000e+02  FES =  98100  μ_F = 0.842  μ_CR = 0.873
```

```
Generation  990: Best = 4.000000e+02  FES =  99100  μ_F = 0.846  μ_CR = 0.877

Optimization complete.
Best Fitness             = 4.000000e+02
Total Function Evals     = 100000
Total Successful Updates = 37951

================================================================================
=
Results:
  Best Fitness           : 4.000000e+02
  Global Optimum         : 4.000000e+02
  Error (|best-optimum|) : 0.000000e+00
  Best Solution (5 dim)  : [ 32.51072529   7.75531311 -56.4575778   77.49269151
19.66855862]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F4.png
```

In [14]:

```python
# ============================================================
# F5 — Shifted and Rotated Rastrigin
# ============================================================
run_and_plot('F5', make_wrapper(simple.f5), 500.0, "Shifted and Rotated Rastri
```

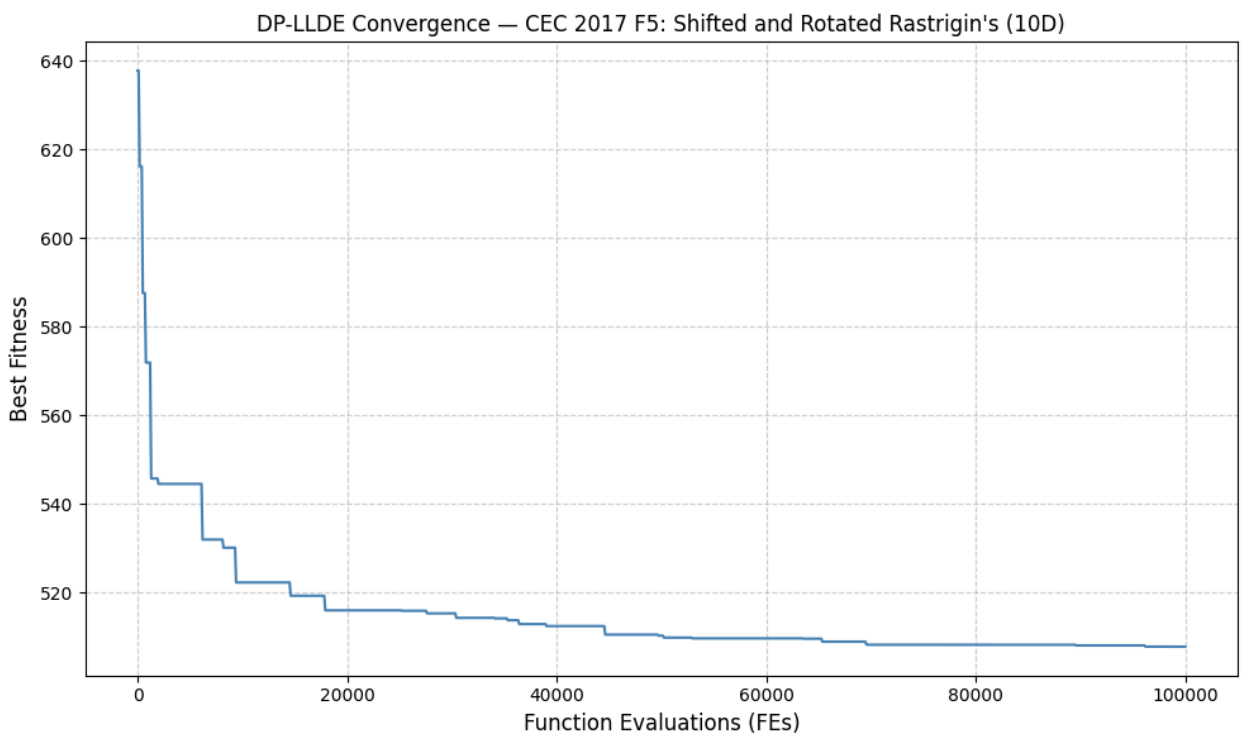
```
================================================================================
=
DP-LLDE on CEC 2017 F5 — Single Run
================================================================================
=
Function       : Shifted and Rotated Rastrigin's (10D)
Bounds         : [-100, 100]
Global optimum : 500.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 6.377184e+02
Generation   10: Best = 5.718597e+02  FES =   1100  μ_F = 0.589  μ_CR = 0.354
Generation   20: Best = 5.444877e+02  FES =   2100  μ_F = 0.647  μ_CR = 0.353
Generation   30: Best = 5.444877e+02  FES =   3100  μ_F = 0.699  μ_CR = 0.325
Generation   40: Best = 5.444877e+02  FES =   4100  μ_F = 0.757  μ_CR = 0.297
Generation   50: Best = 5.444877e+02  FES =   5100  μ_F = 0.779  μ_CR = 0.267
Generation   60: Best = 5.444877e+02  FES =   6100  μ_F = 0.792  μ_CR = 0.236
Generation   70: Best = 5.319262e+02  FES =   7100  μ_F = 0.790  μ_CR = 0.197
Generation   80: Best = 5.319262e+02  FES =   8100  μ_F = 0.822  μ_CR = 0.197
Generation   90: Best = 5.300941e+02  FES =   9100  μ_F = 0.866  μ_CR = 0.161
Generation  100: Best = 5.222443e+02  FES =  10100  μ_F = 0.876  μ_CR = 0.133
Generation  110: Best = 5.222443e+02  FES =  11100  μ_F = 0.868  μ_CR = 0.132
Generation  120: Best = 5.222443e+02  FES =  12100  μ_F = 0.844  μ_CR = 0.124
Generation  130: Best = 5.222443e+02  FES =  13100  μ_F = 0.836  μ_CR = 0.106
Generation  140: Best = 5.222443e+02  FES =  14100  μ_F = 0.818  μ_CR = 0.106
Generation  150: Best = 5.192498e+02  FES =  15100  μ_F = 0.805  μ_CR = 0.098
Generation  160: Best = 5.192498e+02  FES =  16100  μ_F = 0.849  μ_CR = 0.091
Generation  170: Best = 5.192498e+02  FES =  17100  μ_F = 0.842  μ_CR = 0.097
Generation  180: Best = 5.159623e+02  FES =  18100  μ_F = 0.846  μ_CR = 0.074
Generation  190: Best = 5.159623e+02  FES =  19100  μ_F = 0.849  μ_CR = 0.076
Generation  200: Best = 5.159623e+02  FES =  20100  μ_F = 0.865  μ_CR = 0.071
Generation  210: Best = 5.159623e+02  FES =  21100  μ_F = 0.814  μ_CR = 0.091
Generation  220: Best = 5.159623e+02  FES =  22100  μ_F = 0.799  μ_CR = 0.089
Generation  230: Best = 5.159623e+02  FES =  23100  μ_F = 0.838  μ_CR = 0.096
Generation  240: Best = 5.159623e+02  FES =  24100  μ_F = 0.804  μ_CR = 0.102
Generation  250: Best = 5.159623e+02  FES =  25100  μ_F = 0.673  μ_CR = 0.079
Generation  260: Best = 5.158633e+02  FES =  26100  μ_F = 0.722  μ_CR = 0.096
Generation  270: Best = 5.158633e+02  FES =  27100  μ_F = 0.745  μ_CR = 0.089
Generation  280: Best = 5.152809e+02  FES =  28100  μ_F = 0.778  μ_CR = 0.078
Generation  290: Best = 5.152809e+02  FES =  29100  μ_F = 0.840  μ_CR = 0.072
Generation  300: Best = 5.152809e+02  FES =  30100  μ_F = 0.874  μ_CR = 0.114
Generation  310: Best = 5.142620e+02  FES =  31100  μ_F = 0.878  μ_CR = 0.107
Generation  320: Best = 5.142620e+02  FES =  32100  μ_F = 0.809  μ_CR = 0.106
Generation  330: Best = 5.142620e+02  FES =  33100  μ_F = 0.780  μ_CR = 0.109
Generation  340: Best = 5.142620e+02  FES =  34100  μ_F = 0.757  μ_CR = 0.100
Generation  350: Best = 5.141475e+02  FES =  35100  μ_F = 0.699  μ_CR = 0.073
Generation  360: Best = 5.137255e+02  FES =  36100  μ_F = 0.706  μ_CR = 0.074
Generation  370: Best = 5.128655e+02  FES =  37100  μ_F = 0.754  μ_CR = 0.094
Generation  380: Best = 5.128655e+02  FES =  38100  μ_F = 0.752  μ_CR = 0.088
Generation  390: Best = 5.124060e+02  FES =  39100  μ_F = 0.678  μ_CR = 0.083
Generation  400: Best = 5.124060e+02  FES =  40100  μ_F = 0.721  μ_CR = 0.076
Generation  410: Best = 5.124060e+02  FES =  41100  μ_F = 0.750  μ_CR = 0.076
Generation  420: Best = 5.124060e+02  FES =  42100  μ_F = 0.760  μ_CR = 0.088
Generation  430: Best = 5.124060e+02  FES =  43100  μ_F = 0.799  μ_CR = 0.097
Generation  440: Best = 5.124060e+02  FES =  44100  μ_F = 0.810  μ_CR = 0.082
```

```
Generation  450: Best = 5.104783e+02  FES =  45100  μ_F = 0.807  μ_CR = 0.084
Generation  460: Best = 5.104783e+02  FES =  46100  μ_F = 0.786  μ_CR = 0.070
Generation  470: Best = 5.104783e+02  FES =  47100  μ_F = 0.806  μ_CR = 0.063
Generation  480: Best = 5.104783e+02  FES =  48100  μ_F = 0.811  μ_CR = 0.093
Generation  490: Best = 5.104783e+02  FES =  49100  μ_F = 0.813  μ_CR = 0.088
Generation  500: Best = 5.102388e+02  FES =  50100  μ_F = 0.836  μ_CR = 0.094
Generation  510: Best = 5.098097e+02  FES =  51100  μ_F = 0.811  μ_CR = 0.110
Generation  520: Best = 5.098097e+02  FES =  52100  μ_F = 0.796  μ_CR = 0.103
Generation  530: Best = 5.096302e+02  FES =  53100  μ_F = 0.818  μ_CR = 0.092
Generation  540: Best = 5.096302e+02  FES =  54100  μ_F = 0.847  μ_CR = 0.070
Generation  550: Best = 5.096302e+02  FES =  55100  μ_F = 0.841  μ_CR = 0.072
Generation  560: Best = 5.096302e+02  FES =  56100  μ_F = 0.870  μ_CR = 0.083
Generation  570: Best = 5.096302e+02  FES =  57100  μ_F = 0.859  μ_CR = 0.088
Generation  580: Best = 5.096302e+02  FES =  58100  μ_F = 0.834  μ_CR = 0.095
Generation  590: Best = 5.096302e+02  FES =  59100  μ_F = 0.833  μ_CR = 0.084
Generation  600: Best = 5.096302e+02  FES =  60100  μ_F = 0.779  μ_CR = 0.088
Generation  610: Best = 5.096302e+02  FES =  61100  μ_F = 0.694  μ_CR = 0.073
Generation  620: Best = 5.096302e+02  FES =  62100  μ_F = 0.654  μ_CR = 0.089
Generation  630: Best = 5.096302e+02  FES =  63100  μ_F = 0.696  μ_CR = 0.083
Generation  640: Best = 5.095627e+02  FES =  64100  μ_F = 0.758  μ_CR = 0.067
Generation  650: Best = 5.095627e+02  FES =  65100  μ_F = 0.736  μ_CR = 0.072
Generation  660: Best = 5.088886e+02  FES =  66100  μ_F = 0.770  μ_CR = 0.068
Generation  670: Best = 5.088886e+02  FES =  67100  μ_F = 0.796  μ_CR = 0.064
Generation  680: Best = 5.088886e+02  FES =  68100  μ_F = 0.828  μ_CR = 0.080
Generation  690: Best = 5.088886e+02  FES =  69100  μ_F = 0.765  μ_CR = 0.074
Generation  700: Best = 5.081929e+02  FES =  70100  μ_F = 0.687  μ_CR = 0.085
Generation  710: Best = 5.081929e+02  FES =  71100  μ_F = 0.673  μ_CR = 0.075
Generation  720: Best = 5.081929e+02  FES =  72100  μ_F = 0.740  μ_CR = 0.092
Generation  730: Best = 5.081929e+02  FES =  73100  μ_F = 0.761  μ_CR = 0.087
Generation  740: Best = 5.081929e+02  FES =  74100  μ_F = 0.758  μ_CR = 0.092
Generation  750: Best = 5.081929e+02  FES =  75100  μ_F = 0.803  μ_CR = 0.073
Generation  760: Best = 5.081929e+02  FES =  76100  μ_F = 0.835  μ_CR = 0.063
Generation  770: Best = 5.081929e+02  FES =  77100  μ_F = 0.818  μ_CR = 0.058
Generation  780: Best = 5.081929e+02  FES =  78100  μ_F = 0.778  μ_CR = 0.077
Generation  790: Best = 5.081929e+02  FES =  79100  μ_F = 0.742  μ_CR = 0.072
Generation  800: Best = 5.081929e+02  FES =  80100  μ_F = 0.749  μ_CR = 0.067
Generation  810: Best = 5.081929e+02  FES =  81100  μ_F = 0.715  μ_CR = 0.066
Generation  820: Best = 5.081929e+02  FES =  82100  μ_F = 0.719  μ_CR = 0.061
Generation  830: Best = 5.081929e+02  FES =  83100  μ_F = 0.718  μ_CR = 0.066
Generation  840: Best = 5.081929e+02  FES =  84100  μ_F = 0.727  μ_CR = 0.070
Generation  850: Best = 5.081929e+02  FES =  85100  μ_F = 0.648  μ_CR = 0.073
Generation  860: Best = 5.081929e+02  FES =  86100  μ_F = 0.611  μ_CR = 0.093
Generation  870: Best = 5.081929e+02  FES =  87100  μ_F = 0.614  μ_CR = 0.085
Generation  880: Best = 5.081929e+02  FES =  88100  μ_F = 0.649  μ_CR = 0.103
Generation  890: Best = 5.081808e+02  FES =  89100  μ_F = 0.618  μ_CR = 0.094
Generation  900: Best = 5.080459e+02  FES =  90100  μ_F = 0.646  μ_CR = 0.090
Generation  910: Best = 5.080459e+02  FES =  91100  μ_F = 0.688  μ_CR = 0.092
Generation  920: Best = 5.080459e+02  FES =  92100  μ_F = 0.705  μ_CR = 0.098
Generation  930: Best = 5.080459e+02  FES =  93100  μ_F = 0.677  μ_CR = 0.103
Generation  940: Best = 5.080459e+02  FES =  94100  μ_F = 0.698  μ_CR = 0.094
Generation  950: Best = 5.080459e+02  FES =  95100  μ_F = 0.721  μ_CR = 0.103
Generation  960: Best = 5.080459e+02  FES =  96100  μ_F = 0.725  μ_CR = 0.081
Generation  970: Best = 5.077706e+02  FES =  97100  μ_F = 0.709  μ_CR = 0.095
Generation  980: Best = 5.077706e+02  FES =  98100  μ_F = 0.665  μ_CR = 0.103
```

```
Generation  990: Best = 5.077706e+02  FES =  99100  μ_F = 0.592  μ_CR = 0.084

Optimization complete.
Best Fitness             = 5.077587e+02
Total Function Evals     = 100000
Total Successful Updates = 2509

================================================================================
=
Results:
  Best Fitness           : 5.077587e+02
  Global Optimum         : 5.000000e+02
  Error (|best-optimum|) : 7.758743e+00
  Best Solution (5 dim)  : [ -3.68505055  48.83563998 -38.99614924 -48.14528285
33.14622364]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F5.png
```

In [15]:
```
# ============================================================
# F6 — Shifted and Rotated Schaffer's F7
# ============================================================
run_and_plot('F6', make_wrapper(simple.f6), 600.0, "Shifted and Rotated Schaff
```

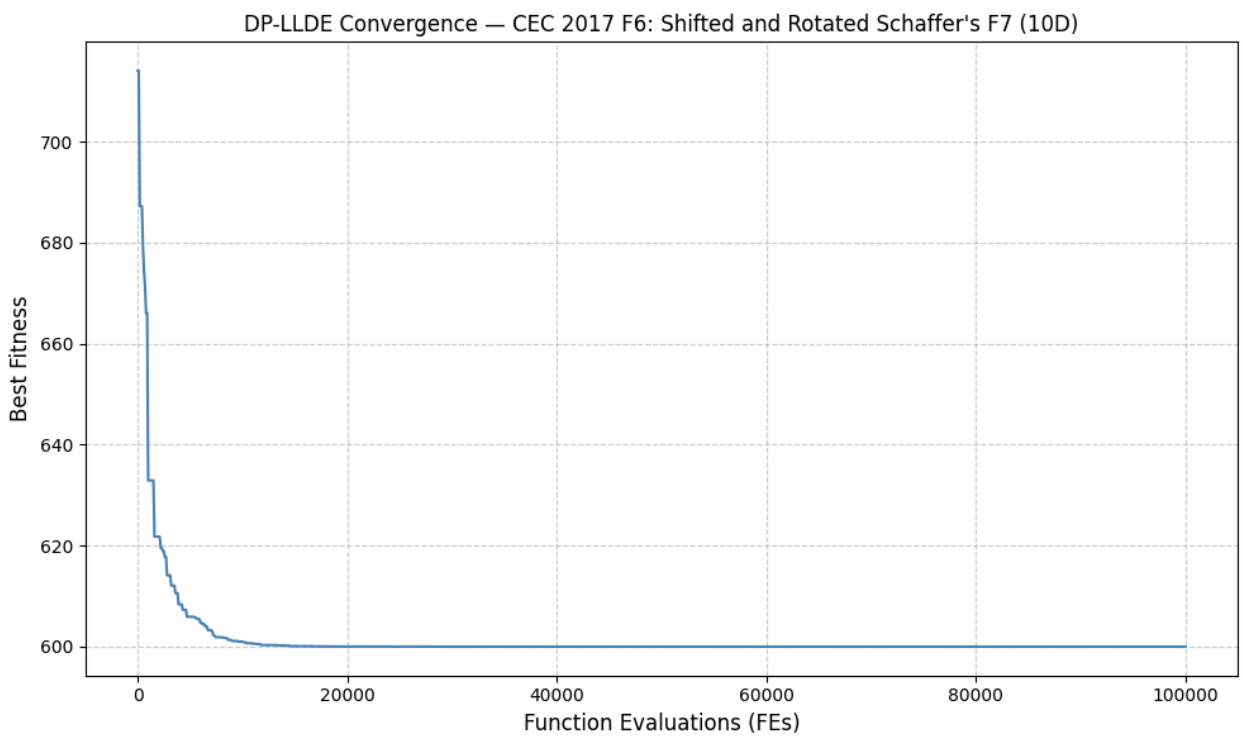
```
================================================================================
=
DP-LLDE on CEC 2017 F6 — Single Run
================================================================================
=
Function       : Shifted and Rotated Schaffer's F7 (10D)
Bounds         : [-100, 100]
Global optimum : 600.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 7.140452e+02
Generation   10: Best = 6.329242e+02  FES =   1100  μ_F = 0.582  μ_CR = 0.352
Generation   20: Best = 6.217817e+02  FES =   2100  μ_F = 0.633  μ_CR = 0.354
Generation   30: Best = 6.141204e+02  FES =   3100  μ_F = 0.660  μ_CR = 0.352
Generation   40: Best = 6.083732e+02  FES =   4100  μ_F = 0.704  μ_CR = 0.344
Generation   50: Best = 6.059138e+02  FES =   5100  μ_F = 0.724  μ_CR = 0.348
Generation   60: Best = 6.046847e+02  FES =   6100  μ_F = 0.746  μ_CR = 0.352
Generation   70: Best = 6.032822e+02  FES =   7100  μ_F = 0.758  μ_CR = 0.353
Generation   80: Best = 6.018303e+02  FES =   8100  μ_F = 0.746  μ_CR = 0.356
Generation   90: Best = 6.011224e+02  FES =   9100  μ_F = 0.748  μ_CR = 0.363
Generation  100: Best = 6.009947e+02  FES =  10100  μ_F = 0.749  μ_CR = 0.370
Generation  110: Best = 6.006051e+02  FES =  11100  μ_F = 0.739  μ_CR = 0.364
Generation  120: Best = 6.003218e+02  FES =  12100  μ_F = 0.744  μ_CR = 0.353
Generation  130: Best = 6.003218e+02  FES =  13100  μ_F = 0.776  μ_CR = 0.335
Generation  140: Best = 6.002512e+02  FES =  14100  μ_F = 0.766  μ_CR = 0.330
Generation  150: Best = 6.001060e+02  FES =  15100  μ_F = 0.759  μ_CR = 0.335
Generation  160: Best = 6.001018e+02  FES =  16100  μ_F = 0.747  μ_CR = 0.340
Generation  170: Best = 6.000515e+02  FES =  17100  μ_F = 0.750  μ_CR = 0.342
Generation  180: Best = 6.000414e+02  FES =  18100  μ_F = 0.732  μ_CR = 0.333
Generation  190: Best = 6.000326e+02  FES =  19100  μ_F = 0.753  μ_CR = 0.325
Generation  200: Best = 6.000239e+02  FES =  20100  μ_F = 0.765  μ_CR = 0.332
Generation  210: Best = 6.000234e+02  FES =  21100  μ_F = 0.787  μ_CR = 0.315
Generation  220: Best = 6.000133e+02  FES =  22100  μ_F = 0.802  μ_CR = 0.313
Generation  230: Best = 6.000115e+02  FES =  23100  μ_F = 0.812  μ_CR = 0.307
Generation  240: Best = 6.000084e+02  FES =  24100  μ_F = 0.796  μ_CR = 0.305
Generation  250: Best = 6.000048e+02  FES =  25100  μ_F = 0.806  μ_CR = 0.293
Generation  260: Best = 6.000048e+02  FES =  26100  μ_F = 0.799  μ_CR = 0.288
Generation  270: Best = 6.000036e+02  FES =  27100  μ_F = 0.806  μ_CR = 0.275
Generation  280: Best = 6.000031e+02  FES =  28100  μ_F = 0.785  μ_CR = 0.268
Generation  290: Best = 6.000023e+02  FES =  29100  μ_F = 0.792  μ_CR = 0.261
Generation  300: Best = 6.000022e+02  FES =  30100  μ_F = 0.762  μ_CR = 0.259
Generation  310: Best = 6.000013e+02  FES =  31100  μ_F = 0.746  μ_CR = 0.268
Generation  320: Best = 6.000009e+02  FES =  32100  μ_F = 0.756  μ_CR = 0.262
Generation  330: Best = 6.000007e+02  FES =  33100  μ_F = 0.776  μ_CR = 0.254
Generation  340: Best = 6.000005e+02  FES =  34100  μ_F = 0.783  μ_CR = 0.235
Generation  350: Best = 6.000005e+02  FES =  35100  μ_F = 0.778  μ_CR = 0.232
Generation  360: Best = 6.000003e+02  FES =  36100  μ_F = 0.774  μ_CR = 0.226
Generation  370: Best = 6.000003e+02  FES =  37100  μ_F = 0.773  μ_CR = 0.218
Generation  380: Best = 6.000003e+02  FES =  38100  μ_F = 0.783  μ_CR = 0.204
Generation  390: Best = 6.000003e+02  FES =  39100  μ_F = 0.775  μ_CR = 0.197
Generation  400: Best = 6.000002e+02  FES =  40100  μ_F = 0.785  μ_CR = 0.186
Generation  410: Best = 6.000002e+02  FES =  41100  μ_F = 0.803  μ_CR = 0.182
Generation  420: Best = 6.000002e+02  FES =  42100  μ_F = 0.804  μ_CR = 0.155
Generation  430: Best = 6.000002e+02  FES =  43100  μ_F = 0.800  μ_CR = 0.146
Generation  440: Best = 6.000001e+02  FES =  44100  μ_F = 0.818  μ_CR = 0.147
```

```
Generation  450: Best = 6.000001e+02  FES =  45100  μ_F = 0.824  μ_CR = 0.133
Generation  460: Best = 6.000001e+02  FES =  46100  μ_F = 0.804  μ_CR = 0.146
Generation  470: Best = 6.000001e+02  FES =  47100  μ_F = 0.795  μ_CR = 0.156
Generation  480: Best = 6.000001e+02  FES =  48100  μ_F = 0.779  μ_CR = 0.144
Generation  490: Best = 6.000001e+02  FES =  49100  μ_F = 0.805  μ_CR = 0.132
Generation  500: Best = 6.000000e+02  FES =  50100  μ_F = 0.786  μ_CR = 0.138
Generation  510: Best = 6.000000e+02  FES =  51100  μ_F = 0.775  μ_CR = 0.135
Generation  520: Best = 6.000000e+02  FES =  52100  μ_F = 0.739  μ_CR = 0.142
Generation  530: Best = 6.000000e+02  FES =  53100  μ_F = 0.746  μ_CR = 0.155
Generation  540: Best = 6.000000e+02  FES =  54100  μ_F = 0.747  μ_CR = 0.149
Generation  550: Best = 6.000000e+02  FES =  55100  μ_F = 0.751  μ_CR = 0.153
Generation  560: Best = 6.000000e+02  FES =  56100  μ_F = 0.793  μ_CR = 0.157
Generation  570: Best = 6.000000e+02  FES =  57100  μ_F = 0.789  μ_CR = 0.153
Generation  580: Best = 6.000000e+02  FES =  58100  μ_F = 0.804  μ_CR = 0.146
Generation  590: Best = 6.000000e+02  FES =  59100  μ_F = 0.815  μ_CR = 0.145
Generation  600: Best = 6.000000e+02  FES =  60100  μ_F = 0.821  μ_CR = 0.139
Generation  610: Best = 6.000000e+02  FES =  61100  μ_F = 0.810  μ_CR = 0.138
Generation  620: Best = 6.000000e+02  FES =  62100  μ_F = 0.821  μ_CR = 0.125
Generation  630: Best = 6.000000e+02  FES =  63100  μ_F = 0.813  μ_CR = 0.117
Generation  640: Best = 6.000000e+02  FES =  64100  μ_F = 0.817  μ_CR = 0.110
Generation  650: Best = 6.000000e+02  FES =  65100  μ_F = 0.819  μ_CR = 0.117
Generation  660: Best = 6.000000e+02  FES =  66100  μ_F = 0.828  μ_CR = 0.119
Generation  670: Best = 6.000000e+02  FES =  67100  μ_F = 0.842  μ_CR = 0.116
Generation  680: Best = 6.000000e+02  FES =  68100  μ_F = 0.858  μ_CR = 0.112
Generation  690: Best = 6.000000e+02  FES =  69100  μ_F = 0.848  μ_CR = 0.114
Generation  700: Best = 6.000000e+02  FES =  70100  μ_F = 0.837  μ_CR = 0.116
Generation  710: Best = 6.000000e+02  FES =  71100  μ_F = 0.824  μ_CR = 0.134
Generation  720: Best = 6.000000e+02  FES =  72100  μ_F = 0.815  μ_CR = 0.115
Generation  730: Best = 6.000000e+02  FES =  73100  μ_F = 0.782  μ_CR = 0.166
Generation  740: Best = 6.000000e+02  FES =  74100  μ_F = 0.737  μ_CR = 0.151
Generation  750: Best = 6.000000e+02  FES =  75100  μ_F = 0.719  μ_CR = 0.159
Generation  760: Best = 6.000000e+02  FES =  76100  μ_F = 0.752  μ_CR = 0.145
Generation  770: Best = 6.000000e+02  FES =  77100  μ_F = 0.783  μ_CR = 0.124
Generation  780: Best = 6.000000e+02  FES =  78100  μ_F = 0.814  μ_CR = 0.132
Generation  790: Best = 6.000000e+02  FES =  79100  μ_F = 0.835  μ_CR = 0.132
Generation  800: Best = 6.000000e+02  FES =  80100  μ_F = 0.830  μ_CR = 0.121
Generation  810: Best = 6.000000e+02  FES =  81100  μ_F = 0.837  μ_CR = 0.120
Generation  820: Best = 6.000000e+02  FES =  82100  μ_F = 0.857  μ_CR = 0.125
Generation  830: Best = 6.000000e+02  FES =  83100  μ_F = 0.847  μ_CR = 0.117
Generation  840: Best = 6.000000e+02  FES =  84100  μ_F = 0.812  μ_CR = 0.107
Generation  850: Best = 6.000000e+02  FES =  85100  μ_F = 0.833  μ_CR = 0.100
Generation  860: Best = 6.000000e+02  FES =  86100  μ_F = 0.816  μ_CR = 0.098
Generation  870: Best = 6.000000e+02  FES =  87100  μ_F = 0.775  μ_CR = 0.106
Generation  880: Best = 6.000000e+02  FES =  88100  μ_F = 0.812  μ_CR = 0.093
Generation  890: Best = 6.000000e+02  FES =  89100  μ_F = 0.791  μ_CR = 0.082
Generation  900: Best = 6.000000e+02  FES =  90100  μ_F = 0.806  μ_CR = 0.083
Generation  910: Best = 6.000000e+02  FES =  91100  μ_F = 0.800  μ_CR = 0.101
Generation  920: Best = 6.000000e+02  FES =  92100  μ_F = 0.774  μ_CR = 0.099
Generation  930: Best = 6.000000e+02  FES =  93100  μ_F = 0.770  μ_CR = 0.109
Generation  940: Best = 6.000000e+02  FES =  94100  μ_F = 0.742  μ_CR = 0.131
Generation  950: Best = 6.000000e+02  FES =  95100  μ_F = 0.755  μ_CR = 0.133
Generation  960: Best = 6.000000e+02  FES =  96100  μ_F = 0.763  μ_CR = 0.130
Generation  970: Best = 6.000000e+02  FES =  97100  μ_F = 0.757  μ_CR = 0.150
Generation  980: Best = 6.000000e+02  FES =  98100  μ_F = 0.782  μ_CR = 0.148
```

```
Generation  990: Best = 6.000000e+02  FES =  99100  μ_F = 0.808  μ_CR = 0.140

Optimization complete.
Best Fitness             = 6.000000e+02
Total Function Evals     = 100000
Total Successful Updates = 11299

================================================================================
=
Results:
  Best Fitness           : 6.000000e+02
  Global Optimum         : 6.000000e+02
  Error (|best-optimum|) : 8.256476e-06
  Best Solution (5 dim)  : [ 79.08940122 -24.57277755 -34.67395995 -44.68161124
75.99788609]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F6.png
```

In [16]:

```python
# ============================================================
# F7 — Shifted and Rotated Lunacek Bi-Rastrigin
# ============================================================
run_and_plot('F7', make_wrapper(simple.f7), 700.0, "Shifted and Rotated Lunace
```

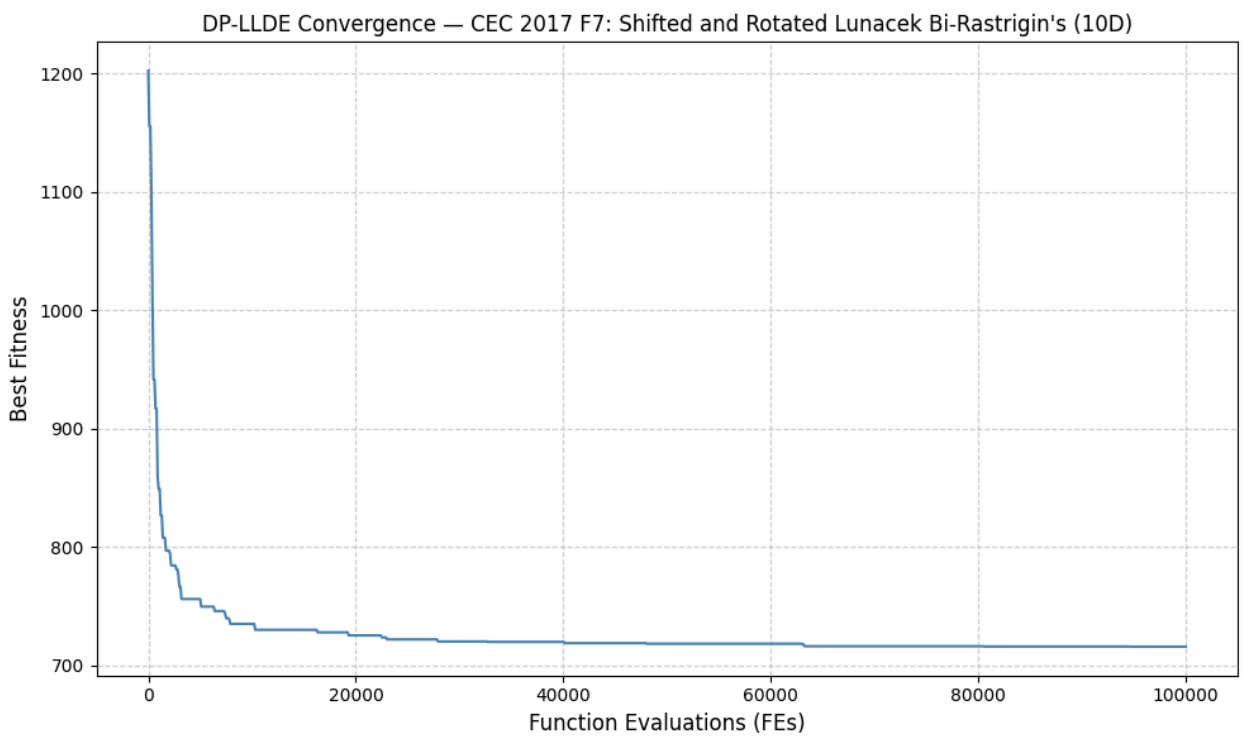
```
================================================================================
=
DP-LLDE on CEC 2017 F7 — Single Run
================================================================================
=
Function       : Shifted and Rotated Lunacek Bi-Rastrigin's (10D)
Bounds         : [-100, 100]
Global optimum : 700.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 1.202264e+03
Generation   10: Best = 8.489179e+02  FES =   1100  μ_F = 0.569  μ_CR = 0.355
Generation   20: Best = 7.968365e+02  FES =   2100  μ_F = 0.623  μ_CR = 0.361
Generation   30: Best = 7.663103e+02  FES =   3100  μ_F = 0.686  μ_CR = 0.344
Generation   40: Best = 7.559472e+02  FES =   4100  μ_F = 0.734  μ_CR = 0.324
Generation   50: Best = 7.559472e+02  FES =   5100  μ_F = 0.753  μ_CR = 0.296
Generation   60: Best = 7.494917e+02  FES =   6100  μ_F = 0.748  μ_CR = 0.272
Generation   70: Best = 7.457571e+02  FES =   7100  μ_F = 0.749  μ_CR = 0.223
Generation   80: Best = 7.349353e+02  FES =   8100  μ_F = 0.765  μ_CR = 0.195
Generation   90: Best = 7.349353e+02  FES =   9100  μ_F = 0.795  μ_CR = 0.157
Generation  100: Best = 7.349353e+02  FES =  10100  μ_F = 0.843  μ_CR = 0.123
Generation  110: Best = 7.298979e+02  FES =  11100  μ_F = 0.846  μ_CR = 0.100
Generation  120: Best = 7.298979e+02  FES =  12100  μ_F = 0.832  μ_CR = 0.093
Generation  130: Best = 7.298979e+02  FES =  13100  μ_F = 0.845  μ_CR = 0.106
Generation  140: Best = 7.298979e+02  FES =  14100  μ_F = 0.805  μ_CR = 0.107
Generation  150: Best = 7.298979e+02  FES =  15100  μ_F = 0.813  μ_CR = 0.130
Generation  160: Best = 7.298979e+02  FES =  16100  μ_F = 0.842  μ_CR = 0.128
Generation  170: Best = 7.277937e+02  FES =  17100  μ_F = 0.824  μ_CR = 0.100
Generation  180: Best = 7.277937e+02  FES =  18100  μ_F = 0.828  μ_CR = 0.087
Generation  190: Best = 7.277937e+02  FES =  19100  μ_F = 0.752  μ_CR = 0.079
Generation  200: Best = 7.252010e+02  FES =  20100  μ_F = 0.747  μ_CR = 0.085
Generation  210: Best = 7.252010e+02  FES =  21100  μ_F = 0.721  μ_CR = 0.087
Generation  220: Best = 7.252010e+02  FES =  22100  μ_F = 0.713  μ_CR = 0.088
Generation  230: Best = 7.218319e+02  FES =  23100  μ_F = 0.774  μ_CR = 0.071
Generation  240: Best = 7.218319e+02  FES =  24100  μ_F = 0.834  μ_CR = 0.066
Generation  250: Best = 7.218319e+02  FES =  25100  μ_F = 0.875  μ_CR = 0.085
Generation  260: Best = 7.218319e+02  FES =  26100  μ_F = 0.765  μ_CR = 0.081
Generation  270: Best = 7.218319e+02  FES =  27100  μ_F = 0.807  μ_CR = 0.087
Generation  280: Best = 7.200815e+02  FES =  28100  μ_F = 0.818  μ_CR = 0.092
Generation  290: Best = 7.200815e+02  FES =  29100  μ_F = 0.863  μ_CR = 0.109
Generation  300: Best = 7.200815e+02  FES =  30100  μ_F = 0.859  μ_CR = 0.093
Generation  310: Best = 7.200815e+02  FES =  31100  μ_F = 0.856  μ_CR = 0.090
Generation  320: Best = 7.200815e+02  FES =  32100  μ_F = 0.863  μ_CR = 0.079
Generation  330: Best = 7.197237e+02  FES =  33100  μ_F = 0.892  μ_CR = 0.105
Generation  340: Best = 7.197237e+02  FES =  34100  μ_F = 0.856  μ_CR = 0.095
Generation  350: Best = 7.197237e+02  FES =  35100  μ_F = 0.841  μ_CR = 0.080
Generation  360: Best = 7.197237e+02  FES =  36100  μ_F = 0.847  μ_CR = 0.068
Generation  370: Best = 7.197237e+02  FES =  37100  μ_F = 0.782  μ_CR = 0.076
Generation  380: Best = 7.197237e+02  FES =  38100  μ_F = 0.797  μ_CR = 0.078
Generation  390: Best = 7.197237e+02  FES =  39100  μ_F = 0.815  μ_CR = 0.096
Generation  400: Best = 7.197237e+02  FES =  40100  μ_F = 0.823  μ_CR = 0.101
Generation  410: Best = 7.186821e+02  FES =  41100  μ_F = 0.768  μ_CR = 0.090
Generation  420: Best = 7.186821e+02  FES =  42100  μ_F = 0.767  μ_CR = 0.091
Generation  430: Best = 7.186821e+02  FES =  43100  μ_F = 0.735  μ_CR = 0.092
Generation  440: Best = 7.186821e+02  FES =  44100  μ_F = 0.721  μ_CR = 0.095
```

```
Generation  450: Best = 7.186821e+02  FES =  45100  μ_F = 0.679  μ_CR = 0.082
Generation  460: Best = 7.186821e+02  FES =  46100  μ_F = 0.650  μ_CR = 0.068
Generation  470: Best = 7.186821e+02  FES =  47100  μ_F = 0.668  μ_CR = 0.083
Generation  480: Best = 7.181744e+02  FES =  48100  μ_F = 0.612  μ_CR = 0.082
Generation  490: Best = 7.181744e+02  FES =  49100  μ_F = 0.656  μ_CR = 0.079
Generation  500: Best = 7.181744e+02  FES =  50100  μ_F = 0.700  μ_CR = 0.077
Generation  510: Best = 7.181744e+02  FES =  51100  μ_F = 0.671  μ_CR = 0.062
Generation  520: Best = 7.181744e+02  FES =  52100  μ_F = 0.657  μ_CR = 0.067
Generation  530: Best = 7.181744e+02  FES =  53100  μ_F = 0.659  μ_CR = 0.065
Generation  540: Best = 7.181744e+02  FES =  54100  μ_F = 0.671  μ_CR = 0.072
Generation  550: Best = 7.181744e+02  FES =  55100  μ_F = 0.631  μ_CR = 0.060
Generation  560: Best = 7.181744e+02  FES =  56100  μ_F = 0.665  μ_CR = 0.062
Generation  570: Best = 7.181744e+02  FES =  57100  μ_F = 0.695  μ_CR = 0.061
Generation  580: Best = 7.181744e+02  FES =  58100  μ_F = 0.683  μ_CR = 0.085
Generation  590: Best = 7.181744e+02  FES =  59100  μ_F = 0.677  μ_CR = 0.072
Generation  600: Best = 7.181744e+02  FES =  60100  μ_F = 0.595  μ_CR = 0.064
Generation  610: Best = 7.181744e+02  FES =  61100  μ_F = 0.576  μ_CR = 0.084
Generation  620: Best = 7.181744e+02  FES =  62100  μ_F = 0.597  μ_CR = 0.082
Generation  630: Best = 7.175763e+02  FES =  63100  μ_F = 0.682  μ_CR = 0.076
Generation  640: Best = 7.160525e+02  FES =  64100  μ_F = 0.608  μ_CR = 0.075
Generation  650: Best = 7.160525e+02  FES =  65100  μ_F = 0.592  μ_CR = 0.072
Generation  660: Best = 7.160525e+02  FES =  66100  μ_F = 0.588  μ_CR = 0.088
Generation  670: Best = 7.160525e+02  FES =  67100  μ_F = 0.614  μ_CR = 0.092
Generation  680: Best = 7.160525e+02  FES =  68100  μ_F = 0.595  μ_CR = 0.097
Generation  690: Best = 7.160525e+02  FES =  69100  μ_F = 0.514  μ_CR = 0.094
Generation  700: Best = 7.160525e+02  FES =  70100  μ_F = 0.577  μ_CR = 0.089
Generation  710: Best = 7.160525e+02  FES =  71100  μ_F = 0.567  μ_CR = 0.086
Generation  720: Best = 7.160525e+02  FES =  72100  μ_F = 0.592  μ_CR = 0.086
Generation  730: Best = 7.160525e+02  FES =  73100  μ_F = 0.604  μ_CR = 0.088
Generation  740: Best = 7.160525e+02  FES =  74100  μ_F = 0.574  μ_CR = 0.101
Generation  750: Best = 7.160525e+02  FES =  75100  μ_F = 0.657  μ_CR = 0.086
Generation  760: Best = 7.160525e+02  FES =  76100  μ_F = 0.611  μ_CR = 0.082
Generation  770: Best = 7.160525e+02  FES =  77100  μ_F = 0.608  μ_CR = 0.075
Generation  780: Best = 7.160525e+02  FES =  78100  μ_F = 0.596  μ_CR = 0.090
Generation  790: Best = 7.160525e+02  FES =  79100  μ_F = 0.671  μ_CR = 0.117
Generation  800: Best = 7.160525e+02  FES =  80100  μ_F = 0.685  μ_CR = 0.128
Generation  810: Best = 7.158029e+02  FES =  81100  μ_F = 0.730  μ_CR = 0.122
Generation  820: Best = 7.158029e+02  FES =  82100  μ_F = 0.766  μ_CR = 0.092
Generation  830: Best = 7.158029e+02  FES =  83100  μ_F = 0.809  μ_CR = 0.089
Generation  840: Best = 7.158029e+02  FES =  84100  μ_F = 0.746  μ_CR = 0.085
Generation  850: Best = 7.158029e+02  FES =  85100  μ_F = 0.767  μ_CR = 0.077
Generation  860: Best = 7.158029e+02  FES =  86100  μ_F = 0.779  μ_CR = 0.076
Generation  870: Best = 7.158029e+02  FES =  87100  μ_F = 0.780  μ_CR = 0.084
Generation  880: Best = 7.158029e+02  FES =  88100  μ_F = 0.772  μ_CR = 0.080
Generation  890: Best = 7.158029e+02  FES =  89100  μ_F = 0.701  μ_CR = 0.062
Generation  900: Best = 7.158029e+02  FES =  90100  μ_F = 0.675  μ_CR = 0.066
Generation  910: Best = 7.158029e+02  FES =  91100  μ_F = 0.618  μ_CR = 0.092
Generation  920: Best = 7.158029e+02  FES =  92100  μ_F = 0.696  μ_CR = 0.072
Generation  930: Best = 7.158029e+02  FES =  93100  μ_F = 0.741  μ_CR = 0.078
Generation  940: Best = 7.158029e+02  FES =  94100  μ_F = 0.758  μ_CR = 0.074
Generation  950: Best = 7.156720e+02  FES =  95100  μ_F = 0.806  μ_CR = 0.074
Generation  960: Best = 7.156720e+02  FES =  96100  μ_F = 0.819  μ_CR = 0.069
Generation  970: Best = 7.156720e+02  FES =  97100  μ_F = 0.819  μ_CR = 0.084
Generation  980: Best = 7.156720e+02  FES =  98100  μ_F = 0.810  μ_CR = 0.063
```

```
Generation  990: Best = 7.156720e+02  FES =  99100  μ_F = 0.829  μ_CR = 0.067

Optimization complete.
Best Fitness             = 7.156720e+02
Total Function Evals     = 100000
Total Successful Updates = 2883

================================================================================
=
Results:
  Best Fitness           : 7.156720e+02
  Global Optimum         : 7.000000e+02
  Error (|best-optimum|) : 1.567204e+01
  Best Solution (5 dim)  : [-19.4421595   21.14425835 -22.3436518  -10.25435416
15.77648641]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F7.png
```

In [17]:
```
# ============================================================
# F8 — Shifted and Rotated Non-Continuous Rastrigin
# ============================================================
run_and_plot('F8', make_wrapper(simple.f8), 800.0, "Shifted and Rotated Non-Co
```

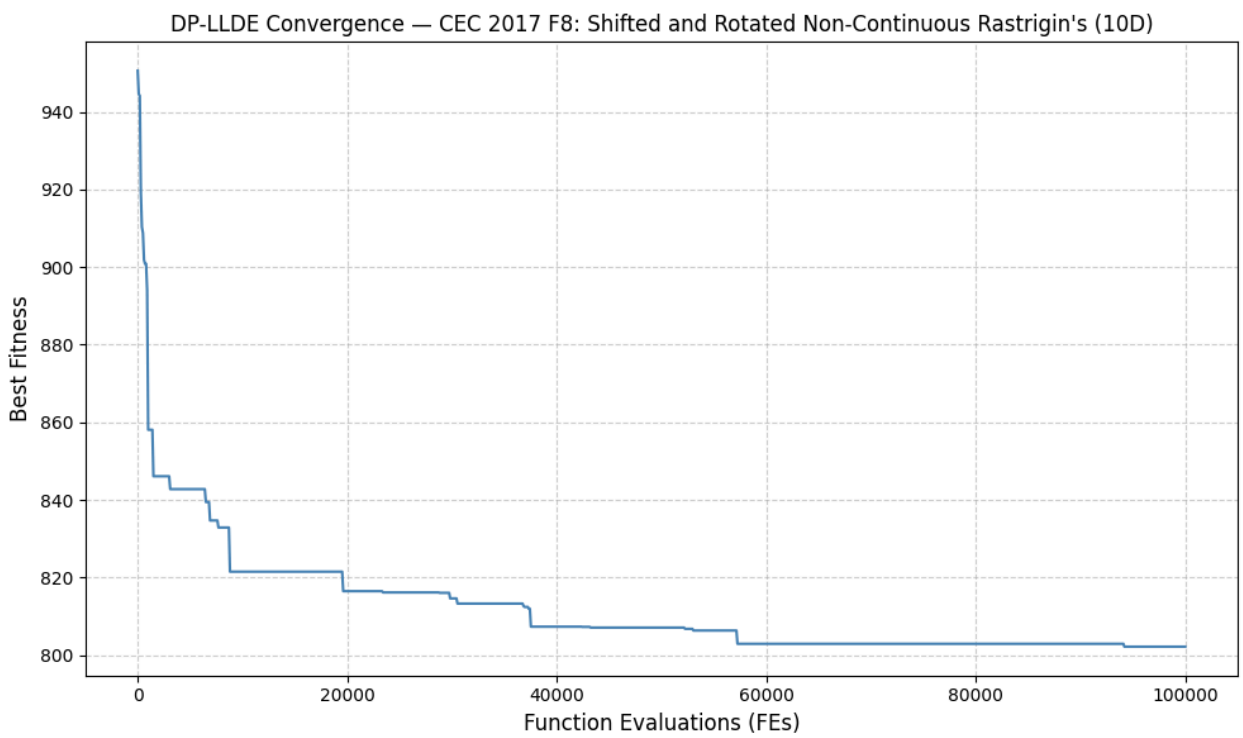
```
================================================================================
=
DP-LLDE on CEC 2017 F8 — Single Run
================================================================================
=
Function       : Shifted and Rotated Non-Continuous Rastrigin's (10D)
Bounds         : [-100, 100]
Global optimum : 800.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 9.505966e+02
Generation   10: Best = 8.580610e+02  FES =   1100  μ_F = 0.592  μ_CR = 0.354
Generation   20: Best = 8.460823e+02  FES =   2100  μ_F = 0.623  μ_CR = 0.353
Generation   30: Best = 8.460823e+02  FES =   3100  μ_F = 0.663  μ_CR = 0.341
Generation   40: Best = 8.427672e+02  FES =   4100  μ_F = 0.724  μ_CR = 0.292
Generation   50: Best = 8.427672e+02  FES =   5100  μ_F = 0.741  μ_CR = 0.281
Generation   60: Best = 8.427672e+02  FES =   6100  μ_F = 0.746  μ_CR = 0.244
Generation   70: Best = 8.346868e+02  FES =   7100  μ_F = 0.760  μ_CR = 0.231
Generation   80: Best = 8.328655e+02  FES =   8100  μ_F = 0.795  μ_CR = 0.187
Generation   90: Best = 8.214897e+02  FES =   9100  μ_F = 0.809  μ_CR = 0.164
Generation  100: Best = 8.214897e+02  FES =  10100  μ_F = 0.769  μ_CR = 0.137
Generation  110: Best = 8.214897e+02  FES =  11100  μ_F = 0.771  μ_CR = 0.131
Generation  120: Best = 8.214897e+02  FES =  12100  μ_F = 0.793  μ_CR = 0.115
Generation  130: Best = 8.214897e+02  FES =  13100  μ_F = 0.791  μ_CR = 0.089
Generation  140: Best = 8.214897e+02  FES =  14100  μ_F = 0.796  μ_CR = 0.088
Generation  150: Best = 8.214897e+02  FES =  15100  μ_F = 0.787  μ_CR = 0.081
Generation  160: Best = 8.214897e+02  FES =  16100  μ_F = 0.822  μ_CR = 0.087
Generation  170: Best = 8.214897e+02  FES =  17100  μ_F = 0.833  μ_CR = 0.079
Generation  180: Best = 8.214897e+02  FES =  18100  μ_F = 0.874  μ_CR = 0.086
Generation  190: Best = 8.214897e+02  FES =  19100  μ_F = 0.868  μ_CR = 0.087
Generation  200: Best = 8.164650e+02  FES =  20100  μ_F = 0.860  μ_CR = 0.103
Generation  210: Best = 8.164650e+02  FES =  21100  μ_F = 0.805  μ_CR = 0.097
Generation  220: Best = 8.164650e+02  FES =  22100  μ_F = 0.746  μ_CR = 0.104
Generation  230: Best = 8.164650e+02  FES =  23100  μ_F = 0.803  μ_CR = 0.094
Generation  240: Best = 8.161379e+02  FES =  24100  μ_F = 0.815  μ_CR = 0.100
Generation  250: Best = 8.161379e+02  FES =  25100  μ_F = 0.773  μ_CR = 0.089
Generation  260: Best = 8.161379e+02  FES =  26100  μ_F = 0.760  μ_CR = 0.105
Generation  270: Best = 8.161379e+02  FES =  27100  μ_F = 0.672  μ_CR = 0.108
Generation  280: Best = 8.161379e+02  FES =  28100  μ_F = 0.665  μ_CR = 0.121
Generation  290: Best = 8.160343e+02  FES =  29100  μ_F = 0.598  μ_CR = 0.105
Generation  300: Best = 8.146015e+02  FES =  30100  μ_F = 0.607  μ_CR = 0.108
Generation  310: Best = 8.132678e+02  FES =  31100  μ_F = 0.699  μ_CR = 0.111
Generation  320: Best = 8.132678e+02  FES =  32100  μ_F = 0.744  μ_CR = 0.104
Generation  330: Best = 8.132678e+02  FES =  33100  μ_F = 0.839  μ_CR = 0.087
Generation  340: Best = 8.132678e+02  FES =  34100  μ_F = 0.818  μ_CR = 0.082
Generation  350: Best = 8.132678e+02  FES =  35100  μ_F = 0.845  μ_CR = 0.111
Generation  360: Best = 8.132678e+02  FES =  36100  μ_F = 0.836  μ_CR = 0.108
Generation  370: Best = 8.123978e+02  FES =  37100  μ_F = 0.745  μ_CR = 0.131
Generation  380: Best = 8.073138e+02  FES =  38100  μ_F = 0.711  μ_CR = 0.121
Generation  390: Best = 8.073138e+02  FES =  39100  μ_F = 0.684  μ_CR = 0.142
Generation  400: Best = 8.073138e+02  FES =  40100  μ_F = 0.731  μ_CR = 0.112
Generation  410: Best = 8.073138e+02  FES =  41100  μ_F = 0.692  μ_CR = 0.086
Generation  420: Best = 8.073138e+02  FES =  42100  μ_F = 0.613  μ_CR = 0.077
Generation  430: Best = 8.072612e+02  FES =  43100  μ_F = 0.657  μ_CR = 0.071
Generation  440: Best = 8.070810e+02  FES =  44100  μ_F = 0.705  μ_CR = 0.113
```

```
Generation  450: Best = 8.070810e+02  FES =  45100  μ_F = 0.735  μ_CR = 0.132
Generation  460: Best = 8.070810e+02  FES =  46100  μ_F = 0.662  μ_CR = 0.088
Generation  470: Best = 8.070810e+02  FES =  47100  μ_F = 0.668  μ_CR = 0.072
Generation  480: Best = 8.070810e+02  FES =  48100  μ_F = 0.632  μ_CR = 0.082
Generation  490: Best = 8.070810e+02  FES =  49100  μ_F = 0.584  μ_CR = 0.082
Generation  500: Best = 8.070810e+02  FES =  50100  μ_F = 0.647  μ_CR = 0.066
Generation  510: Best = 8.070810e+02  FES =  51100  μ_F = 0.692  μ_CR = 0.074
Generation  520: Best = 8.070810e+02  FES =  52100  μ_F = 0.750  μ_CR = 0.070
Generation  530: Best = 8.063230e+02  FES =  53100  μ_F = 0.727  μ_CR = 0.083
Generation  540: Best = 8.063230e+02  FES =  54100  μ_F = 0.796  μ_CR = 0.090
Generation  550: Best = 8.063230e+02  FES =  55100  μ_F = 0.766  μ_CR = 0.087
Generation  560: Best = 8.063230e+02  FES =  56100  μ_F = 0.758  μ_CR = 0.098
Generation  570: Best = 8.063230e+02  FES =  57100  μ_F = 0.741  μ_CR = 0.100
Generation  580: Best = 8.029021e+02  FES =  58100  μ_F = 0.794  μ_CR = 0.095
Generation  590: Best = 8.029021e+02  FES =  59100  μ_F = 0.753  μ_CR = 0.104
Generation  600: Best = 8.029021e+02  FES =  60100  μ_F = 0.758  μ_CR = 0.115
Generation  610: Best = 8.029021e+02  FES =  61100  μ_F = 0.785  μ_CR = 0.108
Generation  620: Best = 8.029021e+02  FES =  62100  μ_F = 0.796  μ_CR = 0.085
Generation  630: Best = 8.029021e+02  FES =  63100  μ_F = 0.723  μ_CR = 0.080
Generation  640: Best = 8.029021e+02  FES =  64100  μ_F = 0.761  μ_CR = 0.080
Generation  650: Best = 8.029021e+02  FES =  65100  μ_F = 0.718  μ_CR = 0.105
Generation  660: Best = 8.029021e+02  FES =  66100  μ_F = 0.729  μ_CR = 0.101
Generation  670: Best = 8.029021e+02  FES =  67100  μ_F = 0.746  μ_CR = 0.087
Generation  680: Best = 8.029021e+02  FES =  68100  μ_F = 0.737  μ_CR = 0.079
Generation  690: Best = 8.029021e+02  FES =  69100  μ_F = 0.777  μ_CR = 0.070
Generation  700: Best = 8.029021e+02  FES =  70100  μ_F = 0.761  μ_CR = 0.080
Generation  710: Best = 8.029021e+02  FES =  71100  μ_F = 0.679  μ_CR = 0.086
Generation  720: Best = 8.029021e+02  FES =  72100  μ_F = 0.739  μ_CR = 0.098
Generation  730: Best = 8.029021e+02  FES =  73100  μ_F = 0.744  μ_CR = 0.087
Generation  740: Best = 8.029021e+02  FES =  74100  μ_F = 0.740  μ_CR = 0.087
Generation  750: Best = 8.029021e+02  FES =  75100  μ_F = 0.644  μ_CR = 0.084
Generation  760: Best = 8.029021e+02  FES =  76100  μ_F = 0.641  μ_CR = 0.074
Generation  770: Best = 8.029021e+02  FES =  77100  μ_F = 0.621  μ_CR = 0.087
Generation  780: Best = 8.029021e+02  FES =  78100  μ_F = 0.621  μ_CR = 0.093
Generation  790: Best = 8.029021e+02  FES =  79100  μ_F = 0.675  μ_CR = 0.089
Generation  800: Best = 8.029021e+02  FES =  80100  μ_F = 0.624  μ_CR = 0.079
Generation  810: Best = 8.029021e+02  FES =  81100  μ_F = 0.592  μ_CR = 0.064
Generation  820: Best = 8.029021e+02  FES =  82100  μ_F = 0.581  μ_CR = 0.081
Generation  830: Best = 8.029021e+02  FES =  83100  μ_F = 0.638  μ_CR = 0.075
Generation  840: Best = 8.029021e+02  FES =  84100  μ_F = 0.616  μ_CR = 0.069
Generation  850: Best = 8.029021e+02  FES =  85100  μ_F = 0.645  μ_CR = 0.055
Generation  860: Best = 8.029021e+02  FES =  86100  μ_F = 0.555  μ_CR = 0.080
Generation  870: Best = 8.029021e+02  FES =  87100  μ_F = 0.569  μ_CR = 0.079
Generation  880: Best = 8.029021e+02  FES =  88100  μ_F = 0.479  μ_CR = 0.067
Generation  890: Best = 8.029021e+02  FES =  89100  μ_F = 0.476  μ_CR = 0.063
Generation  900: Best = 8.029021e+02  FES =  90100  μ_F = 0.472  μ_CR = 0.071
Generation  910: Best = 8.029021e+02  FES =  91100  μ_F = 0.591  μ_CR = 0.077
Generation  920: Best = 8.029021e+02  FES =  92100  μ_F = 0.549  μ_CR = 0.068
Generation  930: Best = 8.029021e+02  FES =  93100  μ_F = 0.573  μ_CR = 0.078
Generation  940: Best = 8.029021e+02  FES =  94100  μ_F = 0.516  μ_CR = 0.082
Generation  950: Best = 8.021592e+02  FES =  95100  μ_F = 0.522  μ_CR = 0.094
Generation  960: Best = 8.021592e+02  FES =  96100  μ_F = 0.391  μ_CR = 0.089
Generation  970: Best = 8.021592e+02  FES =  97100  μ_F = 0.445  μ_CR = 0.076
Generation  980: Best = 8.021592e+02  FES =  98100  μ_F = 0.413  μ_CR = 0.086
```

```
Generation  990: Best = 8.021592e+02  FES =  99100  μ_F = 0.431  μ_CR = 0.093

Optimization complete.
Best Fitness             = 8.021592e+02
Total Function Evals     = 100000
Total Successful Updates = 2609

================================================================================
=
Results:
  Best Fitness           : 8.021592e+02
  Global Optimum         : 8.000000e+02
  Error (|best-optimum|) : 2.159158e+00
  Best Solution (5 dim)  : [ 46.42679495 -66.82508997  32.78046477 -47.67163664
-34.83769159]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F8.png
```

In [18]:

```python
# ============================================================
# F9 — Shifted and Rotated Levy
# ============================================================
run_and_plot('F9', make_wrapper(simple.f9), 900.0, 'Shifted and Rotated Levy')
```

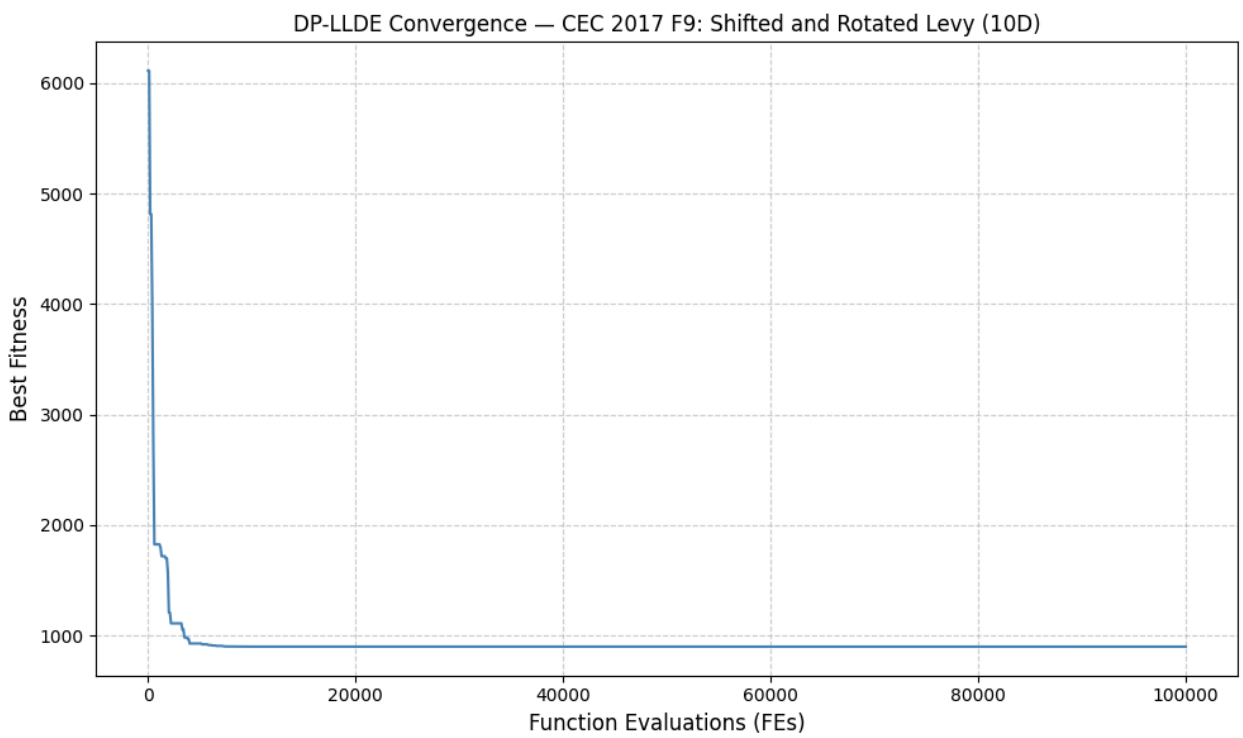
```
================================================================================
=
DP-LLDE on CEC 2017 F9 — Single Run
================================================================================
=
Function       : Shifted and Rotated Levy (10D)
Bounds         : [-100, 100]
Global optimum : 900.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 6.112557e+03
Generation   10: Best = 1.826023e+03  FES =   1100  μ_F = 0.580  μ_CR = 0.351
Generation   20: Best = 1.208896e+03  FES =   2100  μ_F = 0.629  μ_CR = 0.351
Generation   30: Best = 1.111376e+03  FES =   3100  μ_F = 0.654  μ_CR = 0.355
Generation   40: Best = 9.289044e+02  FES =   4100  μ_F = 0.699  μ_CR = 0.359
Generation   50: Best = 9.289044e+02  FES =   5100  μ_F = 0.734  μ_CR = 0.363
Generation   60: Best = 9.146935e+02  FES =   6100  μ_F = 0.757  μ_CR = 0.352
Generation   70: Best = 9.071511e+02  FES =   7100  μ_F = 0.776  μ_CR = 0.367
Generation   80: Best = 9.019421e+02  FES =   8100  μ_F = 0.776  μ_CR = 0.369
Generation   90: Best = 9.010530e+02  FES =   9100  μ_F = 0.777  μ_CR = 0.379
Generation  100: Best = 9.003551e+02  FES =  10100  μ_F = 0.783  μ_CR = 0.377
Generation  110: Best = 9.001822e+02  FES =  11100  μ_F = 0.787  μ_CR = 0.377
Generation  120: Best = 9.000804e+02  FES =  12100  μ_F = 0.778  μ_CR = 0.365
Generation  130: Best = 9.000326e+02  FES =  13100  μ_F = 0.780  μ_CR = 0.368
Generation  140: Best = 9.000056e+02  FES =  14100  μ_F = 0.784  μ_CR = 0.371
Generation  150: Best = 9.000034e+02  FES =  15100  μ_F = 0.792  μ_CR = 0.360
Generation  160: Best = 9.000008e+02  FES =  16100  μ_F = 0.789  μ_CR = 0.354
Generation  170: Best = 9.000002e+02  FES =  17100  μ_F = 0.786  μ_CR = 0.356
Generation  180: Best = 9.000001e+02  FES =  18100  μ_F = 0.788  μ_CR = 0.349
Generation  190: Best = 9.000000e+02  FES =  19100  μ_F = 0.791  μ_CR = 0.346
Generation  200: Best = 9.000000e+02  FES =  20100  μ_F = 0.795  μ_CR = 0.349
Generation  210: Best = 9.000000e+02  FES =  21100  μ_F = 0.796  μ_CR = 0.349
Generation  220: Best = 9.000000e+02  FES =  22100  μ_F = 0.799  μ_CR = 0.356
Generation  230: Best = 9.000000e+02  FES =  23100  μ_F = 0.817  μ_CR = 0.357
Generation  240: Best = 9.000000e+02  FES =  24100  μ_F = 0.835  μ_CR = 0.357
Generation  250: Best = 9.000000e+02  FES =  25100  μ_F = 0.830  μ_CR = 0.355
Generation  260: Best = 9.000000e+02  FES =  26100  μ_F = 0.832  μ_CR = 0.362
Generation  270: Best = 9.000000e+02  FES =  27100  μ_F = 0.831  μ_CR = 0.358
Generation  280: Best = 9.000000e+02  FES =  28100  μ_F = 0.831  μ_CR = 0.350
Generation  290: Best = 9.000000e+02  FES =  29100  μ_F = 0.829  μ_CR = 0.349
Generation  300: Best = 9.000000e+02  FES =  30100  μ_F = 0.826  μ_CR = 0.362
Generation  310: Best = 9.000000e+02  FES =  31100  μ_F = 0.828  μ_CR = 0.363
Generation  320: Best = 9.000000e+02  FES =  32100  μ_F = 0.817  μ_CR = 0.368
Generation  330: Best = 9.000000e+02  FES =  33100  μ_F = 0.826  μ_CR = 0.366
Generation  340: Best = 9.000000e+02  FES =  34100  μ_F = 0.832  μ_CR = 0.360
Generation  350: Best = 9.000000e+02  FES =  35100  μ_F = 0.843  μ_CR = 0.359
Generation  360: Best = 9.000000e+02  FES =  36100  μ_F = 0.838  μ_CR = 0.351
Generation  370: Best = 9.000000e+02  FES =  37100  μ_F = 0.841  μ_CR = 0.350
Generation  380: Best = 9.000000e+02  FES =  38100  μ_F = 0.848  μ_CR = 0.350
Generation  390: Best = 9.000000e+02  FES =  39100  μ_F = 0.836  μ_CR = 0.331
Generation  400: Best = 9.000000e+02  FES =  40100  μ_F = 0.839  μ_CR = 0.315
Generation  410: Best = 9.000000e+02  FES =  41100  μ_F = 0.855  μ_CR = 0.317
Generation  420: Best = 9.000000e+02  FES =  42100  μ_F = 0.866  μ_CR = 0.312
Generation  430: Best = 9.000000e+02  FES =  43100  μ_F = 0.866  μ_CR = 0.310
Generation  440: Best = 9.000000e+02  FES =  44100  μ_F = 0.865  μ_CR = 0.309
```

```
Generation  450: Best = 9.000000e+02  FES =  45100  μ_F = 0.867  μ_CR = 0.307
Generation  460: Best = 9.000000e+02  FES =  46100  μ_F = 0.873  μ_CR = 0.306
Generation  470: Best = 9.000000e+02  FES =  47100  μ_F = 0.872  μ_CR = 0.303
Generation  480: Best = 9.000000e+02  FES =  48100  μ_F = 0.876  μ_CR = 0.305
Generation  490: Best = 9.000000e+02  FES =  49100  μ_F = 0.877  μ_CR = 0.309
Generation  500: Best = 9.000000e+02  FES =  50100  μ_F = 0.878  μ_CR = 0.307
Generation  510: Best = 9.000000e+02  FES =  51100  μ_F = 0.872  μ_CR = 0.304
Generation  520: Best = 9.000000e+02  FES =  52100  μ_F = 0.864  μ_CR = 0.297
Generation  530: Best = 9.000000e+02  FES =  53100  μ_F = 0.867  μ_CR = 0.295
Generation  540: Best = 9.000000e+02  FES =  54100  μ_F = 0.867  μ_CR = 0.294
Generation  550: Best = 9.000000e+02  FES =  55100  μ_F = 0.866  μ_CR = 0.289
Generation  560: Best = 9.000000e+02  FES =  56100  μ_F = 0.869  μ_CR = 0.285
Generation  570: Best = 9.000000e+02  FES =  57100  μ_F = 0.872  μ_CR = 0.278
Generation  580: Best = 9.000000e+02  FES =  58100  μ_F = 0.867  μ_CR = 0.270
Generation  590: Best = 9.000000e+02  FES =  59100  μ_F = 0.874  μ_CR = 0.271
Generation  600: Best = 9.000000e+02  FES =  60100  μ_F = 0.875  μ_CR = 0.277
Generation  610: Best = 9.000000e+02  FES =  61100  μ_F = 0.874  μ_CR = 0.272
Generation  620: Best = 9.000000e+02  FES =  62100  μ_F = 0.875  μ_CR = 0.276
Generation  630: Best = 9.000000e+02  FES =  63100  μ_F = 0.865  μ_CR = 0.269
Generation  640: Best = 9.000000e+02  FES =  64100  μ_F = 0.865  μ_CR = 0.269
Generation  650: Best = 9.000000e+02  FES =  65100  μ_F = 0.866  μ_CR = 0.269
Generation  660: Best = 9.000000e+02  FES =  66100  μ_F = 0.867  μ_CR = 0.269
Generation  670: Best = 9.000000e+02  FES =  67100  μ_F = 0.876  μ_CR = 0.267
Generation  680: Best = 9.000000e+02  FES =  68100  μ_F = 0.879  μ_CR = 0.263
Generation  690: Best = 9.000000e+02  FES =  69100  μ_F = 0.877  μ_CR = 0.256
Generation  700: Best = 9.000000e+02  FES =  70100  μ_F = 0.876  μ_CR = 0.258
Generation  710: Best = 9.000000e+02  FES =  71100  μ_F = 0.873  μ_CR = 0.257
Generation  720: Best = 9.000000e+02  FES =  72100  μ_F = 0.873  μ_CR = 0.253
Generation  730: Best = 9.000000e+02  FES =  73100  μ_F = 0.875  μ_CR = 0.257
Generation  740: Best = 9.000000e+02  FES =  74100  μ_F = 0.874  μ_CR = 0.253
Generation  750: Best = 9.000000e+02  FES =  75100  μ_F = 0.881  μ_CR = 0.254
Generation  760: Best = 9.000000e+02  FES =  76100  μ_F = 0.878  μ_CR = 0.251
Generation  770: Best = 9.000000e+02  FES =  77100  μ_F = 0.871  μ_CR = 0.245
Generation  780: Best = 9.000000e+02  FES =  78100  μ_F = 0.870  μ_CR = 0.241
Generation  790: Best = 9.000000e+02  FES =  79100  μ_F = 0.874  μ_CR = 0.234
Generation  800: Best = 9.000000e+02  FES =  80100  μ_F = 0.878  μ_CR = 0.233
Generation  810: Best = 9.000000e+02  FES =  81100  μ_F = 0.876  μ_CR = 0.229
Generation  820: Best = 9.000000e+02  FES =  82100  μ_F = 0.879  μ_CR = 0.228
Generation  830: Best = 9.000000e+02  FES =  83100  μ_F = 0.876  μ_CR = 0.227
Generation  840: Best = 9.000000e+02  FES =  84100  μ_F = 0.874  μ_CR = 0.224
Generation  850: Best = 9.000000e+02  FES =  85100  μ_F = 0.874  μ_CR = 0.223
Generation  860: Best = 9.000000e+02  FES =  86100  μ_F = 0.875  μ_CR = 0.225
Generation  870: Best = 9.000000e+02  FES =  87100  μ_F = 0.876  μ_CR = 0.221
Generation  880: Best = 9.000000e+02  FES =  88100  μ_F = 0.874  μ_CR = 0.220
Generation  890: Best = 9.000000e+02  FES =  89100  μ_F = 0.872  μ_CR = 0.220
Generation  900: Best = 9.000000e+02  FES =  90100  μ_F = 0.869  μ_CR = 0.222
Generation  910: Best = 9.000000e+02  FES =  91100  μ_F = 0.872  μ_CR = 0.223
Generation  920: Best = 9.000000e+02  FES =  92100  μ_F = 0.872  μ_CR = 0.221
Generation  930: Best = 9.000000e+02  FES =  93100  μ_F = 0.874  μ_CR = 0.219
Generation  940: Best = 9.000000e+02  FES =  94100  μ_F = 0.877  μ_CR = 0.217
Generation  950: Best = 9.000000e+02  FES =  95100  μ_F = 0.878  μ_CR = 0.224
Generation  960: Best = 9.000000e+02  FES =  96100  μ_F = 0.881  μ_CR = 0.226
Generation  970: Best = 9.000000e+02  FES =  97100  μ_F = 0.877  μ_CR = 0.229
Generation  980: Best = 9.000000e+02  FES =  98100  μ_F = 0.884  μ_CR = 0.227
```

```
Generation  990: Best = 9.000000e+02  FES =  99100  μ_F = 0.881  μ_CR = 0.231

Optimization complete.
Best Fitness             = 9.000000e+02
Total Function Evals     = 100000
Total Successful Updates = 60927

================================================================================
=
Results:
  Best Fitness           : 9.000000e+02
  Global Optimum         : 9.000000e+02
  Error (|best-optimum|) : 0.000000e+00
  Best Solution (5 dim)  : [-23.59313971   0.12032404  20.8326407   52.54992609
-18.22787181]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F9.png
```

In [19]:
```
# ============================================================
# F10 — Shifted and Rotated Schwefel
# ============================================================
run_and_plot('F10', make_wrapper(simple.f10), 1000.0, "Shifted and Rotated Sch
```

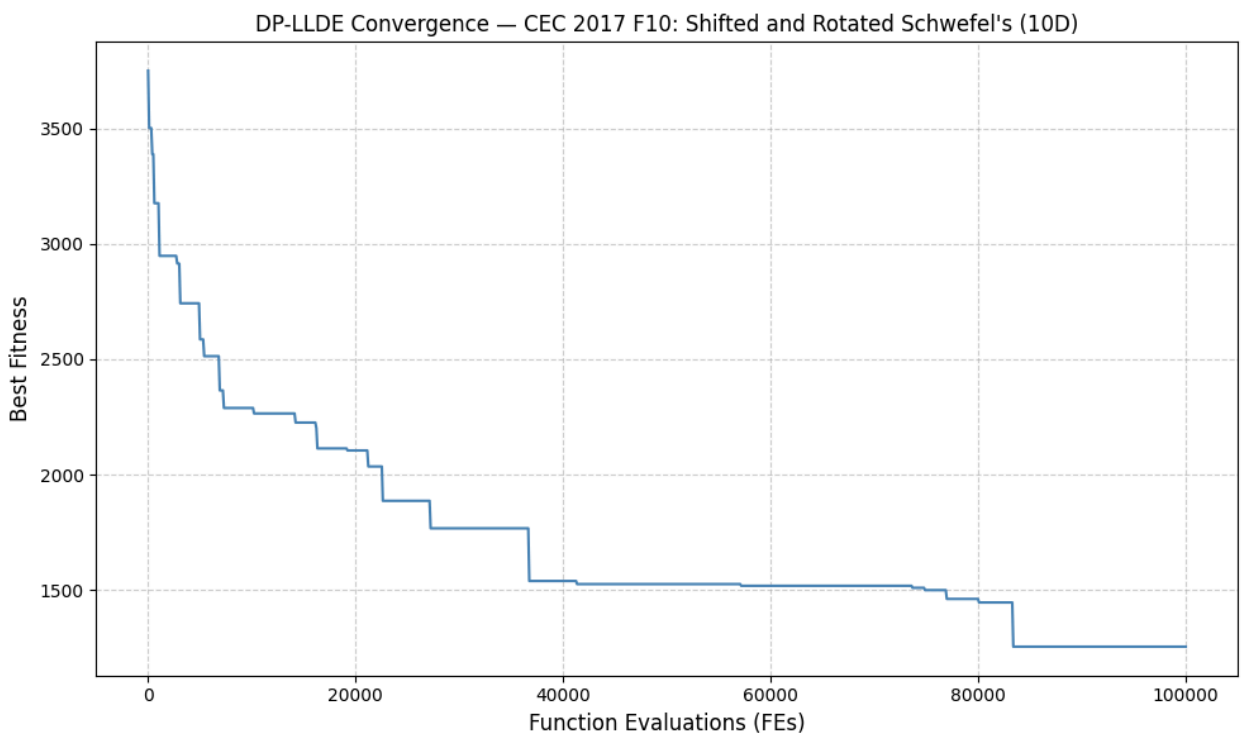
```
================================================================================
=
DP-LLDE on CEC 2017 F10 — Single Run
================================================================================
=
Function       : Shifted and Rotated Schwefel's (10D)
Bounds         : [-100, 100]
Global optimum : 1000.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 3.749402e+03
Generation   10: Best = 3.175657e+03  FES =   1100  μ_F = 0.596  μ_CR = 0.346
Generation   20: Best = 2.947616e+03  FES =   2100  μ_F = 0.658  μ_CR = 0.306
Generation   30: Best = 2.913715e+03  FES =   3100  μ_F = 0.701  μ_CR = 0.290
Generation   40: Best = 2.742290e+03  FES =   4100  μ_F = 0.741  μ_CR = 0.271
Generation   50: Best = 2.585973e+03  FES =   5100  μ_F = 0.766  μ_CR = 0.235
Generation   60: Best = 2.513276e+03  FES =   6100  μ_F = 0.790  μ_CR = 0.223
Generation   70: Best = 2.364959e+03  FES =   7100  μ_F = 0.808  μ_CR = 0.204
Generation   80: Best = 2.288892e+03  FES =   8100  μ_F = 0.817  μ_CR = 0.160
Generation   90: Best = 2.288892e+03  FES =   9100  μ_F = 0.807  μ_CR = 0.143
Generation  100: Best = 2.288892e+03  FES =  10100  μ_F = 0.813  μ_CR = 0.123
Generation  110: Best = 2.265185e+03  FES =  11100  μ_F = 0.829  μ_CR = 0.105
Generation  120: Best = 2.265185e+03  FES =  12100  μ_F = 0.820  μ_CR = 0.106
Generation  130: Best = 2.265185e+03  FES =  13100  μ_F = 0.850  μ_CR = 0.111
Generation  140: Best = 2.265185e+03  FES =  14100  μ_F = 0.859  μ_CR = 0.105
Generation  150: Best = 2.225682e+03  FES =  15100  μ_F = 0.869  μ_CR = 0.094
Generation  160: Best = 2.225682e+03  FES =  16100  μ_F = 0.814  μ_CR = 0.080
Generation  170: Best = 2.113754e+03  FES =  17100  μ_F = 0.846  μ_CR = 0.077
Generation  180: Best = 2.113754e+03  FES =  18100  μ_F = 0.824  μ_CR = 0.073
Generation  190: Best = 2.113754e+03  FES =  19100  μ_F = 0.783  μ_CR = 0.077
Generation  200: Best = 2.104982e+03  FES =  20100  μ_F = 0.838  μ_CR = 0.086
Generation  210: Best = 2.104982e+03  FES =  21100  μ_F = 0.843  μ_CR = 0.079
Generation  220: Best = 2.035576e+03  FES =  22100  μ_F = 0.851  μ_CR = 0.083
Generation  230: Best = 1.886799e+03  FES =  23100  μ_F = 0.802  μ_CR = 0.097
Generation  240: Best = 1.886799e+03  FES =  24100  μ_F = 0.765  μ_CR = 0.101
Generation  250: Best = 1.886799e+03  FES =  25100  μ_F = 0.817  μ_CR = 0.085
Generation  260: Best = 1.886799e+03  FES =  26100  μ_F = 0.837  μ_CR = 0.089
Generation  270: Best = 1.886799e+03  FES =  27100  μ_F = 0.801  μ_CR = 0.088
Generation  280: Best = 1.767751e+03  FES =  28100  μ_F = 0.758  μ_CR = 0.075
Generation  290: Best = 1.767751e+03  FES =  29100  μ_F = 0.801  μ_CR = 0.086
Generation  300: Best = 1.767751e+03  FES =  30100  μ_F = 0.838  μ_CR = 0.082
Generation  310: Best = 1.767751e+03  FES =  31100  μ_F = 0.848  μ_CR = 0.095
Generation  320: Best = 1.767751e+03  FES =  32100  μ_F = 0.851  μ_CR = 0.089
Generation  330: Best = 1.767751e+03  FES =  33100  μ_F = 0.874  μ_CR = 0.081
Generation  340: Best = 1.767751e+03  FES =  34100  μ_F = 0.838  μ_CR = 0.076
Generation  350: Best = 1.767751e+03  FES =  35100  μ_F = 0.850  μ_CR = 0.089
Generation  360: Best = 1.767751e+03  FES =  36100  μ_F = 0.795  μ_CR = 0.087
Generation  370: Best = 1.539752e+03  FES =  37100  μ_F = 0.719  μ_CR = 0.082
Generation  380: Best = 1.539752e+03  FES =  38100  μ_F = 0.755  μ_CR = 0.073
Generation  390: Best = 1.539752e+03  FES =  39100  μ_F = 0.757  μ_CR = 0.100
Generation  400: Best = 1.539752e+03  FES =  40100  μ_F = 0.794  μ_CR = 0.097
Generation  410: Best = 1.539752e+03  FES =  41100  μ_F = 0.847  μ_CR = 0.079
Generation  420: Best = 1.526298e+03  FES =  42100  μ_F = 0.819  μ_CR = 0.070
Generation  430: Best = 1.526298e+03  FES =  43100  μ_F = 0.847  μ_CR = 0.052
Generation  440: Best = 1.526298e+03  FES =  44100  μ_F = 0.858  μ_CR = 0.077
```

```
Generation  450: Best = 1.526298e+03  FES =  45100  μ_F = 0.846  μ_CR = 0.101
Generation  460: Best = 1.526298e+03  FES =  46100  μ_F = 0.794  μ_CR = 0.079
Generation  470: Best = 1.526298e+03  FES =  47100  μ_F = 0.789  μ_CR = 0.071
Generation  480: Best = 1.526298e+03  FES =  48100  μ_F = 0.789  μ_CR = 0.067
Generation  490: Best = 1.526298e+03  FES =  49100  μ_F = 0.785  μ_CR = 0.084
Generation  500: Best = 1.526298e+03  FES =  50100  μ_F = 0.719  μ_CR = 0.088
Generation  510: Best = 1.526298e+03  FES =  51100  μ_F = 0.675  μ_CR = 0.081
Generation  520: Best = 1.526298e+03  FES =  52100  μ_F = 0.734  μ_CR = 0.069
Generation  530: Best = 1.526298e+03  FES =  53100  μ_F = 0.759  μ_CR = 0.067
Generation  540: Best = 1.526298e+03  FES =  54100  μ_F = 0.728  μ_CR = 0.063
Generation  550: Best = 1.526298e+03  FES =  55100  μ_F = 0.607  μ_CR = 0.084
Generation  560: Best = 1.526298e+03  FES =  56100  μ_F = 0.667  μ_CR = 0.070
Generation  570: Best = 1.526298e+03  FES =  57100  μ_F = 0.679  μ_CR = 0.091
Generation  580: Best = 1.518799e+03  FES =  58100  μ_F = 0.709  μ_CR = 0.087
Generation  590: Best = 1.518799e+03  FES =  59100  μ_F = 0.674  μ_CR = 0.079
Generation  600: Best = 1.518799e+03  FES =  60100  μ_F = 0.674  μ_CR = 0.076
Generation  610: Best = 1.518799e+03  FES =  61100  μ_F = 0.679  μ_CR = 0.085
Generation  620: Best = 1.518799e+03  FES =  62100  μ_F = 0.650  μ_CR = 0.090
Generation  630: Best = 1.518799e+03  FES =  63100  μ_F = 0.676  μ_CR = 0.075
Generation  640: Best = 1.518799e+03  FES =  64100  μ_F = 0.693  μ_CR = 0.081
Generation  650: Best = 1.518799e+03  FES =  65100  μ_F = 0.756  μ_CR = 0.070
Generation  660: Best = 1.518799e+03  FES =  66100  μ_F = 0.749  μ_CR = 0.073
Generation  670: Best = 1.518799e+03  FES =  67100  μ_F = 0.673  μ_CR = 0.072
Generation  680: Best = 1.518799e+03  FES =  68100  μ_F = 0.612  μ_CR = 0.062
Generation  690: Best = 1.518799e+03  FES =  69100  μ_F = 0.580  μ_CR = 0.068
Generation  700: Best = 1.518799e+03  FES =  70100  μ_F = 0.502  μ_CR = 0.070
Generation  710: Best = 1.518799e+03  FES =  71100  μ_F = 0.553  μ_CR = 0.085
Generation  720: Best = 1.518799e+03  FES =  72100  μ_F = 0.642  μ_CR = 0.086
Generation  730: Best = 1.518799e+03  FES =  73100  μ_F = 0.668  μ_CR = 0.076
Generation  740: Best = 1.510132e+03  FES =  74100  μ_F = 0.697  μ_CR = 0.072
Generation  750: Best = 1.500124e+03  FES =  75100  μ_F = 0.777  μ_CR = 0.067
Generation  760: Best = 1.500124e+03  FES =  76100  μ_F = 0.694  μ_CR = 0.076
Generation  770: Best = 1.461971e+03  FES =  77100  μ_F = 0.688  μ_CR = 0.074
Generation  780: Best = 1.461971e+03  FES =  78100  μ_F = 0.649  μ_CR = 0.061
Generation  790: Best = 1.461971e+03  FES =  79100  μ_F = 0.647  μ_CR = 0.056
Generation  800: Best = 1.446690e+03  FES =  80100  μ_F = 0.648  μ_CR = 0.056
Generation  810: Best = 1.446690e+03  FES =  81100  μ_F = 0.696  μ_CR = 0.061
Generation  820: Best = 1.446690e+03  FES =  82100  μ_F = 0.687  μ_CR = 0.065
Generation  830: Best = 1.446690e+03  FES =  83100  μ_F = 0.580  μ_CR = 0.086
Generation  840: Best = 1.255191e+03  FES =  84100  μ_F = 0.609  μ_CR = 0.087
Generation  850: Best = 1.255191e+03  FES =  85100  μ_F = 0.639  μ_CR = 0.105
Generation  860: Best = 1.255191e+03  FES =  86100  μ_F = 0.671  μ_CR = 0.129
Generation  870: Best = 1.255191e+03  FES =  87100  μ_F = 0.602  μ_CR = 0.129
Generation  880: Best = 1.255191e+03  FES =  88100  μ_F = 0.570  μ_CR = 0.115
Generation  890: Best = 1.255191e+03  FES =  89100  μ_F = 0.554  μ_CR = 0.127
Generation  900: Best = 1.255191e+03  FES =  90100  μ_F = 0.567  μ_CR = 0.115
Generation  910: Best = 1.255191e+03  FES =  91100  μ_F = 0.557  μ_CR = 0.117
Generation  920: Best = 1.255191e+03  FES =  92100  μ_F = 0.558  μ_CR = 0.098
Generation  930: Best = 1.255191e+03  FES =  93100  μ_F = 0.588  μ_CR = 0.090
Generation  940: Best = 1.255191e+03  FES =  94100  μ_F = 0.598  μ_CR = 0.080
Generation  950: Best = 1.255191e+03  FES =  95100  μ_F = 0.682  μ_CR = 0.080
Generation  960: Best = 1.255191e+03  FES =  96100  μ_F = 0.719  μ_CR = 0.068
Generation  970: Best = 1.255191e+03  FES =  97100  μ_F = 0.665  μ_CR = 0.059
Generation  980: Best = 1.255191e+03  FES =  98100  μ_F = 0.637  μ_CR = 0.082
```

```
Generation  990: Best = 1.255191e+03  FES =  99100  μ_F = 0.650  μ_CR = 0.087

Optimization complete.
Best Fitness             = 1.255191e+03
Total Function Evals     = 100000
Total Successful Updates = 1940

================================================================================
=
Results:
  Best Fitness           : 1.255191e+03
  Global Optimum         : 1.000000e+03
  Error (|best-optimum|) : 2.551913e+02
  Best Solution (5 dim)  : [-43.15355658 -26.57819885 -15.99487061  66.88277793
64.55001444]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F10.png
```

6.3 Hybrid Functions (F11-F20)

In [20]:
```
# ============================================================
# F11 — Hybrid Function 1 (Zakharov; Rosenbrock; Rastrigin)
# ============================================================
run_and_plot('F11', make_wrapper(hybrid.f11), 1100.0, 'Hybrid: Zakharov; Roser
```

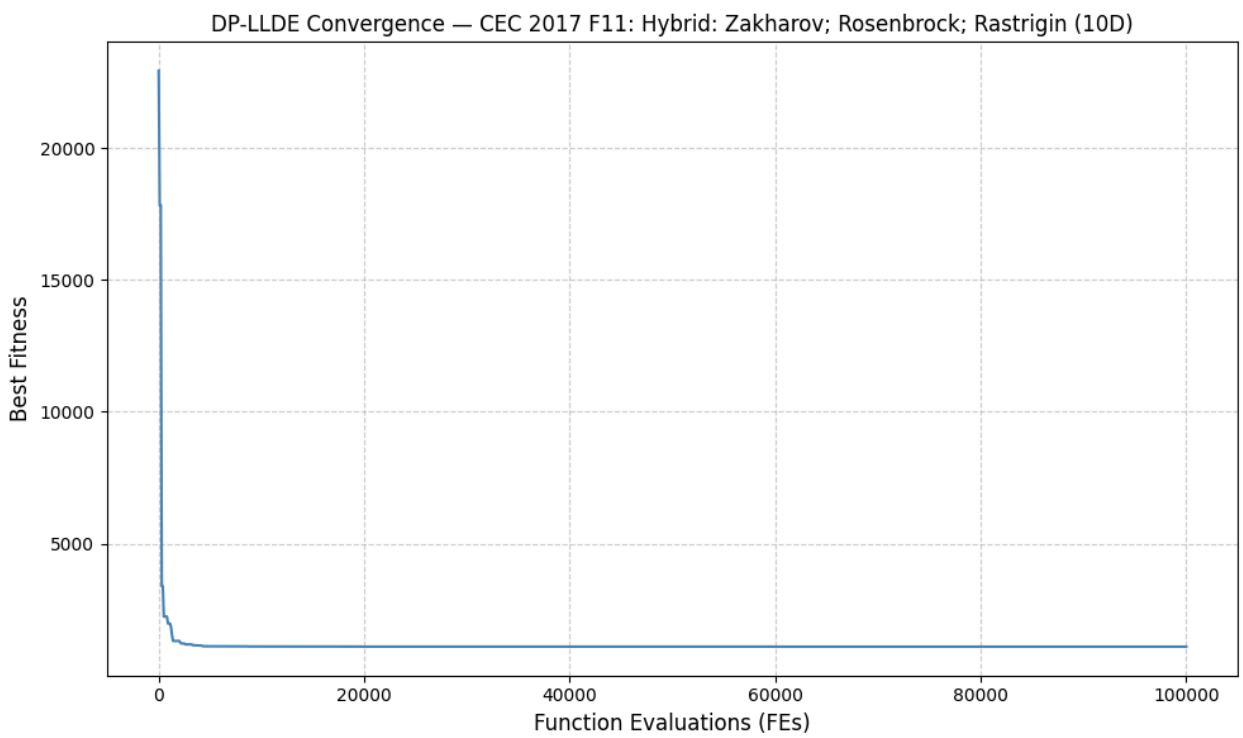
```
================================================================================
=
DP-LLDE on CEC 2017 F11 — Single Run
================================================================================
=
Function       : Hybrid: Zakharov; Rosenbrock; Rastrigin (10D)
Bounds         : [-100, 100]
Global optimum : 1100.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 2.292232e+04
Generation   10: Best = 1.980741e+03  FES =   1100  μ_F = 0.575  μ_CR = 0.350
Generation   20: Best = 1.320330e+03  FES =   2100  μ_F = 0.630  μ_CR = 0.361
Generation   30: Best = 1.189729e+03  FES =   3100  μ_F = 0.679  μ_CR = 0.370
Generation   40: Best = 1.149063e+03  FES =   4100  μ_F = 0.710  μ_CR = 0.369
Generation   50: Best = 1.119630e+03  FES =   5100  μ_F = 0.729  μ_CR = 0.371
Generation   60: Best = 1.119630e+03  FES =   6100  μ_F = 0.725  μ_CR = 0.371
Generation   70: Best = 1.118005e+03  FES =   7100  μ_F = 0.740  μ_CR = 0.373
Generation   80: Best = 1.114734e+03  FES =   8100  μ_F = 0.774  μ_CR = 0.372
Generation   90: Best = 1.110838e+03  FES =   9100  μ_F = 0.782  μ_CR = 0.348
Generation  100: Best = 1.110838e+03  FES =  10100  μ_F = 0.813  μ_CR = 0.333
Generation  110: Best = 1.110838e+03  FES =  11100  μ_F = 0.785  μ_CR = 0.303
Generation  120: Best = 1.109125e+03  FES =  12100  μ_F = 0.799  μ_CR = 0.294
Generation  130: Best = 1.107906e+03  FES =  13100  μ_F = 0.823  μ_CR = 0.269
Generation  140: Best = 1.107906e+03  FES =  14100  μ_F = 0.819  μ_CR = 0.234
Generation  150: Best = 1.107906e+03  FES =  15100  μ_F = 0.825  μ_CR = 0.186
Generation  160: Best = 1.107601e+03  FES =  16100  μ_F = 0.832  μ_CR = 0.168
Generation  170: Best = 1.107266e+03  FES =  17100  μ_F = 0.854  μ_CR = 0.154
Generation  180: Best = 1.107266e+03  FES =  18100  μ_F = 0.854  μ_CR = 0.147
Generation  190: Best = 1.107202e+03  FES =  19100  μ_F = 0.869  μ_CR = 0.127
Generation  200: Best = 1.104523e+03  FES =  20100  μ_F = 0.880  μ_CR = 0.133
Generation  210: Best = 1.104523e+03  FES =  21100  μ_F = 0.874  μ_CR = 0.121
Generation  220: Best = 1.104420e+03  FES =  22100  μ_F = 0.878  μ_CR = 0.120
Generation  230: Best = 1.104312e+03  FES =  23100  μ_F = 0.876  μ_CR = 0.116
Generation  240: Best = 1.104312e+03  FES =  24100  μ_F = 0.867  μ_CR = 0.120
Generation  250: Best = 1.104312e+03  FES =  25100  μ_F = 0.859  μ_CR = 0.098
Generation  260: Best = 1.104312e+03  FES =  26100  μ_F = 0.824  μ_CR = 0.107
Generation  270: Best = 1.104312e+03  FES =  27100  μ_F = 0.811  μ_CR = 0.118
Generation  280: Best = 1.104312e+03  FES =  28100  μ_F = 0.838  μ_CR = 0.116
Generation  290: Best = 1.104241e+03  FES =  29100  μ_F = 0.864  μ_CR = 0.118
Generation  300: Best = 1.104241e+03  FES =  30100  μ_F = 0.848  μ_CR = 0.137
Generation  310: Best = 1.104241e+03  FES =  31100  μ_F = 0.846  μ_CR = 0.123
Generation  320: Best = 1.104241e+03  FES =  32100  μ_F = 0.854  μ_CR = 0.113
Generation  330: Best = 1.104241e+03  FES =  33100  μ_F = 0.840  μ_CR = 0.095
Generation  340: Best = 1.104241e+03  FES =  34100  μ_F = 0.807  μ_CR = 0.097
Generation  350: Best = 1.104235e+03  FES =  35100  μ_F = 0.801  μ_CR = 0.092
Generation  360: Best = 1.104235e+03  FES =  36100  μ_F = 0.805  μ_CR = 0.098
Generation  370: Best = 1.104235e+03  FES =  37100  μ_F = 0.819  μ_CR = 0.111
Generation  380: Best = 1.104235e+03  FES =  38100  μ_F = 0.815  μ_CR = 0.099
Generation  390: Best = 1.104071e+03  FES =  39100  μ_F = 0.788  μ_CR = 0.093
Generation  400: Best = 1.104071e+03  FES =  40100  μ_F = 0.806  μ_CR = 0.103
Generation  410: Best = 1.104071e+03  FES =  41100  μ_F = 0.824  μ_CR = 0.113
Generation  420: Best = 1.103953e+03  FES =  42100  μ_F = 0.792  μ_CR = 0.112
Generation  430: Best = 1.103953e+03  FES =  43100  μ_F = 0.845  μ_CR = 0.089
Generation  440: Best = 1.103953e+03  FES =  44100  μ_F = 0.821  μ_CR = 0.094
```

```
Generation  450: Best = 1.103953e+03  FES =  45100  μ_F = 0.856  μ_CR = 0.088
Generation  460: Best = 1.103953e+03  FES =  46100  μ_F = 0.834  μ_CR = 0.115
Generation  470: Best = 1.103953e+03  FES =  47100  μ_F = 0.842  μ_CR = 0.102
Generation  480: Best = 1.103953e+03  FES =  48100  μ_F = 0.827  μ_CR = 0.092
Generation  490: Best = 1.103953e+03  FES =  49100  μ_F = 0.858  μ_CR = 0.098
Generation  500: Best = 1.103926e+03  FES =  50100  μ_F = 0.854  μ_CR = 0.114
Generation  510: Best = 1.103899e+03  FES =  51100  μ_F = 0.862  μ_CR = 0.119
Generation  520: Best = 1.103892e+03  FES =  52100  μ_F = 0.840  μ_CR = 0.115
Generation  530: Best = 1.103886e+03  FES =  53100  μ_F = 0.820  μ_CR = 0.114
Generation  540: Best = 1.103556e+03  FES =  54100  μ_F = 0.813  μ_CR = 0.096
Generation  550: Best = 1.103553e+03  FES =  55100  μ_F = 0.809  μ_CR = 0.099
Generation  560: Best = 1.103553e+03  FES =  56100  μ_F = 0.838  μ_CR = 0.104
Generation  570: Best = 1.103544e+03  FES =  57100  μ_F = 0.797  μ_CR = 0.110
Generation  580: Best = 1.103450e+03  FES =  58100  μ_F = 0.801  μ_CR = 0.095
Generation  590: Best = 1.103450e+03  FES =  59100  μ_F = 0.823  μ_CR = 0.093
Generation  600: Best = 1.103450e+03  FES =  60100  μ_F = 0.790  μ_CR = 0.158
Generation  610: Best = 1.103446e+03  FES =  61100  μ_F = 0.808  μ_CR = 0.146
Generation  620: Best = 1.103446e+03  FES =  62100  μ_F = 0.832  μ_CR = 0.129
Generation  630: Best = 1.103446e+03  FES =  63100  μ_F = 0.837  μ_CR = 0.110
Generation  640: Best = 1.103443e+03  FES =  64100  μ_F = 0.823  μ_CR = 0.122
Generation  650: Best = 1.103443e+03  FES =  65100  μ_F = 0.818  μ_CR = 0.117
Generation  660: Best = 1.103443e+03  FES =  66100  μ_F = 0.815  μ_CR = 0.096
Generation  670: Best = 1.103443e+03  FES =  67100  μ_F = 0.843  μ_CR = 0.099
Generation  680: Best = 1.103443e+03  FES =  68100  μ_F = 0.876  μ_CR = 0.109
Generation  690: Best = 1.103365e+03  FES =  69100  μ_F = 0.855  μ_CR = 0.104
Generation  700: Best = 1.103365e+03  FES =  70100  μ_F = 0.861  μ_CR = 0.103
Generation  710: Best = 1.103327e+03  FES =  71100  μ_F = 0.860  μ_CR = 0.097
Generation  720: Best = 1.103327e+03  FES =  72100  μ_F = 0.857  μ_CR = 0.081
Generation  730: Best = 1.103327e+03  FES =  73100  μ_F = 0.820  μ_CR = 0.098
Generation  740: Best = 1.103327e+03  FES =  74100  μ_F = 0.785  μ_CR = 0.115
Generation  750: Best = 1.103325e+03  FES =  75100  μ_F = 0.806  μ_CR = 0.111
Generation  760: Best = 1.103324e+03  FES =  76100  μ_F = 0.814  μ_CR = 0.105
Generation  770: Best = 1.103324e+03  FES =  77100  μ_F = 0.829  μ_CR = 0.101
Generation  780: Best = 1.103301e+03  FES =  78100  μ_F = 0.801  μ_CR = 0.096
Generation  790: Best = 1.103301e+03  FES =  79100  μ_F = 0.835  μ_CR = 0.092
Generation  800: Best = 1.103301e+03  FES =  80100  μ_F = 0.816  μ_CR = 0.089
Generation  810: Best = 1.103301e+03  FES =  81100  μ_F = 0.817  μ_CR = 0.078
Generation  820: Best = 1.103301e+03  FES =  82100  μ_F = 0.812  μ_CR = 0.091
Generation  830: Best = 1.103301e+03  FES =  83100  μ_F = 0.788  μ_CR = 0.090
Generation  840: Best = 1.103301e+03  FES =  84100  μ_F = 0.836  μ_CR = 0.109
Generation  850: Best = 1.103301e+03  FES =  85100  μ_F = 0.849  μ_CR = 0.101
Generation  860: Best = 1.103301e+03  FES =  86100  μ_F = 0.829  μ_CR = 0.088
Generation  870: Best = 1.103140e+03  FES =  87100  μ_F = 0.877  μ_CR = 0.098
Generation  880: Best = 1.103140e+03  FES =  88100  μ_F = 0.870  μ_CR = 0.111
Generation  890: Best = 1.103131e+03  FES =  89100  μ_F = 0.864  μ_CR = 0.122
Generation  900: Best = 1.103131e+03  FES =  90100  μ_F = 0.824  μ_CR = 0.112
Generation  910: Best = 1.103131e+03  FES =  91100  μ_F = 0.763  μ_CR = 0.110
Generation  920: Best = 1.103131e+03  FES =  92100  μ_F = 0.768  μ_CR = 0.110
Generation  930: Best = 1.103097e+03  FES =  93100  μ_F = 0.798  μ_CR = 0.094
Generation  940: Best = 1.103097e+03  FES =  94100  μ_F = 0.825  μ_CR = 0.098
Generation  950: Best = 1.103097e+03  FES =  95100  μ_F = 0.829  μ_CR = 0.096
Generation  960: Best = 1.103097e+03  FES =  96100  μ_F = 0.815  μ_CR = 0.095
Generation  970: Best = 1.103088e+03  FES =  97100  μ_F = 0.818  μ_CR = 0.102
Generation  980: Best = 1.103083e+03  FES =  98100  μ_F = 0.810  μ_CR = 0.094
```

```
Generation  990: Best = 1.103074e+03  FES =  99100  μ_F = 0.804  μ_CR = 0.087

Optimization complete.
Best Fitness             = 1.103074e+03
Total Function Evals     = 100000
Total Successful Updates = 6374

================================================================================
=
Results:
  Best Fitness           : 1.103074e+03
  Global Optimum         : 1.100000e+03
  Error (|best-optimum|) : 3.073868e+00
  Best Solution (5 dim)  : [-16.804931   -42.74588947 -58.60748467  16.98714206
19.51988077]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F11.png
```

In [21]:

```
# ============================================================
# F12 — Hybrid Function 2 (Elliptic; Schwefel; Bent Cigar)
# ============================================================
run_and_plot('F12', make_wrapper(hybrid.f12), 1200.0, 'Hybrid: Elliptic; Schwe
```

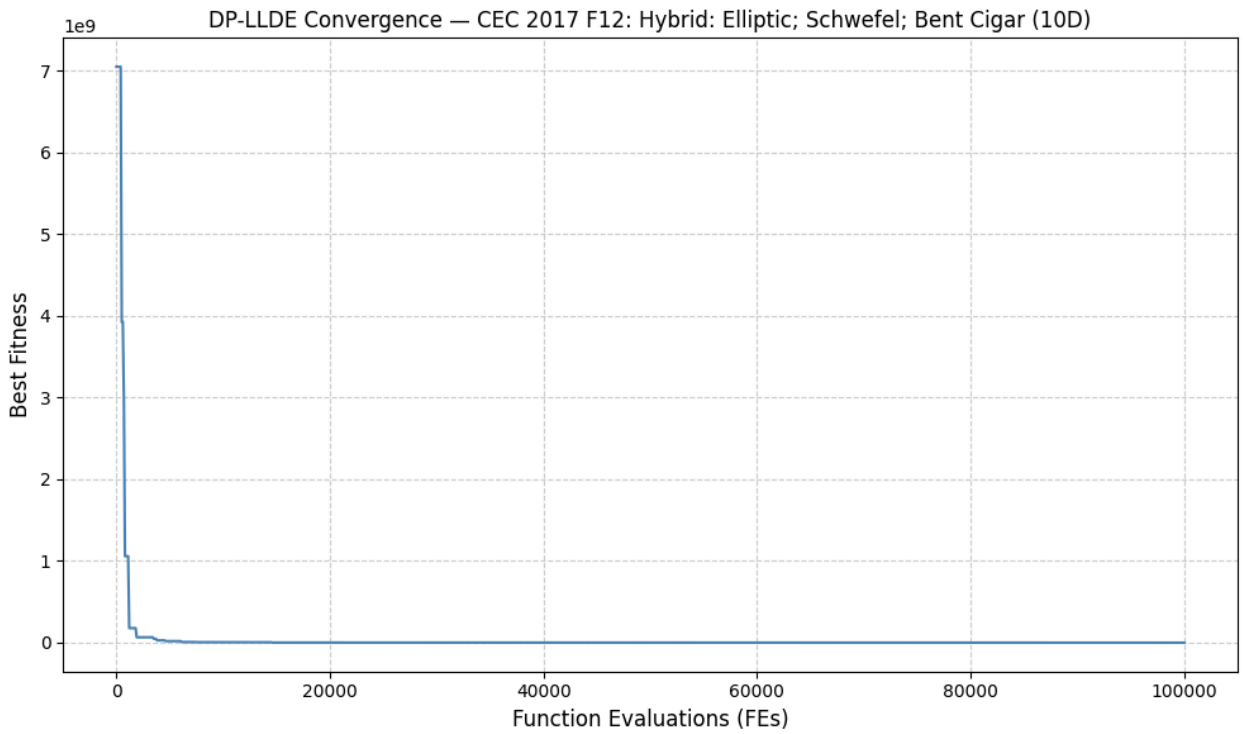
```
================================================================================
=
DP-LLDE on CEC 2017 F12 — Single Run
================================================================================
=
Function       : Hybrid: Elliptic; Schwefel; Bent Cigar (10D)
Bounds         : [-100, 100]
Global optimum : 1200.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 7.052719e+09
Generation   10: Best = 1.059061e+09  FES =   1100  μ_F = 0.565  μ_CR = 0.353
Generation   20: Best = 6.608687e+07  FES =   2100  μ_F = 0.631  μ_CR = 0.360
Generation   30: Best = 6.608687e+07  FES =   3100  μ_F = 0.679  μ_CR = 0.374
Generation   40: Best = 2.843259e+07  FES =   4100  μ_F = 0.703  μ_CR = 0.347
Generation   50: Best = 1.915405e+07  FES =   5100  μ_F = 0.726  μ_CR = 0.371
Generation   60: Best = 1.915405e+07  FES =   6100  μ_F = 0.725  μ_CR = 0.353
Generation   70: Best = 8.275568e+06  FES =   7100  μ_F = 0.759  μ_CR = 0.346
Generation   80: Best = 6.104102e+06  FES =   8100  μ_F = 0.795  μ_CR = 0.333
Generation   90: Best = 6.104102e+06  FES =   9100  μ_F = 0.793  μ_CR = 0.357
Generation  100: Best = 5.547196e+06  FES =  10100  μ_F = 0.839  μ_CR = 0.328
Generation  110: Best = 5.547196e+06  FES =  11100  μ_F = 0.829  μ_CR = 0.323
Generation  120: Best = 5.547196e+06  FES =  12100  μ_F = 0.824  μ_CR = 0.275
Generation  130: Best = 4.811944e+06  FES =  13100  μ_F = 0.795  μ_CR = 0.204
Generation  140: Best = 4.811944e+06  FES =  14100  μ_F = 0.811  μ_CR = 0.198
Generation  150: Best = 1.547158e+06  FES =  15100  μ_F = 0.829  μ_CR = 0.159
Generation  160: Best = 1.547158e+06  FES =  16100  μ_F = 0.820  μ_CR = 0.126
Generation  170: Best = 1.547158e+06  FES =  17100  μ_F = 0.836  μ_CR = 0.117
Generation  180: Best = 1.165061e+06  FES =  18100  μ_F = 0.861  μ_CR = 0.133
Generation  190: Best = 1.164651e+06  FES =  19100  μ_F = 0.863  μ_CR = 0.124
Generation  200: Best = 8.449174e+05  FES =  20100  μ_F = 0.855  μ_CR = 0.123
Generation  210: Best = 6.056583e+05  FES =  21100  μ_F = 0.829  μ_CR = 0.127
Generation  220: Best = 6.052765e+05  FES =  22100  μ_F = 0.810  μ_CR = 0.139
Generation  230: Best = 2.240713e+05  FES =  23100  μ_F = 0.840  μ_CR = 0.123
Generation  240: Best = 2.239475e+05  FES =  24100  μ_F = 0.873  μ_CR = 0.137
Generation  250: Best = 2.239475e+05  FES =  25100  μ_F = 0.850  μ_CR = 0.122
Generation  260: Best = 2.239475e+05  FES =  26100  μ_F = 0.847  μ_CR = 0.125
Generation  270: Best = 2.239475e+05  FES =  27100  μ_F = 0.864  μ_CR = 0.123
Generation  280: Best = 2.239475e+05  FES =  28100  μ_F = 0.834  μ_CR = 0.128
Generation  290: Best = 2.239475e+05  FES =  29100  μ_F = 0.824  μ_CR = 0.093
Generation  300: Best = 2.239475e+05  FES =  30100  μ_F = 0.848  μ_CR = 0.100
Generation  310: Best = 2.239475e+05  FES =  31100  μ_F = 0.884  μ_CR = 0.132
Generation  320: Best = 2.239475e+05  FES =  32100  μ_F = 0.856  μ_CR = 0.149
Generation  330: Best = 2.239475e+05  FES =  33100  μ_F = 0.868  μ_CR = 0.137
Generation  340: Best = 2.239475e+05  FES =  34100  μ_F = 0.794  μ_CR = 0.137
Generation  350: Best = 1.599721e+05  FES =  35100  μ_F = 0.807  μ_CR = 0.134
Generation  360: Best = 1.589212e+05  FES =  36100  μ_F = 0.766  μ_CR = 0.113
Generation  370: Best = 1.589212e+05  FES =  37100  μ_F = 0.765  μ_CR = 0.121
Generation  380: Best = 1.589212e+05  FES =  38100  μ_F = 0.789  μ_CR = 0.120
Generation  390: Best = 1.589212e+05  FES =  39100  μ_F = 0.800  μ_CR = 0.131
Generation  400: Best = 7.998984e+04  FES =  40100  μ_F = 0.792  μ_CR = 0.145
Generation  410: Best = 7.998984e+04  FES =  41100  μ_F = 0.813  μ_CR = 0.126
Generation  420: Best = 7.972512e+04  FES =  42100  μ_F = 0.831  μ_CR = 0.132
Generation  430: Best = 7.972512e+04  FES =  43100  μ_F = 0.839  μ_CR = 0.139
Generation  440: Best = 7.972512e+04  FES =  44100  μ_F = 0.871  μ_CR = 0.147
```

```
Generation  450: Best = 7.966353e+04  FES =  45100  μ_F = 0.881  μ_CR = 0.138
Generation  460: Best = 7.966353e+04  FES =  46100  μ_F = 0.887  μ_CR = 0.143
Generation  470: Best = 7.932045e+04  FES =  47100  μ_F = 0.903  μ_CR = 0.141
Generation  480: Best = 7.932045e+04  FES =  48100  μ_F = 0.932  μ_CR = 0.117
Generation  490: Best = 7.932045e+04  FES =  49100  μ_F = 0.875  μ_CR = 0.140
Generation  500: Best = 7.932045e+04  FES =  50100  μ_F = 0.847  μ_CR = 0.156
Generation  510: Best = 7.932045e+04  FES =  51100  μ_F = 0.819  μ_CR = 0.146
Generation  520: Best = 7.932045e+04  FES =  52100  μ_F = 0.792  μ_CR = 0.132
Generation  530: Best = 7.932045e+04  FES =  53100  μ_F = 0.774  μ_CR = 0.105
Generation  540: Best = 5.649964e+04  FES =  54100  μ_F = 0.792  μ_CR = 0.117
Generation  550: Best = 5.649964e+04  FES =  55100  μ_F = 0.763  μ_CR = 0.116
Generation  560: Best = 5.649964e+04  FES =  56100  μ_F = 0.827  μ_CR = 0.108
Generation  570: Best = 5.649964e+04  FES =  57100  μ_F = 0.833  μ_CR = 0.115
Generation  580: Best = 5.649964e+04  FES =  58100  μ_F = 0.850  μ_CR = 0.111
Generation  590: Best = 5.649964e+04  FES =  59100  μ_F = 0.857  μ_CR = 0.100
Generation  600: Best = 5.649964e+04  FES =  60100  μ_F = 0.827  μ_CR = 0.140
Generation  610: Best = 5.649964e+04  FES =  61100  μ_F = 0.776  μ_CR = 0.165
Generation  620: Best = 5.649964e+04  FES =  62100  μ_F = 0.751  μ_CR = 0.163
Generation  630: Best = 5.649964e+04  FES =  63100  μ_F = 0.784  μ_CR = 0.147
Generation  640: Best = 5.649964e+04  FES =  64100  μ_F = 0.784  μ_CR = 0.169
Generation  650: Best = 5.649964e+04  FES =  65100  μ_F = 0.834  μ_CR = 0.154
Generation  660: Best = 5.649964e+04  FES =  66100  μ_F = 0.850  μ_CR = 0.150
Generation  670: Best = 5.649964e+04  FES =  67100  μ_F = 0.825  μ_CR = 0.122
Generation  680: Best = 5.649964e+04  FES =  68100  μ_F = 0.854  μ_CR = 0.099
Generation  690: Best = 5.333209e+04  FES =  69100  μ_F = 0.831  μ_CR = 0.100
Generation  700: Best = 5.333209e+04  FES =  70100  μ_F = 0.826  μ_CR = 0.097
Generation  710: Best = 5.333209e+04  FES =  71100  μ_F = 0.757  μ_CR = 0.091
Generation  720: Best = 2.844138e+04  FES =  72100  μ_F = 0.667  μ_CR = 0.086
Generation  730: Best = 2.844138e+04  FES =  73100  μ_F = 0.600  μ_CR = 0.108
Generation  740: Best = 2.844138e+04  FES =  74100  μ_F = 0.655  μ_CR = 0.097
Generation  750: Best = 2.844138e+04  FES =  75100  μ_F = 0.675  μ_CR = 0.132
Generation  760: Best = 2.844138e+04  FES =  76100  μ_F = 0.683  μ_CR = 0.123
Generation  770: Best = 2.696595e+04  FES =  77100  μ_F = 0.788  μ_CR = 0.122
Generation  780: Best = 2.696595e+04  FES =  78100  μ_F = 0.769  μ_CR = 0.144
Generation  790: Best = 2.696595e+04  FES =  79100  μ_F = 0.782  μ_CR = 0.189
Generation  800: Best = 2.696595e+04  FES =  80100  μ_F = 0.833  μ_CR = 0.214
Generation  810: Best = 2.696595e+04  FES =  81100  μ_F = 0.822  μ_CR = 0.269
Generation  820: Best = 2.696595e+04  FES =  82100  μ_F = 0.769  μ_CR = 0.269
Generation  830: Best = 2.696595e+04  FES =  83100  μ_F = 0.774  μ_CR = 0.286
Generation  840: Best = 2.696595e+04  FES =  84100  μ_F = 0.745  μ_CR = 0.260
Generation  850: Best = 2.696595e+04  FES =  85100  μ_F = 0.746  μ_CR = 0.220
Generation  860: Best = 2.696595e+04  FES =  86100  μ_F = 0.748  μ_CR = 0.190
Generation  870: Best = 2.696595e+04  FES =  87100  μ_F = 0.732  μ_CR = 0.173
Generation  880: Best = 2.696595e+04  FES =  88100  μ_F = 0.760  μ_CR = 0.137
Generation  890: Best = 2.696595e+04  FES =  89100  μ_F = 0.761  μ_CR = 0.112
Generation  900: Best = 2.696595e+04  FES =  90100  μ_F = 0.764  μ_CR = 0.100
Generation  910: Best = 2.696595e+04  FES =  91100  μ_F = 0.767  μ_CR = 0.082
Generation  920: Best = 2.696595e+04  FES =  92100  μ_F = 0.789  μ_CR = 0.108
Generation  930: Best = 2.696595e+04  FES =  93100  μ_F = 0.825  μ_CR = 0.086
Generation  940: Best = 2.696595e+04  FES =  94100  μ_F = 0.798  μ_CR = 0.090
Generation  950: Best = 2.323369e+04  FES =  95100  μ_F = 0.844  μ_CR = 0.123
Generation  960: Best = 2.323369e+04  FES =  96100  μ_F = 0.759  μ_CR = 0.134
Generation  970: Best = 2.235439e+04  FES =  97100  μ_F = 0.755  μ_CR = 0.159
Generation  980: Best = 2.235439e+04  FES =  98100  μ_F = 0.772  μ_CR = 0.194
```

```
Generation  990: Best = 2.235439e+04  FES =  99100  μ_F = 0.771  μ_CR = 0.211

Optimization complete.
Best Fitness             = 2.235439e+04
Total Function Evals     = 100000
Total Successful Updates = 3789

================================================================================
=
Results:
  Best Fitness           : 2.235439e+04
  Global Optimum         : 1.200000e+03
  Error (|best-optimum|) : 2.115439e+04
  Best Solution (5 dim)  : [ 54.06662483  40.81039297 -75.45989825  32.68873387
-43.99926311]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F12.png
```

In [22]:

```python
# ============================================================
# F13 — Hybrid Function 3 (Bent Cigar; Rosenbrock; Bi-Rastrigin)
# ============================================================
run_and_plot('F13', make_wrapper(hybrid.f13), 1300.0, 'Hybrid: Bent Cigar; Ros
```

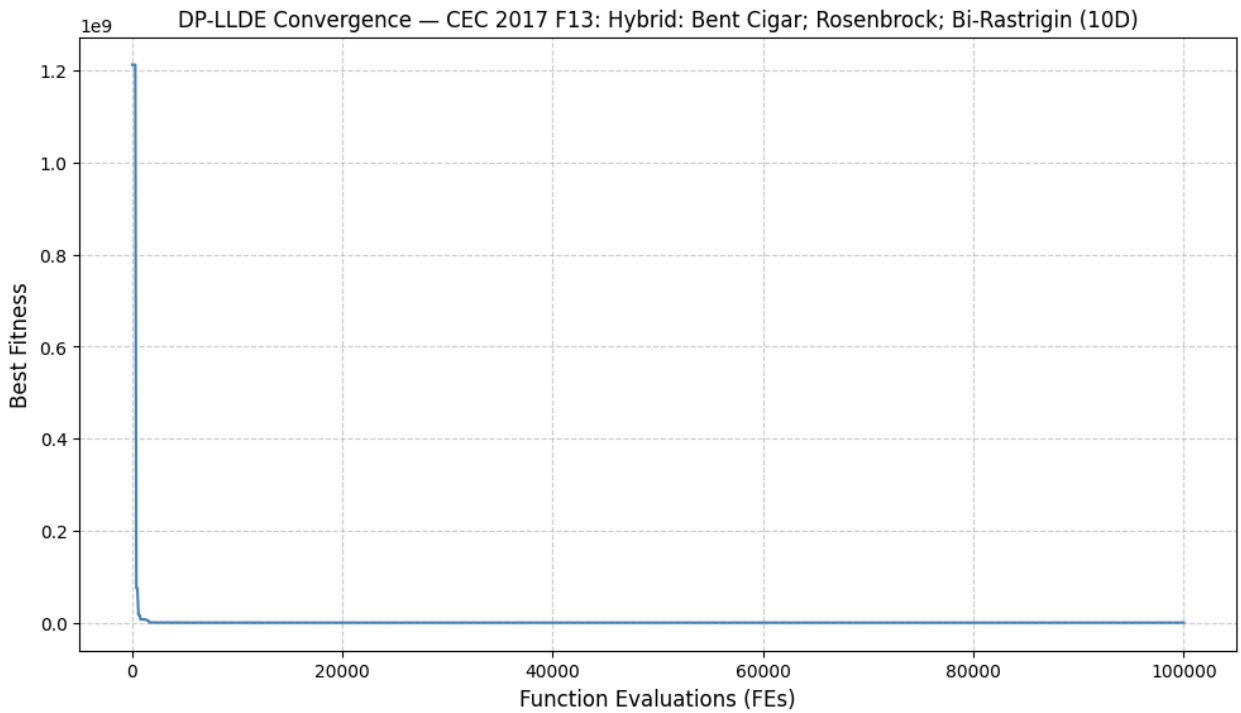
```
================================================================================
=
DP-LLDE on CEC 2017 F13 — Single Run
================================================================================
=
Function       : Hybrid: Bent Cigar; Rosenbrock; Bi-Rastrigin (10D)
Bounds         : [-100, 100]
Global optimum : 1300.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 1.212051e+09
Generation   10: Best = 6.999643e+06  FES =   1100  μ_F = 0.583  μ_CR = 0.356
Generation   20: Best = 3.196527e+05  FES =   2100  μ_F = 0.649  μ_CR = 0.359
Generation   30: Best = 2.666906e+04  FES =   3100  μ_F = 0.698  μ_CR = 0.368
Generation   40: Best = 2.655381e+04  FES =   4100  μ_F = 0.746  μ_CR = 0.355
Generation   50: Best = 1.512009e+04  FES =   5100  μ_F = 0.757  μ_CR = 0.348
Generation   60: Best = 1.508186e+04  FES =   6100  μ_F = 0.774  μ_CR = 0.335
Generation   70: Best = 9.382440e+03  FES =   7100  μ_F = 0.780  μ_CR = 0.359
Generation   80: Best = 5.834007e+03  FES =   8100  μ_F = 0.787  μ_CR = 0.372
Generation   90: Best = 2.566618e+03  FES =   9100  μ_F = 0.812  μ_CR = 0.390
Generation  100: Best = 1.661843e+03  FES =  10100  μ_F = 0.833  μ_CR = 0.406
Generation  110: Best = 1.597920e+03  FES =  11100  μ_F = 0.831  μ_CR = 0.405
Generation  120: Best = 1.597920e+03  FES =  12100  μ_F = 0.831  μ_CR = 0.404
Generation  130: Best = 1.597920e+03  FES =  13100  μ_F = 0.851  μ_CR = 0.403
Generation  140: Best = 1.441840e+03  FES =  14100  μ_F = 0.863  μ_CR = 0.418
Generation  150: Best = 1.441840e+03  FES =  15100  μ_F = 0.880  μ_CR = 0.410
Generation  160: Best = 1.441840e+03  FES =  16100  μ_F = 0.868  μ_CR = 0.406
Generation  170: Best = 1.427166e+03  FES =  17100  μ_F = 0.850  μ_CR = 0.404
Generation  180: Best = 1.387028e+03  FES =  18100  μ_F = 0.854  μ_CR = 0.415
Generation  190: Best = 1.387028e+03  FES =  19100  μ_F = 0.875  μ_CR = 0.418
Generation  200: Best = 1.383632e+03  FES =  20100  μ_F = 0.878  μ_CR = 0.445
Generation  210: Best = 1.351505e+03  FES =  21100  μ_F = 0.875  μ_CR = 0.443
Generation  220: Best = 1.343378e+03  FES =  22100  μ_F = 0.871  μ_CR = 0.439
Generation  230: Best = 1.343378e+03  FES =  23100  μ_F = 0.855  μ_CR = 0.446
Generation  240: Best = 1.324862e+03  FES =  24100  μ_F = 0.846  μ_CR = 0.408
Generation  250: Best = 1.324862e+03  FES =  25100  μ_F = 0.854  μ_CR = 0.381
Generation  260: Best = 1.324862e+03  FES =  26100  μ_F = 0.839  μ_CR = 0.324
Generation  270: Best = 1.324862e+03  FES =  27100  μ_F = 0.819  μ_CR = 0.308
Generation  280: Best = 1.324671e+03  FES =  28100  μ_F = 0.819  μ_CR = 0.244
Generation  290: Best = 1.324671e+03  FES =  29100  μ_F = 0.816  μ_CR = 0.197
Generation  300: Best = 1.323628e+03  FES =  30100  μ_F = 0.819  μ_CR = 0.186
Generation  310: Best = 1.321272e+03  FES =  31100  μ_F = 0.827  μ_CR = 0.165
Generation  320: Best = 1.320885e+03  FES =  32100  μ_F = 0.844  μ_CR = 0.148
Generation  330: Best = 1.320885e+03  FES =  33100  μ_F = 0.851  μ_CR = 0.117
Generation  340: Best = 1.320885e+03  FES =  34100  μ_F = 0.854  μ_CR = 0.123
Generation  350: Best = 1.320548e+03  FES =  35100  μ_F = 0.825  μ_CR = 0.123
Generation  360: Best = 1.320548e+03  FES =  36100  μ_F = 0.855  μ_CR = 0.114
Generation  370: Best = 1.318136e+03  FES =  37100  μ_F = 0.878  μ_CR = 0.100
Generation  380: Best = 1.317585e+03  FES =  38100  μ_F = 0.762  μ_CR = 0.106
Generation  390: Best = 1.315471e+03  FES =  39100  μ_F = 0.760  μ_CR = 0.120
Generation  400: Best = 1.315471e+03  FES =  40100  μ_F = 0.739  μ_CR = 0.099
Generation  410: Best = 1.315471e+03  FES =  41100  μ_F = 0.768  μ_CR = 0.089
Generation  420: Best = 1.315419e+03  FES =  42100  μ_F = 0.839  μ_CR = 0.097
Generation  430: Best = 1.315419e+03  FES =  43100  μ_F = 0.827  μ_CR = 0.093
Generation  440: Best = 1.315011e+03  FES =  44100  μ_F = 0.813  μ_CR = 0.081
```

```
Generation  450: Best = 1.315011e+03  FES =  45100  μ_F = 0.788  μ_CR = 0.092
Generation  460: Best = 1.315011e+03  FES =  46100  μ_F = 0.833  μ_CR = 0.080
Generation  470: Best = 1.314776e+03  FES =  47100  μ_F = 0.842  μ_CR = 0.081
Generation  480: Best = 1.314776e+03  FES =  48100  μ_F = 0.867  μ_CR = 0.090
Generation  490: Best = 1.314776e+03  FES =  49100  μ_F = 0.811  μ_CR = 0.114
Generation  500: Best = 1.314776e+03  FES =  50100  μ_F = 0.813  μ_CR = 0.105
Generation  510: Best = 1.314776e+03  FES =  51100  μ_F = 0.838  μ_CR = 0.131
Generation  520: Best = 1.313223e+03  FES =  52100  μ_F = 0.849  μ_CR = 0.107
Generation  530: Best = 1.313223e+03  FES =  53100  μ_F = 0.854  μ_CR = 0.098
Generation  540: Best = 1.313199e+03  FES =  54100  μ_F = 0.862  μ_CR = 0.091
Generation  550: Best = 1.313061e+03  FES =  55100  μ_F = 0.859  μ_CR = 0.090
Generation  560: Best = 1.312755e+03  FES =  56100  μ_F = 0.857  μ_CR = 0.085
Generation  570: Best = 1.312253e+03  FES =  57100  μ_F = 0.785  μ_CR = 0.085
Generation  580: Best = 1.312253e+03  FES =  58100  μ_F = 0.737  μ_CR = 0.082
Generation  590: Best = 1.312253e+03  FES =  59100  μ_F = 0.789  μ_CR = 0.106
Generation  600: Best = 1.312253e+03  FES =  60100  μ_F = 0.761  μ_CR = 0.120
Generation  610: Best = 1.312253e+03  FES =  61100  μ_F = 0.702  μ_CR = 0.089
Generation  620: Best = 1.311347e+03  FES =  62100  μ_F = 0.721  μ_CR = 0.105
Generation  630: Best = 1.311347e+03  FES =  63100  μ_F = 0.717  μ_CR = 0.095
Generation  640: Best = 1.311347e+03  FES =  64100  μ_F = 0.688  μ_CR = 0.093
Generation  650: Best = 1.311347e+03  FES =  65100  μ_F = 0.672  μ_CR = 0.092
Generation  660: Best = 1.311347e+03  FES =  66100  μ_F = 0.642  μ_CR = 0.084
Generation  670: Best = 1.311347e+03  FES =  67100  μ_F = 0.682  μ_CR = 0.082
Generation  680: Best = 1.311347e+03  FES =  68100  μ_F = 0.691  μ_CR = 0.101
Generation  690: Best = 1.311347e+03  FES =  69100  μ_F = 0.617  μ_CR = 0.081
Generation  700: Best = 1.311347e+03  FES =  70100  μ_F = 0.643  μ_CR = 0.086
Generation  710: Best = 1.308634e+03  FES =  71100  μ_F = 0.632  μ_CR = 0.098
Generation  720: Best = 1.308634e+03  FES =  72100  μ_F = 0.672  μ_CR = 0.113
Generation  730: Best = 1.308634e+03  FES =  73100  μ_F = 0.691  μ_CR = 0.104
Generation  740: Best = 1.308634e+03  FES =  74100  μ_F = 0.725  μ_CR = 0.100
Generation  750: Best = 1.308634e+03  FES =  75100  μ_F = 0.676  μ_CR = 0.110
Generation  760: Best = 1.308634e+03  FES =  76100  μ_F = 0.655  μ_CR = 0.107
Generation  770: Best = 1.308634e+03  FES =  77100  μ_F = 0.656  μ_CR = 0.087
Generation  780: Best = 1.308587e+03  FES =  78100  μ_F = 0.762  μ_CR = 0.087
Generation  790: Best = 1.308578e+03  FES =  79100  μ_F = 0.716  μ_CR = 0.081
Generation  800: Best = 1.308574e+03  FES =  80100  μ_F = 0.726  μ_CR = 0.081
Generation  810: Best = 1.308574e+03  FES =  81100  μ_F = 0.751  μ_CR = 0.083
Generation  820: Best = 1.308574e+03  FES =  82100  μ_F = 0.796  μ_CR = 0.077
Generation  830: Best = 1.308574e+03  FES =  83100  μ_F = 0.745  μ_CR = 0.068
Generation  840: Best = 1.308574e+03  FES =  84100  μ_F = 0.743  μ_CR = 0.107
Generation  850: Best = 1.308574e+03  FES =  85100  μ_F = 0.760  μ_CR = 0.080
Generation  860: Best = 1.308548e+03  FES =  86100  μ_F = 0.782  μ_CR = 0.090
Generation  870: Best = 1.308548e+03  FES =  87100  μ_F = 0.799  μ_CR = 0.085
Generation  880: Best = 1.308548e+03  FES =  88100  μ_F = 0.768  μ_CR = 0.078
Generation  890: Best = 1.308548e+03  FES =  89100  μ_F = 0.788  μ_CR = 0.085
Generation  900: Best = 1.308548e+03  FES =  90100  μ_F = 0.797  μ_CR = 0.085
Generation  910: Best = 1.308533e+03  FES =  91100  μ_F = 0.820  μ_CR = 0.072
Generation  920: Best = 1.308533e+03  FES =  92100  μ_F = 0.817  μ_CR = 0.085
Generation  930: Best = 1.308533e+03  FES =  93100  μ_F = 0.835  μ_CR = 0.091
Generation  940: Best = 1.308533e+03  FES =  94100  μ_F = 0.841  μ_CR = 0.090
Generation  950: Best = 1.308532e+03  FES =  95100  μ_F = 0.839  μ_CR = 0.108
Generation  960: Best = 1.308532e+03  FES =  96100  μ_F = 0.809  μ_CR = 0.099
Generation  970: Best = 1.308415e+03  FES =  97100  μ_F = 0.775  μ_CR = 0.086
Generation  980: Best = 1.308415e+03  FES =  98100  μ_F = 0.832  μ_CR = 0.101
```

```
Generation  990: Best = 1.308415e+03  FES =  99100  μ_F = 0.808  μ_CR = 0.114

Optimization complete.
Best Fitness             = 1.308351e+03
Total Function Evals     = 100000
Total Successful Updates = 6016

================================================================================
=
Results:
  Best Fitness           : 1.308351e+03
  Global Optimum         : 1.300000e+03
  Error (|best-optimum|) : 8.350809e+00
  Best Solution (5 dim)  : [  4.58772193 -46.19504428  -6.2117988    8.45623551
73.08949467]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F13.png
```

In [23]:
```
# ============================================================
# F14 — Hybrid Function 4 (Elliptic; Ackley; Schaffer; Rastrigin)
# ============================================================
run_and_plot('F14', make_wrapper(hybrid.f14), 1400.0, 'Hybrid: Elliptic; Ackle
```

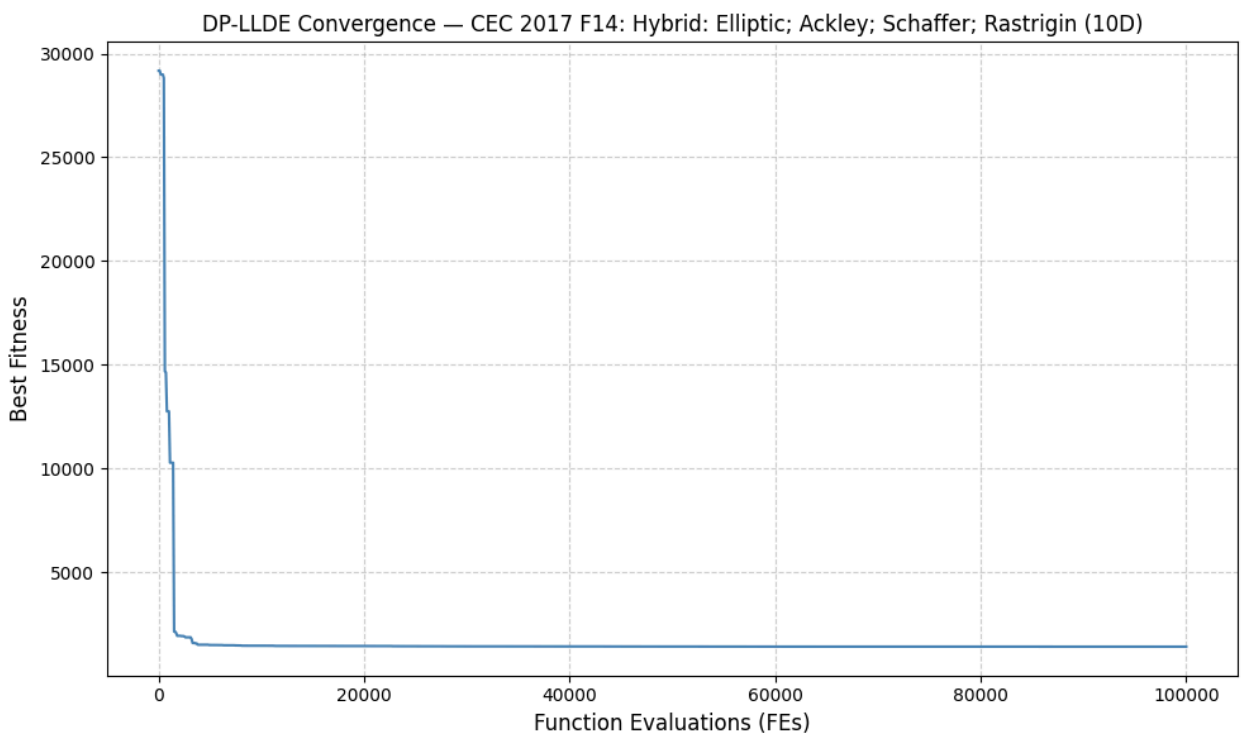
```
================================================================================
=
DP-LLDE on CEC 2017 F14 — Single Run
================================================================================
=
Function       : Hybrid: Elliptic; Ackley; Schaffer; Rastrigin (10D)
Bounds         : [-100, 100]
Global optimum : 1400.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 2.917787e+04
Generation   10: Best = 1.274997e+04  FES =   1100  μ_F = 0.576  μ_CR = 0.352
Generation   20: Best = 1.924486e+03  FES =   2100  μ_F = 0.649  μ_CR = 0.370
Generation   30: Best = 1.851633e+03  FES =   3100  μ_F = 0.698  μ_CR = 0.373
Generation   40: Best = 1.494228e+03  FES =   4100  μ_F = 0.736  μ_CR = 0.354
Generation   50: Best = 1.482490e+03  FES =   5100  μ_F = 0.783  μ_CR = 0.354
Generation   60: Best = 1.480321e+03  FES =   6100  μ_F = 0.802  μ_CR = 0.357
Generation   70: Best = 1.469565e+03  FES =   7100  μ_F = 0.812  μ_CR = 0.369
Generation   80: Best = 1.457143e+03  FES =   8100  μ_F = 0.815  μ_CR = 0.380
Generation   90: Best = 1.447809e+03  FES =   9100  μ_F = 0.831  μ_CR = 0.369
Generation  100: Best = 1.447809e+03  FES =  10100  μ_F = 0.847  μ_CR = 0.354
Generation  110: Best = 1.444889e+03  FES =  11100  μ_F = 0.847  μ_CR = 0.347
Generation  120: Best = 1.436315e+03  FES =  12100  μ_F = 0.854  μ_CR = 0.322
Generation  130: Best = 1.433063e+03  FES =  13100  μ_F = 0.881  μ_CR = 0.313
Generation  140: Best = 1.433063e+03  FES =  14100  μ_F = 0.874  μ_CR = 0.282
Generation  150: Best = 1.433063e+03  FES =  15100  μ_F = 0.867  μ_CR = 0.232
Generation  160: Best = 1.433063e+03  FES =  16100  μ_F = 0.864  μ_CR = 0.206
Generation  170: Best = 1.432879e+03  FES =  17100  μ_F = 0.852  μ_CR = 0.162
Generation  180: Best = 1.431868e+03  FES =  18100  μ_F = 0.857  μ_CR = 0.138
Generation  190: Best = 1.431868e+03  FES =  19100  μ_F = 0.841  μ_CR = 0.143
Generation  200: Best = 1.431300e+03  FES =  20100  μ_F = 0.845  μ_CR = 0.123
Generation  210: Best = 1.429547e+03  FES =  21100  μ_F = 0.810  μ_CR = 0.110
Generation  220: Best = 1.429547e+03  FES =  22100  μ_F = 0.825  μ_CR = 0.106
Generation  230: Best = 1.423752e+03  FES =  23100  μ_F = 0.800  μ_CR = 0.113
Generation  240: Best = 1.423752e+03  FES =  24100  μ_F = 0.816  μ_CR = 0.106
Generation  250: Best = 1.417848e+03  FES =  25100  μ_F = 0.830  μ_CR = 0.109
Generation  260: Best = 1.417848e+03  FES =  26100  μ_F = 0.848  μ_CR = 0.105
Generation  270: Best = 1.417848e+03  FES =  27100  μ_F = 0.871  μ_CR = 0.105
Generation  280: Best = 1.416323e+03  FES =  28100  μ_F = 0.862  μ_CR = 0.115
Generation  290: Best = 1.416323e+03  FES =  29100  μ_F = 0.865  μ_CR = 0.111
Generation  300: Best = 1.416262e+03  FES =  30100  μ_F = 0.847  μ_CR = 0.095
Generation  310: Best = 1.416262e+03  FES =  31100  μ_F = 0.850  μ_CR = 0.093
Generation  320: Best = 1.414876e+03  FES =  32100  μ_F = 0.875  μ_CR = 0.109
Generation  330: Best = 1.414619e+03  FES =  33100  μ_F = 0.861  μ_CR = 0.109
Generation  340: Best = 1.413409e+03  FES =  34100  μ_F = 0.874  μ_CR = 0.117
Generation  350: Best = 1.412548e+03  FES =  35100  μ_F = 0.876  μ_CR = 0.104
Generation  360: Best = 1.411787e+03  FES =  36100  μ_F = 0.860  μ_CR = 0.102
Generation  370: Best = 1.411787e+03  FES =  37100  μ_F = 0.844  μ_CR = 0.089
Generation  380: Best = 1.411787e+03  FES =  38100  μ_F = 0.830  μ_CR = 0.081
Generation  390: Best = 1.410813e+03  FES =  39100  μ_F = 0.822  μ_CR = 0.095
Generation  400: Best = 1.410813e+03  FES =  40100  μ_F = 0.840  μ_CR = 0.085
Generation  410: Best = 1.410813e+03  FES =  41100  μ_F = 0.853  μ_CR = 0.089
Generation  420: Best = 1.410646e+03  FES =  42100  μ_F = 0.858  μ_CR = 0.079
Generation  430: Best = 1.410646e+03  FES =  43100  μ_F = 0.851  μ_CR = 0.078
Generation  440: Best = 1.410646e+03  FES =  44100  μ_F = 0.859  μ_CR = 0.087
```

```
Generation  450: Best = 1.409197e+03  FES =  45100  μ_F = 0.863  μ_CR = 0.084
Generation  460: Best = 1.407317e+03  FES =  46100  μ_F = 0.826  μ_CR = 0.097
Generation  470: Best = 1.407317e+03  FES =  47100  μ_F = 0.820  μ_CR = 0.085
Generation  480: Best = 1.407317e+03  FES =  48100  μ_F = 0.835  μ_CR = 0.084
Generation  490: Best = 1.407317e+03  FES =  49100  μ_F = 0.866  μ_CR = 0.107
Generation  500: Best = 1.406716e+03  FES =  50100  μ_F = 0.846  μ_CR = 0.115
Generation  510: Best = 1.406716e+03  FES =  51100  μ_F = 0.821  μ_CR = 0.107
Generation  520: Best = 1.404970e+03  FES =  52100  μ_F = 0.818  μ_CR = 0.108
Generation  530: Best = 1.404970e+03  FES =  53100  μ_F = 0.845  μ_CR = 0.106
Generation  540: Best = 1.404970e+03  FES =  54100  μ_F = 0.855  μ_CR = 0.124
Generation  550: Best = 1.404289e+03  FES =  55100  μ_F = 0.833  μ_CR = 0.129
Generation  560: Best = 1.404289e+03  FES =  56100  μ_F = 0.841  μ_CR = 0.111
Generation  570: Best = 1.404248e+03  FES =  57100  μ_F = 0.811  μ_CR = 0.104
Generation  580: Best = 1.402869e+03  FES =  58100  μ_F = 0.845  μ_CR = 0.121
Generation  590: Best = 1.402869e+03  FES =  59100  μ_F = 0.854  μ_CR = 0.110
Generation  600: Best = 1.402869e+03  FES =  60100  μ_F = 0.858  μ_CR = 0.093
Generation  610: Best = 1.402869e+03  FES =  61100  μ_F = 0.852  μ_CR = 0.101
Generation  620: Best = 1.401883e+03  FES =  62100  μ_F = 0.851  μ_CR = 0.100
Generation  630: Best = 1.401755e+03  FES =  63100  μ_F = 0.863  μ_CR = 0.104
Generation  640: Best = 1.401755e+03  FES =  64100  μ_F = 0.851  μ_CR = 0.092
Generation  650: Best = 1.401738e+03  FES =  65100  μ_F = 0.834  μ_CR = 0.096
Generation  660: Best = 1.400641e+03  FES =  66100  μ_F = 0.833  μ_CR = 0.109
Generation  670: Best = 1.400641e+03  FES =  67100  μ_F = 0.818  μ_CR = 0.104
Generation  680: Best = 1.400641e+03  FES =  68100  μ_F = 0.818  μ_CR = 0.106
Generation  690: Best = 1.400641e+03  FES =  69100  μ_F = 0.788  μ_CR = 0.101
Generation  700: Best = 1.400516e+03  FES =  70100  μ_F = 0.796  μ_CR = 0.091
Generation  710: Best = 1.400516e+03  FES =  71100  μ_F = 0.811  μ_CR = 0.092
Generation  720: Best = 1.400516e+03  FES =  72100  μ_F = 0.801  μ_CR = 0.092
Generation  730: Best = 1.400516e+03  FES =  73100  μ_F = 0.780  μ_CR = 0.106
Generation  740: Best = 1.400516e+03  FES =  74100  μ_F = 0.815  μ_CR = 0.109
Generation  750: Best = 1.400516e+03  FES =  75100  μ_F = 0.757  μ_CR = 0.108
Generation  760: Best = 1.400516e+03  FES =  76100  μ_F = 0.758  μ_CR = 0.117
Generation  770: Best = 1.400516e+03  FES =  77100  μ_F = 0.750  μ_CR = 0.105
Generation  780: Best = 1.400516e+03  FES =  78100  μ_F = 0.748  μ_CR = 0.095
Generation  790: Best = 1.400516e+03  FES =  79100  μ_F = 0.728  μ_CR = 0.098
Generation  800: Best = 1.400331e+03  FES =  80100  μ_F = 0.775  μ_CR = 0.089
Generation  810: Best = 1.400331e+03  FES =  81100  μ_F = 0.796  μ_CR = 0.072
Generation  820: Best = 1.400310e+03  FES =  82100  μ_F = 0.827  μ_CR = 0.085
Generation  830: Best = 1.400262e+03  FES =  83100  μ_F = 0.815  μ_CR = 0.092
Generation  840: Best = 1.400254e+03  FES =  84100  μ_F = 0.781  μ_CR = 0.096
Generation  850: Best = 1.400254e+03  FES =  85100  μ_F = 0.759  μ_CR = 0.105
Generation  860: Best = 1.400254e+03  FES =  86100  μ_F = 0.805  μ_CR = 0.110
Generation  870: Best = 1.400254e+03  FES =  87100  μ_F = 0.795  μ_CR = 0.114
Generation  880: Best = 1.400254e+03  FES =  88100  μ_F = 0.826  μ_CR = 0.107
Generation  890: Best = 1.400254e+03  FES =  89100  μ_F = 0.837  μ_CR = 0.113
Generation  900: Best = 1.400253e+03  FES =  90100  μ_F = 0.862  μ_CR = 0.111
Generation  910: Best = 1.400253e+03  FES =  91100  μ_F = 0.852  μ_CR = 0.089
Generation  920: Best = 1.400253e+03  FES =  92100  μ_F = 0.794  μ_CR = 0.101
Generation  930: Best = 1.400250e+03  FES =  93100  μ_F = 0.791  μ_CR = 0.101
Generation  940: Best = 1.400250e+03  FES =  94100  μ_F = 0.773  μ_CR = 0.082
Generation  950: Best = 1.400250e+03  FES =  95100  μ_F = 0.809  μ_CR = 0.097
Generation  960: Best = 1.400250e+03  FES =  96100  μ_F = 0.821  μ_CR = 0.105
Generation  970: Best = 1.400250e+03  FES =  97100  μ_F = 0.816  μ_CR = 0.098
Generation  980: Best = 1.400250e+03  FES =  98100  μ_F = 0.811  μ_CR = 0.094
```

```
Generation  990: Best = 1.400250e+03  FES =  99100  μ_F = 0.806  μ_CR = 0.093

Optimization complete.
Best Fitness             = 1.400232e+03
Total Function Evals     = 100000
Total Successful Updates = 7528

================================================================================
=
Results:
  Best Fitness           : 1.400232e+03
  Global Optimum         : 1.400000e+03
  Error (|best-optimum|) : 2.320530e-01
  Best Solution (5 dim)  : [-36.59972124 -70.80342463  66.33661178 -39.9191231
62.20194457]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F14.png
```

In [24]:

```
# ==============================================================
# F15 — Hybrid Function 5 (Bent Cigar; HGBat; Rastrigin; Rosenbrock)
# ==============================================================
run_and_plot('F15', make_wrapper(hybrid.f15), 1500.0, 'Hybrid: Bent Cigar; HGE
```

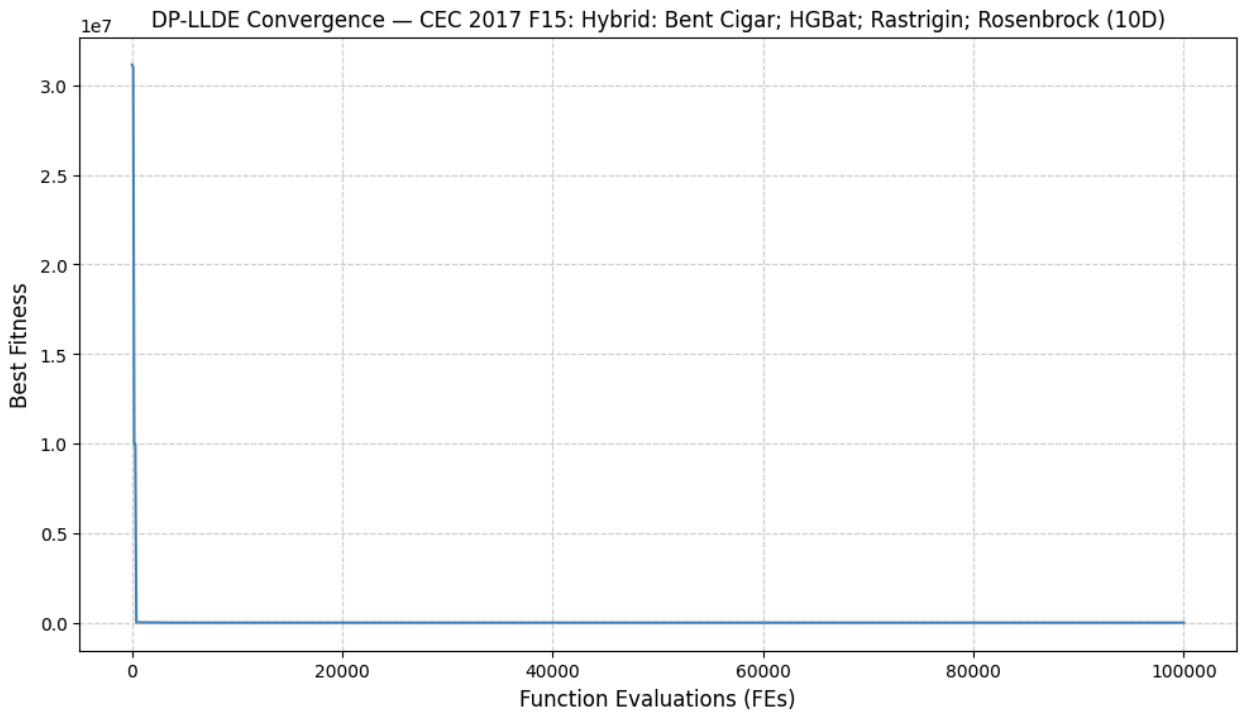
```
================================================================================
=
DP-LLDE on CEC 2017 F15 — Single Run
================================================================================
=
Function       : Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock (10D)
Bounds         : [-100, 100]
Global optimum : 1500.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 3.114909e+07
Generation   10: Best = 1.623632e+04  FES =   1100  μ_F = 0.592  μ_CR = 0.353
Generation   20: Best = 1.005621e+04  FES =   2100  μ_F = 0.655  μ_CR = 0.365
Generation   30: Best = 3.234705e+03  FES =   3100  μ_F = 0.711  μ_CR = 0.380
Generation   40: Best = 2.745474e+03  FES =   4100  μ_F = 0.758  μ_CR = 0.383
Generation   50: Best = 1.885532e+03  FES =   5100  μ_F = 0.791  μ_CR = 0.396
Generation   60: Best = 1.604902e+03  FES =   6100  μ_F = 0.811  μ_CR = 0.398
Generation   70: Best = 1.584215e+03  FES =   7100  μ_F = 0.827  μ_CR = 0.398
Generation   80: Best = 1.543533e+03  FES =   8100  μ_F = 0.822  μ_CR = 0.408
Generation   90: Best = 1.528394e+03  FES =   9100  μ_F = 0.835  μ_CR = 0.423
Generation  100: Best = 1.522753e+03  FES =  10100  μ_F = 0.841  μ_CR = 0.417
Generation  110: Best = 1.517115e+03  FES =  11100  μ_F = 0.810  μ_CR = 0.419
Generation  120: Best = 1.511546e+03  FES =  12100  μ_F = 0.837  μ_CR = 0.409
Generation  130: Best = 1.510350e+03  FES =  13100  μ_F = 0.847  μ_CR = 0.366
Generation  140: Best = 1.508441e+03  FES =  14100  μ_F = 0.842  μ_CR = 0.358
Generation  150: Best = 1.508441e+03  FES =  15100  μ_F = 0.825  μ_CR = 0.325
Generation  160: Best = 1.508441e+03  FES =  16100  μ_F = 0.835  μ_CR = 0.300
Generation  170: Best = 1.508441e+03  FES =  17100  μ_F = 0.816  μ_CR = 0.244
Generation  180: Best = 1.507415e+03  FES =  18100  μ_F = 0.824  μ_CR = 0.216
Generation  190: Best = 1.507180e+03  FES =  19100  μ_F = 0.828  μ_CR = 0.188
Generation  200: Best = 1.506109e+03  FES =  20100  μ_F = 0.818  μ_CR = 0.161
Generation  210: Best = 1.505412e+03  FES =  21100  μ_F = 0.821  μ_CR = 0.152
Generation  220: Best = 1.505412e+03  FES =  22100  μ_F = 0.816  μ_CR = 0.150
Generation  230: Best = 1.505106e+03  FES =  23100  μ_F = 0.820  μ_CR = 0.108
Generation  240: Best = 1.504688e+03  FES =  24100  μ_F = 0.824  μ_CR = 0.107
Generation  250: Best = 1.503973e+03  FES =  25100  μ_F = 0.810  μ_CR = 0.100
Generation  260: Best = 1.503618e+03  FES =  26100  μ_F = 0.761  μ_CR = 0.112
Generation  270: Best = 1.503618e+03  FES =  27100  μ_F = 0.782  μ_CR = 0.117
Generation  280: Best = 1.503526e+03  FES =  28100  μ_F = 0.805  μ_CR = 0.115
Generation  290: Best = 1.502951e+03  FES =  29100  μ_F = 0.838  μ_CR = 0.112
Generation  300: Best = 1.502817e+03  FES =  30100  μ_F = 0.851  μ_CR = 0.126
Generation  310: Best = 1.502797e+03  FES =  31100  μ_F = 0.854  μ_CR = 0.119
Generation  320: Best = 1.502642e+03  FES =  32100  μ_F = 0.843  μ_CR = 0.107
Generation  330: Best = 1.502642e+03  FES =  33100  μ_F = 0.855  μ_CR = 0.108
Generation  340: Best = 1.502369e+03  FES =  34100  μ_F = 0.850  μ_CR = 0.089
Generation  350: Best = 1.502369e+03  FES =  35100  μ_F = 0.865  μ_CR = 0.100
Generation  360: Best = 1.502369e+03  FES =  36100  μ_F = 0.835  μ_CR = 0.099
Generation  370: Best = 1.502369e+03  FES =  37100  μ_F = 0.844  μ_CR = 0.110
Generation  380: Best = 1.502302e+03  FES =  38100  μ_F = 0.830  μ_CR = 0.100
Generation  390: Best = 1.502270e+03  FES =  39100  μ_F = 0.832  μ_CR = 0.108
Generation  400: Best = 1.502270e+03  FES =  40100  μ_F = 0.836  μ_CR = 0.107
Generation  410: Best = 1.502270e+03  FES =  41100  μ_F = 0.859  μ_CR = 0.112
Generation  420: Best = 1.502269e+03  FES =  42100  μ_F = 0.857  μ_CR = 0.103
Generation  430: Best = 1.502264e+03  FES =  43100  μ_F = 0.833  μ_CR = 0.102
Generation  440: Best = 1.502218e+03  FES =  44100  μ_F = 0.836  μ_CR = 0.110
```

```
Generation  450: Best = 1.502218e+03  FES =  45100  μ_F = 0.835  μ_CR = 0.119
Generation  460: Best = 1.501839e+03  FES =  46100  μ_F = 0.814  μ_CR = 0.104
Generation  470: Best = 1.501824e+03  FES =  47100  μ_F = 0.795  μ_CR = 0.097
Generation  480: Best = 1.501707e+03  FES =  48100  μ_F = 0.798  μ_CR = 0.104
Generation  490: Best = 1.501682e+03  FES =  49100  μ_F = 0.792  μ_CR = 0.096
Generation  500: Best = 1.501564e+03  FES =  50100  μ_F = 0.780  μ_CR = 0.097
Generation  510: Best = 1.501564e+03  FES =  51100  μ_F = 0.788  μ_CR = 0.120
Generation  520: Best = 1.501546e+03  FES =  52100  μ_F = 0.772  μ_CR = 0.108
Generation  530: Best = 1.501546e+03  FES =  53100  μ_F = 0.822  μ_CR = 0.090
Generation  540: Best = 1.501546e+03  FES =  54100  μ_F = 0.854  μ_CR = 0.091
Generation  550: Best = 1.501546e+03  FES =  55100  μ_F = 0.882  μ_CR = 0.105
Generation  560: Best = 1.501532e+03  FES =  56100  μ_F = 0.870  μ_CR = 0.122
Generation  570: Best = 1.501382e+03  FES =  57100  μ_F = 0.828  μ_CR = 0.088
Generation  580: Best = 1.501382e+03  FES =  58100  μ_F = 0.851  μ_CR = 0.111
Generation  590: Best = 1.501356e+03  FES =  59100  μ_F = 0.867  μ_CR = 0.106
Generation  600: Best = 1.501310e+03  FES =  60100  μ_F = 0.854  μ_CR = 0.111
Generation  610: Best = 1.501310e+03  FES =  61100  μ_F = 0.860  μ_CR = 0.095
Generation  620: Best = 1.501266e+03  FES =  62100  μ_F = 0.835  μ_CR = 0.096
Generation  630: Best = 1.501266e+03  FES =  63100  μ_F = 0.883  μ_CR = 0.113
Generation  640: Best = 1.501266e+03  FES =  64100  μ_F = 0.819  μ_CR = 0.114
Generation  650: Best = 1.501266e+03  FES =  65100  μ_F = 0.785  μ_CR = 0.121
Generation  660: Best = 1.501262e+03  FES =  66100  μ_F = 0.854  μ_CR = 0.104
Generation  670: Best = 1.501248e+03  FES =  67100  μ_F = 0.884  μ_CR = 0.112
Generation  680: Best = 1.501228e+03  FES =  68100  μ_F = 0.858  μ_CR = 0.104
Generation  690: Best = 1.501228e+03  FES =  69100  μ_F = 0.870  μ_CR = 0.110
Generation  700: Best = 1.501228e+03  FES =  70100  μ_F = 0.840  μ_CR = 0.121
Generation  710: Best = 1.501228e+03  FES =  71100  μ_F = 0.812  μ_CR = 0.102
Generation  720: Best = 1.501228e+03  FES =  72100  μ_F = 0.796  μ_CR = 0.107
Generation  730: Best = 1.501088e+03  FES =  73100  μ_F = 0.773  μ_CR = 0.105
Generation  740: Best = 1.501085e+03  FES =  74100  μ_F = 0.765  μ_CR = 0.095
Generation  750: Best = 1.501085e+03  FES =  75100  μ_F = 0.793  μ_CR = 0.100
Generation  760: Best = 1.501085e+03  FES =  76100  μ_F = 0.773  μ_CR = 0.101
Generation  770: Best = 1.501085e+03  FES =  77100  μ_F = 0.778  μ_CR = 0.125
Generation  780: Best = 1.501085e+03  FES =  78100  μ_F = 0.767  μ_CR = 0.119
Generation  790: Best = 1.501085e+03  FES =  79100  μ_F = 0.769  μ_CR = 0.115
Generation  800: Best = 1.501083e+03  FES =  80100  μ_F = 0.820  μ_CR = 0.117
Generation  810: Best = 1.501062e+03  FES =  81100  μ_F = 0.816  μ_CR = 0.111
Generation  820: Best = 1.501037e+03  FES =  82100  μ_F = 0.806  μ_CR = 0.116
Generation  830: Best = 1.501021e+03  FES =  83100  μ_F = 0.828  μ_CR = 0.106
Generation  840: Best = 1.501021e+03  FES =  84100  μ_F = 0.822  μ_CR = 0.109
Generation  850: Best = 1.501020e+03  FES =  85100  μ_F = 0.799  μ_CR = 0.117
Generation  860: Best = 1.501020e+03  FES =  86100  μ_F = 0.797  μ_CR = 0.105
Generation  870: Best = 1.500989e+03  FES =  87100  μ_F = 0.781  μ_CR = 0.101
Generation  880: Best = 1.500989e+03  FES =  88100  μ_F = 0.784  μ_CR = 0.120
Generation  890: Best = 1.500974e+03  FES =  89100  μ_F = 0.810  μ_CR = 0.130
Generation  900: Best = 1.500974e+03  FES =  90100  μ_F = 0.829  μ_CR = 0.138
Generation  910: Best = 1.500974e+03  FES =  91100  μ_F = 0.831  μ_CR = 0.114
Generation  920: Best = 1.500955e+03  FES =  92100  μ_F = 0.842  μ_CR = 0.113
Generation  930: Best = 1.500955e+03  FES =  93100  μ_F = 0.814  μ_CR = 0.121
Generation  940: Best = 1.500953e+03  FES =  94100  μ_F = 0.786  μ_CR = 0.107
Generation  950: Best = 1.500952e+03  FES =  95100  μ_F = 0.794  μ_CR = 0.123
Generation  960: Best = 1.500952e+03  FES =  96100  μ_F = 0.818  μ_CR = 0.113
Generation  970: Best = 1.500950e+03  FES =  97100  μ_F = 0.830  μ_CR = 0.096
Generation  980: Best = 1.500943e+03  FES =  98100  μ_F = 0.836  μ_CR = 0.115
```

```
Generation  990: Best = 1.500889e+03  FES =  99100  μ_F = 0.848  μ_CR = 0.101

Optimization complete.
Best Fitness             = 1.500884e+03
Total Function Evals     = 100000
Total Successful Updates = 7674

================================================================================
=
Results:
  Best Fitness           : 1.500884e+03
  Global Optimum         : 1.500000e+03
  Error (|best-optimum|) : 8.843735e-01
  Best Solution (5 dim)  : [  9.64821155  -1.83373884  65.5339865  -42.12176062
56.63168725]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F15.png
```

In [25]:
```
# ============================================================
# F16 — Hybrid Function 6 (Schaffer; HGBat; Rosenbrock; Schwefel)
# ============================================================
run_and_plot('F16', make_wrapper(hybrid.f16), 1600.0, 'Hybrid: Schaffer; HGBat
```

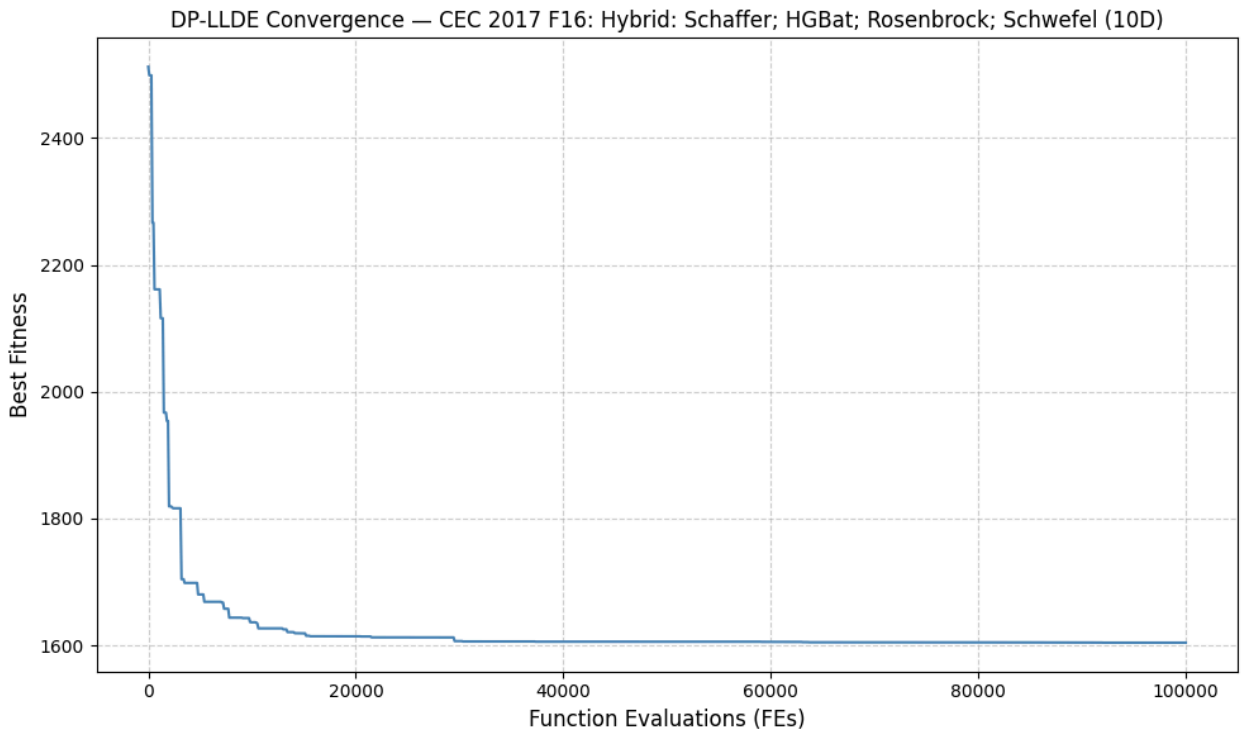
```
================================================================================
=
DP-LLDE on CEC 2017 F16 — Single Run
================================================================================
=
Function       : Hybrid: Schaffer; HGBat; Rosenbrock; Schwefel (10D)
Bounds         : [-100, 100]
Global optimum : 1600.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 2.512498e+03
Generation   10: Best = 2.161482e+03  FES =   1100  μ_F = 0.591  μ_CR = 0.354
Generation   20: Best = 1.819270e+03  FES =   2100  μ_F = 0.668  μ_CR = 0.351
Generation   30: Best = 1.816277e+03  FES =   3100  μ_F = 0.724  μ_CR = 0.349
Generation   40: Best = 1.698526e+03  FES =   4100  μ_F = 0.777  μ_CR = 0.317
Generation   50: Best = 1.680387e+03  FES =   5100  μ_F = 0.822  μ_CR = 0.298
Generation   60: Best = 1.668786e+03  FES =   6100  μ_F = 0.798  μ_CR = 0.273
Generation   70: Best = 1.668786e+03  FES =   7100  μ_F = 0.810  μ_CR = 0.243
Generation   80: Best = 1.643801e+03  FES =   8100  μ_F = 0.832  μ_CR = 0.205
Generation   90: Best = 1.643728e+03  FES =   9100  μ_F = 0.858  μ_CR = 0.179
Generation  100: Best = 1.636484e+03  FES =  10100  μ_F = 0.847  μ_CR = 0.151
Generation  110: Best = 1.626918e+03  FES =  11100  μ_F = 0.842  μ_CR = 0.134
Generation  120: Best = 1.626918e+03  FES =  12100  μ_F = 0.829  μ_CR = 0.134
Generation  130: Best = 1.625303e+03  FES =  13100  μ_F = 0.818  μ_CR = 0.139
Generation  140: Best = 1.620913e+03  FES =  14100  μ_F = 0.858  μ_CR = 0.134
Generation  150: Best = 1.619003e+03  FES =  15100  μ_F = 0.881  μ_CR = 0.121
Generation  160: Best = 1.614471e+03  FES =  16100  μ_F = 0.864  μ_CR = 0.126
Generation  170: Best = 1.614468e+03  FES =  17100  μ_F = 0.847  μ_CR = 0.126
Generation  180: Best = 1.614468e+03  FES =  18100  μ_F = 0.854  μ_CR = 0.132
Generation  190: Best = 1.614468e+03  FES =  19100  μ_F = 0.841  μ_CR = 0.125
Generation  200: Best = 1.614387e+03  FES =  20100  μ_F = 0.854  μ_CR = 0.101
Generation  210: Best = 1.614083e+03  FES =  21100  μ_F = 0.851  μ_CR = 0.096
Generation  220: Best = 1.612740e+03  FES =  22100  μ_F = 0.820  μ_CR = 0.097
Generation  230: Best = 1.612591e+03  FES =  23100  μ_F = 0.842  μ_CR = 0.113
Generation  240: Best = 1.612591e+03  FES =  24100  μ_F = 0.850  μ_CR = 0.125
Generation  250: Best = 1.612591e+03  FES =  25100  μ_F = 0.830  μ_CR = 0.113
Generation  260: Best = 1.612591e+03  FES =  26100  μ_F = 0.826  μ_CR = 0.115
Generation  270: Best = 1.612591e+03  FES =  27100  μ_F = 0.830  μ_CR = 0.119
Generation  280: Best = 1.612591e+03  FES =  28100  μ_F = 0.844  μ_CR = 0.125
Generation  290: Best = 1.612591e+03  FES =  29100  μ_F = 0.844  μ_CR = 0.130
Generation  300: Best = 1.606807e+03  FES =  30100  μ_F = 0.862  μ_CR = 0.130
Generation  310: Best = 1.606200e+03  FES =  31100  μ_F = 0.860  μ_CR = 0.117
Generation  320: Best = 1.606200e+03  FES =  32100  μ_F = 0.857  μ_CR = 0.121
Generation  330: Best = 1.606200e+03  FES =  33100  μ_F = 0.849  μ_CR = 0.121
Generation  340: Best = 1.606200e+03  FES =  34100  μ_F = 0.815  μ_CR = 0.127
Generation  350: Best = 1.606200e+03  FES =  35100  μ_F = 0.829  μ_CR = 0.110
Generation  360: Best = 1.606200e+03  FES =  36100  μ_F = 0.832  μ_CR = 0.111
Generation  370: Best = 1.606200e+03  FES =  37100  μ_F = 0.829  μ_CR = 0.122
Generation  380: Best = 1.605910e+03  FES =  38100  μ_F = 0.826  μ_CR = 0.103
Generation  390: Best = 1.605910e+03  FES =  39100  μ_F = 0.856  μ_CR = 0.103
Generation  400: Best = 1.605909e+03  FES =  40100  μ_F = 0.838  μ_CR = 0.094
Generation  410: Best = 1.605867e+03  FES =  41100  μ_F = 0.836  μ_CR = 0.098
Generation  420: Best = 1.605867e+03  FES =  42100  μ_F = 0.828  μ_CR = 0.107
Generation  430: Best = 1.605867e+03  FES =  43100  μ_F = 0.852  μ_CR = 0.099
Generation  440: Best = 1.605867e+03  FES =  44100  μ_F = 0.825  μ_CR = 0.089
```

```
Generation  450: Best = 1.605864e+03  FES =  45100  μ_F = 0.830  μ_CR = 0.093
Generation  460: Best = 1.605864e+03  FES =  46100  μ_F = 0.827  μ_CR = 0.113
Generation  470: Best = 1.605864e+03  FES =  47100  μ_F = 0.873  μ_CR = 0.109
Generation  480: Best = 1.605864e+03  FES =  48100  μ_F = 0.890  μ_CR = 0.109
Generation  490: Best = 1.605864e+03  FES =  49100  μ_F = 0.901  μ_CR = 0.093
Generation  500: Best = 1.605864e+03  FES =  50100  μ_F = 0.884  μ_CR = 0.100
Generation  510: Best = 1.605864e+03  FES =  51100  μ_F = 0.871  μ_CR = 0.094
Generation  520: Best = 1.605864e+03  FES =  52100  μ_F = 0.898  μ_CR = 0.081
Generation  530: Best = 1.605864e+03  FES =  53100  μ_F = 0.794  μ_CR = 0.109
Generation  540: Best = 1.605864e+03  FES =  54100  μ_F = 0.758  μ_CR = 0.092
Generation  550: Best = 1.605864e+03  FES =  55100  μ_F = 0.765  μ_CR = 0.086
Generation  560: Best = 1.605819e+03  FES =  56100  μ_F = 0.790  μ_CR = 0.080
Generation  570: Best = 1.605819e+03  FES =  57100  μ_F = 0.824  μ_CR = 0.083
Generation  580: Best = 1.605819e+03  FES =  58100  μ_F = 0.771  μ_CR = 0.099
Generation  590: Best = 1.605646e+03  FES =  59100  μ_F = 0.761  μ_CR = 0.105
Generation  600: Best = 1.605646e+03  FES =  60100  μ_F = 0.750  μ_CR = 0.114
Generation  610: Best = 1.605500e+03  FES =  61100  μ_F = 0.744  μ_CR = 0.096
Generation  620: Best = 1.605500e+03  FES =  62100  μ_F = 0.752  μ_CR = 0.089
Generation  630: Best = 1.605220e+03  FES =  63100  μ_F = 0.760  μ_CR = 0.104
Generation  640: Best = 1.604941e+03  FES =  64100  μ_F = 0.759  μ_CR = 0.095
Generation  650: Best = 1.604941e+03  FES =  65100  μ_F = 0.763  μ_CR = 0.089
Generation  660: Best = 1.604941e+03  FES =  66100  μ_F = 0.756  μ_CR = 0.091
Generation  670: Best = 1.604941e+03  FES =  67100  μ_F = 0.763  μ_CR = 0.090
Generation  680: Best = 1.604941e+03  FES =  68100  μ_F = 0.757  μ_CR = 0.090
Generation  690: Best = 1.604941e+03  FES =  69100  μ_F = 0.781  μ_CR = 0.110
Generation  700: Best = 1.604941e+03  FES =  70100  μ_F = 0.774  μ_CR = 0.117
Generation  710: Best = 1.604941e+03  FES =  71100  μ_F = 0.769  μ_CR = 0.108
Generation  720: Best = 1.604926e+03  FES =  72100  μ_F = 0.801  μ_CR = 0.145
Generation  730: Best = 1.604926e+03  FES =  73100  μ_F = 0.756  μ_CR = 0.139
Generation  740: Best = 1.604926e+03  FES =  74100  μ_F = 0.798  μ_CR = 0.120
Generation  750: Best = 1.604926e+03  FES =  75100  μ_F = 0.780  μ_CR = 0.122
Generation  760: Best = 1.604926e+03  FES =  76100  μ_F = 0.816  μ_CR = 0.113
Generation  770: Best = 1.604765e+03  FES =  77100  μ_F = 0.761  μ_CR = 0.103
Generation  780: Best = 1.604765e+03  FES =  78100  μ_F = 0.749  μ_CR = 0.091
Generation  790: Best = 1.604765e+03  FES =  79100  μ_F = 0.760  μ_CR = 0.094
Generation  800: Best = 1.604738e+03  FES =  80100  μ_F = 0.786  μ_CR = 0.086
Generation  810: Best = 1.604738e+03  FES =  81100  μ_F = 0.756  μ_CR = 0.085
Generation  820: Best = 1.604738e+03  FES =  82100  μ_F = 0.772  μ_CR = 0.094
Generation  830: Best = 1.604738e+03  FES =  83100  μ_F = 0.791  μ_CR = 0.097
Generation  840: Best = 1.604733e+03  FES =  84100  μ_F = 0.790  μ_CR = 0.098
Generation  850: Best = 1.604733e+03  FES =  85100  μ_F = 0.767  μ_CR = 0.091
Generation  860: Best = 1.604697e+03  FES =  86100  μ_F = 0.727  μ_CR = 0.099
Generation  870: Best = 1.604697e+03  FES =  87100  μ_F = 0.762  μ_CR = 0.085
Generation  880: Best = 1.604697e+03  FES =  88100  μ_F = 0.814  μ_CR = 0.128
Generation  890: Best = 1.604697e+03  FES =  89100  μ_F = 0.796  μ_CR = 0.125
Generation  900: Best = 1.604697e+03  FES =  90100  μ_F = 0.742  μ_CR = 0.124
Generation  910: Best = 1.604697e+03  FES =  91100  μ_F = 0.729  μ_CR = 0.140
Generation  920: Best = 1.604338e+03  FES =  92100  μ_F = 0.806  μ_CR = 0.141
Generation  930: Best = 1.604338e+03  FES =  93100  μ_F = 0.692  μ_CR = 0.143
Generation  940: Best = 1.604338e+03  FES =  94100  μ_F = 0.668  μ_CR = 0.129
Generation  950: Best = 1.604338e+03  FES =  95100  μ_F = 0.655  μ_CR = 0.117
Generation  960: Best = 1.604338e+03  FES =  96100  μ_F = 0.624  μ_CR = 0.133
Generation  970: Best = 1.604338e+03  FES =  97100  μ_F = 0.639  μ_CR = 0.127
Generation  980: Best = 1.604330e+03  FES =  98100  μ_F = 0.635  μ_CR = 0.113
```

```
Generation  990: Best = 1.604330e+03  FES =  99100  μ_F = 0.703  μ_CR = 0.107

Optimization complete.
Best Fitness             = 1.604330e+03
Total Function Evals     = 100000
Total Successful Updates = 5451

================================================================================
=
Results:
  Best Fitness           : 1.604330e+03
  Global Optimum         : 1.600000e+03
  Error (|best-optimum|) : 4.329604e+00
  Best Solution (5 dim)  : [ 36.70788172  55.97164433  63.22592553 -68.42887985
-35.15704479]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F16.png
```

In [26]:

```
# ============================================================
# F17 — Hybrid Function 7 (Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Ra
# ============================================================
run_and_plot('F17', make_wrapper(hybrid.f17), 1700.0, 'Hybrid: Katsuura; Ackle
```

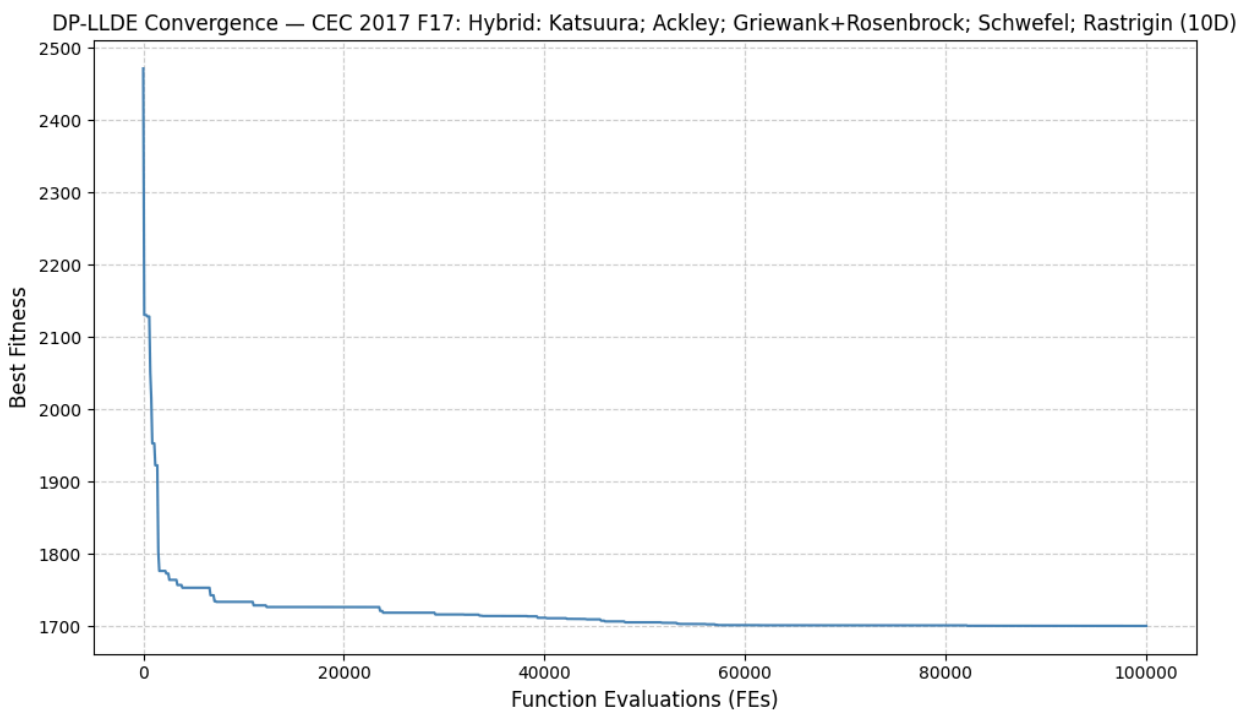
```
================================================================================
=
DP-LLDE on CEC 2017 F17 — Single Run
================================================================================
=
Function       : Hybrid: Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastr
igin (10D)
Bounds         : [-100, 100]
Global optimum : 1700.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 2.471803e+03
Generation   10: Best = 1.953086e+03  FES =   1100  μ_F = 0.585  μ_CR = 0.351
Generation   20: Best = 1.776588e+03  FES =   2100  μ_F = 0.642  μ_CR = 0.335
Generation   30: Best = 1.764218e+03  FES =   3100  μ_F = 0.680  μ_CR = 0.326
Generation   40: Best = 1.753302e+03  FES =   4100  μ_F = 0.728  μ_CR = 0.294
Generation   50: Best = 1.753302e+03  FES =   5100  μ_F = 0.745  μ_CR = 0.273
Generation   60: Best = 1.753302e+03  FES =   6100  μ_F = 0.749  μ_CR = 0.250
Generation   70: Best = 1.742853e+03  FES =   7100  μ_F = 0.768  μ_CR = 0.230
Generation   80: Best = 1.733781e+03  FES =   8100  μ_F = 0.802  μ_CR = 0.203
Generation   90: Best = 1.733781e+03  FES =   9100  μ_F = 0.810  μ_CR = 0.183
Generation  100: Best = 1.733781e+03  FES =  10100  μ_F = 0.822  μ_CR = 0.168
Generation  110: Best = 1.728884e+03  FES =  11100  μ_F = 0.841  μ_CR = 0.172
Generation  120: Best = 1.728884e+03  FES =  12100  μ_F = 0.863  μ_CR = 0.140
Generation  130: Best = 1.726669e+03  FES =  13100  μ_F = 0.864  μ_CR = 0.132
Generation  140: Best = 1.726669e+03  FES =  14100  μ_F = 0.867  μ_CR = 0.123
Generation  150: Best = 1.726669e+03  FES =  15100  μ_F = 0.870  μ_CR = 0.130
Generation  160: Best = 1.726669e+03  FES =  16100  μ_F = 0.865  μ_CR = 0.125
Generation  170: Best = 1.726669e+03  FES =  17100  μ_F = 0.870  μ_CR = 0.137
Generation  180: Best = 1.726669e+03  FES =  18100  μ_F = 0.852  μ_CR = 0.129
Generation  190: Best = 1.726669e+03  FES =  19100  μ_F = 0.833  μ_CR = 0.129
Generation  200: Best = 1.726669e+03  FES =  20100  μ_F = 0.846  μ_CR = 0.130
Generation  210: Best = 1.726669e+03  FES =  21100  μ_F = 0.861  μ_CR = 0.110
Generation  220: Best = 1.726560e+03  FES =  22100  μ_F = 0.872  μ_CR = 0.115
Generation  230: Best = 1.726560e+03  FES =  23100  μ_F = 0.852  μ_CR = 0.119
Generation  240: Best = 1.718776e+03  FES =  24100  μ_F = 0.847  μ_CR = 0.120
Generation  250: Best = 1.718776e+03  FES =  25100  μ_F = 0.853  μ_CR = 0.112
Generation  260: Best = 1.718776e+03  FES =  26100  μ_F = 0.823  μ_CR = 0.118
Generation  270: Best = 1.718776e+03  FES =  27100  μ_F = 0.821  μ_CR = 0.124
Generation  280: Best = 1.718776e+03  FES =  28100  μ_F = 0.807  μ_CR = 0.108
Generation  290: Best = 1.718776e+03  FES =  29100  μ_F = 0.822  μ_CR = 0.110
Generation  300: Best = 1.716328e+03  FES =  30100  μ_F = 0.850  μ_CR = 0.100
Generation  310: Best = 1.716328e+03  FES =  31100  μ_F = 0.883  μ_CR = 0.089
Generation  320: Best = 1.716081e+03  FES =  32100  μ_F = 0.887  μ_CR = 0.088
Generation  330: Best = 1.716081e+03  FES =  33100  μ_F = 0.873  μ_CR = 0.089
Generation  340: Best = 1.714258e+03  FES =  34100  μ_F = 0.867  μ_CR = 0.084
Generation  350: Best = 1.714214e+03  FES =  35100  μ_F = 0.866  μ_CR = 0.093
Generation  360: Best = 1.714214e+03  FES =  36100  μ_F = 0.884  μ_CR = 0.125
Generation  370: Best = 1.714214e+03  FES =  37100  μ_F = 0.885  μ_CR = 0.115
Generation  380: Best = 1.714214e+03  FES =  38100  μ_F = 0.884  μ_CR = 0.108
Generation  390: Best = 1.713740e+03  FES =  39100  μ_F = 0.873  μ_CR = 0.098
Generation  400: Best = 1.711756e+03  FES =  40100  μ_F = 0.880  μ_CR = 0.099
Generation  410: Best = 1.711150e+03  FES =  41100  μ_F = 0.880  μ_CR = 0.100
Generation  420: Best = 1.711150e+03  FES =  42100  μ_F = 0.864  μ_CR = 0.105
Generation  430: Best = 1.710260e+03  FES =  43100  μ_F = 0.850  μ_CR = 0.097
```

```
Generation  440: Best = 1.710083e+03  FES =  44100  μ_F = 0.880  μ_CR = 0.094
Generation  450: Best = 1.709576e+03  FES =  45100  μ_F = 0.878  μ_CR = 0.098
Generation  460: Best = 1.706716e+03  FES =  46100  μ_F = 0.887  μ_CR = 0.078
Generation  470: Best = 1.706716e+03  FES =  47100  μ_F = 0.875  μ_CR = 0.085
Generation  480: Best = 1.705454e+03  FES =  48100  μ_F = 0.854  μ_CR = 0.081
Generation  490: Best = 1.705454e+03  FES =  49100  μ_F = 0.852  μ_CR = 0.082
Generation  500: Best = 1.705454e+03  FES =  50100  μ_F = 0.824  μ_CR = 0.096
Generation  510: Best = 1.705454e+03  FES =  51100  μ_F = 0.834  μ_CR = 0.091
Generation  520: Best = 1.704810e+03  FES =  52100  μ_F = 0.852  μ_CR = 0.088
Generation  530: Best = 1.704692e+03  FES =  53100  μ_F = 0.819  μ_CR = 0.085
Generation  540: Best = 1.703205e+03  FES =  54100  μ_F = 0.800  μ_CR = 0.086
Generation  550: Best = 1.703099e+03  FES =  55100  μ_F = 0.780  μ_CR = 0.072
Generation  560: Best = 1.703099e+03  FES =  56100  μ_F = 0.807  μ_CR = 0.085
Generation  570: Best = 1.701936e+03  FES =  57100  μ_F = 0.810  μ_CR = 0.081
Generation  580: Best = 1.701484e+03  FES =  58100  μ_F = 0.819  μ_CR = 0.082
Generation  590: Best = 1.701484e+03  FES =  59100  μ_F = 0.799  μ_CR = 0.086
Generation  600: Best = 1.701484e+03  FES =  60100  μ_F = 0.791  μ_CR = 0.084
Generation  610: Best = 1.701484e+03  FES =  61100  μ_F = 0.825  μ_CR = 0.088
Generation  620: Best = 1.701361e+03  FES =  62100  μ_F = 0.806  μ_CR = 0.078
Generation  630: Best = 1.701361e+03  FES =  63100  μ_F = 0.805  μ_CR = 0.087
Generation  640: Best = 1.701347e+03  FES =  64100  μ_F = 0.837  μ_CR = 0.082
Generation  650: Best = 1.701342e+03  FES =  65100  μ_F = 0.824  μ_CR = 0.079
Generation  660: Best = 1.701342e+03  FES =  66100  μ_F = 0.803  μ_CR = 0.076
Generation  670: Best = 1.701342e+03  FES =  67100  μ_F = 0.780  μ_CR = 0.074
Generation  680: Best = 1.701342e+03  FES =  68100  μ_F = 0.781  μ_CR = 0.088
Generation  690: Best = 1.701283e+03  FES =  69100  μ_F = 0.783  μ_CR = 0.090
Generation  700: Best = 1.701283e+03  FES =  70100  μ_F = 0.808  μ_CR = 0.093
Generation  710: Best = 1.701264e+03  FES =  71100  μ_F = 0.805  μ_CR = 0.091
Generation  720: Best = 1.701261e+03  FES =  72100  μ_F = 0.834  μ_CR = 0.083
Generation  730: Best = 1.701261e+03  FES =  73100  μ_F = 0.812  μ_CR = 0.085
Generation  740: Best = 1.701256e+03  FES =  74100  μ_F = 0.835  μ_CR = 0.090
Generation  750: Best = 1.701256e+03  FES =  75100  μ_F = 0.825  μ_CR = 0.088
Generation  760: Best = 1.701256e+03  FES =  76100  μ_F = 0.839  μ_CR = 0.090
Generation  770: Best = 1.701256e+03  FES =  77100  μ_F = 0.818  μ_CR = 0.092
Generation  780: Best = 1.701256e+03  FES =  78100  μ_F = 0.807  μ_CR = 0.092
Generation  790: Best = 1.701256e+03  FES =  79100  μ_F = 0.824  μ_CR = 0.091
Generation  800: Best = 1.701256e+03  FES =  80100  μ_F = 0.832  μ_CR = 0.093
Generation  810: Best = 1.701186e+03  FES =  81100  μ_F = 0.850  μ_CR = 0.083
Generation  820: Best = 1.701185e+03  FES =  82100  μ_F = 0.849  μ_CR = 0.088
Generation  830: Best = 1.700652e+03  FES =  83100  μ_F = 0.832  μ_CR = 0.087
Generation  840: Best = 1.700631e+03  FES =  84100  μ_F = 0.857  μ_CR = 0.081
Generation  850: Best = 1.700631e+03  FES =  85100  μ_F = 0.886  μ_CR = 0.076
Generation  860: Best = 1.700494e+03  FES =  86100  μ_F = 0.876  μ_CR = 0.092
Generation  870: Best = 1.700494e+03  FES =  87100  μ_F = 0.888  μ_CR = 0.082
Generation  880: Best = 1.700494e+03  FES =  88100  μ_F = 0.869  μ_CR = 0.076
Generation  890: Best = 1.700490e+03  FES =  89100  μ_F = 0.844  μ_CR = 0.085
Generation  900: Best = 1.700481e+03  FES =  90100  μ_F = 0.843  μ_CR = 0.075
Generation  910: Best = 1.700481e+03  FES =  91100  μ_F = 0.833  μ_CR = 0.086
Generation  920: Best = 1.700473e+03  FES =  92100  μ_F = 0.821  μ_CR = 0.086
Generation  930: Best = 1.700473e+03  FES =  93100  μ_F = 0.827  μ_CR = 0.087
Generation  940: Best = 1.700473e+03  FES =  94100  μ_F = 0.807  μ_CR = 0.084
Generation  950: Best = 1.700473e+03  FES =  95100  μ_F = 0.833  μ_CR = 0.077
Generation  960: Best = 1.700473e+03  FES =  96100  μ_F = 0.851  μ_CR = 0.095
Generation  970: Best = 1.700473e+03  FES =  97100  μ_F = 0.879  μ_CR = 0.091
```

```
Generation  980: Best = 1.700472e+03  FES =  98100  μ_F = 0.846  μ_CR = 0.080
Generation  990: Best = 1.700472e+03  FES =  99100  μ_F = 0.849  μ_CR = 0.085

Optimization complete.
Best Fitness             = 1.700472e+03
Total Function Evals     = 100000
Total Successful Updates = 6480

================================================================================
=
Results:
  Best Fitness           : 1.700472e+03
  Global Optimum         : 1.700000e+03
  Error (|best-optimum|) : 4.721823e-01
  Best Solution (5 dim)  : [-53.29934568  -7.16014975 -21.96010488  15.44488631
-53.05437443]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F17.png
```

In [27]:
```
# ============================================================
# F18 — Hybrid Function 8 (Elliptic; Ackley; Rastrigin; HGBat; Discus)
# ============================================================
run_and_plot('F18', make_wrapper(hybrid.f18), 1800.0, 'Hybrid: Elliptic; Ackle
```

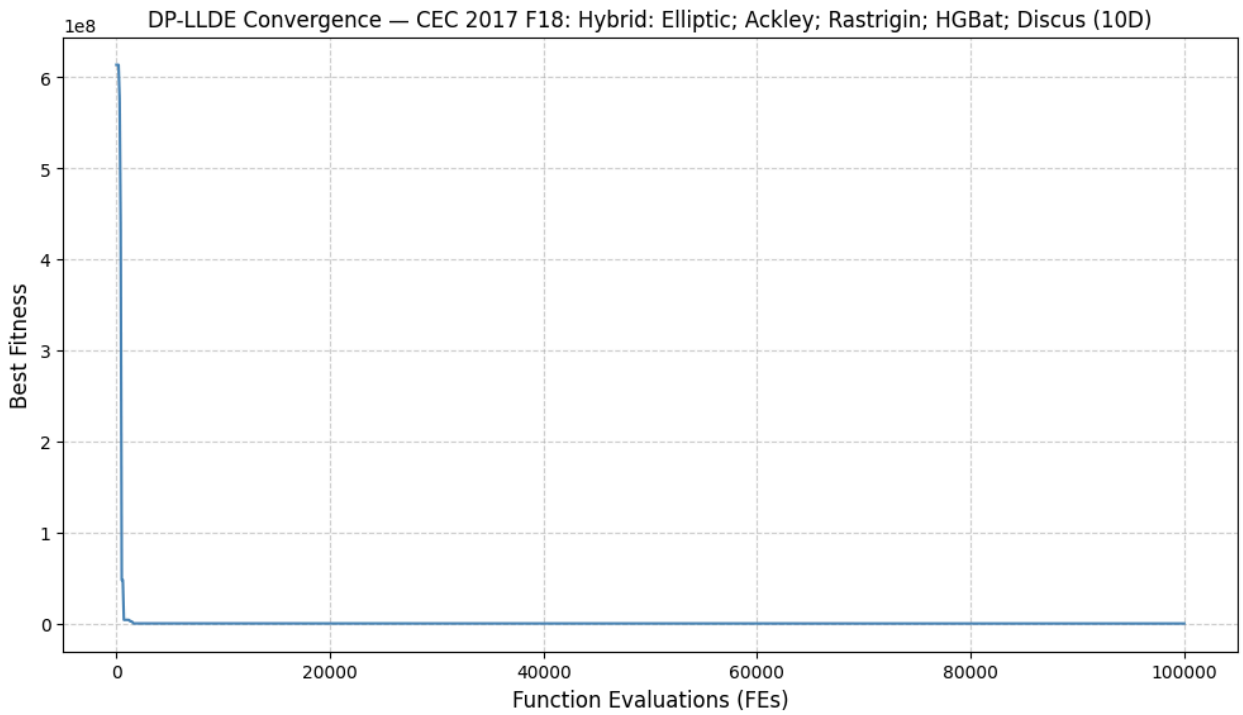
```
================================================================================
=
DP-LLDE on CEC 2017 F18 — Single Run
================================================================================
=
Function       : Hybrid: Elliptic; Ackley; Rastrigin; HGBat; Discus (10D)
Bounds         : [-100, 100]
Global optimum : 1800.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 6.135676e+08
Generation   10: Best = 4.006051e+06  FES =   1100  μ_F = 0.588  μ_CR = 0.353
Generation   20: Best = 1.066501e+05  FES =   2100  μ_F = 0.659  μ_CR = 0.361
Generation   30: Best = 5.583760e+04  FES =   3100  μ_F = 0.698  μ_CR = 0.361
Generation   40: Best = 5.582227e+04  FES =   4100  μ_F = 0.750  μ_CR = 0.351
Generation   50: Best = 3.463127e+04  FES =   5100  μ_F = 0.787  μ_CR = 0.354
Generation   60: Best = 3.363763e+04  FES =   6100  μ_F = 0.797  μ_CR = 0.358
Generation   70: Best = 3.363763e+04  FES =   7100  μ_F = 0.833  μ_CR = 0.348
Generation   80: Best = 3.104436e+04  FES =   8100  μ_F = 0.850  μ_CR = 0.337
Generation   90: Best = 2.949859e+04  FES =   9100  μ_F = 0.844  μ_CR = 0.335
Generation  100: Best = 2.849977e+04  FES =  10100  μ_F = 0.851  μ_CR = 0.329
Generation  110: Best = 2.655074e+04  FES =  11100  μ_F = 0.846  μ_CR = 0.322
Generation  120: Best = 2.608014e+04  FES =  12100  μ_F = 0.848  μ_CR = 0.312
Generation  130: Best = 2.005670e+04  FES =  13100  μ_F = 0.843  μ_CR = 0.310
Generation  140: Best = 1.379856e+04  FES =  14100  μ_F = 0.847  μ_CR = 0.306
Generation  150: Best = 2.976655e+03  FES =  15100  μ_F = 0.843  μ_CR = 0.298
Generation  160: Best = 2.049125e+03  FES =  16100  μ_F = 0.848  μ_CR = 0.299
Generation  170: Best = 1.942320e+03  FES =  17100  μ_F = 0.838  μ_CR = 0.294
Generation  180: Best = 1.937785e+03  FES =  18100  μ_F = 0.825  μ_CR = 0.282
Generation  190: Best = 1.937785e+03  FES =  19100  μ_F = 0.846  μ_CR = 0.272
Generation  200: Best = 1.913420e+03  FES =  20100  μ_F = 0.843  μ_CR = 0.282
Generation  210: Best = 1.883938e+03  FES =  21100  μ_F = 0.835  μ_CR = 0.278
Generation  220: Best = 1.838539e+03  FES =  22100  μ_F = 0.841  μ_CR = 0.279
Generation  230: Best = 1.838020e+03  FES =  23100  μ_F = 0.847  μ_CR = 0.273
Generation  240: Best = 1.838020e+03  FES =  24100  μ_F = 0.845  μ_CR = 0.261
Generation  250: Best = 1.834649e+03  FES =  25100  μ_F = 0.837  μ_CR = 0.257
Generation  260: Best = 1.834649e+03  FES =  26100  μ_F = 0.859  μ_CR = 0.255
Generation  270: Best = 1.834649e+03  FES =  27100  μ_F = 0.860  μ_CR = 0.246
Generation  280: Best = 1.830248e+03  FES =  28100  μ_F = 0.854  μ_CR = 0.228
Generation  290: Best = 1.830248e+03  FES =  29100  μ_F = 0.853  μ_CR = 0.219
Generation  300: Best = 1.830248e+03  FES =  30100  μ_F = 0.840  μ_CR = 0.207
Generation  310: Best = 1.830248e+03  FES =  31100  μ_F = 0.834  μ_CR = 0.211
Generation  320: Best = 1.826664e+03  FES =  32100  μ_F = 0.847  μ_CR = 0.208
Generation  330: Best = 1.826664e+03  FES =  33100  μ_F = 0.868  μ_CR = 0.198
Generation  340: Best = 1.825121e+03  FES =  34100  μ_F = 0.870  μ_CR = 0.172
Generation  350: Best = 1.825121e+03  FES =  35100  μ_F = 0.861  μ_CR = 0.151
Generation  360: Best = 1.825121e+03  FES =  36100  μ_F = 0.864  μ_CR = 0.149
Generation  370: Best = 1.825022e+03  FES =  37100  μ_F = 0.840  μ_CR = 0.141
Generation  380: Best = 1.825022e+03  FES =  38100  μ_F = 0.871  μ_CR = 0.138
Generation  390: Best = 1.823928e+03  FES =  39100  μ_F = 0.869  μ_CR = 0.146
Generation  400: Best = 1.823802e+03  FES =  40100  μ_F = 0.861  μ_CR = 0.142
Generation  410: Best = 1.823802e+03  FES =  41100  μ_F = 0.832  μ_CR = 0.134
Generation  420: Best = 1.823802e+03  FES =  42100  μ_F = 0.848  μ_CR = 0.139
Generation  430: Best = 1.821883e+03  FES =  43100  μ_F = 0.819  μ_CR = 0.162
Generation  440: Best = 1.816390e+03  FES =  44100  μ_F = 0.801  μ_CR = 0.162
```

```
Generation  450: Best = 1.816390e+03  FES =  45100  μ_F = 0.808  μ_CR = 0.169
Generation  460: Best = 1.816390e+03  FES =  46100  μ_F = 0.816  μ_CR = 0.189
Generation  470: Best = 1.816390e+03  FES =  47100  μ_F = 0.783  μ_CR = 0.160
Generation  480: Best = 1.816390e+03  FES =  48100  μ_F = 0.806  μ_CR = 0.145
Generation  490: Best = 1.816390e+03  FES =  49100  μ_F = 0.834  μ_CR = 0.142
Generation  500: Best = 1.816194e+03  FES =  50100  μ_F = 0.769  μ_CR = 0.137
Generation  510: Best = 1.816194e+03  FES =  51100  μ_F = 0.768  μ_CR = 0.150
Generation  520: Best = 1.816194e+03  FES =  52100  μ_F = 0.788  μ_CR = 0.137
Generation  530: Best = 1.815864e+03  FES =  53100  μ_F = 0.810  μ_CR = 0.113
Generation  540: Best = 1.812294e+03  FES =  54100  μ_F = 0.771  μ_CR = 0.124
Generation  550: Best = 1.804103e+03  FES =  55100  μ_F = 0.776  μ_CR = 0.113
Generation  560: Best = 1.804103e+03  FES =  56100  μ_F = 0.785  μ_CR = 0.102
Generation  570: Best = 1.804103e+03  FES =  57100  μ_F = 0.811  μ_CR = 0.104
Generation  580: Best = 1.804103e+03  FES =  58100  μ_F = 0.842  μ_CR = 0.098
Generation  590: Best = 1.803056e+03  FES =  59100  μ_F = 0.867  μ_CR = 0.103
Generation  600: Best = 1.803056e+03  FES =  60100  μ_F = 0.862  μ_CR = 0.114
Generation  610: Best = 1.803056e+03  FES =  61100  μ_F = 0.841  μ_CR = 0.125
Generation  620: Best = 1.801610e+03  FES =  62100  μ_F = 0.810  μ_CR = 0.132
Generation  630: Best = 1.801574e+03  FES =  63100  μ_F = 0.814  μ_CR = 0.129
Generation  640: Best = 1.801574e+03  FES =  64100  μ_F = 0.837  μ_CR = 0.117
Generation  650: Best = 1.801574e+03  FES =  65100  μ_F = 0.801  μ_CR = 0.132
Generation  660: Best = 1.801574e+03  FES =  66100  μ_F = 0.796  μ_CR = 0.114
Generation  670: Best = 1.801574e+03  FES =  67100  μ_F = 0.801  μ_CR = 0.116
Generation  680: Best = 1.801574e+03  FES =  68100  μ_F = 0.807  μ_CR = 0.108
Generation  690: Best = 1.801574e+03  FES =  69100  μ_F = 0.816  μ_CR = 0.101
Generation  700: Best = 1.801574e+03  FES =  70100  μ_F = 0.814  μ_CR = 0.109
Generation  710: Best = 1.801574e+03  FES =  71100  μ_F = 0.783  μ_CR = 0.117
Generation  720: Best = 1.801574e+03  FES =  72100  μ_F = 0.767  μ_CR = 0.123
Generation  730: Best = 1.801574e+03  FES =  73100  μ_F = 0.764  μ_CR = 0.112
Generation  740: Best = 1.801574e+03  FES =  74100  μ_F = 0.779  μ_CR = 0.121
Generation  750: Best = 1.801574e+03  FES =  75100  μ_F = 0.768  μ_CR = 0.121
Generation  760: Best = 1.801466e+03  FES =  76100  μ_F = 0.841  μ_CR = 0.138
Generation  770: Best = 1.801466e+03  FES =  77100  μ_F = 0.847  μ_CR = 0.133
Generation  780: Best = 1.801466e+03  FES =  78100  μ_F = 0.831  μ_CR = 0.126
Generation  790: Best = 1.801466e+03  FES =  79100  μ_F = 0.858  μ_CR = 0.132
Generation  800: Best = 1.801068e+03  FES =  80100  μ_F = 0.869  μ_CR = 0.137
Generation  810: Best = 1.801068e+03  FES =  81100  μ_F = 0.840  μ_CR = 0.128
Generation  820: Best = 1.801068e+03  FES =  82100  μ_F = 0.746  μ_CR = 0.124
Generation  830: Best = 1.801052e+03  FES =  83100  μ_F = 0.761  μ_CR = 0.122
Generation  840: Best = 1.801052e+03  FES =  84100  μ_F = 0.774  μ_CR = 0.132
Generation  850: Best = 1.801052e+03  FES =  85100  μ_F = 0.770  μ_CR = 0.142
Generation  860: Best = 1.801052e+03  FES =  86100  μ_F = 0.776  μ_CR = 0.119
Generation  870: Best = 1.801052e+03  FES =  87100  μ_F = 0.726  μ_CR = 0.116
Generation  880: Best = 1.801052e+03  FES =  88100  μ_F = 0.721  μ_CR = 0.111
Generation  890: Best = 1.801026e+03  FES =  89100  μ_F = 0.768  μ_CR = 0.117
Generation  900: Best = 1.801026e+03  FES =  90100  μ_F = 0.762  μ_CR = 0.135
Generation  910: Best = 1.801026e+03  FES =  91100  μ_F = 0.793  μ_CR = 0.127
Generation  920: Best = 1.800987e+03  FES =  92100  μ_F = 0.781  μ_CR = 0.147
Generation  930: Best = 1.800935e+03  FES =  93100  μ_F = 0.807  μ_CR = 0.160
Generation  940: Best = 1.800932e+03  FES =  94100  μ_F = 0.824  μ_CR = 0.135
Generation  950: Best = 1.800662e+03  FES =  95100  μ_F = 0.830  μ_CR = 0.131
Generation  960: Best = 1.800584e+03  FES =  96100  μ_F = 0.854  μ_CR = 0.104
Generation  970: Best = 1.800584e+03  FES =  97100  μ_F = 0.865  μ_CR = 0.090
Generation  980: Best = 1.800584e+03  FES =  98100  μ_F = 0.866  μ_CR = 0.101
```

```
Generation  990: Best = 1.800578e+03  FES =  99100  μ_F = 0.854  μ_CR = 0.119

Optimization complete.
Best Fitness             = 1.800463e+03
Total Function Evals     = 100000
Total Successful Updates = 7924

================================================================================
=
Results:
  Best Fitness           : 1.800463e+03
  Global Optimum         : 1.800000e+03
  Error (|best-optimum|) : 4.630561e-01
  Best Solution (5 dim)  : [ 18.10811332   1.85064469  17.89087427 -47.17219635
-11.39935867]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F18.png
```

In [28]:
```
# ============================================================
# F19 — Hybrid Function 9 (Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierst
# ============================================================
run_and_plot('F19', make_wrapper(hybrid.f19), 1900.0, 'Hybrid: Bent Cigar; Ras
```

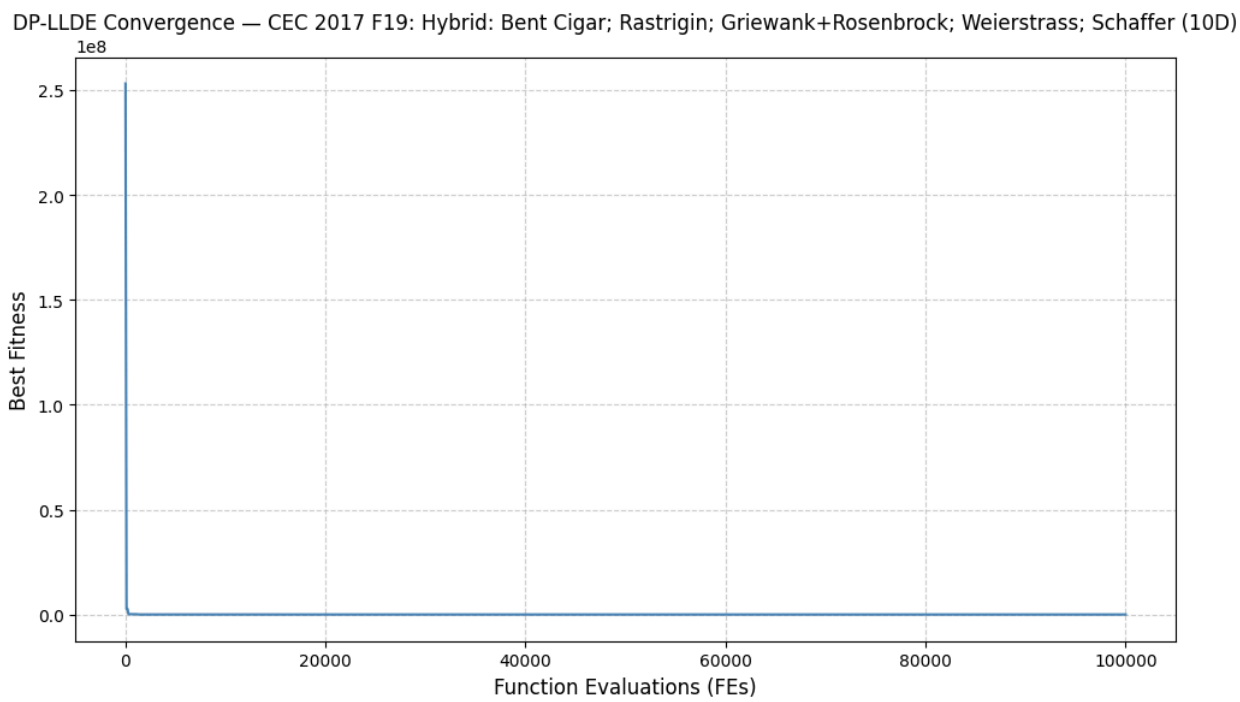
```
================================================================================
=
DP-LLDE on CEC 2017 F19 — Single Run
================================================================================
=
Function       : Hybrid: Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstras
s; Schaffer (10D)
Bounds         : [-100, 100]
Global optimum : 1900.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 2.530389e+08
Generation   10: Best = 1.269618e+05  FES =   1100  μ_F = 0.593  μ_CR = 0.352
Generation   20: Best = 7.732173e+03  FES =   2100  μ_F = 0.655  μ_CR = 0.358
Generation   30: Best = 5.115634e+03  FES =   3100  μ_F = 0.713  μ_CR = 0.370
Generation   40: Best = 2.245612e+03  FES =   4100  μ_F = 0.752  μ_CR = 0.360
Generation   50: Best = 2.054358e+03  FES =   5100  μ_F = 0.788  μ_CR = 0.363
Generation   60: Best = 1.954333e+03  FES =   6100  μ_F = 0.807  μ_CR = 0.360
Generation   70: Best = 1.927934e+03  FES =   7100  μ_F = 0.829  μ_CR = 0.362
Generation   80: Best = 1.924950e+03  FES =   8100  μ_F = 0.843  μ_CR = 0.356
Generation   90: Best = 1.917941e+03  FES =   9100  μ_F = 0.835  μ_CR = 0.369
Generation  100: Best = 1.907903e+03  FES =  10100  μ_F = 0.846  μ_CR = 0.378
Generation  110: Best = 1.907062e+03  FES =  11100  μ_F = 0.837  μ_CR = 0.384
Generation  120: Best = 1.907062e+03  FES =  12100  μ_F = 0.833  μ_CR = 0.360
Generation  130: Best = 1.907062e+03  FES =  13100  μ_F = 0.859  μ_CR = 0.345
Generation  140: Best = 1.907062e+03  FES =  14100  μ_F = 0.882  μ_CR = 0.340
Generation  150: Best = 1.904349e+03  FES =  15100  μ_F = 0.887  μ_CR = 0.311
Generation  160: Best = 1.904349e+03  FES =  16100  μ_F = 0.879  μ_CR = 0.286
Generation  170: Best = 1.904349e+03  FES =  17100  μ_F = 0.861  μ_CR = 0.258
Generation  180: Best = 1.904349e+03  FES =  18100  μ_F = 0.854  μ_CR = 0.224
Generation  190: Best = 1.903530e+03  FES =  19100  μ_F = 0.855  μ_CR = 0.201
Generation  200: Best = 1.903294e+03  FES =  20100  μ_F = 0.866  μ_CR = 0.173
Generation  210: Best = 1.902293e+03  FES =  21100  μ_F = 0.867  μ_CR = 0.175
Generation  220: Best = 1.902293e+03  FES =  22100  μ_F = 0.840  μ_CR = 0.163
Generation  230: Best = 1.902293e+03  FES =  23100  μ_F = 0.840  μ_CR = 0.158
Generation  240: Best = 1.902292e+03  FES =  24100  μ_F = 0.820  μ_CR = 0.152
Generation  250: Best = 1.902117e+03  FES =  25100  μ_F = 0.846  μ_CR = 0.139
Generation  260: Best = 1.901949e+03  FES =  26100  μ_F = 0.828  μ_CR = 0.137
Generation  270: Best = 1.901949e+03  FES =  27100  μ_F = 0.813  μ_CR = 0.134
Generation  280: Best = 1.901949e+03  FES =  28100  μ_F = 0.845  μ_CR = 0.132
Generation  290: Best = 1.901949e+03  FES =  29100  μ_F = 0.824  μ_CR = 0.128
Generation  300: Best = 1.901949e+03  FES =  30100  μ_F = 0.830  μ_CR = 0.126
Generation  310: Best = 1.901583e+03  FES =  31100  μ_F = 0.875  μ_CR = 0.137
Generation  320: Best = 1.901536e+03  FES =  32100  μ_F = 0.865  μ_CR = 0.139
Generation  330: Best = 1.901424e+03  FES =  33100  μ_F = 0.874  μ_CR = 0.139
Generation  340: Best = 1.901107e+03  FES =  34100  μ_F = 0.879  μ_CR = 0.131
Generation  350: Best = 1.901107e+03  FES =  35100  μ_F = 0.874  μ_CR = 0.118
Generation  360: Best = 1.900834e+03  FES =  36100  μ_F = 0.831  μ_CR = 0.123
Generation  370: Best = 1.900834e+03  FES =  37100  μ_F = 0.843  μ_CR = 0.117
Generation  380: Best = 1.900833e+03  FES =  38100  μ_F = 0.857  μ_CR = 0.105
Generation  390: Best = 1.900832e+03  FES =  39100  μ_F = 0.840  μ_CR = 0.121
Generation  400: Best = 1.900643e+03  FES =  40100  μ_F = 0.852  μ_CR = 0.108
Generation  410: Best = 1.900643e+03  FES =  41100  μ_F = 0.858  μ_CR = 0.101
Generation  420: Best = 1.900625e+03  FES =  42100  μ_F = 0.853  μ_CR = 0.122
Generation  430: Best = 1.900625e+03  FES =  43100  μ_F = 0.851  μ_CR = 0.121
```

```
Generation  440: Best = 1.900625e+03  FES =  44100  μ_F = 0.859  μ_CR = 0.120
Generation  450: Best = 1.900558e+03  FES =  45100  μ_F = 0.863  μ_CR = 0.118
Generation  460: Best = 1.900533e+03  FES =  46100  μ_F = 0.847  μ_CR = 0.120
Generation  470: Best = 1.900533e+03  FES =  47100  μ_F = 0.848  μ_CR = 0.121
Generation  480: Best = 1.900449e+03  FES =  48100  μ_F = 0.855  μ_CR = 0.128
Generation  490: Best = 1.900444e+03  FES =  49100  μ_F = 0.853  μ_CR = 0.127
Generation  500: Best = 1.900297e+03  FES =  50100  μ_F = 0.831  μ_CR = 0.125
Generation  510: Best = 1.900291e+03  FES =  51100  μ_F = 0.831  μ_CR = 0.128
Generation  520: Best = 1.900274e+03  FES =  52100  μ_F = 0.831  μ_CR = 0.109
Generation  530: Best = 1.900220e+03  FES =  53100  μ_F = 0.824  μ_CR = 0.108
Generation  540: Best = 1.900220e+03  FES =  54100  μ_F = 0.822  μ_CR = 0.097
Generation  550: Best = 1.900195e+03  FES =  55100  μ_F = 0.802  μ_CR = 0.099
Generation  560: Best = 1.900193e+03  FES =  56100  μ_F = 0.812  μ_CR = 0.093
Generation  570: Best = 1.900104e+03  FES =  57100  μ_F = 0.806  μ_CR = 0.093
Generation  580: Best = 1.900104e+03  FES =  58100  μ_F = 0.844  μ_CR = 0.092
Generation  590: Best = 1.900085e+03  FES =  59100  μ_F = 0.831  μ_CR = 0.093
Generation  600: Best = 1.900085e+03  FES =  60100  μ_F = 0.817  μ_CR = 0.092
Generation  610: Best = 1.900085e+03  FES =  61100  μ_F = 0.818  μ_CR = 0.103
Generation  620: Best = 1.900085e+03  FES =  62100  μ_F = 0.815  μ_CR = 0.104
Generation  630: Best = 1.900085e+03  FES =  63100  μ_F = 0.815  μ_CR = 0.100
Generation  640: Best = 1.900085e+03  FES =  64100  μ_F = 0.810  μ_CR = 0.094
Generation  650: Best = 1.900078e+03  FES =  65100  μ_F = 0.814  μ_CR = 0.102
Generation  660: Best = 1.900071e+03  FES =  66100  μ_F = 0.799  μ_CR = 0.106
Generation  670: Best = 1.900071e+03  FES =  67100  μ_F = 0.809  μ_CR = 0.105
Generation  680: Best = 1.900071e+03  FES =  68100  μ_F = 0.820  μ_CR = 0.099
Generation  690: Best = 1.900049e+03  FES =  69100  μ_F = 0.857  μ_CR = 0.104
Generation  700: Best = 1.900049e+03  FES =  70100  μ_F = 0.841  μ_CR = 0.094
Generation  710: Best = 1.900049e+03  FES =  71100  μ_F = 0.848  μ_CR = 0.095
Generation  720: Best = 1.900035e+03  FES =  72100  μ_F = 0.835  μ_CR = 0.104
Generation  730: Best = 1.900035e+03  FES =  73100  μ_F = 0.837  μ_CR = 0.093
Generation  740: Best = 1.900033e+03  FES =  74100  μ_F = 0.818  μ_CR = 0.088
Generation  750: Best = 1.900033e+03  FES =  75100  μ_F = 0.788  μ_CR = 0.094
Generation  760: Best = 1.900033e+03  FES =  76100  μ_F = 0.813  μ_CR = 0.098
Generation  770: Best = 1.900033e+03  FES =  77100  μ_F = 0.786  μ_CR = 0.089
Generation  780: Best = 1.900033e+03  FES =  78100  μ_F = 0.787  μ_CR = 0.107
Generation  790: Best = 1.900031e+03  FES =  79100  μ_F = 0.803  μ_CR = 0.093
Generation  800: Best = 1.900029e+03  FES =  80100  μ_F = 0.784  μ_CR = 0.096
Generation  810: Best = 1.900028e+03  FES =  81100  μ_F = 0.746  μ_CR = 0.096
Generation  820: Best = 1.900014e+03  FES =  82100  μ_F = 0.765  μ_CR = 0.099
Generation  830: Best = 1.900014e+03  FES =  83100  μ_F = 0.778  μ_CR = 0.123
Generation  840: Best = 1.900014e+03  FES =  84100  μ_F = 0.764  μ_CR = 0.117
Generation  850: Best = 1.900014e+03  FES =  85100  μ_F = 0.787  μ_CR = 0.110
Generation  860: Best = 1.900013e+03  FES =  86100  μ_F = 0.779  μ_CR = 0.094
Generation  870: Best = 1.900013e+03  FES =  87100  μ_F = 0.785  μ_CR = 0.095
Generation  880: Best = 1.900013e+03  FES =  88100  μ_F = 0.810  μ_CR = 0.101
Generation  890: Best = 1.900013e+03  FES =  89100  μ_F = 0.814  μ_CR = 0.106
Generation  900: Best = 1.900013e+03  FES =  90100  μ_F = 0.818  μ_CR = 0.120
Generation  910: Best = 1.900013e+03  FES =  91100  μ_F = 0.795  μ_CR = 0.114
Generation  920: Best = 1.900013e+03  FES =  92100  μ_F = 0.809  μ_CR = 0.094
Generation  930: Best = 1.900013e+03  FES =  93100  μ_F = 0.790  μ_CR = 0.101
Generation  940: Best = 1.900013e+03  FES =  94100  μ_F = 0.807  μ_CR = 0.081
Generation  950: Best = 1.900013e+03  FES =  95100  μ_F = 0.826  μ_CR = 0.103
Generation  960: Best = 1.900013e+03  FES =  96100  μ_F = 0.824  μ_CR = 0.105
Generation  970: Best = 1.900013e+03  FES =  97100  μ_F = 0.814  μ_CR = 0.107
```

```
Generation  980: Best = 1.900012e+03  FES =  98100  μ_F = 0.830  μ_CR = 0.101
Generation  990: Best = 1.900012e+03  FES =  99100  μ_F = 0.847  μ_CR = 0.091

Optimization complete.
Best Fitness             = 1.900012e+03
Total Function Evals     = 100000
Total Successful Updates = 9126

================================================================================
=
Results:
  Best Fitness           : 1.900012e+03
  Global Optimum         : 1.900000e+03
  Error (|best-optimum|) : 1.161373e-02
  Best Solution (5 dim)  : [ 79.28050993   6.94736859 -35.04039679  35.2397432
25.11487396]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F19.png
```

In [29]:
```
# ============================================================
# F20 — Hybrid Function 10 (HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; S
# ============================================================
run_and_plot('F20', make_wrapper(hybrid.f20), 2000.0, 'Hybrid: HappyCat; Katsu
```

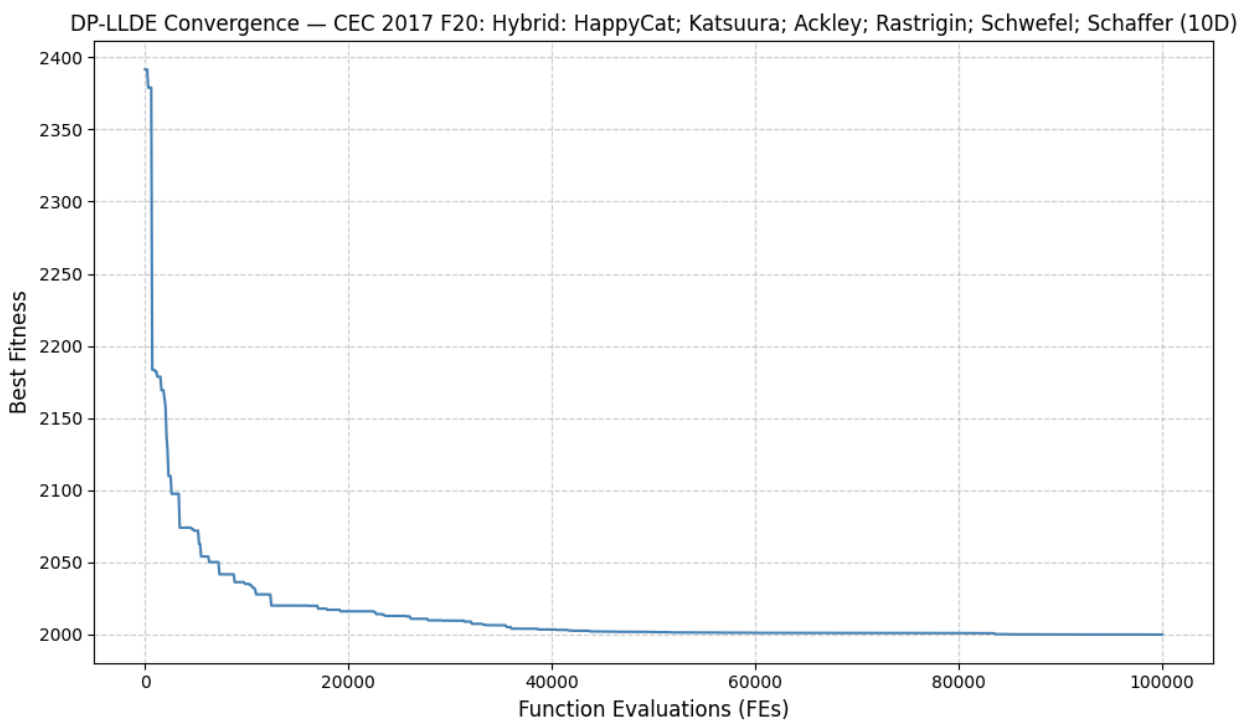
```
================================================================================
=
DP-LLDE on CEC 2017 F20 — Single Run
================================================================================
=
Function       : Hybrid: HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaf
fer (10D)
Bounds         : [-100, 100]
Global optimum : 2000.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 2.391561e+03
Generation   10: Best = 2.182241e+03  FES =   1100  μ_F = 0.580  μ_CR = 0.341
Generation   20: Best = 2.158179e+03  FES =   2100  μ_F = 0.642  μ_CR = 0.332
Generation   30: Best = 2.097516e+03  FES =   3100  μ_F = 0.694  μ_CR = 0.335
Generation   40: Best = 2.074059e+03  FES =   4100  μ_F = 0.723  μ_CR = 0.308
Generation   50: Best = 2.072007e+03  FES =   5100  μ_F = 0.769  μ_CR = 0.290
Generation   60: Best = 2.054131e+03  FES =   6100  μ_F = 0.773  μ_CR = 0.267
Generation   70: Best = 2.050249e+03  FES =   7100  μ_F = 0.799  μ_CR = 0.256
Generation   80: Best = 2.041765e+03  FES =   8100  μ_F = 0.814  μ_CR = 0.232
Generation   90: Best = 2.036352e+03  FES =   9100  μ_F = 0.822  μ_CR = 0.208
Generation  100: Best = 2.035137e+03  FES =  10100  μ_F = 0.797  μ_CR = 0.202
Generation  110: Best = 2.027887e+03  FES =  11100  μ_F = 0.805  μ_CR = 0.173
Generation  120: Best = 2.027829e+03  FES =  12100  μ_F = 0.812  μ_CR = 0.145
Generation  130: Best = 2.020106e+03  FES =  13100  μ_F = 0.823  μ_CR = 0.133
Generation  140: Best = 2.020081e+03  FES =  14100  μ_F = 0.824  μ_CR = 0.131
Generation  150: Best = 2.020081e+03  FES =  15100  μ_F = 0.853  μ_CR = 0.115
Generation  160: Best = 2.020081e+03  FES =  16100  μ_F = 0.856  μ_CR = 0.111
Generation  170: Best = 2.017983e+03  FES =  17100  μ_F = 0.867  μ_CR = 0.107
Generation  180: Best = 2.017199e+03  FES =  18100  μ_F = 0.868  μ_CR = 0.113
Generation  190: Best = 2.017199e+03  FES =  19100  μ_F = 0.853  μ_CR = 0.102
Generation  200: Best = 2.016123e+03  FES =  20100  μ_F = 0.846  μ_CR = 0.102
Generation  210: Best = 2.016123e+03  FES =  21100  μ_F = 0.850  μ_CR = 0.098
Generation  220: Best = 2.016123e+03  FES =  22100  μ_F = 0.866  μ_CR = 0.103
Generation  230: Best = 2.014211e+03  FES =  23100  μ_F = 0.882  μ_CR = 0.101
Generation  240: Best = 2.012949e+03  FES =  24100  μ_F = 0.867  μ_CR = 0.087
Generation  250: Best = 2.012949e+03  FES =  25100  μ_F = 0.827  μ_CR = 0.090
Generation  260: Best = 2.012527e+03  FES =  26100  μ_F = 0.828  μ_CR = 0.097
Generation  270: Best = 2.010946e+03  FES =  27100  μ_F = 0.819  μ_CR = 0.098
Generation  280: Best = 2.009888e+03  FES =  28100  μ_F = 0.832  μ_CR = 0.115
Generation  290: Best = 2.009888e+03  FES =  29100  μ_F = 0.824  μ_CR = 0.126
Generation  300: Best = 2.009696e+03  FES =  30100  μ_F = 0.810  μ_CR = 0.113
Generation  310: Best = 2.009631e+03  FES =  31100  μ_F = 0.836  μ_CR = 0.125
Generation  320: Best = 2.009000e+03  FES =  32100  μ_F = 0.821  μ_CR = 0.129
Generation  330: Best = 2.007461e+03  FES =  33100  μ_F = 0.844  μ_CR = 0.122
Generation  340: Best = 2.006537e+03  FES =  34100  μ_F = 0.866  μ_CR = 0.114
Generation  350: Best = 2.006537e+03  FES =  35100  μ_F = 0.875  μ_CR = 0.108
Generation  360: Best = 2.004075e+03  FES =  36100  μ_F = 0.882  μ_CR = 0.108
Generation  370: Best = 2.004075e+03  FES =  37100  μ_F = 0.859  μ_CR = 0.107
Generation  380: Best = 2.004075e+03  FES =  38100  μ_F = 0.861  μ_CR = 0.099
Generation  390: Best = 2.003522e+03  FES =  39100  μ_F = 0.846  μ_CR = 0.097
Generation  400: Best = 2.003488e+03  FES =  40100  μ_F = 0.840  μ_CR = 0.102
Generation  410: Best = 2.003235e+03  FES =  41100  μ_F = 0.837  μ_CR = 0.112
Generation  420: Best = 2.002673e+03  FES =  42100  μ_F = 0.834  μ_CR = 0.108
Generation  430: Best = 2.002658e+03  FES =  43100  μ_F = 0.819  μ_CR = 0.109
```

```
Generation  440: Best = 2.002199e+03  FES =  44100  μ_F = 0.829  μ_CR = 0.104
Generation  450: Best = 2.002072e+03  FES =  45100  μ_F = 0.834  μ_CR = 0.110
Generation  460: Best = 2.002029e+03  FES =  46100  μ_F = 0.836  μ_CR = 0.101
Generation  470: Best = 2.001918e+03  FES =  47100  μ_F = 0.825  μ_CR = 0.101
Generation  480: Best = 2.001858e+03  FES =  48100  μ_F = 0.837  μ_CR = 0.105
Generation  490: Best = 2.001846e+03  FES =  49100  μ_F = 0.851  μ_CR = 0.107
Generation  500: Best = 2.001721e+03  FES =  50100  μ_F = 0.847  μ_CR = 0.112
Generation  510: Best = 2.001686e+03  FES =  51100  μ_F = 0.829  μ_CR = 0.110
Generation  520: Best = 2.001459e+03  FES =  52100  μ_F = 0.835  μ_CR = 0.109
Generation  530: Best = 2.001432e+03  FES =  53100  μ_F = 0.834  μ_CR = 0.107
Generation  540: Best = 2.001432e+03  FES =  54100  μ_F = 0.846  μ_CR = 0.110
Generation  550: Best = 2.001432e+03  FES =  55100  μ_F = 0.843  μ_CR = 0.097
Generation  560: Best = 2.001389e+03  FES =  56100  μ_F = 0.846  μ_CR = 0.104
Generation  570: Best = 2.001311e+03  FES =  57100  μ_F = 0.833  μ_CR = 0.098
Generation  580: Best = 2.001311e+03  FES =  58100  μ_F = 0.855  μ_CR = 0.114
Generation  590: Best = 2.001311e+03  FES =  59100  μ_F = 0.859  μ_CR = 0.109
Generation  600: Best = 2.001189e+03  FES =  60100  μ_F = 0.851  μ_CR = 0.108
Generation  610: Best = 2.001178e+03  FES =  61100  μ_F = 0.862  μ_CR = 0.107
Generation  620: Best = 2.001178e+03  FES =  62100  μ_F = 0.858  μ_CR = 0.100
Generation  630: Best = 2.001178e+03  FES =  63100  μ_F = 0.845  μ_CR = 0.111
Generation  640: Best = 2.001133e+03  FES =  64100  μ_F = 0.851  μ_CR = 0.100
Generation  650: Best = 2.001133e+03  FES =  65100  μ_F = 0.839  μ_CR = 0.114
Generation  660: Best = 2.001133e+03  FES =  66100  μ_F = 0.845  μ_CR = 0.116
Generation  670: Best = 2.001101e+03  FES =  67100  μ_F = 0.848  μ_CR = 0.112
Generation  680: Best = 2.001101e+03  FES =  68100  μ_F = 0.856  μ_CR = 0.123
Generation  690: Best = 2.001100e+03  FES =  69100  μ_F = 0.856  μ_CR = 0.118
Generation  700: Best = 2.001100e+03  FES =  70100  μ_F = 0.837  μ_CR = 0.119
Generation  710: Best = 2.001099e+03  FES =  71100  μ_F = 0.853  μ_CR = 0.120
Generation  720: Best = 2.001069e+03  FES =  72100  μ_F = 0.859  μ_CR = 0.119
Generation  730: Best = 2.001066e+03  FES =  73100  μ_F = 0.872  μ_CR = 0.112
Generation  740: Best = 2.001064e+03  FES =  74100  μ_F = 0.870  μ_CR = 0.111
Generation  750: Best = 2.001051e+03  FES =  75100  μ_F = 0.875  μ_CR = 0.111
Generation  760: Best = 2.001051e+03  FES =  76100  μ_F = 0.877  μ_CR = 0.119
Generation  770: Best = 2.001037e+03  FES =  77100  μ_F = 0.866  μ_CR = 0.116
Generation  780: Best = 2.001036e+03  FES =  78100  μ_F = 0.863  μ_CR = 0.121
Generation  790: Best = 2.001030e+03  FES =  79100  μ_F = 0.859  μ_CR = 0.111
Generation  800: Best = 2.001029e+03  FES =  80100  μ_F = 0.871  μ_CR = 0.110
Generation  810: Best = 2.001029e+03  FES =  81100  μ_F = 0.860  μ_CR = 0.104
Generation  820: Best = 2.001027e+03  FES =  82100  μ_F = 0.860  μ_CR = 0.105
Generation  830: Best = 2.001023e+03  FES =  83100  μ_F = 0.847  μ_CR = 0.108
Generation  840: Best = 2.000327e+03  FES =  84100  μ_F = 0.847  μ_CR = 0.114
Generation  850: Best = 2.000184e+03  FES =  85100  μ_F = 0.842  μ_CR = 0.122
Generation  860: Best = 2.000180e+03  FES =  86100  μ_F = 0.846  μ_CR = 0.114
Generation  870: Best = 2.000179e+03  FES =  87100  μ_F = 0.852  μ_CR = 0.114
Generation  880: Best = 2.000094e+03  FES =  88100  μ_F = 0.857  μ_CR = 0.112
Generation  890: Best = 2.000094e+03  FES =  89100  μ_F = 0.850  μ_CR = 0.124
Generation  900: Best = 2.000094e+03  FES =  90100  μ_F = 0.852  μ_CR = 0.122
Generation  910: Best = 2.000087e+03  FES =  91100  μ_F = 0.844  μ_CR = 0.119
Generation  920: Best = 2.000075e+03  FES =  92100  μ_F = 0.838  μ_CR = 0.120
Generation  930: Best = 2.000024e+03  FES =  93100  μ_F = 0.828  μ_CR = 0.115
Generation  940: Best = 2.000024e+03  FES =  94100  μ_F = 0.837  μ_CR = 0.120
Generation  950: Best = 2.000024e+03  FES =  95100  μ_F = 0.851  μ_CR = 0.122
Generation  960: Best = 2.000024e+03  FES =  96100  μ_F = 0.849  μ_CR = 0.116
Generation  970: Best = 2.000018e+03  FES =  97100  μ_F = 0.855  μ_CR = 0.114
```

```
Generation  980: Best = 2.000016e+03  FES =  98100  μ_F = 0.858  μ_CR = 0.119
Generation  990: Best = 2.000014e+03  FES =  99100  μ_F = 0.863  μ_CR = 0.129

Optimization complete.
Best Fitness             = 2.000011e+03
Total Function Evals     = 100000
Total Successful Updates = 12560

================================================================================
=
Results:
  Best Fitness           : 2.000011e+03
  Global Optimum         : 2.000000e+03
  Error (|best-optimum|) : 1.148618e-02
  Best Solution (5 dim)  : [-30.59082933 -55.82105055 -29.5845103   61.04514996
69.96487003]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F20.png
```

6.4 Composite Functions (F21-F30)

In [30]:
```
# ==============================================================
# F21 — Composition Function 1 (Rosenbrock; Elliptic; Rastrigin)
# ==============================================================
run_and_plot('F21', make_wrapper(composition.f21), 2100.0, 'Composition: Roser
```

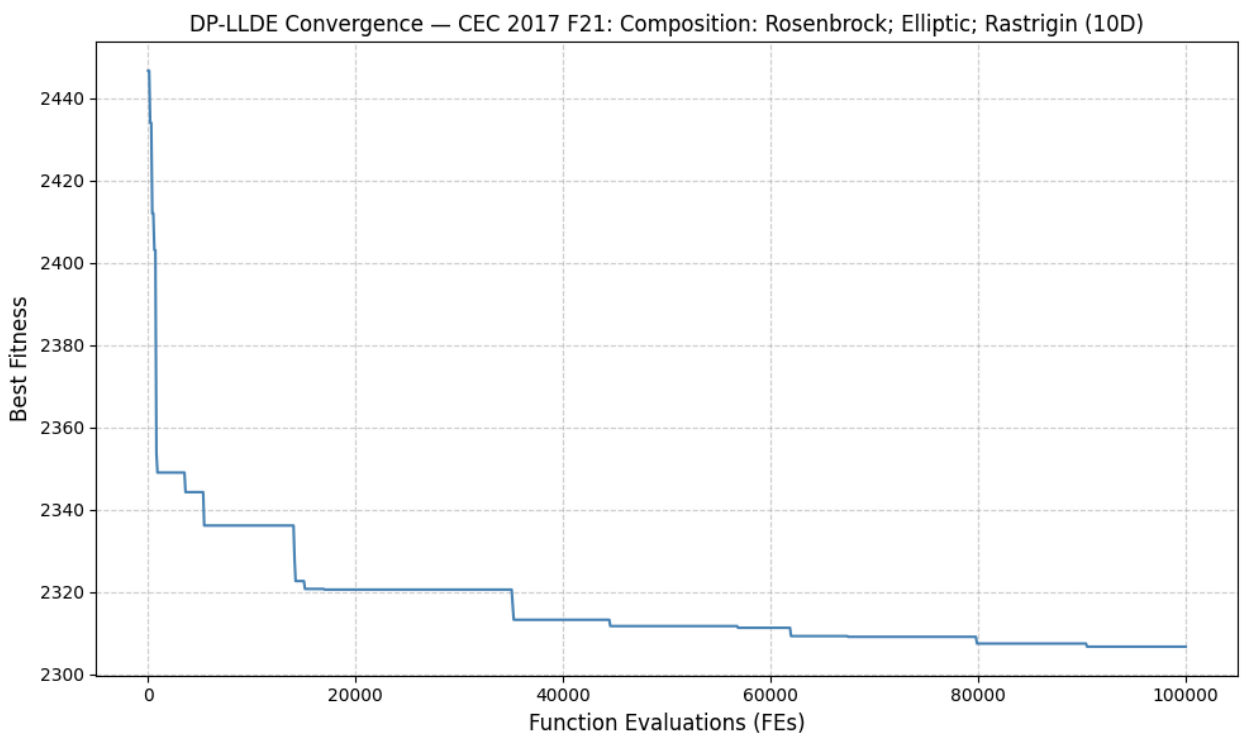
```
================================================================================
=
DP-LLDE on CEC 2017 F21 — Single Run
================================================================================
=
Function       : Composition: Rosenbrock; Elliptic; Rastrigin (10D)
Bounds         : [-100, 100]
Global optimum : 2100.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 2.446690e+03
Generation   10: Best = 2.349057e+03  FES =   1100  μ_F = 0.579  μ_CR = 0.349
Generation   20: Best = 2.349057e+03  FES =   2100  μ_F = 0.644  μ_CR = 0.352
Generation   30: Best = 2.349057e+03  FES =   3100  μ_F = 0.666  μ_CR = 0.334
Generation   40: Best = 2.344305e+03  FES =   4100  μ_F = 0.751  μ_CR = 0.316
Generation   50: Best = 2.344305e+03  FES =   5100  μ_F = 0.763  μ_CR = 0.303
Generation   60: Best = 2.336191e+03  FES =   6100  μ_F = 0.748  μ_CR = 0.268
Generation   70: Best = 2.336191e+03  FES =   7100  μ_F = 0.740  μ_CR = 0.235
Generation   80: Best = 2.336191e+03  FES =   8100  μ_F = 0.763  μ_CR = 0.205
Generation   90: Best = 2.336191e+03  FES =   9100  μ_F = 0.760  μ_CR = 0.161
Generation  100: Best = 2.336191e+03  FES =  10100  μ_F = 0.792  μ_CR = 0.134
Generation  110: Best = 2.336191e+03  FES =  11100  μ_F = 0.841  μ_CR = 0.125
Generation  120: Best = 2.336191e+03  FES =  12100  μ_F = 0.837  μ_CR = 0.121
Generation  130: Best = 2.336191e+03  FES =  13100  μ_F = 0.807  μ_CR = 0.091
Generation  140: Best = 2.336191e+03  FES =  14100  μ_F = 0.787  μ_CR = 0.090
Generation  150: Best = 2.322702e+03  FES =  15100  μ_F = 0.782  μ_CR = 0.110
Generation  160: Best = 2.320787e+03  FES =  16100  μ_F = 0.797  μ_CR = 0.121
Generation  170: Best = 2.320623e+03  FES =  17100  μ_F = 0.802  μ_CR = 0.118
Generation  180: Best = 2.320623e+03  FES =  18100  μ_F = 0.750  μ_CR = 0.108
Generation  190: Best = 2.320623e+03  FES =  19100  μ_F = 0.803  μ_CR = 0.107
Generation  200: Best = 2.320623e+03  FES =  20100  μ_F = 0.827  μ_CR = 0.086
Generation  210: Best = 2.320623e+03  FES =  21100  μ_F = 0.777  μ_CR = 0.096
Generation  220: Best = 2.320623e+03  FES =  22100  μ_F = 0.803  μ_CR = 0.072
Generation  230: Best = 2.320623e+03  FES =  23100  μ_F = 0.826  μ_CR = 0.074
Generation  240: Best = 2.320623e+03  FES =  24100  μ_F = 0.822  μ_CR = 0.079
Generation  250: Best = 2.320623e+03  FES =  25100  μ_F = 0.863  μ_CR = 0.082
Generation  260: Best = 2.320623e+03  FES =  26100  μ_F = 0.844  μ_CR = 0.066
Generation  270: Best = 2.320623e+03  FES =  27100  μ_F = 0.879  μ_CR = 0.068
Generation  280: Best = 2.320623e+03  FES =  28100  μ_F = 0.830  μ_CR = 0.086
Generation  290: Best = 2.320623e+03  FES =  29100  μ_F = 0.849  μ_CR = 0.103
Generation  300: Best = 2.320623e+03  FES =  30100  μ_F = 0.860  μ_CR = 0.086
Generation  310: Best = 2.320623e+03  FES =  31100  μ_F = 0.870  μ_CR = 0.096
Generation  320: Best = 2.320623e+03  FES =  32100  μ_F = 0.836  μ_CR = 0.103
Generation  330: Best = 2.320623e+03  FES =  33100  μ_F = 0.822  μ_CR = 0.098
Generation  340: Best = 2.320623e+03  FES =  34100  μ_F = 0.787  μ_CR = 0.091
Generation  350: Best = 2.320623e+03  FES =  35100  μ_F = 0.785  μ_CR = 0.082
Generation  360: Best = 2.313291e+03  FES =  36100  μ_F = 0.808  μ_CR = 0.071
Generation  370: Best = 2.313291e+03  FES =  37100  μ_F = 0.803  μ_CR = 0.078
Generation  380: Best = 2.313291e+03  FES =  38100  μ_F = 0.752  μ_CR = 0.080
Generation  390: Best = 2.313291e+03  FES =  39100  μ_F = 0.792  μ_CR = 0.083
Generation  400: Best = 2.313291e+03  FES =  40100  μ_F = 0.758  μ_CR = 0.101
Generation  410: Best = 2.313291e+03  FES =  41100  μ_F = 0.777  μ_CR = 0.107
Generation  420: Best = 2.313291e+03  FES =  42100  μ_F = 0.785  μ_CR = 0.116
Generation  430: Best = 2.313291e+03  FES =  43100  μ_F = 0.792  μ_CR = 0.103
Generation  440: Best = 2.313291e+03  FES =  44100  μ_F = 0.815  μ_CR = 0.093
```

```
Generation  450: Best = 2.311762e+03  FES =  45100  μ_F = 0.846  μ_CR = 0.067
Generation  460: Best = 2.311762e+03  FES =  46100  μ_F = 0.854  μ_CR = 0.079
Generation  470: Best = 2.311762e+03  FES =  47100  μ_F = 0.878  μ_CR = 0.078
Generation  480: Best = 2.311762e+03  FES =  48100  μ_F = 0.899  μ_CR = 0.082
Generation  490: Best = 2.311762e+03  FES =  49100  μ_F = 0.898  μ_CR = 0.096
Generation  500: Best = 2.311762e+03  FES =  50100  μ_F = 0.865  μ_CR = 0.094
Generation  510: Best = 2.311762e+03  FES =  51100  μ_F = 0.889  μ_CR = 0.088
Generation  520: Best = 2.311762e+03  FES =  52100  μ_F = 0.890  μ_CR = 0.096
Generation  530: Best = 2.311762e+03  FES =  53100  μ_F = 0.871  μ_CR = 0.066
Generation  540: Best = 2.311762e+03  FES =  54100  μ_F = 0.823  μ_CR = 0.086
Generation  550: Best = 2.311762e+03  FES =  55100  μ_F = 0.789  μ_CR = 0.098
Generation  560: Best = 2.311762e+03  FES =  56100  μ_F = 0.754  μ_CR = 0.086
Generation  570: Best = 2.311349e+03  FES =  57100  μ_F = 0.712  μ_CR = 0.077
Generation  580: Best = 2.311349e+03  FES =  58100  μ_F = 0.740  μ_CR = 0.083
Generation  590: Best = 2.311349e+03  FES =  59100  μ_F = 0.685  μ_CR = 0.092
Generation  600: Best = 2.311349e+03  FES =  60100  μ_F = 0.601  μ_CR = 0.076
Generation  610: Best = 2.311349e+03  FES =  61100  μ_F = 0.605  μ_CR = 0.064
Generation  620: Best = 2.309328e+03  FES =  62100  μ_F = 0.717  μ_CR = 0.070
Generation  630: Best = 2.309328e+03  FES =  63100  μ_F = 0.657  μ_CR = 0.073
Generation  640: Best = 2.309328e+03  FES =  64100  μ_F = 0.721  μ_CR = 0.082
Generation  650: Best = 2.309328e+03  FES =  65100  μ_F = 0.710  μ_CR = 0.085
Generation  660: Best = 2.309328e+03  FES =  66100  μ_F = 0.627  μ_CR = 0.081
Generation  670: Best = 2.309328e+03  FES =  67100  μ_F = 0.619  μ_CR = 0.059
Generation  680: Best = 2.309147e+03  FES =  68100  μ_F = 0.593  μ_CR = 0.055
Generation  690: Best = 2.309147e+03  FES =  69100  μ_F = 0.550  μ_CR = 0.052
Generation  700: Best = 2.309147e+03  FES =  70100  μ_F = 0.561  μ_CR = 0.054
Generation  710: Best = 2.309147e+03  FES =  71100  μ_F = 0.612  μ_CR = 0.058
Generation  720: Best = 2.309147e+03  FES =  72100  μ_F = 0.608  μ_CR = 0.076
Generation  730: Best = 2.309147e+03  FES =  73100  μ_F = 0.684  μ_CR = 0.090
Generation  740: Best = 2.309147e+03  FES =  74100  μ_F = 0.615  μ_CR = 0.115
Generation  750: Best = 2.309147e+03  FES =  75100  μ_F = 0.695  μ_CR = 0.101
Generation  760: Best = 2.309147e+03  FES =  76100  μ_F = 0.725  μ_CR = 0.089
Generation  770: Best = 2.309147e+03  FES =  77100  μ_F = 0.701  μ_CR = 0.080
Generation  780: Best = 2.309147e+03  FES =  78100  μ_F = 0.646  μ_CR = 0.080
Generation  790: Best = 2.309147e+03  FES =  79100  μ_F = 0.671  μ_CR = 0.081
Generation  800: Best = 2.307513e+03  FES =  80100  μ_F = 0.621  μ_CR = 0.076
Generation  810: Best = 2.307513e+03  FES =  81100  μ_F = 0.695  μ_CR = 0.090
Generation  820: Best = 2.307513e+03  FES =  82100  μ_F = 0.661  μ_CR = 0.099
Generation  830: Best = 2.307513e+03  FES =  83100  μ_F = 0.694  μ_CR = 0.079
Generation  840: Best = 2.307513e+03  FES =  84100  μ_F = 0.693  μ_CR = 0.086
Generation  850: Best = 2.307513e+03  FES =  85100  μ_F = 0.684  μ_CR = 0.103
Generation  860: Best = 2.307513e+03  FES =  86100  μ_F = 0.664  μ_CR = 0.084
Generation  870: Best = 2.307513e+03  FES =  87100  μ_F = 0.671  μ_CR = 0.070
Generation  880: Best = 2.307513e+03  FES =  88100  μ_F = 0.719  μ_CR = 0.084
Generation  890: Best = 2.307513e+03  FES =  89100  μ_F = 0.690  μ_CR = 0.073
Generation  900: Best = 2.307513e+03  FES =  90100  μ_F = 0.642  μ_CR = 0.067
Generation  910: Best = 2.306755e+03  FES =  91100  μ_F = 0.657  μ_CR = 0.074
Generation  920: Best = 2.306755e+03  FES =  92100  μ_F = 0.661  μ_CR = 0.094
Generation  930: Best = 2.306755e+03  FES =  93100  μ_F = 0.600  μ_CR = 0.088
Generation  940: Best = 2.306755e+03  FES =  94100  μ_F = 0.523  μ_CR = 0.078
Generation  950: Best = 2.306755e+03  FES =  95100  μ_F = 0.501  μ_CR = 0.069
Generation  960: Best = 2.306755e+03  FES =  96100  μ_F = 0.428  μ_CR = 0.064
Generation  970: Best = 2.306755e+03  FES =  97100  μ_F = 0.468  μ_CR = 0.077
Generation  980: Best = 2.306755e+03  FES =  98100  μ_F = 0.498  μ_CR = 0.084
```

```
Generation  990: Best = 2.306755e+03  FES =  99100  μ_F = 0.579  μ_CR = 0.100

Optimization complete.
Best Fitness             = 2.306755e+03
Total Function Evals     = 100000
Total Successful Updates = 2272

================================================================================
=
Results:
  Best Fitness           : 2.306755e+03
  Global Optimum         : 2.100000e+03
  Error (|best-optimum|) : 2.067551e+02
  Best Solution (5 dim)  : [ -6.5445353  -44.08252675  34.38848659  52.552515
48.77772676]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F21.png
```

In [31]:

```python
# ==============================================================
# F22 — Composition Function 2 (Rastrigin; Griewank; Schwefel)
# ==============================================================
run_and_plot('F22', make_wrapper(composition.f22), 2200.0, 'Composition: Rastr
```

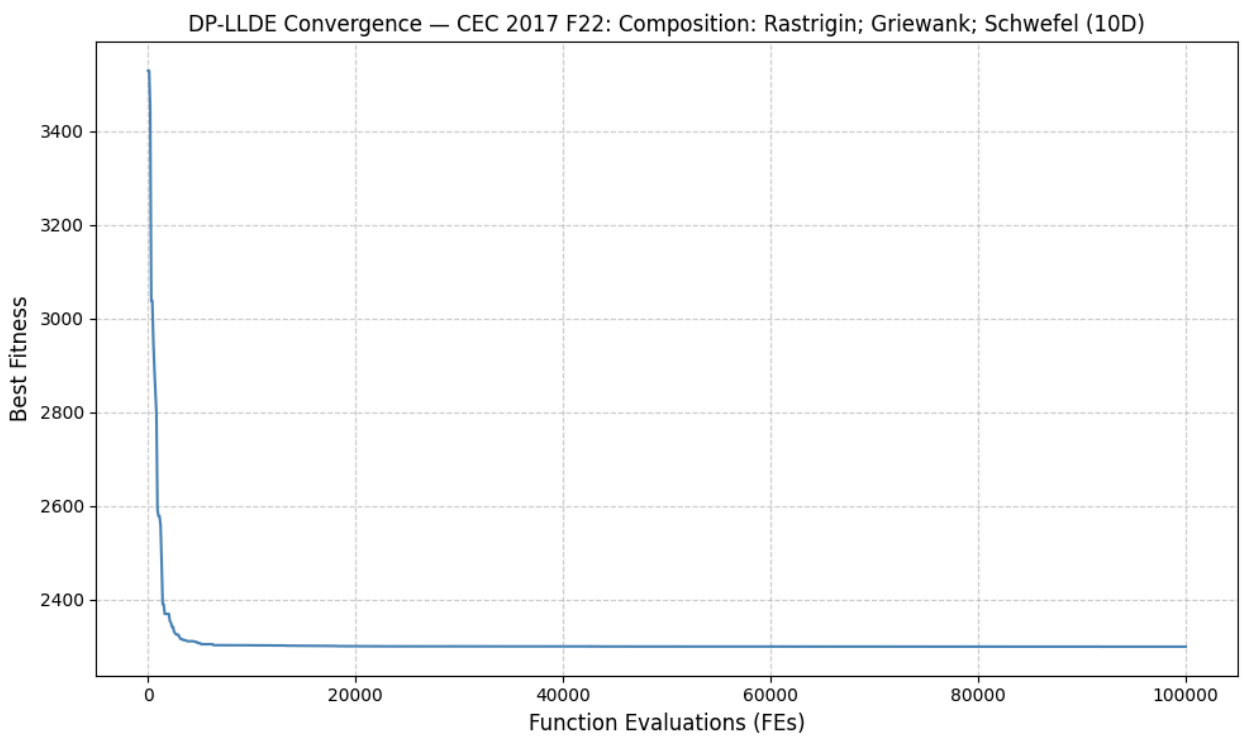
```
================================================================================
=
DP-LLDE on CEC 2017 F22 — Single Run
================================================================================
=
Function       : Composition: Rastrigin; Griewank; Schwefel (10D)
Bounds         : [-100, 100]
Global optimum : 2200.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 3.528003e+03
Generation   10: Best = 2.578427e+03  FES =   1100  μ_F = 0.574  μ_CR = 0.351
Generation   20: Best = 2.370023e+03  FES =   2100  μ_F = 0.620  μ_CR = 0.356
Generation   30: Best = 2.321176e+03  FES =   3100  μ_F = 0.661  μ_CR = 0.365
Generation   40: Best = 2.311879e+03  FES =   4100  μ_F = 0.704  μ_CR = 0.363
Generation   50: Best = 2.307778e+03  FES =   5100  μ_F = 0.741  μ_CR = 0.366
Generation   60: Best = 2.305415e+03  FES =   6100  μ_F = 0.761  μ_CR = 0.362
Generation   70: Best = 2.303457e+03  FES =   7100  μ_F = 0.765  μ_CR = 0.357
Generation   80: Best = 2.303224e+03  FES =   8100  μ_F = 0.762  μ_CR = 0.362
Generation   90: Best = 2.303224e+03  FES =   9100  μ_F = 0.790  μ_CR = 0.354
Generation  100: Best = 2.303091e+03  FES =  10100  μ_F = 0.810  μ_CR = 0.327
Generation  110: Best = 2.303091e+03  FES =  11100  μ_F = 0.813  μ_CR = 0.297
Generation  120: Best = 2.302790e+03  FES =  12100  μ_F = 0.790  μ_CR = 0.265
Generation  130: Best = 2.302716e+03  FES =  13100  μ_F = 0.822  μ_CR = 0.227
Generation  140: Best = 2.302281e+03  FES =  14100  μ_F = 0.847  μ_CR = 0.192
Generation  150: Best = 2.302039e+03  FES =  15100  μ_F = 0.855  μ_CR = 0.160
Generation  160: Best = 2.302028e+03  FES =  16100  μ_F = 0.864  μ_CR = 0.156
Generation  170: Best = 2.301702e+03  FES =  17100  μ_F = 0.866  μ_CR = 0.117
Generation  180: Best = 2.301702e+03  FES =  18100  μ_F = 0.837  μ_CR = 0.112
Generation  190: Best = 2.301144e+03  FES =  19100  μ_F = 0.858  μ_CR = 0.112
Generation  200: Best = 2.301144e+03  FES =  20100  μ_F = 0.858  μ_CR = 0.100
Generation  210: Best = 2.301144e+03  FES =  21100  μ_F = 0.853  μ_CR = 0.086
Generation  220: Best = 2.301058e+03  FES =  22100  μ_F = 0.830  μ_CR = 0.091
Generation  230: Best = 2.300967e+03  FES =  23100  μ_F = 0.823  μ_CR = 0.098
Generation  240: Best = 2.300950e+03  FES =  24100  μ_F = 0.803  μ_CR = 0.100
Generation  250: Best = 2.300949e+03  FES =  25100  μ_F = 0.799  μ_CR = 0.104
Generation  260: Best = 2.300949e+03  FES =  26100  μ_F = 0.817  μ_CR = 0.100
Generation  270: Best = 2.300949e+03  FES =  27100  μ_F = 0.837  μ_CR = 0.093
Generation  280: Best = 2.300929e+03  FES =  28100  μ_F = 0.828  μ_CR = 0.104
Generation  290: Best = 2.300929e+03  FES =  29100  μ_F = 0.863  μ_CR = 0.112
Generation  300: Best = 2.300921e+03  FES =  30100  μ_F = 0.863  μ_CR = 0.102
Generation  310: Best = 2.300897e+03  FES =  31100  μ_F = 0.869  μ_CR = 0.092
Generation  320: Best = 2.300897e+03  FES =  32100  μ_F = 0.864  μ_CR = 0.100
Generation  330: Best = 2.300886e+03  FES =  33100  μ_F = 0.870  μ_CR = 0.087
Generation  340: Best = 2.300886e+03  FES =  34100  μ_F = 0.866  μ_CR = 0.092
Generation  350: Best = 2.300870e+03  FES =  35100  μ_F = 0.878  μ_CR = 0.118
Generation  360: Best = 2.300867e+03  FES =  36100  μ_F = 0.883  μ_CR = 0.111
Generation  370: Best = 2.300864e+03  FES =  37100  μ_F = 0.881  μ_CR = 0.104
Generation  380: Best = 2.300860e+03  FES =  38100  μ_F = 0.876  μ_CR = 0.095
Generation  390: Best = 2.300857e+03  FES =  39100  μ_F = 0.853  μ_CR = 0.087
Generation  400: Best = 2.300857e+03  FES =  40100  μ_F = 0.825  μ_CR = 0.101
Generation  410: Best = 2.300857e+03  FES =  41100  μ_F = 0.813  μ_CR = 0.098
Generation  420: Best = 2.300857e+03  FES =  42100  μ_F = 0.843  μ_CR = 0.090
Generation  430: Best = 2.300761e+03  FES =  43100  μ_F = 0.833  μ_CR = 0.087
Generation  440: Best = 2.300528e+03  FES =  44100  μ_F = 0.855  μ_CR = 0.098
```

```
Generation  450: Best = 2.300528e+03  FES =  45100  μ_F = 0.848  μ_CR = 0.095
Generation  460: Best = 2.300513e+03  FES =  46100  μ_F = 0.810  μ_CR = 0.105
Generation  470: Best = 2.300510e+03  FES =  47100  μ_F = 0.820  μ_CR = 0.116
Generation  480: Best = 2.300508e+03  FES =  48100  μ_F = 0.830  μ_CR = 0.109
Generation  490: Best = 2.300508e+03  FES =  49100  μ_F = 0.825  μ_CR = 0.116
Generation  500: Best = 2.300502e+03  FES =  50100  μ_F = 0.827  μ_CR = 0.106
Generation  510: Best = 2.300502e+03  FES =  51100  μ_F = 0.831  μ_CR = 0.097
Generation  520: Best = 2.300502e+03  FES =  52100  μ_F = 0.866  μ_CR = 0.087
Generation  530: Best = 2.300502e+03  FES =  53100  μ_F = 0.862  μ_CR = 0.088
Generation  540: Best = 2.300422e+03  FES =  54100  μ_F = 0.813  μ_CR = 0.074
Generation  550: Best = 2.300422e+03  FES =  55100  μ_F = 0.802  μ_CR = 0.071
Generation  560: Best = 2.300411e+03  FES =  56100  μ_F = 0.746  μ_CR = 0.087
Generation  570: Best = 2.300411e+03  FES =  57100  μ_F = 0.801  μ_CR = 0.095
Generation  580: Best = 2.300411e+03  FES =  58100  μ_F = 0.771  μ_CR = 0.086
Generation  590: Best = 2.300411e+03  FES =  59100  μ_F = 0.740  μ_CR = 0.109
Generation  600: Best = 2.300411e+03  FES =  60100  μ_F = 0.760  μ_CR = 0.100
Generation  610: Best = 2.300400e+03  FES =  61100  μ_F = 0.759  μ_CR = 0.097
Generation  620: Best = 2.300392e+03  FES =  62100  μ_F = 0.742  μ_CR = 0.108
Generation  630: Best = 2.300365e+03  FES =  63100  μ_F = 0.803  μ_CR = 0.096
Generation  640: Best = 2.300365e+03  FES =  64100  μ_F = 0.791  μ_CR = 0.112
Generation  650: Best = 2.300345e+03  FES =  65100  μ_F = 0.776  μ_CR = 0.090
Generation  660: Best = 2.300327e+03  FES =  66100  μ_F = 0.797  μ_CR = 0.100
Generation  670: Best = 2.300327e+03  FES =  67100  μ_F = 0.840  μ_CR = 0.096
Generation  680: Best = 2.300327e+03  FES =  68100  μ_F = 0.850  μ_CR = 0.120
Generation  690: Best = 2.300327e+03  FES =  69100  μ_F = 0.815  μ_CR = 0.115
Generation  700: Best = 2.300327e+03  FES =  70100  μ_F = 0.811  μ_CR = 0.098
Generation  710: Best = 2.300290e+03  FES =  71100  μ_F = 0.778  μ_CR = 0.096
Generation  720: Best = 2.300289e+03  FES =  72100  μ_F = 0.772  μ_CR = 0.102
Generation  730: Best = 2.300289e+03  FES =  73100  μ_F = 0.770  μ_CR = 0.096
Generation  740: Best = 2.300288e+03  FES =  74100  μ_F = 0.813  μ_CR = 0.082
Generation  750: Best = 2.300288e+03  FES =  75100  μ_F = 0.770  μ_CR = 0.089
Generation  760: Best = 2.300288e+03  FES =  76100  μ_F = 0.732  μ_CR = 0.115
Generation  770: Best = 2.300288e+03  FES =  77100  μ_F = 0.711  μ_CR = 0.092
Generation  780: Best = 2.300288e+03  FES =  78100  μ_F = 0.717  μ_CR = 0.079
Generation  790: Best = 2.300288e+03  FES =  79100  μ_F = 0.728  μ_CR = 0.099
Generation  800: Best = 2.300275e+03  FES =  80100  μ_F = 0.727  μ_CR = 0.084
Generation  810: Best = 2.300275e+03  FES =  81100  μ_F = 0.767  μ_CR = 0.085
Generation  820: Best = 2.300275e+03  FES =  82100  μ_F = 0.796  μ_CR = 0.073
Generation  830: Best = 2.300275e+03  FES =  83100  μ_F = 0.801  μ_CR = 0.085
Generation  840: Best = 2.300275e+03  FES =  84100  μ_F = 0.743  μ_CR = 0.082
Generation  850: Best = 2.300250e+03  FES =  85100  μ_F = 0.758  μ_CR = 0.081
Generation  860: Best = 2.300250e+03  FES =  86100  μ_F = 0.722  μ_CR = 0.085
Generation  870: Best = 2.300250e+03  FES =  87100  μ_F = 0.719  μ_CR = 0.094
Generation  880: Best = 2.300250e+03  FES =  88100  μ_F = 0.733  μ_CR = 0.093
Generation  890: Best = 2.300250e+03  FES =  89100  μ_F = 0.712  μ_CR = 0.114
Generation  900: Best = 2.300250e+03  FES =  90100  μ_F = 0.682  μ_CR = 0.122
Generation  910: Best = 2.300250e+03  FES =  91100  μ_F = 0.648  μ_CR = 0.095
Generation  920: Best = 2.300250e+03  FES =  92100  μ_F = 0.690  μ_CR = 0.113
Generation  930: Best = 2.300237e+03  FES =  93100  μ_F = 0.650  μ_CR = 0.109
Generation  940: Best = 2.300237e+03  FES =  94100  μ_F = 0.659  μ_CR = 0.107
Generation  950: Best = 2.300237e+03  FES =  95100  μ_F = 0.679  μ_CR = 0.102
Generation  960: Best = 2.300237e+03  FES =  96100  μ_F = 0.671  μ_CR = 0.106
Generation  970: Best = 2.300237e+03  FES =  97100  μ_F = 0.713  μ_CR = 0.113
Generation  980: Best = 2.300237e+03  FES =  98100  μ_F = 0.676  μ_CR = 0.091
```

```
Generation  990: Best = 2.300180e+03  FES =  99100  μ_F = 0.665  μ_CR = 0.089

Optimization complete.
Best Fitness             = 2.300180e+03
Total Function Evals     = 100000
Total Successful Updates = 6928

================================================================================
=
Results:
  Best Fitness           : 2.300180e+03
  Global Optimum         : 2.200000e+03
  Error (|best-optimum|) : 1.001800e+02
  Best Solution (5 dim)  : [-52.40596509  11.27138108  23.01676907  35.13242966
30.4893901 ]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F22.png
```

In [32]:

```
# ============================================================
# F23 — Composition Function 3 (Rosenbrock; Ackley; Schwefel; Rastrigin)
# ============================================================
run_and_plot('F23', make_wrapper(composition.f23), 2300.0, 'Composition: Roser
```

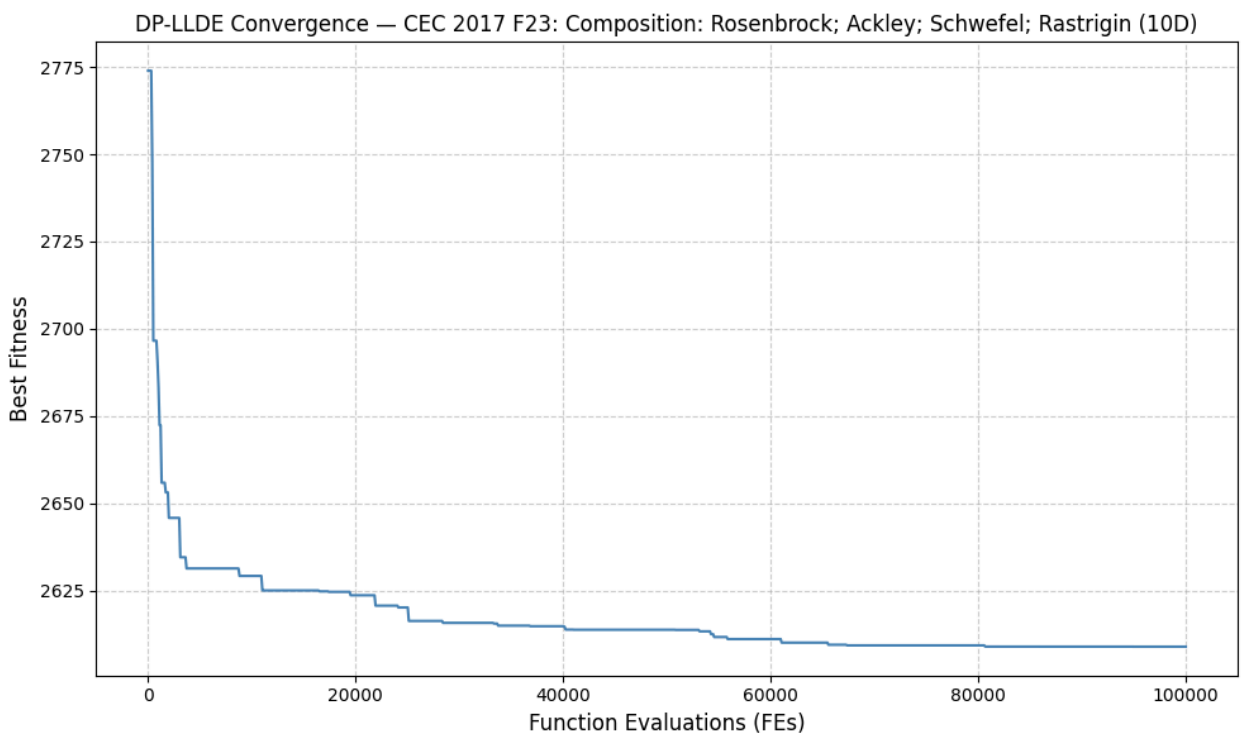
```
================================================================================
=
DP-LLDE on CEC 2017 F23 — Single Run
================================================================================
=
Function       : Composition: Rosenbrock; Ackley; Schwefel; Rastrigin (10D)
Bounds         : [-100, 100]
Global optimum : 2300.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 2.773921e+03
Generation   10: Best = 2.683384e+03  FES =   1100  μ_F = 0.597  μ_CR = 0.355
Generation   20: Best = 2.645769e+03  FES =   2100  μ_F = 0.666  μ_CR = 0.352
Generation   30: Best = 2.645769e+03  FES =   3100  μ_F = 0.738  μ_CR = 0.342
Generation   40: Best = 2.631283e+03  FES =   4100  μ_F = 0.784  μ_CR = 0.300
Generation   50: Best = 2.631283e+03  FES =   5100  μ_F = 0.832  μ_CR = 0.269
Generation   60: Best = 2.631283e+03  FES =   6100  μ_F = 0.828  μ_CR = 0.224
Generation   70: Best = 2.631283e+03  FES =   7100  μ_F = 0.825  μ_CR = 0.237
Generation   80: Best = 2.631283e+03  FES =   8100  μ_F = 0.844  μ_CR = 0.192
Generation   90: Best = 2.629149e+03  FES =   9100  μ_F = 0.859  μ_CR = 0.148
Generation  100: Best = 2.629149e+03  FES =  10100  μ_F = 0.876  μ_CR = 0.127
Generation  110: Best = 2.625020e+03  FES =  11100  μ_F = 0.851  μ_CR = 0.140
Generation  120: Best = 2.624978e+03  FES =  12100  μ_F = 0.848  μ_CR = 0.112
Generation  130: Best = 2.624978e+03  FES =  13100  μ_F = 0.807  μ_CR = 0.118
Generation  140: Best = 2.624978e+03  FES =  14100  μ_F = 0.726  μ_CR = 0.102
Generation  150: Best = 2.624978e+03  FES =  15100  μ_F = 0.744  μ_CR = 0.083
Generation  160: Best = 2.624978e+03  FES =  16100  μ_F = 0.764  μ_CR = 0.089
Generation  170: Best = 2.624726e+03  FES =  17100  μ_F = 0.785  μ_CR = 0.096
Generation  180: Best = 2.624576e+03  FES =  18100  μ_F = 0.800  μ_CR = 0.083
Generation  190: Best = 2.624576e+03  FES =  19100  μ_F = 0.831  μ_CR = 0.092
Generation  200: Best = 2.623596e+03  FES =  20100  μ_F = 0.865  μ_CR = 0.076
Generation  210: Best = 2.623596e+03  FES =  21100  μ_F = 0.807  μ_CR = 0.084
Generation  220: Best = 2.620625e+03  FES =  22100  μ_F = 0.680  μ_CR = 0.066
Generation  230: Best = 2.620625e+03  FES =  23100  μ_F = 0.681  μ_CR = 0.070
Generation  240: Best = 2.620625e+03  FES =  24100  μ_F = 0.741  μ_CR = 0.074
Generation  250: Best = 2.620104e+03  FES =  25100  μ_F = 0.744  μ_CR = 0.095
Generation  260: Best = 2.616242e+03  FES =  26100  μ_F = 0.731  μ_CR = 0.079
Generation  270: Best = 2.616242e+03  FES =  27100  μ_F = 0.729  μ_CR = 0.071
Generation  280: Best = 2.616242e+03  FES =  28100  μ_F = 0.785  μ_CR = 0.073
Generation  290: Best = 2.615690e+03  FES =  29100  μ_F = 0.722  μ_CR = 0.070
Generation  300: Best = 2.615690e+03  FES =  30100  μ_F = 0.705  μ_CR = 0.068
Generation  310: Best = 2.615690e+03  FES =  31100  μ_F = 0.764  μ_CR = 0.105
Generation  320: Best = 2.615690e+03  FES =  32100  μ_F = 0.784  μ_CR = 0.091
Generation  330: Best = 2.615690e+03  FES =  33100  μ_F = 0.736  μ_CR = 0.080
Generation  340: Best = 2.614892e+03  FES =  34100  μ_F = 0.738  μ_CR = 0.071
Generation  350: Best = 2.614892e+03  FES =  35100  μ_F = 0.750  μ_CR = 0.081
Generation  360: Best = 2.614892e+03  FES =  36100  μ_F = 0.672  μ_CR = 0.087
Generation  370: Best = 2.614734e+03  FES =  37100  μ_F = 0.626  μ_CR = 0.078
Generation  380: Best = 2.614734e+03  FES =  38100  μ_F = 0.640  μ_CR = 0.071
Generation  390: Best = 2.614734e+03  FES =  39100  μ_F = 0.692  μ_CR = 0.065
Generation  400: Best = 2.614734e+03  FES =  40100  μ_F = 0.746  μ_CR = 0.064
Generation  410: Best = 2.613723e+03  FES =  41100  μ_F = 0.785  μ_CR = 0.064
Generation  420: Best = 2.613723e+03  FES =  42100  μ_F = 0.781  μ_CR = 0.061
Generation  430: Best = 2.613723e+03  FES =  43100  μ_F = 0.798  μ_CR = 0.075
Generation  440: Best = 2.613723e+03  FES =  44100  μ_F = 0.776  μ_CR = 0.066
```

```
Generation  450: Best = 2.613723e+03  FES =  45100  μ_F = 0.796  μ_CR = 0.071
Generation  460: Best = 2.613723e+03  FES =  46100  μ_F = 0.825  μ_CR = 0.077
Generation  470: Best = 2.613723e+03  FES =  47100  μ_F = 0.794  μ_CR = 0.075
Generation  480: Best = 2.613723e+03  FES =  48100  μ_F = 0.818  μ_CR = 0.063
Generation  490: Best = 2.613723e+03  FES =  49100  μ_F = 0.829  μ_CR = 0.073
Generation  500: Best = 2.613723e+03  FES =  50100  μ_F = 0.835  μ_CR = 0.100
Generation  510: Best = 2.613677e+03  FES =  51100  μ_F = 0.816  μ_CR = 0.086
Generation  520: Best = 2.613677e+03  FES =  52100  μ_F = 0.732  μ_CR = 0.102
Generation  530: Best = 2.613677e+03  FES =  53100  μ_F = 0.786  μ_CR = 0.101
Generation  540: Best = 2.613259e+03  FES =  54100  μ_F = 0.840  μ_CR = 0.081
Generation  550: Best = 2.611631e+03  FES =  55100  μ_F = 0.819  μ_CR = 0.069
Generation  560: Best = 2.611043e+03  FES =  56100  μ_F = 0.821  μ_CR = 0.077
Generation  570: Best = 2.611043e+03  FES =  57100  μ_F = 0.822  μ_CR = 0.085
Generation  580: Best = 2.611043e+03  FES =  58100  μ_F = 0.845  μ_CR = 0.075
Generation  590: Best = 2.611043e+03  FES =  59100  μ_F = 0.716  μ_CR = 0.076
Generation  600: Best = 2.611043e+03  FES =  60100  μ_F = 0.686  μ_CR = 0.088
Generation  610: Best = 2.610028e+03  FES =  61100  μ_F = 0.732  μ_CR = 0.109
Generation  620: Best = 2.610028e+03  FES =  62100  μ_F = 0.722  μ_CR = 0.084
Generation  630: Best = 2.610028e+03  FES =  63100  μ_F = 0.766  μ_CR = 0.079
Generation  640: Best = 2.610028e+03  FES =  64100  μ_F = 0.742  μ_CR = 0.091
Generation  650: Best = 2.610028e+03  FES =  65100  μ_F = 0.671  μ_CR = 0.084
Generation  660: Best = 2.609422e+03  FES =  66100  μ_F = 0.714  μ_CR = 0.085
Generation  670: Best = 2.609422e+03  FES =  67100  μ_F = 0.624  μ_CR = 0.096
Generation  680: Best = 2.609240e+03  FES =  68100  μ_F = 0.622  μ_CR = 0.088
Generation  690: Best = 2.609240e+03  FES =  69100  μ_F = 0.600  μ_CR = 0.101
Generation  700: Best = 2.609240e+03  FES =  70100  μ_F = 0.637  μ_CR = 0.084
Generation  710: Best = 2.609240e+03  FES =  71100  μ_F = 0.609  μ_CR = 0.092
Generation  720: Best = 2.609240e+03  FES =  72100  μ_F = 0.579  μ_CR = 0.082
Generation  730: Best = 2.609240e+03  FES =  73100  μ_F = 0.647  μ_CR = 0.094
Generation  740: Best = 2.609240e+03  FES =  74100  μ_F = 0.627  μ_CR = 0.115
Generation  750: Best = 2.609240e+03  FES =  75100  μ_F = 0.644  μ_CR = 0.089
Generation  760: Best = 2.609240e+03  FES =  76100  μ_F = 0.610  μ_CR = 0.110
Generation  770: Best = 2.609240e+03  FES =  77100  μ_F = 0.629  μ_CR = 0.087
Generation  780: Best = 2.609240e+03  FES =  78100  μ_F = 0.672  μ_CR = 0.096
Generation  790: Best = 2.609240e+03  FES =  79100  μ_F = 0.740  μ_CR = 0.087
Generation  800: Best = 2.609240e+03  FES =  80100  μ_F = 0.765  μ_CR = 0.079
Generation  810: Best = 2.608888e+03  FES =  81100  μ_F = 0.737  μ_CR = 0.083
Generation  820: Best = 2.608888e+03  FES =  82100  μ_F = 0.790  μ_CR = 0.073
Generation  830: Best = 2.608888e+03  FES =  83100  μ_F = 0.833  μ_CR = 0.080
Generation  840: Best = 2.608888e+03  FES =  84100  μ_F = 0.857  μ_CR = 0.070
Generation  850: Best = 2.608888e+03  FES =  85100  μ_F = 0.753  μ_CR = 0.068
Generation  860: Best = 2.608888e+03  FES =  86100  μ_F = 0.669  μ_CR = 0.086
Generation  870: Best = 2.608888e+03  FES =  87100  μ_F = 0.739  μ_CR = 0.071
Generation  880: Best = 2.608888e+03  FES =  88100  μ_F = 0.743  μ_CR = 0.075
Generation  890: Best = 2.608888e+03  FES =  89100  μ_F = 0.680  μ_CR = 0.108
Generation  900: Best = 2.608888e+03  FES =  90100  μ_F = 0.716  μ_CR = 0.108
Generation  910: Best = 2.608872e+03  FES =  91100  μ_F = 0.751  μ_CR = 0.087
Generation  920: Best = 2.608872e+03  FES =  92100  μ_F = 0.787  μ_CR = 0.066
Generation  930: Best = 2.608872e+03  FES =  93100  μ_F = 0.753  μ_CR = 0.095
Generation  940: Best = 2.608872e+03  FES =  94100  μ_F = 0.803  μ_CR = 0.087
Generation  950: Best = 2.608872e+03  FES =  95100  μ_F = 0.821  μ_CR = 0.065
Generation  960: Best = 2.608872e+03  FES =  96100  μ_F = 0.844  μ_CR = 0.075
Generation  970: Best = 2.608872e+03  FES =  97100  μ_F = 0.825  μ_CR = 0.085
Generation  980: Best = 2.608872e+03  FES =  98100  μ_F = 0.760  μ_CR = 0.074
```

```
Generation  990: Best = 2.608872e+03  FES =  99100  μ_F = 0.729  μ_CR = 0.076

Optimization complete.
Best Fitness             = 2.608867e+03
Total Function Evals     = 100000
Total Successful Updates = 2661

================================================================================
=
Results:
  Best Fitness           : 2.608867e+03
  Global Optimum         : 2.300000e+03
  Error (|best-optimum|) : 3.088673e+02
  Best Solution (5 dim)  : [-84.30719475 -78.54180499 -60.48625863  85.20259576
-55.51996901]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F23.png
```

In [33]:

```
# ==============================================================
# F24 — Composition Function 4 (Ackley; Elliptic; Griewank; Rastrigin)
# ==============================================================
run_and_plot('F24', make_wrapper(composition.f24), 2400.0, 'Composition: Ackle
```

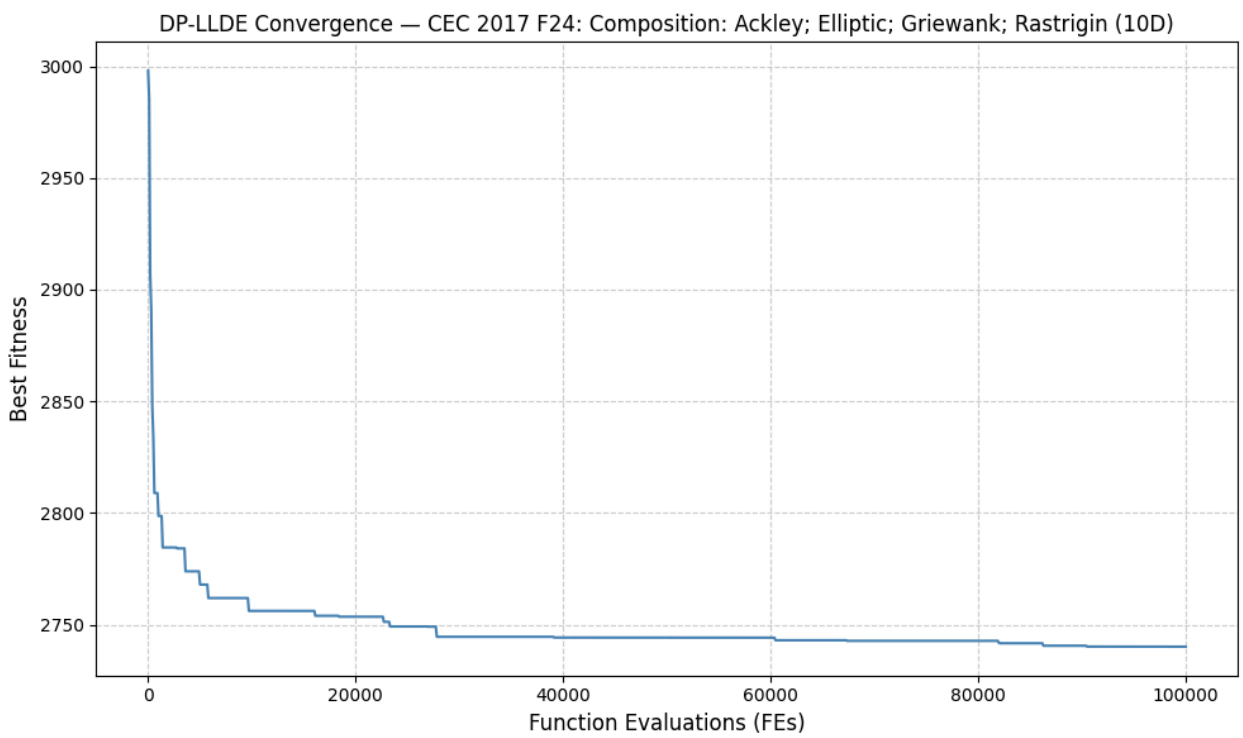
```
================================================================================
=
DP-LLDE on CEC 2017 F24 — Single Run
================================================================================
=
Function       : Composition: Ackley; Elliptic; Griewank; Rastrigin (10D)
Bounds         : [-100, 100]
Global optimum : 2400.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 2.997998e+03
Generation   10: Best = 2.798695e+03  FES =   1100  μ_F = 0.601  μ_CR = 0.349
Generation   20: Best = 2.784612e+03  FES =   2100  μ_F = 0.656  μ_CR = 0.356
Generation   30: Best = 2.784204e+03  FES =   3100  μ_F = 0.685  μ_CR = 0.366
Generation   40: Best = 2.773931e+03  FES =   4100  μ_F = 0.743  μ_CR = 0.349
Generation   50: Best = 2.767987e+03  FES =   5100  μ_F = 0.765  μ_CR = 0.315
Generation   60: Best = 2.761958e+03  FES =   6100  μ_F = 0.768  μ_CR = 0.273
Generation   70: Best = 2.761958e+03  FES =   7100  μ_F = 0.812  μ_CR = 0.240
Generation   80: Best = 2.761958e+03  FES =   8100  μ_F = 0.806  μ_CR = 0.201
Generation   90: Best = 2.761958e+03  FES =   9100  μ_F = 0.706  μ_CR = 0.216
Generation  100: Best = 2.756200e+03  FES =  10100  μ_F = 0.737  μ_CR = 0.153
Generation  110: Best = 2.756200e+03  FES =  11100  μ_F = 0.778  μ_CR = 0.145
Generation  120: Best = 2.756200e+03  FES =  12100  μ_F = 0.796  μ_CR = 0.130
Generation  130: Best = 2.756200e+03  FES =  13100  μ_F = 0.825  μ_CR = 0.105
Generation  140: Best = 2.756200e+03  FES =  14100  μ_F = 0.805  μ_CR = 0.108
Generation  150: Best = 2.756200e+03  FES =  15100  μ_F = 0.786  μ_CR = 0.104
Generation  160: Best = 2.756195e+03  FES =  16100  μ_F = 0.780  μ_CR = 0.073
Generation  170: Best = 2.754027e+03  FES =  17100  μ_F = 0.841  μ_CR = 0.095
Generation  180: Best = 2.754027e+03  FES =  18100  μ_F = 0.823  μ_CR = 0.088
Generation  190: Best = 2.753608e+03  FES =  19100  μ_F = 0.787  μ_CR = 0.080
Generation  200: Best = 2.753608e+03  FES =  20100  μ_F = 0.770  μ_CR = 0.079
Generation  210: Best = 2.753602e+03  FES =  21100  μ_F = 0.738  μ_CR = 0.079
Generation  220: Best = 2.753602e+03  FES =  22100  μ_F = 0.746  μ_CR = 0.084
Generation  230: Best = 2.751323e+03  FES =  23100  μ_F = 0.702  μ_CR = 0.087
Generation  240: Best = 2.749233e+03  FES =  24100  μ_F = 0.721  μ_CR = 0.085
Generation  250: Best = 2.749233e+03  FES =  25100  μ_F = 0.677  μ_CR = 0.078
Generation  260: Best = 2.749233e+03  FES =  26100  μ_F = 0.706  μ_CR = 0.072
Generation  270: Best = 2.749153e+03  FES =  27100  μ_F = 0.718  μ_CR = 0.085
Generation  280: Best = 2.744605e+03  FES =  28100  μ_F = 0.741  μ_CR = 0.083
Generation  290: Best = 2.744605e+03  FES =  29100  μ_F = 0.807  μ_CR = 0.084
Generation  300: Best = 2.744605e+03  FES =  30100  μ_F = 0.809  μ_CR = 0.089
Generation  310: Best = 2.744605e+03  FES =  31100  μ_F = 0.872  μ_CR = 0.082
Generation  320: Best = 2.744605e+03  FES =  32100  μ_F = 0.848  μ_CR = 0.065
Generation  330: Best = 2.744605e+03  FES =  33100  μ_F = 0.848  μ_CR = 0.057
Generation  340: Best = 2.744605e+03  FES =  34100  μ_F = 0.798  μ_CR = 0.060
Generation  350: Best = 2.744605e+03  FES =  35100  μ_F = 0.706  μ_CR = 0.083
Generation  360: Best = 2.744605e+03  FES =  36100  μ_F = 0.687  μ_CR = 0.072
Generation  370: Best = 2.744605e+03  FES =  37100  μ_F = 0.740  μ_CR = 0.077
Generation  380: Best = 2.744605e+03  FES =  38100  μ_F = 0.800  μ_CR = 0.079
Generation  390: Best = 2.744605e+03  FES =  39100  μ_F = 0.775  μ_CR = 0.087
Generation  400: Best = 2.744292e+03  FES =  40100  μ_F = 0.808  μ_CR = 0.093
Generation  410: Best = 2.744292e+03  FES =  41100  μ_F = 0.763  μ_CR = 0.080
Generation  420: Best = 2.744292e+03  FES =  42100  μ_F = 0.777  μ_CR = 0.101
Generation  430: Best = 2.744292e+03  FES =  43100  μ_F = 0.795  μ_CR = 0.066
Generation  440: Best = 2.744292e+03  FES =  44100  μ_F = 0.777  μ_CR = 0.076
```

```
Generation  450: Best = 2.744292e+03  FES =  45100  μ_F = 0.818  μ_CR = 0.082
Generation  460: Best = 2.744292e+03  FES =  46100  μ_F = 0.756  μ_CR = 0.057
Generation  470: Best = 2.744292e+03  FES =  47100  μ_F = 0.815  μ_CR = 0.065
Generation  480: Best = 2.744292e+03  FES =  48100  μ_F = 0.782  μ_CR = 0.069
Generation  490: Best = 2.744292e+03  FES =  49100  μ_F = 0.766  μ_CR = 0.079
Generation  500: Best = 2.744292e+03  FES =  50100  μ_F = 0.854  μ_CR = 0.071
Generation  510: Best = 2.744292e+03  FES =  51100  μ_F = 0.799  μ_CR = 0.076
Generation  520: Best = 2.744292e+03  FES =  52100  μ_F = 0.800  μ_CR = 0.087
Generation  530: Best = 2.744292e+03  FES =  53100  μ_F = 0.762  μ_CR = 0.077
Generation  540: Best = 2.744292e+03  FES =  54100  μ_F = 0.795  μ_CR = 0.081
Generation  550: Best = 2.744240e+03  FES =  55100  μ_F = 0.759  μ_CR = 0.086
Generation  560: Best = 2.744240e+03  FES =  56100  μ_F = 0.706  μ_CR = 0.064
Generation  570: Best = 2.744240e+03  FES =  57100  μ_F = 0.720  μ_CR = 0.068
Generation  580: Best = 2.744240e+03  FES =  58100  μ_F = 0.692  μ_CR = 0.055
Generation  590: Best = 2.744240e+03  FES =  59100  μ_F = 0.714  μ_CR = 0.067
Generation  600: Best = 2.744240e+03  FES =  60100  μ_F = 0.703  μ_CR = 0.062
Generation  610: Best = 2.743071e+03  FES =  61100  μ_F = 0.676  μ_CR = 0.068
Generation  620: Best = 2.743071e+03  FES =  62100  μ_F = 0.622  μ_CR = 0.065
Generation  630: Best = 2.743071e+03  FES =  63100  μ_F = 0.626  μ_CR = 0.074
Generation  640: Best = 2.743071e+03  FES =  64100  μ_F = 0.610  μ_CR = 0.067
Generation  650: Best = 2.743071e+03  FES =  65100  μ_F = 0.619  μ_CR = 0.060
Generation  660: Best = 2.743071e+03  FES =  66100  μ_F = 0.639  μ_CR = 0.076
Generation  670: Best = 2.743071e+03  FES =  67100  μ_F = 0.724  μ_CR = 0.094
Generation  680: Best = 2.742810e+03  FES =  68100  μ_F = 0.663  μ_CR = 0.083
Generation  690: Best = 2.742810e+03  FES =  69100  μ_F = 0.667  μ_CR = 0.095
Generation  700: Best = 2.742810e+03  FES =  70100  μ_F = 0.621  μ_CR = 0.089
Generation  710: Best = 2.742810e+03  FES =  71100  μ_F = 0.634  μ_CR = 0.077
Generation  720: Best = 2.742810e+03  FES =  72100  μ_F = 0.633  μ_CR = 0.090
Generation  730: Best = 2.742810e+03  FES =  73100  μ_F = 0.667  μ_CR = 0.081
Generation  740: Best = 2.742810e+03  FES =  74100  μ_F = 0.675  μ_CR = 0.087
Generation  750: Best = 2.742810e+03  FES =  75100  μ_F = 0.689  μ_CR = 0.072
Generation  760: Best = 2.742810e+03  FES =  76100  μ_F = 0.701  μ_CR = 0.076
Generation  770: Best = 2.742810e+03  FES =  77100  μ_F = 0.672  μ_CR = 0.095
Generation  780: Best = 2.742810e+03  FES =  78100  μ_F = 0.696  μ_CR = 0.085
Generation  790: Best = 2.742810e+03  FES =  79100  μ_F = 0.776  μ_CR = 0.095
Generation  800: Best = 2.742810e+03  FES =  80100  μ_F = 0.770  μ_CR = 0.075
Generation  810: Best = 2.742810e+03  FES =  81100  μ_F = 0.825  μ_CR = 0.091
Generation  820: Best = 2.741740e+03  FES =  82100  μ_F = 0.796  μ_CR = 0.075
Generation  830: Best = 2.741740e+03  FES =  83100  μ_F = 0.788  μ_CR = 0.059
Generation  840: Best = 2.741740e+03  FES =  84100  μ_F = 0.658  μ_CR = 0.075
Generation  850: Best = 2.741740e+03  FES =  85100  μ_F = 0.677  μ_CR = 0.073
Generation  860: Best = 2.741740e+03  FES =  86100  μ_F = 0.732  μ_CR = 0.063
Generation  870: Best = 2.740620e+03  FES =  87100  μ_F = 0.577  μ_CR = 0.064
Generation  880: Best = 2.740620e+03  FES =  88100  μ_F = 0.528  μ_CR = 0.085
Generation  890: Best = 2.740620e+03  FES =  89100  μ_F = 0.553  μ_CR = 0.085
Generation  900: Best = 2.740620e+03  FES =  90100  μ_F = 0.561  μ_CR = 0.074
Generation  910: Best = 2.740244e+03  FES =  91100  μ_F = 0.542  μ_CR = 0.070
Generation  920: Best = 2.740244e+03  FES =  92100  μ_F = 0.536  μ_CR = 0.073
Generation  930: Best = 2.740244e+03  FES =  93100  μ_F = 0.633  μ_CR = 0.069
Generation  940: Best = 2.740196e+03  FES =  94100  μ_F = 0.562  μ_CR = 0.065
Generation  950: Best = 2.740196e+03  FES =  95100  μ_F = 0.542  μ_CR = 0.061
Generation  960: Best = 2.740196e+03  FES =  96100  μ_F = 0.556  μ_CR = 0.060
Generation  970: Best = 2.740196e+03  FES =  97100  μ_F = 0.472  μ_CR = 0.070
Generation  980: Best = 2.740196e+03  FES =  98100  μ_F = 0.468  μ_CR = 0.079
```

```
Generation  990: Best = 2.740196e+03  FES =  99100  μ_F = 0.442  μ_CR = 0.074

Optimization complete.
Best Fitness             = 2.740196e+03
Total Function Evals     = 100000
Total Successful Updates = 2446

================================================================================
=
Results:
  Best Fitness           : 2.740196e+03
  Global Optimum         : 2.400000e+03
  Error (|best-optimum|) : 3.401957e+02
  Best Solution (5 dim)  : [ 32.53966463  56.52695991 -77.66875047 -98.32099
69.45936237]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F24.png
```

In [34]:

```
# ============================================================
# F25 — Composition Function 5 (Rastrigin; HappyCat; Ackley; Discus; Rosenbroc
# ============================================================
run_and_plot('F25', make_wrapper(composition.f25), 2500.0, 'Composition: Rastr
```

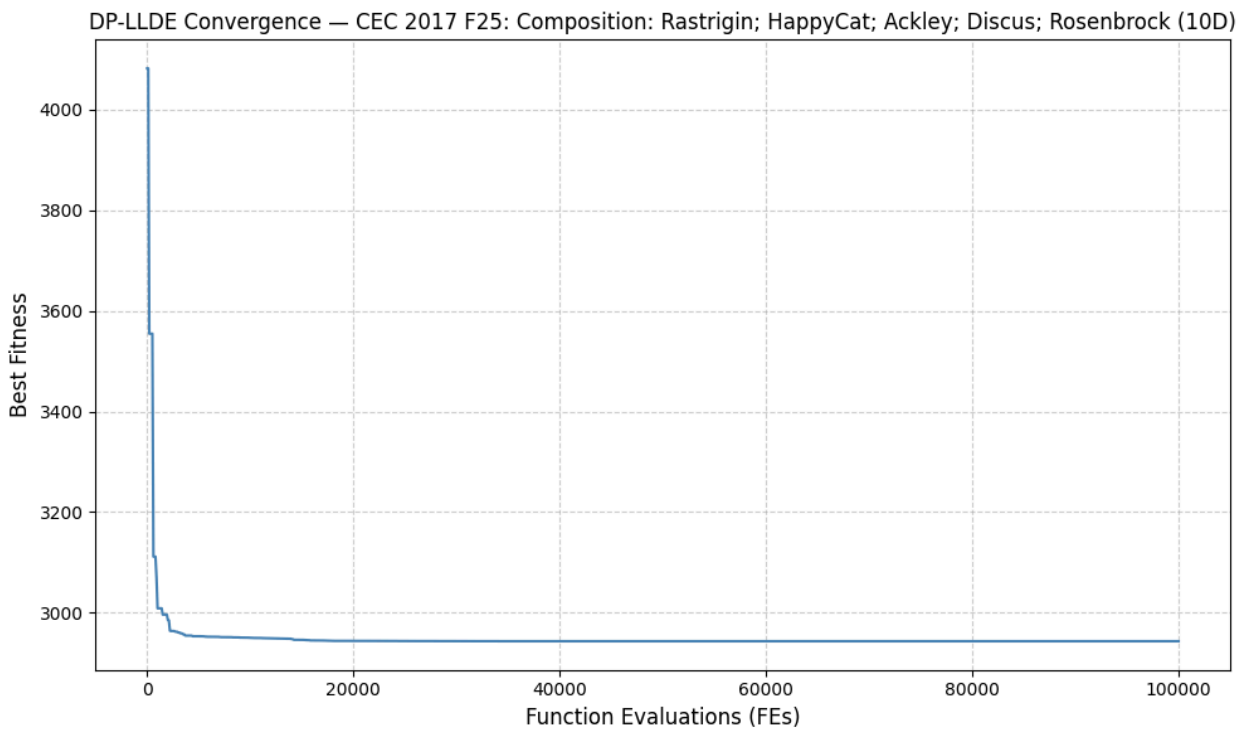
```
================================================================================
=
DP-LLDE on CEC 2017 F25 — Single Run
================================================================================
=
Function       : Composition: Rastrigin; HappyCat; Ackley; Discus; Rosenbrock
(10D)
Bounds         : [-100, 100]
Global optimum : 2500.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 4.081798e+03
Generation   10: Best = 3.008847e+03  FES =   1100  μ_F = 0.587  μ_CR = 0.359
Generation   20: Best = 2.985759e+03  FES =   2100  μ_F = 0.641  μ_CR = 0.361
Generation   30: Best = 2.960889e+03  FES =   3100  μ_F = 0.673  μ_CR = 0.364
Generation   40: Best = 2.954670e+03  FES =   4100  μ_F = 0.689  μ_CR = 0.362
Generation   50: Best = 2.953379e+03  FES =   5100  μ_F = 0.724  μ_CR = 0.362
Generation   60: Best = 2.952419e+03  FES =   6100  μ_F = 0.743  μ_CR = 0.361
Generation   70: Best = 2.951961e+03  FES =   7100  μ_F = 0.751  μ_CR = 0.370
Generation   80: Best = 2.951506e+03  FES =   8100  μ_F = 0.749  μ_CR = 0.367
Generation   90: Best = 2.950771e+03  FES =   9100  μ_F = 0.753  μ_CR = 0.396
Generation  100: Best = 2.950336e+03  FES =  10100  μ_F = 0.765  μ_CR = 0.408
Generation  110: Best = 2.949878e+03  FES =  11100  μ_F = 0.760  μ_CR = 0.428
Generation  120: Best = 2.949286e+03  FES =  12100  μ_F = 0.762  μ_CR = 0.437
Generation  130: Best = 2.948914e+03  FES =  13100  μ_F = 0.780  μ_CR = 0.450
Generation  140: Best = 2.947726e+03  FES =  14100  μ_F = 0.808  μ_CR = 0.464
Generation  150: Best = 2.946197e+03  FES =  15100  μ_F = 0.787  μ_CR = 0.498
Generation  160: Best = 2.945117e+03  FES =  16100  μ_F = 0.760  μ_CR = 0.522
Generation  170: Best = 2.944901e+03  FES =  17100  μ_F = 0.733  μ_CR = 0.537
Generation  180: Best = 2.944217e+03  FES =  18100  μ_F = 0.754  μ_CR = 0.562
Generation  190: Best = 2.944217e+03  FES =  19100  μ_F = 0.775  μ_CR = 0.564
Generation  200: Best = 2.944132e+03  FES =  20100  μ_F = 0.752  μ_CR = 0.570
Generation  210: Best = 2.944132e+03  FES =  21100  μ_F = 0.747  μ_CR = 0.575
Generation  220: Best = 2.944132e+03  FES =  22100  μ_F = 0.744  μ_CR = 0.569
Generation  230: Best = 2.943982e+03  FES =  23100  μ_F = 0.728  μ_CR = 0.592
Generation  240: Best = 2.943982e+03  FES =  24100  μ_F = 0.700  μ_CR = 0.580
Generation  250: Best = 2.943877e+03  FES =  25100  μ_F = 0.687  μ_CR = 0.566
Generation  260: Best = 2.943877e+03  FES =  26100  μ_F = 0.678  μ_CR = 0.613
Generation  270: Best = 2.943877e+03  FES =  27100  μ_F = 0.687  μ_CR = 0.614
Generation  280: Best = 2.943877e+03  FES =  28100  μ_F = 0.683  μ_CR = 0.655
Generation  290: Best = 2.943777e+03  FES =  29100  μ_F = 0.679  μ_CR = 0.690
Generation  300: Best = 2.943777e+03  FES =  30100  μ_F = 0.651  μ_CR = 0.722
Generation  310: Best = 2.943777e+03  FES =  31100  μ_F = 0.652  μ_CR = 0.741
Generation  320: Best = 2.943599e+03  FES =  32100  μ_F = 0.625  μ_CR = 0.760
Generation  330: Best = 2.943570e+03  FES =  33100  μ_F = 0.655  μ_CR = 0.759
Generation  340: Best = 2.943543e+03  FES =  34100  μ_F = 0.662  μ_CR = 0.751
Generation  350: Best = 2.943514e+03  FES =  35100  μ_F = 0.663  μ_CR = 0.758
Generation  360: Best = 2.943510e+03  FES =  36100  μ_F = 0.691  μ_CR = 0.756
Generation  370: Best = 2.943510e+03  FES =  37100  μ_F = 0.724  μ_CR = 0.768
Generation  380: Best = 2.943510e+03  FES =  38100  μ_F = 0.740  μ_CR = 0.776
Generation  390: Best = 2.943510e+03  FES =  39100  μ_F = 0.760  μ_CR = 0.774
Generation  400: Best = 2.943510e+03  FES =  40100  μ_F = 0.769  μ_CR = 0.773
Generation  410: Best = 2.943510e+03  FES =  41100  μ_F = 0.787  μ_CR = 0.778
Generation  420: Best = 2.943510e+03  FES =  42100  μ_F = 0.804  μ_CR = 0.785
Generation  430: Best = 2.943510e+03  FES =  43100  μ_F = 0.817  μ_CR = 0.791
```

```
Generation  440: Best = 2.943510e+03  FES =  44100  μ_F = 0.821  μ_CR = 0.784
Generation  450: Best = 2.943510e+03  FES =  45100  μ_F = 0.831  μ_CR = 0.784
Generation  460: Best = 2.943510e+03  FES =  46100  μ_F = 0.846  μ_CR = 0.779
Generation  470: Best = 2.943510e+03  FES =  47100  μ_F = 0.849  μ_CR = 0.782
Generation  480: Best = 2.943510e+03  FES =  48100  μ_F = 0.856  μ_CR = 0.785
Generation  490: Best = 2.943510e+03  FES =  49100  μ_F = 0.855  μ_CR = 0.789
Generation  500: Best = 2.943510e+03  FES =  50100  μ_F = 0.860  μ_CR = 0.784
Generation  510: Best = 2.943510e+03  FES =  51100  μ_F = 0.860  μ_CR = 0.782
Generation  520: Best = 2.943510e+03  FES =  52100  μ_F = 0.858  μ_CR = 0.777
Generation  530: Best = 2.943510e+03  FES =  53100  μ_F = 0.854  μ_CR = 0.777
Generation  540: Best = 2.943510e+03  FES =  54100  μ_F = 0.848  μ_CR = 0.777
Generation  550: Best = 2.943510e+03  FES =  55100  μ_F = 0.846  μ_CR = 0.776
Generation  560: Best = 2.943510e+03  FES =  56100  μ_F = 0.846  μ_CR = 0.773
Generation  570: Best = 2.943510e+03  FES =  57100  μ_F = 0.851  μ_CR = 0.768
Generation  580: Best = 2.943510e+03  FES =  58100  μ_F = 0.845  μ_CR = 0.769
Generation  590: Best = 2.943510e+03  FES =  59100  μ_F = 0.854  μ_CR = 0.765
Generation  600: Best = 2.943510e+03  FES =  60100  μ_F = 0.860  μ_CR = 0.760
Generation  610: Best = 2.943510e+03  FES =  61100  μ_F = 0.861  μ_CR = 0.750
Generation  620: Best = 2.943510e+03  FES =  62100  μ_F = 0.857  μ_CR = 0.755
Generation  630: Best = 2.943510e+03  FES =  63100  μ_F = 0.854  μ_CR = 0.751
Generation  640: Best = 2.943510e+03  FES =  64100  μ_F = 0.852  μ_CR = 0.751
Generation  650: Best = 2.943510e+03  FES =  65100  μ_F = 0.850  μ_CR = 0.751
Generation  660: Best = 2.943510e+03  FES =  66100  μ_F = 0.854  μ_CR = 0.754
Generation  670: Best = 2.943510e+03  FES =  67100  μ_F = 0.861  μ_CR = 0.750
Generation  680: Best = 2.943510e+03  FES =  68100  μ_F = 0.861  μ_CR = 0.750
Generation  690: Best = 2.943510e+03  FES =  69100  μ_F = 0.847  μ_CR = 0.749
Generation  700: Best = 2.943510e+03  FES =  70100  μ_F = 0.841  μ_CR = 0.745
Generation  710: Best = 2.943510e+03  FES =  71100  μ_F = 0.840  μ_CR = 0.744
Generation  720: Best = 2.943510e+03  FES =  72100  μ_F = 0.845  μ_CR = 0.743
Generation  730: Best = 2.943510e+03  FES =  73100  μ_F = 0.848  μ_CR = 0.750
Generation  740: Best = 2.943510e+03  FES =  74100  μ_F = 0.850  μ_CR = 0.752
Generation  750: Best = 2.943510e+03  FES =  75100  μ_F = 0.856  μ_CR = 0.749
Generation  760: Best = 2.943510e+03  FES =  76100  μ_F = 0.856  μ_CR = 0.750
Generation  770: Best = 2.943510e+03  FES =  77100  μ_F = 0.855  μ_CR = 0.748
Generation  780: Best = 2.943510e+03  FES =  78100  μ_F = 0.848  μ_CR = 0.744
Generation  790: Best = 2.943510e+03  FES =  79100  μ_F = 0.849  μ_CR = 0.742
Generation  800: Best = 2.943510e+03  FES =  80100  μ_F = 0.851  μ_CR = 0.739
Generation  810: Best = 2.943510e+03  FES =  81100  μ_F = 0.855  μ_CR = 0.740
Generation  820: Best = 2.943510e+03  FES =  82100  μ_F = 0.855  μ_CR = 0.738
Generation  830: Best = 2.943510e+03  FES =  83100  μ_F = 0.850  μ_CR = 0.739
Generation  840: Best = 2.943510e+03  FES =  84100  μ_F = 0.846  μ_CR = 0.736
Generation  850: Best = 2.943510e+03  FES =  85100  μ_F = 0.849  μ_CR = 0.736
Generation  860: Best = 2.943510e+03  FES =  86100  μ_F = 0.860  μ_CR = 0.732
Generation  870: Best = 2.943510e+03  FES =  87100  μ_F = 0.857  μ_CR = 0.728
Generation  880: Best = 2.943510e+03  FES =  88100  μ_F = 0.856  μ_CR = 0.723
Generation  890: Best = 2.943510e+03  FES =  89100  μ_F = 0.856  μ_CR = 0.721
Generation  900: Best = 2.943510e+03  FES =  90100  μ_F = 0.855  μ_CR = 0.717
Generation  910: Best = 2.943510e+03  FES =  91100  μ_F = 0.860  μ_CR = 0.713
Generation  920: Best = 2.943510e+03  FES =  92100  μ_F = 0.859  μ_CR = 0.709
Generation  930: Best = 2.943510e+03  FES =  93100  μ_F = 0.862  μ_CR = 0.698
Generation  940: Best = 2.943510e+03  FES =  94100  μ_F = 0.863  μ_CR = 0.690
Generation  950: Best = 2.943510e+03  FES =  95100  μ_F = 0.865  μ_CR = 0.693
Generation  960: Best = 2.943510e+03  FES =  96100  μ_F = 0.862  μ_CR = 0.694
Generation  970: Best = 2.943510e+03  FES =  97100  μ_F = 0.853  μ_CR = 0.691
```

```
Generation  980: Best = 2.943510e+03  FES =  98100  μ_F = 0.856  μ_CR = 0.693
Generation  990: Best = 2.943510e+03  FES =  99100  μ_F = 0.856  μ_CR = 0.700

Optimization complete.
Best Fitness             = 2.943510e+03
Total Function Evals     = 100000
Total Successful Updates = 52868

================================================================================
=
Results:
  Best Fitness           : 2.943510e+03
  Global Optimum         : 2.500000e+03
  Error (|best-optimum|) : 4.435096e+02
  Best Solution (5 dim)  : [ 13.94152211  72.64031873 -50.83715062 -16.59633903
46.1974453 ]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F25.png
```

In [35]:

```
# ============================================================
# F26 — Composition Function 6 (Schaffer; Schwefel; Griewank; Rosenbrock; Rast
# ============================================================
run_and_plot('F26', make_wrapper(composition.f26), 2600.0, 'Composition: Schaf
```

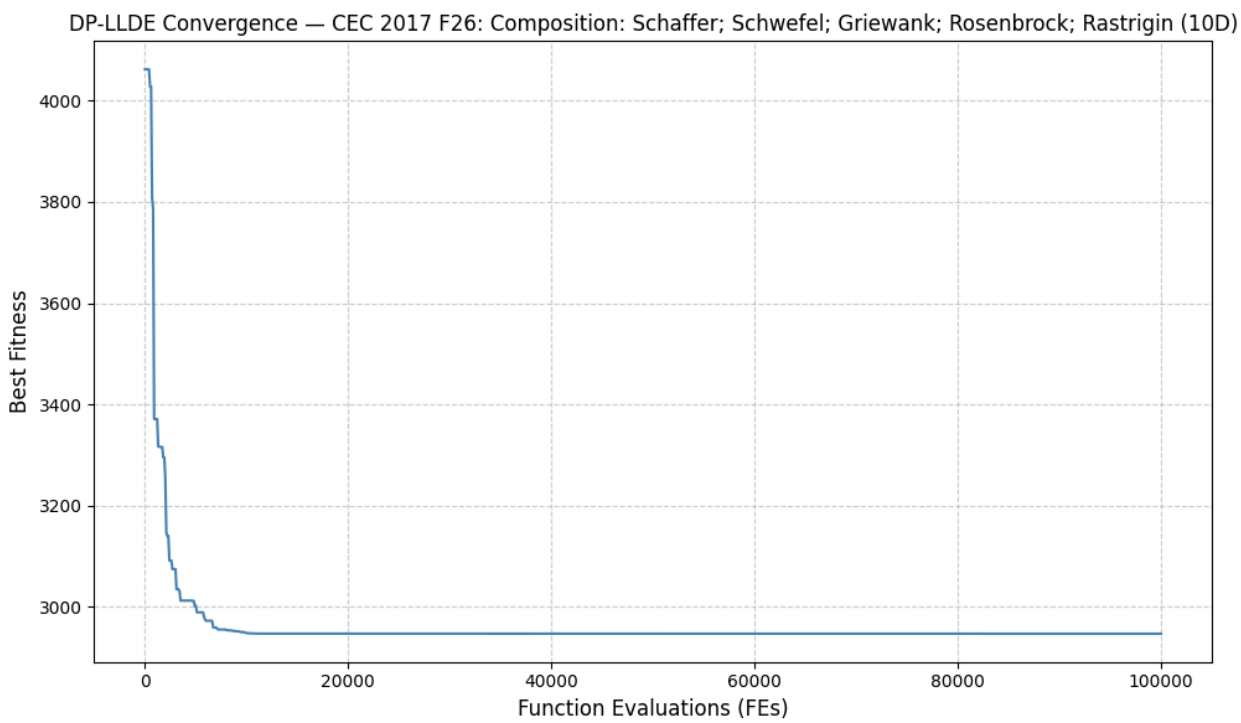
```
================================================================================
=
DP-LLDE on CEC 2017 F26 — Single Run
================================================================================
=
Function       : Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrig
in (10D)
Bounds         : [-100, 100]
Global optimum : 2600.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 4.061684e+03
Generation   10: Best = 3.370966e+03  FES =   1100  μ_F = 0.574  μ_CR = 0.351
Generation   20: Best = 3.255291e+03  FES =   2100  μ_F = 0.626  μ_CR = 0.352
Generation   30: Best = 3.074601e+03  FES =   3100  μ_F = 0.672  μ_CR = 0.367
Generation   40: Best = 3.012500e+03  FES =   4100  μ_F = 0.694  μ_CR = 0.359
Generation   50: Best = 3.002179e+03  FES =   5100  μ_F = 0.732  μ_CR = 0.370
Generation   60: Best = 2.972709e+03  FES =   6100  μ_F = 0.760  μ_CR = 0.369
Generation   70: Best = 2.959383e+03  FES =   7100  μ_F = 0.768  μ_CR = 0.371
Generation   80: Best = 2.954255e+03  FES =   8100  μ_F = 0.774  μ_CR = 0.371
Generation   90: Best = 2.952082e+03  FES =   9100  μ_F = 0.778  μ_CR = 0.373
Generation  100: Best = 2.948785e+03  FES =  10100  μ_F = 0.781  μ_CR = 0.371
Generation  110: Best = 2.947475e+03  FES =  11100  μ_F = 0.785  μ_CR = 0.371
Generation  120: Best = 2.947235e+03  FES =  12100  μ_F = 0.798  μ_CR = 0.370
Generation  130: Best = 2.947160e+03  FES =  13100  μ_F = 0.807  μ_CR = 0.368
Generation  140: Best = 2.947156e+03  FES =  14100  μ_F = 0.821  μ_CR = 0.366
Generation  150: Best = 2.947150e+03  FES =  15100  μ_F = 0.825  μ_CR = 0.369
Generation  160: Best = 2.947149e+03  FES =  16100  μ_F = 0.819  μ_CR = 0.377
Generation  170: Best = 2.947149e+03  FES =  17100  μ_F = 0.812  μ_CR = 0.383
Generation  180: Best = 2.947149e+03  FES =  18100  μ_F = 0.822  μ_CR = 0.381
Generation  190: Best = 2.947149e+03  FES =  19100  μ_F = 0.818  μ_CR = 0.384
Generation  200: Best = 2.947149e+03  FES =  20100  μ_F = 0.820  μ_CR = 0.391
Generation  210: Best = 2.947149e+03  FES =  21100  μ_F = 0.834  μ_CR = 0.392
Generation  220: Best = 2.947149e+03  FES =  22100  μ_F = 0.833  μ_CR = 0.391
Generation  230: Best = 2.947149e+03  FES =  23100  μ_F = 0.831  μ_CR = 0.401
Generation  240: Best = 2.947149e+03  FES =  24100  μ_F = 0.842  μ_CR = 0.405
Generation  250: Best = 2.947149e+03  FES =  25100  μ_F = 0.846  μ_CR = 0.409
Generation  260: Best = 2.947149e+03  FES =  26100  μ_F = 0.837  μ_CR = 0.416
Generation  270: Best = 2.947149e+03  FES =  27100  μ_F = 0.829  μ_CR = 0.409
Generation  280: Best = 2.947149e+03  FES =  28100  μ_F = 0.828  μ_CR = 0.413
Generation  290: Best = 2.947149e+03  FES =  29100  μ_F = 0.831  μ_CR = 0.415
Generation  300: Best = 2.947149e+03  FES =  30100  μ_F = 0.826  μ_CR = 0.425
Generation  310: Best = 2.947149e+03  FES =  31100  μ_F = 0.840  μ_CR = 0.421
Generation  320: Best = 2.947149e+03  FES =  32100  μ_F = 0.835  μ_CR = 0.425
Generation  330: Best = 2.947149e+03  FES =  33100  μ_F = 0.846  μ_CR = 0.425
Generation  340: Best = 2.947149e+03  FES =  34100  μ_F = 0.856  μ_CR = 0.420
Generation  350: Best = 2.947149e+03  FES =  35100  μ_F = 0.868  μ_CR = 0.422
Generation  360: Best = 2.947149e+03  FES =  36100  μ_F = 0.869  μ_CR = 0.415
Generation  370: Best = 2.947149e+03  FES =  37100  μ_F = 0.867  μ_CR = 0.420
Generation  380: Best = 2.947149e+03  FES =  38100  μ_F = 0.865  μ_CR = 0.419
Generation  390: Best = 2.947149e+03  FES =  39100  μ_F = 0.858  μ_CR = 0.398
Generation  400: Best = 2.947149e+03  FES =  40100  μ_F = 0.855  μ_CR = 0.397
Generation  410: Best = 2.947149e+03  FES =  41100  μ_F = 0.860  μ_CR = 0.395
Generation  420: Best = 2.947149e+03  FES =  42100  μ_F = 0.873  μ_CR = 0.393
Generation  430: Best = 2.947149e+03  FES =  43100  μ_F = 0.874  μ_CR = 0.391
```

```
Generation  440: Best = 2.947149e+03  FES =  44100  μ_F = 0.864  μ_CR = 0.388
Generation  450: Best = 2.947149e+03  FES =  45100  μ_F = 0.860  μ_CR = 0.385
Generation  460: Best = 2.947149e+03  FES =  46100  μ_F = 0.870  μ_CR = 0.381
Generation  470: Best = 2.947149e+03  FES =  47100  μ_F = 0.870  μ_CR = 0.391
Generation  480: Best = 2.947149e+03  FES =  48100  μ_F = 0.865  μ_CR = 0.393
Generation  490: Best = 2.947149e+03  FES =  49100  μ_F = 0.863  μ_CR = 0.390
Generation  500: Best = 2.947149e+03  FES =  50100  μ_F = 0.865  μ_CR = 0.385
Generation  510: Best = 2.947149e+03  FES =  51100  μ_F = 0.869  μ_CR = 0.384
Generation  520: Best = 2.947149e+03  FES =  52100  μ_F = 0.861  μ_CR = 0.377
Generation  530: Best = 2.947149e+03  FES =  53100  μ_F = 0.870  μ_CR = 0.373
Generation  540: Best = 2.947149e+03  FES =  54100  μ_F = 0.865  μ_CR = 0.378
Generation  550: Best = 2.947149e+03  FES =  55100  μ_F = 0.866  μ_CR = 0.377
Generation  560: Best = 2.947149e+03  FES =  56100  μ_F = 0.869  μ_CR = 0.371
Generation  570: Best = 2.947149e+03  FES =  57100  μ_F = 0.873  μ_CR = 0.362
Generation  580: Best = 2.947149e+03  FES =  58100  μ_F = 0.877  μ_CR = 0.365
Generation  590: Best = 2.947149e+03  FES =  59100  μ_F = 0.881  μ_CR = 0.367
Generation  600: Best = 2.947149e+03  FES =  60100  μ_F = 0.881  μ_CR = 0.361
Generation  610: Best = 2.947149e+03  FES =  61100  μ_F = 0.875  μ_CR = 0.355
Generation  620: Best = 2.947149e+03  FES =  62100  μ_F = 0.868  μ_CR = 0.360
Generation  630: Best = 2.947149e+03  FES =  63100  μ_F = 0.858  μ_CR = 0.353
Generation  640: Best = 2.947149e+03  FES =  64100  μ_F = 0.861  μ_CR = 0.348
Generation  650: Best = 2.947149e+03  FES =  65100  μ_F = 0.864  μ_CR = 0.352
Generation  660: Best = 2.947149e+03  FES =  66100  μ_F = 0.871  μ_CR = 0.348
Generation  670: Best = 2.947149e+03  FES =  67100  μ_F = 0.874  μ_CR = 0.349
Generation  680: Best = 2.947149e+03  FES =  68100  μ_F = 0.873  μ_CR = 0.343
Generation  690: Best = 2.947149e+03  FES =  69100  μ_F = 0.871  μ_CR = 0.334
Generation  700: Best = 2.947149e+03  FES =  70100  μ_F = 0.869  μ_CR = 0.335
Generation  710: Best = 2.947149e+03  FES =  71100  μ_F = 0.858  μ_CR = 0.328
Generation  720: Best = 2.947149e+03  FES =  72100  μ_F = 0.864  μ_CR = 0.326
Generation  730: Best = 2.947149e+03  FES =  73100  μ_F = 0.863  μ_CR = 0.334
Generation  740: Best = 2.947149e+03  FES =  74100  μ_F = 0.866  μ_CR = 0.331
Generation  750: Best = 2.947149e+03  FES =  75100  μ_F = 0.876  μ_CR = 0.326
Generation  760: Best = 2.947149e+03  FES =  76100  μ_F = 0.874  μ_CR = 0.332
Generation  770: Best = 2.947149e+03  FES =  77100  μ_F = 0.872  μ_CR = 0.332
Generation  780: Best = 2.947149e+03  FES =  78100  μ_F = 0.866  μ_CR = 0.327
Generation  790: Best = 2.947149e+03  FES =  79100  μ_F = 0.867  μ_CR = 0.329
Generation  800: Best = 2.947149e+03  FES =  80100  μ_F = 0.868  μ_CR = 0.324
Generation  810: Best = 2.947149e+03  FES =  81100  μ_F = 0.866  μ_CR = 0.327
Generation  820: Best = 2.947149e+03  FES =  82100  μ_F = 0.875  μ_CR = 0.322
Generation  830: Best = 2.947149e+03  FES =  83100  μ_F = 0.869  μ_CR = 0.316
Generation  840: Best = 2.947149e+03  FES =  84100  μ_F = 0.852  μ_CR = 0.314
Generation  850: Best = 2.947149e+03  FES =  85100  μ_F = 0.863  μ_CR = 0.307
Generation  860: Best = 2.947149e+03  FES =  86100  μ_F = 0.868  μ_CR = 0.300
Generation  870: Best = 2.947149e+03  FES =  87100  μ_F = 0.872  μ_CR = 0.299
Generation  880: Best = 2.947149e+03  FES =  88100  μ_F = 0.876  μ_CR = 0.287
Generation  890: Best = 2.947149e+03  FES =  89100  μ_F = 0.874  μ_CR = 0.289
Generation  900: Best = 2.947149e+03  FES =  90100  μ_F = 0.866  μ_CR = 0.281
Generation  910: Best = 2.947149e+03  FES =  91100  μ_F = 0.868  μ_CR = 0.272
Generation  920: Best = 2.947149e+03  FES =  92100  μ_F = 0.868  μ_CR = 0.262
Generation  930: Best = 2.947149e+03  FES =  93100  μ_F = 0.871  μ_CR = 0.250
Generation  940: Best = 2.947149e+03  FES =  94100  μ_F = 0.882  μ_CR = 0.245
Generation  950: Best = 2.947149e+03  FES =  95100  μ_F = 0.885  μ_CR = 0.240
Generation  960: Best = 2.947149e+03  FES =  96100  μ_F = 0.890  μ_CR = 0.239
Generation  970: Best = 2.947149e+03  FES =  97100  μ_F = 0.887  μ_CR = 0.233
```

```
Generation  980: Best = 2.947149e+03  FES =  98100  μ_F = 0.888  μ_CR = 0.221
Generation  990: Best = 2.947149e+03  FES =  99100  μ_F = 0.882  μ_CR = 0.216

Optimization complete.
Best Fitness             = 2.947149e+03
Total Function Evals     = 100000
Total Successful Updates = 35208

================================================================================
=
Results:
  Best Fitness           : 2.947149e+03
  Global Optimum         : 2.600000e+03
  Error (|best-optimum|) : 3.471489e+02
  Best Solution (5 dim)  : [ 11.83478635 -21.14995206  83.06289177 -67.32963001
32.86364372]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F26.png
```

In [36]:

```
# ============================================================
# F27 — Composition Function 7 (HGBat; Rastrigin; Schwefel; Bent Cigar; Ellipt
# ============================================================
run_and_plot('F27', make_wrapper(composition.f27), 2700.0, 'Composition: HGBat
```

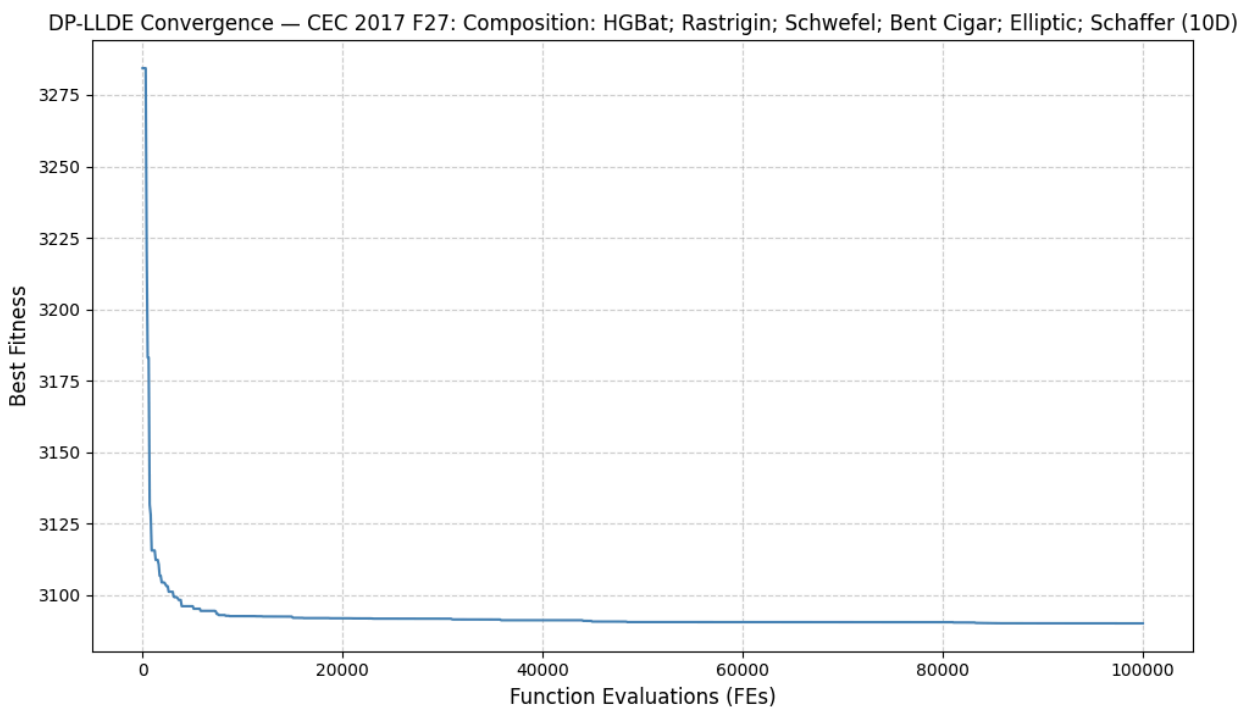
```
================================================================================
=
DP-LLDE on CEC 2017 F27 — Single Run
================================================================================
=
Function       : Composition: HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic;
Schaffer (10D)
Bounds         : [-100, 100]
Global optimum : 2700.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 3.284356e+03
Generation   10: Best = 3.115546e+03  FES =   1100  μ_F = 0.590  μ_CR = 0.349
Generation   20: Best = 3.104377e+03  FES =   2100  μ_F = 0.666  μ_CR = 0.353
Generation   30: Best = 3.101096e+03  FES =   3100  μ_F = 0.721  μ_CR = 0.353
Generation   40: Best = 3.096050e+03  FES =   4100  μ_F = 0.753  μ_CR = 0.327
Generation   50: Best = 3.096050e+03  FES =   5100  μ_F = 0.789  μ_CR = 0.303
Generation   60: Best = 3.094411e+03  FES =   6100  μ_F = 0.796  μ_CR = 0.287
Generation   70: Best = 3.094411e+03  FES =   7100  μ_F = 0.817  μ_CR = 0.267
Generation   80: Best = 3.092938e+03  FES =   8100  μ_F = 0.834  μ_CR = 0.239
Generation   90: Best = 3.092580e+03  FES =   9100  μ_F = 0.848  μ_CR = 0.206
Generation  100: Best = 3.092580e+03  FES =  10100  μ_F = 0.850  μ_CR = 0.156
Generation  110: Best = 3.092565e+03  FES =  11100  μ_F = 0.843  μ_CR = 0.134
Generation  120: Best = 3.092443e+03  FES =  12100  μ_F = 0.830  μ_CR = 0.114
Generation  130: Best = 3.092442e+03  FES =  13100  μ_F = 0.825  μ_CR = 0.118
Generation  140: Best = 3.092408e+03  FES =  14100  μ_F = 0.823  μ_CR = 0.105
Generation  150: Best = 3.091990e+03  FES =  15100  μ_F = 0.826  μ_CR = 0.098
Generation  160: Best = 3.091955e+03  FES =  16100  μ_F = 0.827  μ_CR = 0.106
Generation  170: Best = 3.091884e+03  FES =  17100  μ_F = 0.840  μ_CR = 0.111
Generation  180: Best = 3.091884e+03  FES =  18100  μ_F = 0.842  μ_CR = 0.105
Generation  190: Best = 3.091833e+03  FES =  19100  μ_F = 0.837  μ_CR = 0.105
Generation  200: Best = 3.091813e+03  FES =  20100  μ_F = 0.859  μ_CR = 0.100
Generation  210: Best = 3.091782e+03  FES =  21100  μ_F = 0.864  μ_CR = 0.087
Generation  220: Best = 3.091766e+03  FES =  22100  μ_F = 0.867  μ_CR = 0.090
Generation  230: Best = 3.091728e+03  FES =  23100  μ_F = 0.876  μ_CR = 0.097
Generation  240: Best = 3.091688e+03  FES =  24100  μ_F = 0.873  μ_CR = 0.091
Generation  250: Best = 3.091688e+03  FES =  25100  μ_F = 0.862  μ_CR = 0.088
Generation  260: Best = 3.091684e+03  FES =  26100  μ_F = 0.856  μ_CR = 0.090
Generation  270: Best = 3.091684e+03  FES =  27100  μ_F = 0.881  μ_CR = 0.089
Generation  280: Best = 3.091684e+03  FES =  28100  μ_F = 0.872  μ_CR = 0.096
Generation  290: Best = 3.091676e+03  FES =  29100  μ_F = 0.865  μ_CR = 0.104
Generation  300: Best = 3.091676e+03  FES =  30100  μ_F = 0.864  μ_CR = 0.108
Generation  310: Best = 3.091411e+03  FES =  31100  μ_F = 0.879  μ_CR = 0.097
Generation  320: Best = 3.091392e+03  FES =  32100  μ_F = 0.867  μ_CR = 0.106
Generation  330: Best = 3.091392e+03  FES =  33100  μ_F = 0.861  μ_CR = 0.104
Generation  340: Best = 3.091384e+03  FES =  34100  μ_F = 0.856  μ_CR = 0.093
Generation  350: Best = 3.091384e+03  FES =  35100  μ_F = 0.860  μ_CR = 0.093
Generation  360: Best = 3.091137e+03  FES =  36100  μ_F = 0.868  μ_CR = 0.090
Generation  370: Best = 3.091133e+03  FES =  37100  μ_F = 0.857  μ_CR = 0.096
Generation  380: Best = 3.091133e+03  FES =  38100  μ_F = 0.857  μ_CR = 0.096
Generation  390: Best = 3.091133e+03  FES =  39100  μ_F = 0.878  μ_CR = 0.091
Generation  400: Best = 3.091131e+03  FES =  40100  μ_F = 0.863  μ_CR = 0.090
Generation  410: Best = 3.091131e+03  FES =  41100  μ_F = 0.867  μ_CR = 0.094
Generation  420: Best = 3.091131e+03  FES =  42100  μ_F = 0.875  μ_CR = 0.096
Generation  430: Best = 3.091131e+03  FES =  43100  μ_F = 0.868  μ_CR = 0.089
```

```
Generation  440: Best = 3.090924e+03  FES =  44100  μ_F = 0.875  μ_CR = 0.087
Generation  450: Best = 3.090672e+03  FES =  45100  μ_F = 0.885  μ_CR = 0.101
Generation  460: Best = 3.090650e+03  FES =  46100  μ_F = 0.883  μ_CR = 0.102
Generation  470: Best = 3.090649e+03  FES =  47100  μ_F = 0.884  μ_CR = 0.105
Generation  480: Best = 3.090649e+03  FES =  48100  μ_F = 0.880  μ_CR = 0.095
Generation  490: Best = 3.090490e+03  FES =  49100  μ_F = 0.888  μ_CR = 0.088
Generation  500: Best = 3.090473e+03  FES =  50100  μ_F = 0.892  μ_CR = 0.106
Generation  510: Best = 3.090473e+03  FES =  51100  μ_F = 0.886  μ_CR = 0.101
Generation  520: Best = 3.090473e+03  FES =  52100  μ_F = 0.873  μ_CR = 0.089
Generation  530: Best = 3.090473e+03  FES =  53100  μ_F = 0.864  μ_CR = 0.092
Generation  540: Best = 3.090473e+03  FES =  54100  μ_F = 0.854  μ_CR = 0.086
Generation  550: Best = 3.090473e+03  FES =  55100  μ_F = 0.852  μ_CR = 0.088
Generation  560: Best = 3.090473e+03  FES =  56100  μ_F = 0.890  μ_CR = 0.099
Generation  570: Best = 3.090473e+03  FES =  57100  μ_F = 0.875  μ_CR = 0.090
Generation  580: Best = 3.090473e+03  FES =  58100  μ_F = 0.843  μ_CR = 0.104
Generation  590: Best = 3.090473e+03  FES =  59100  μ_F = 0.846  μ_CR = 0.100
Generation  600: Best = 3.090473e+03  FES =  60100  μ_F = 0.852  μ_CR = 0.098
Generation  610: Best = 3.090473e+03  FES =  61100  μ_F = 0.859  μ_CR = 0.098
Generation  620: Best = 3.090473e+03  FES =  62100  μ_F = 0.869  μ_CR = 0.092
Generation  630: Best = 3.090473e+03  FES =  63100  μ_F = 0.875  μ_CR = 0.088
Generation  640: Best = 3.090473e+03  FES =  64100  μ_F = 0.853  μ_CR = 0.099
Generation  650: Best = 3.090473e+03  FES =  65100  μ_F = 0.831  μ_CR = 0.093
Generation  660: Best = 3.090473e+03  FES =  66100  μ_F = 0.849  μ_CR = 0.085
Generation  670: Best = 3.090473e+03  FES =  67100  μ_F = 0.863  μ_CR = 0.091
Generation  680: Best = 3.090473e+03  FES =  68100  μ_F = 0.864  μ_CR = 0.089
Generation  690: Best = 3.090455e+03  FES =  69100  μ_F = 0.855  μ_CR = 0.095
Generation  700: Best = 3.090455e+03  FES =  70100  μ_F = 0.869  μ_CR = 0.096
Generation  710: Best = 3.090455e+03  FES =  71100  μ_F = 0.875  μ_CR = 0.083
Generation  720: Best = 3.090452e+03  FES =  72100  μ_F = 0.891  μ_CR = 0.079
Generation  730: Best = 3.090452e+03  FES =  73100  μ_F = 0.865  μ_CR = 0.087
Generation  740: Best = 3.090452e+03  FES =  74100  μ_F = 0.860  μ_CR = 0.083
Generation  750: Best = 3.090452e+03  FES =  75100  μ_F = 0.839  μ_CR = 0.082
Generation  760: Best = 3.090452e+03  FES =  76100  μ_F = 0.841  μ_CR = 0.086
Generation  770: Best = 3.090452e+03  FES =  77100  μ_F = 0.843  μ_CR = 0.083
Generation  780: Best = 3.090452e+03  FES =  78100  μ_F = 0.860  μ_CR = 0.090
Generation  790: Best = 3.090452e+03  FES =  79100  μ_F = 0.855  μ_CR = 0.093
Generation  800: Best = 3.090452e+03  FES =  80100  μ_F = 0.854  μ_CR = 0.091
Generation  810: Best = 3.090356e+03  FES =  81100  μ_F = 0.855  μ_CR = 0.091
Generation  820: Best = 3.090341e+03  FES =  82100  μ_F = 0.879  μ_CR = 0.084
Generation  830: Best = 3.090339e+03  FES =  83100  μ_F = 0.854  μ_CR = 0.080
Generation  840: Best = 3.090155e+03  FES =  84100  μ_F = 0.856  μ_CR = 0.092
Generation  850: Best = 3.090104e+03  FES =  85100  μ_F = 0.868  μ_CR = 0.097
Generation  860: Best = 3.090081e+03  FES =  86100  μ_F = 0.873  μ_CR = 0.108
Generation  870: Best = 3.090073e+03  FES =  87100  μ_F = 0.875  μ_CR = 0.110
Generation  880: Best = 3.090067e+03  FES =  88100  μ_F = 0.868  μ_CR = 0.106
Generation  890: Best = 3.090063e+03  FES =  89100  μ_F = 0.855  μ_CR = 0.116
Generation  900: Best = 3.090062e+03  FES =  90100  μ_F = 0.856  μ_CR = 0.110
Generation  910: Best = 3.090058e+03  FES =  91100  μ_F = 0.867  μ_CR = 0.109
Generation  920: Best = 3.090055e+03  FES =  92100  μ_F = 0.860  μ_CR = 0.100
Generation  930: Best = 3.090051e+03  FES =  93100  μ_F = 0.851  μ_CR = 0.105
Generation  940: Best = 3.090051e+03  FES =  94100  μ_F = 0.865  μ_CR = 0.102
Generation  950: Best = 3.090051e+03  FES =  95100  μ_F = 0.860  μ_CR = 0.102
Generation  960: Best = 3.090049e+03  FES =  96100  μ_F = 0.862  μ_CR = 0.103
Generation  970: Best = 3.090048e+03  FES =  97100  μ_F = 0.862  μ_CR = 0.111
```

```
Generation  980: Best = 3.090048e+03  FES =  98100  μ_F = 0.861  μ_CR = 0.115
Generation  990: Best = 3.090047e+03  FES =  99100  μ_F = 0.878  μ_CR = 0.111

Optimization complete.
Best Fitness             = 3.090047e+03
Total Function Evals     = 100000
Total Successful Updates = 13102

================================================================================
=
Results:
  Best Fitness           : 3.090047e+03
  Global Optimum         : 2.700000e+03
  Error (|best-optimum|) : 3.900475e+02
  Best Solution (5 dim)  : [ -80.03305082 -100.          -97.7232938   -92.2213
0008  100.        ]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F27.png
```

In [37]:
```
# ============================================================
# F28 — Composition Function 8 (Ackley; Griewank; Discus; Rosenbrock; HappyCat
# ============================================================
run_and_plot('F28', make_wrapper(composition.f28), 2800.0, 'Composition: Ackle
```

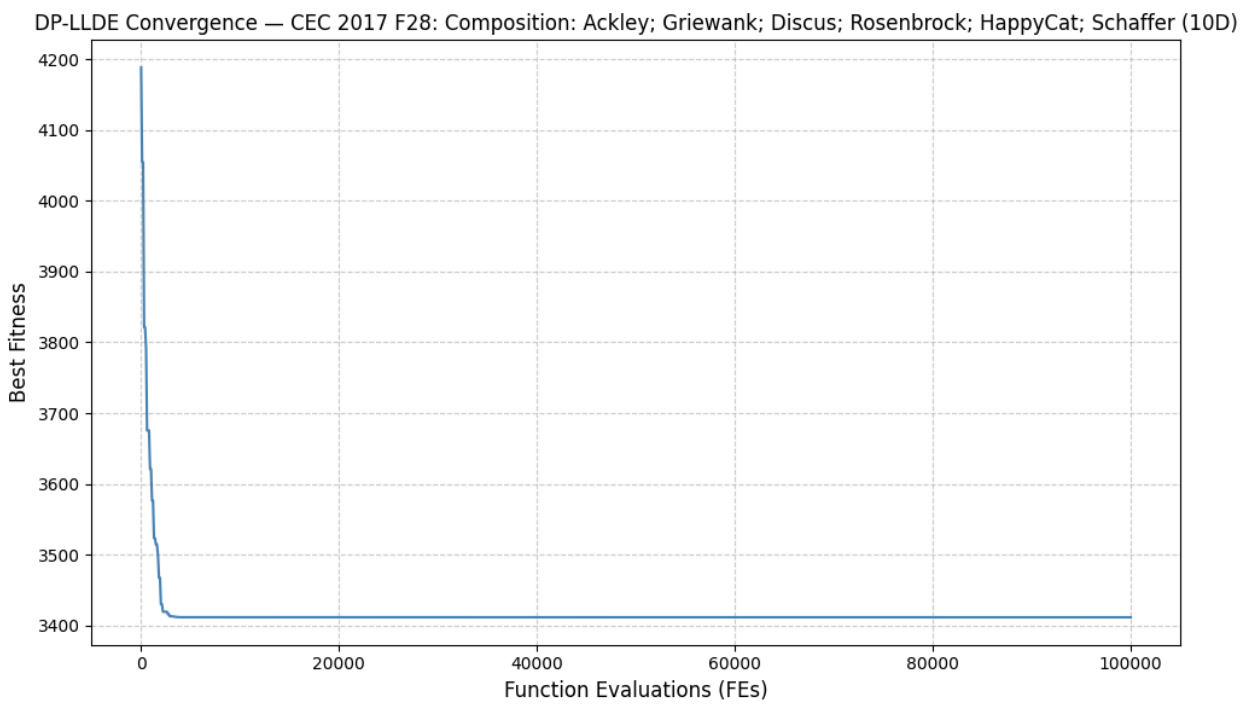
```
================================================================================
=
DP-LLDE on CEC 2017 F28 — Single Run
================================================================================
=
Function       : Composition: Ackley; Griewank; Discus; Rosenbrock; HappyCat; S
chaffer (10D)
Bounds         : [-100, 100]
Global optimum : 2800.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 4.188047e+03
Generation   10: Best = 3.621205e+03  FES =   1100  μ_F = 0.570  μ_CR = 0.357
Generation   20: Best = 3.430504e+03  FES =   2100  μ_F = 0.638  μ_CR = 0.358
Generation   30: Best = 3.413267e+03  FES =   3100  μ_F = 0.687  μ_CR = 0.368
Generation   40: Best = 3.411949e+03  FES =   4100  μ_F = 0.731  μ_CR = 0.359
Generation   50: Best = 3.411847e+03  FES =   5100  μ_F = 0.769  μ_CR = 0.364
Generation   60: Best = 3.411823e+03  FES =   6100  μ_F = 0.781  μ_CR = 0.372
Generation   70: Best = 3.411822e+03  FES =   7100  μ_F = 0.798  μ_CR = 0.373
Generation   80: Best = 3.411822e+03  FES =   8100  μ_F = 0.803  μ_CR = 0.374
Generation   90: Best = 3.411822e+03  FES =   9100  μ_F = 0.820  μ_CR = 0.375
Generation  100: Best = 3.411822e+03  FES =  10100  μ_F = 0.828  μ_CR = 0.380
Generation  110: Best = 3.411822e+03  FES =  11100  μ_F = 0.821  μ_CR = 0.380
Generation  120: Best = 3.411822e+03  FES =  12100  μ_F = 0.834  μ_CR = 0.377
Generation  130: Best = 3.411822e+03  FES =  13100  μ_F = 0.845  μ_CR = 0.374
Generation  140: Best = 3.411822e+03  FES =  14100  μ_F = 0.854  μ_CR = 0.372
Generation  150: Best = 3.411822e+03  FES =  15100  μ_F = 0.853  μ_CR = 0.371
Generation  160: Best = 3.411822e+03  FES =  16100  μ_F = 0.839  μ_CR = 0.370
Generation  170: Best = 3.411822e+03  FES =  17100  μ_F = 0.835  μ_CR = 0.381
Generation  180: Best = 3.411822e+03  FES =  18100  μ_F = 0.839  μ_CR = 0.375
Generation  190: Best = 3.411822e+03  FES =  19100  μ_F = 0.841  μ_CR = 0.373
Generation  200: Best = 3.411822e+03  FES =  20100  μ_F = 0.847  μ_CR = 0.381
Generation  210: Best = 3.411822e+03  FES =  21100  μ_F = 0.859  μ_CR = 0.377
Generation  220: Best = 3.411822e+03  FES =  22100  μ_F = 0.865  μ_CR = 0.376
Generation  230: Best = 3.411822e+03  FES =  23100  μ_F = 0.858  μ_CR = 0.367
Generation  240: Best = 3.411822e+03  FES =  24100  μ_F = 0.865  μ_CR = 0.363
Generation  250: Best = 3.411822e+03  FES =  25100  μ_F = 0.872  μ_CR = 0.360
Generation  260: Best = 3.411822e+03  FES =  26100  μ_F = 0.873  μ_CR = 0.361
Generation  270: Best = 3.411822e+03  FES =  27100  μ_F = 0.875  μ_CR = 0.357
Generation  280: Best = 3.411822e+03  FES =  28100  μ_F = 0.875  μ_CR = 0.353
Generation  290: Best = 3.411822e+03  FES =  29100  μ_F = 0.883  μ_CR = 0.354
Generation  300: Best = 3.411822e+03  FES =  30100  μ_F = 0.879  μ_CR = 0.356
Generation  310: Best = 3.411822e+03  FES =  31100  μ_F = 0.884  μ_CR = 0.355
Generation  320: Best = 3.411822e+03  FES =  32100  μ_F = 0.877  μ_CR = 0.361
Generation  330: Best = 3.411822e+03  FES =  33100  μ_F = 0.880  μ_CR = 0.358
Generation  340: Best = 3.411822e+03  FES =  34100  μ_F = 0.884  μ_CR = 0.352
Generation  350: Best = 3.411822e+03  FES =  35100  μ_F = 0.891  μ_CR = 0.351
Generation  360: Best = 3.411822e+03  FES =  36100  μ_F = 0.884  μ_CR = 0.348
Generation  370: Best = 3.411822e+03  FES =  37100  μ_F = 0.888  μ_CR = 0.350
Generation  380: Best = 3.411822e+03  FES =  38100  μ_F = 0.884  μ_CR = 0.358
Generation  390: Best = 3.411822e+03  FES =  39100  μ_F = 0.879  μ_CR = 0.351
Generation  400: Best = 3.411822e+03  FES =  40100  μ_F = 0.878  μ_CR = 0.344
Generation  410: Best = 3.411822e+03  FES =  41100  μ_F = 0.889  μ_CR = 0.347
Generation  420: Best = 3.411822e+03  FES =  42100  μ_F = 0.894  μ_CR = 0.347
Generation  430: Best = 3.411822e+03  FES =  43100  μ_F = 0.894  μ_CR = 0.350
```

```
Generation  440: Best = 3.411822e+03  FES =  44100  μ_F = 0.889  μ_CR = 0.348
Generation  450: Best = 3.411822e+03  FES =  45100  μ_F = 0.889  μ_CR = 0.346
Generation  460: Best = 3.411822e+03  FES =  46100  μ_F = 0.896  μ_CR = 0.345
Generation  470: Best = 3.411822e+03  FES =  47100  μ_F = 0.897  μ_CR = 0.348
Generation  480: Best = 3.411822e+03  FES =  48100  μ_F = 0.896  μ_CR = 0.349
Generation  490: Best = 3.411822e+03  FES =  49100  μ_F = 0.896  μ_CR = 0.353
Generation  500: Best = 3.411822e+03  FES =  50100  μ_F = 0.896  μ_CR = 0.355
Generation  510: Best = 3.411822e+03  FES =  51100  μ_F = 0.893  μ_CR = 0.354
Generation  520: Best = 3.411822e+03  FES =  52100  μ_F = 0.889  μ_CR = 0.350
Generation  530: Best = 3.411822e+03  FES =  53100  μ_F = 0.890  μ_CR = 0.350
Generation  540: Best = 3.411822e+03  FES =  54100  μ_F = 0.889  μ_CR = 0.348
Generation  550: Best = 3.411822e+03  FES =  55100  μ_F = 0.885  μ_CR = 0.344
Generation  560: Best = 3.411822e+03  FES =  56100  μ_F = 0.884  μ_CR = 0.340
Generation  570: Best = 3.411822e+03  FES =  57100  μ_F = 0.886  μ_CR = 0.338
Generation  580: Best = 3.411822e+03  FES =  58100  μ_F = 0.882  μ_CR = 0.334
Generation  590: Best = 3.411822e+03  FES =  59100  μ_F = 0.892  μ_CR = 0.328
Generation  600: Best = 3.411822e+03  FES =  60100  μ_F = 0.886  μ_CR = 0.328
Generation  610: Best = 3.411822e+03  FES =  61100  μ_F = 0.886  μ_CR = 0.318
Generation  620: Best = 3.411822e+03  FES =  62100  μ_F = 0.883  μ_CR = 0.320
Generation  630: Best = 3.411822e+03  FES =  63100  μ_F = 0.883  μ_CR = 0.318
Generation  640: Best = 3.411822e+03  FES =  64100  μ_F = 0.885  μ_CR = 0.310
Generation  650: Best = 3.411822e+03  FES =  65100  μ_F = 0.883  μ_CR = 0.307
Generation  660: Best = 3.411822e+03  FES =  66100  μ_F = 0.885  μ_CR = 0.308
Generation  670: Best = 3.411822e+03  FES =  67100  μ_F = 0.889  μ_CR = 0.302
Generation  680: Best = 3.411822e+03  FES =  68100  μ_F = 0.893  μ_CR = 0.295
Generation  690: Best = 3.411822e+03  FES =  69100  μ_F = 0.889  μ_CR = 0.286
Generation  700: Best = 3.411822e+03  FES =  70100  μ_F = 0.889  μ_CR = 0.286
Generation  710: Best = 3.411822e+03  FES =  71100  μ_F = 0.885  μ_CR = 0.280
Generation  720: Best = 3.411822e+03  FES =  72100  μ_F = 0.874  μ_CR = 0.273
Generation  730: Best = 3.411822e+03  FES =  73100  μ_F = 0.879  μ_CR = 0.277
Generation  740: Best = 3.411822e+03  FES =  74100  μ_F = 0.875  μ_CR = 0.266
Generation  750: Best = 3.411822e+03  FES =  75100  μ_F = 0.878  μ_CR = 0.266
Generation  760: Best = 3.411822e+03  FES =  76100  μ_F = 0.876  μ_CR = 0.258
Generation  770: Best = 3.411822e+03  FES =  77100  μ_F = 0.875  μ_CR = 0.248
Generation  780: Best = 3.411822e+03  FES =  78100  μ_F = 0.876  μ_CR = 0.238
Generation  790: Best = 3.411822e+03  FES =  79100  μ_F = 0.882  μ_CR = 0.229
Generation  800: Best = 3.411822e+03  FES =  80100  μ_F = 0.884  μ_CR = 0.215
Generation  810: Best = 3.411822e+03  FES =  81100  μ_F = 0.888  μ_CR = 0.217
Generation  820: Best = 3.411822e+03  FES =  82100  μ_F = 0.887  μ_CR = 0.213
Generation  830: Best = 3.411822e+03  FES =  83100  μ_F = 0.888  μ_CR = 0.197
Generation  840: Best = 3.411822e+03  FES =  84100  μ_F = 0.890  μ_CR = 0.190
Generation  850: Best = 3.411822e+03  FES =  85100  μ_F = 0.889  μ_CR = 0.182
Generation  860: Best = 3.411822e+03  FES =  86100  μ_F = 0.886  μ_CR = 0.181
Generation  870: Best = 3.411822e+03  FES =  87100  μ_F = 0.884  μ_CR = 0.176
Generation  880: Best = 3.411822e+03  FES =  88100  μ_F = 0.883  μ_CR = 0.172
Generation  890: Best = 3.411822e+03  FES =  89100  μ_F = 0.885  μ_CR = 0.164
Generation  900: Best = 3.411822e+03  FES =  90100  μ_F = 0.879  μ_CR = 0.161
Generation  910: Best = 3.411822e+03  FES =  91100  μ_F = 0.879  μ_CR = 0.153
Generation  920: Best = 3.411822e+03  FES =  92100  μ_F = 0.882  μ_CR = 0.154
Generation  930: Best = 3.411822e+03  FES =  93100  μ_F = 0.882  μ_CR = 0.157
Generation  940: Best = 3.411822e+03  FES =  94100  μ_F = 0.880  μ_CR = 0.152
Generation  950: Best = 3.411822e+03  FES =  95100  μ_F = 0.883  μ_CR = 0.145
Generation  960: Best = 3.411822e+03  FES =  96100  μ_F = 0.881  μ_CR = 0.143
Generation  970: Best = 3.411822e+03  FES =  97100  μ_F = 0.883  μ_CR = 0.137
```

```
Generation  980: Best = 3.411822e+03  FES =  98100  μ_F = 0.885  μ_CR = 0.136
Generation  990: Best = 3.411822e+03  FES =  99100  μ_F = 0.892  μ_CR = 0.144

Optimization complete.
Best Fitness             = 3.411822e+03
Total Function Evals     = 100000
Total Successful Updates = 57913

================================================================================
=
Results:
  Best Fitness           : 3.411822e+03
  Global Optimum         : 2.800000e+03
  Error (|best-optimum|) : 6.118217e+02
  Best Solution (5 dim)  : [  47.76790668 -100.          100.           96.5113
0875  -38.79309804]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F28.png
```

In [38]:
```python
# ============================================================
# F29 — Composition Function 9 (F15; F16; F17)
# ============================================================
run_and_plot('F29', make_wrapper(composition.f29), 2900.0, 'Composition: F15;
```

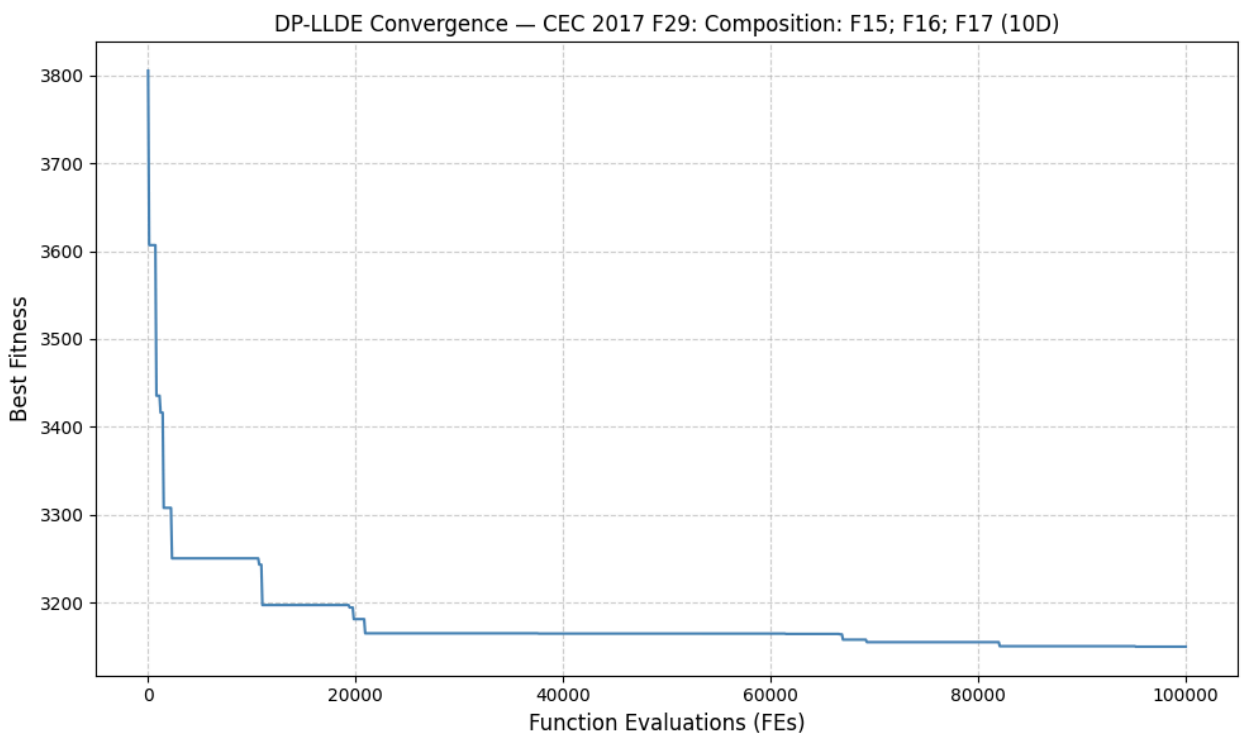
```
================================================================================
=
DP-LLDE on CEC 2017 F29 — Single Run
================================================================================
=
Function       : Composition: F15; F16; F17 (10D)
Bounds         : [-100, 100]
Global optimum : 2900.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 3.805371e+03
Generation   10: Best = 3.435360e+03  FES =   1100  μ_F = 0.588  μ_CR = 0.348
Generation   20: Best = 3.307691e+03  FES =   2100  μ_F = 0.650  μ_CR = 0.330
Generation   30: Best = 3.250180e+03  FES =   3100  μ_F = 0.687  μ_CR = 0.323
Generation   40: Best = 3.250180e+03  FES =   4100  μ_F = 0.770  μ_CR = 0.324
Generation   50: Best = 3.250180e+03  FES =   5100  μ_F = 0.768  μ_CR = 0.278
Generation   60: Best = 3.250180e+03  FES =   6100  μ_F = 0.792  μ_CR = 0.251
Generation   70: Best = 3.250180e+03  FES =   7100  μ_F = 0.788  μ_CR = 0.229
Generation   80: Best = 3.250180e+03  FES =   8100  μ_F = 0.762  μ_CR = 0.218
Generation   90: Best = 3.250180e+03  FES =   9100  μ_F = 0.809  μ_CR = 0.169
Generation  100: Best = 3.250180e+03  FES =  10100  μ_F = 0.800  μ_CR = 0.119
Generation  110: Best = 3.197041e+03  FES =  11100  μ_F = 0.830  μ_CR = 0.135
Generation  120: Best = 3.197041e+03  FES =  12100  μ_F = 0.823  μ_CR = 0.130
Generation  130: Best = 3.197041e+03  FES =  13100  μ_F = 0.887  μ_CR = 0.110
Generation  140: Best = 3.197041e+03  FES =  14100  μ_F = 0.866  μ_CR = 0.115
Generation  150: Best = 3.197041e+03  FES =  15100  μ_F = 0.840  μ_CR = 0.130
Generation  160: Best = 3.197041e+03  FES =  16100  μ_F = 0.831  μ_CR = 0.116
Generation  170: Best = 3.197041e+03  FES =  17100  μ_F = 0.840  μ_CR = 0.123
Generation  180: Best = 3.197041e+03  FES =  18100  μ_F = 0.885  μ_CR = 0.141
Generation  190: Best = 3.197041e+03  FES =  19100  μ_F = 0.829  μ_CR = 0.128
Generation  200: Best = 3.181028e+03  FES =  20100  μ_F = 0.723  μ_CR = 0.135
Generation  210: Best = 3.164849e+03  FES =  21100  μ_F = 0.841  μ_CR = 0.119
Generation  220: Best = 3.164849e+03  FES =  22100  μ_F = 0.831  μ_CR = 0.109
Generation  230: Best = 3.164849e+03  FES =  23100  μ_F = 0.804  μ_CR = 0.073
Generation  240: Best = 3.164849e+03  FES =  24100  μ_F = 0.757  μ_CR = 0.089
Generation  250: Best = 3.164849e+03  FES =  25100  μ_F = 0.767  μ_CR = 0.083
Generation  260: Best = 3.164849e+03  FES =  26100  μ_F = 0.771  μ_CR = 0.074
Generation  270: Best = 3.164849e+03  FES =  27100  μ_F = 0.797  μ_CR = 0.076
Generation  280: Best = 3.164849e+03  FES =  28100  μ_F = 0.744  μ_CR = 0.089
Generation  290: Best = 3.164849e+03  FES =  29100  μ_F = 0.750  μ_CR = 0.086
Generation  300: Best = 3.164849e+03  FES =  30100  μ_F = 0.730  μ_CR = 0.089
Generation  310: Best = 3.164849e+03  FES =  31100  μ_F = 0.693  μ_CR = 0.099
Generation  320: Best = 3.164849e+03  FES =  32100  μ_F = 0.683  μ_CR = 0.099
Generation  330: Best = 3.164849e+03  FES =  33100  μ_F = 0.653  μ_CR = 0.104
Generation  340: Best = 3.164849e+03  FES =  34100  μ_F = 0.640  μ_CR = 0.094
Generation  350: Best = 3.164849e+03  FES =  35100  μ_F = 0.633  μ_CR = 0.096
Generation  360: Best = 3.164849e+03  FES =  36100  μ_F = 0.651  μ_CR = 0.110
Generation  370: Best = 3.164849e+03  FES =  37100  μ_F = 0.686  μ_CR = 0.102
Generation  380: Best = 3.164548e+03  FES =  38100  μ_F = 0.647  μ_CR = 0.089
Generation  390: Best = 3.164548e+03  FES =  39100  μ_F = 0.637  μ_CR = 0.076
Generation  400: Best = 3.164548e+03  FES =  40100  μ_F = 0.661  μ_CR = 0.055
Generation  410: Best = 3.164548e+03  FES =  41100  μ_F = 0.643  μ_CR = 0.069
Generation  420: Best = 3.164548e+03  FES =  42100  μ_F = 0.703  μ_CR = 0.088
Generation  430: Best = 3.164548e+03  FES =  43100  μ_F = 0.745  μ_CR = 0.089
Generation  440: Best = 3.164533e+03  FES =  44100  μ_F = 0.782  μ_CR = 0.078
```

```
Generation  450: Best = 3.164533e+03  FES =  45100  μ_F = 0.788  μ_CR = 0.098
Generation  460: Best = 3.164533e+03  FES =  46100  μ_F = 0.794  μ_CR = 0.112
Generation  470: Best = 3.164533e+03  FES =  47100  μ_F = 0.723  μ_CR = 0.117
Generation  480: Best = 3.164533e+03  FES =  48100  μ_F = 0.772  μ_CR = 0.119
Generation  490: Best = 3.164533e+03  FES =  49100  μ_F = 0.776  μ_CR = 0.125
Generation  500: Best = 3.164533e+03  FES =  50100  μ_F = 0.774  μ_CR = 0.113
Generation  510: Best = 3.164533e+03  FES =  51100  μ_F = 0.772  μ_CR = 0.104
Generation  520: Best = 3.164533e+03  FES =  52100  μ_F = 0.776  μ_CR = 0.111
Generation  530: Best = 3.164533e+03  FES =  53100  μ_F = 0.809  μ_CR = 0.105
Generation  540: Best = 3.164533e+03  FES =  54100  μ_F = 0.789  μ_CR = 0.095
Generation  550: Best = 3.164533e+03  FES =  55100  μ_F = 0.817  μ_CR = 0.079
Generation  560: Best = 3.164533e+03  FES =  56100  μ_F = 0.783  μ_CR = 0.074
Generation  570: Best = 3.164533e+03  FES =  57100  μ_F = 0.739  μ_CR = 0.083
Generation  580: Best = 3.164533e+03  FES =  58100  μ_F = 0.750  μ_CR = 0.086
Generation  590: Best = 3.164533e+03  FES =  59100  μ_F = 0.684  μ_CR = 0.101
Generation  600: Best = 3.164533e+03  FES =  60100  μ_F = 0.664  μ_CR = 0.103
Generation  610: Best = 3.164533e+03  FES =  61100  μ_F = 0.668  μ_CR = 0.089
Generation  620: Best = 3.164160e+03  FES =  62100  μ_F = 0.673  μ_CR = 0.102
Generation  630: Best = 3.164160e+03  FES =  63100  μ_F = 0.676  μ_CR = 0.087
Generation  640: Best = 3.164160e+03  FES =  64100  μ_F = 0.664  μ_CR = 0.096
Generation  650: Best = 3.164160e+03  FES =  65100  μ_F = 0.664  μ_CR = 0.071
Generation  660: Best = 3.164160e+03  FES =  66100  μ_F = 0.651  μ_CR = 0.057
Generation  670: Best = 3.157627e+03  FES =  67100  μ_F = 0.613  μ_CR = 0.048
Generation  680: Best = 3.157627e+03  FES =  68100  μ_F = 0.603  μ_CR = 0.070
Generation  690: Best = 3.157627e+03  FES =  69100  μ_F = 0.573  μ_CR = 0.086
Generation  700: Best = 3.154785e+03  FES =  70100  μ_F = 0.553  μ_CR = 0.087
Generation  710: Best = 3.154785e+03  FES =  71100  μ_F = 0.545  μ_CR = 0.086
Generation  720: Best = 3.154785e+03  FES =  72100  μ_F = 0.594  μ_CR = 0.089
Generation  730: Best = 3.154785e+03  FES =  73100  μ_F = 0.483  μ_CR = 0.102
Generation  740: Best = 3.154785e+03  FES =  74100  μ_F = 0.511  μ_CR = 0.084
Generation  750: Best = 3.154785e+03  FES =  75100  μ_F = 0.577  μ_CR = 0.093
Generation  760: Best = 3.154785e+03  FES =  76100  μ_F = 0.593  μ_CR = 0.095
Generation  770: Best = 3.154785e+03  FES =  77100  μ_F = 0.585  μ_CR = 0.115
Generation  780: Best = 3.154785e+03  FES =  78100  μ_F = 0.594  μ_CR = 0.101
Generation  790: Best = 3.154785e+03  FES =  79100  μ_F = 0.597  μ_CR = 0.106
Generation  800: Best = 3.154785e+03  FES =  80100  μ_F = 0.565  μ_CR = 0.089
Generation  810: Best = 3.154785e+03  FES =  81100  μ_F = 0.632  μ_CR = 0.104
Generation  820: Best = 3.150178e+03  FES =  82100  μ_F = 0.695  μ_CR = 0.076
Generation  830: Best = 3.150178e+03  FES =  83100  μ_F = 0.700  μ_CR = 0.087
Generation  840: Best = 3.150178e+03  FES =  84100  μ_F = 0.716  μ_CR = 0.080
Generation  850: Best = 3.150178e+03  FES =  85100  μ_F = 0.690  μ_CR = 0.070
Generation  860: Best = 3.150178e+03  FES =  86100  μ_F = 0.687  μ_CR = 0.076
Generation  870: Best = 3.150178e+03  FES =  87100  μ_F = 0.719  μ_CR = 0.068
Generation  880: Best = 3.150178e+03  FES =  88100  μ_F = 0.606  μ_CR = 0.068
Generation  890: Best = 3.150178e+03  FES =  89100  μ_F = 0.608  μ_CR = 0.070
Generation  900: Best = 3.150178e+03  FES =  90100  μ_F = 0.526  μ_CR = 0.071
Generation  910: Best = 3.150178e+03  FES =  91100  μ_F = 0.488  μ_CR = 0.086
Generation  920: Best = 3.150178e+03  FES =  92100  μ_F = 0.454  μ_CR = 0.095
Generation  930: Best = 3.150178e+03  FES =  93100  μ_F = 0.423  μ_CR = 0.098
Generation  940: Best = 3.150178e+03  FES =  94100  μ_F = 0.491  μ_CR = 0.073
Generation  950: Best = 3.150178e+03  FES =  95100  μ_F = 0.587  μ_CR = 0.072
Generation  960: Best = 3.149627e+03  FES =  96100  μ_F = 0.575  μ_CR = 0.085
Generation  970: Best = 3.149627e+03  FES =  97100  μ_F = 0.577  μ_CR = 0.085
Generation  980: Best = 3.149627e+03  FES =  98100  μ_F = 0.527  μ_CR = 0.104
```

```
Generation  990: Best = 3.149627e+03  FES =  99100  μ_F = 0.501  μ_CR = 0.113

Optimization complete.
Best Fitness             = 3.149627e+03
Total Function Evals     = 100000
Total Successful Updates = 1783

================================================================================
=
Results:
  Best Fitness           : 3.149627e+03
  Global Optimum         : 2.900000e+03
  Error (|best-optimum|) : 2.496270e+02
  Best Solution (5 dim)  : [ -6.01942355 -77.938201   -74.43843126  78.80520065
-46.18072443]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F29.png
```

In [39]:

```python
# ============================================================
# F30 — Composition Function 10 (F15; F18; F19)
# ============================================================
run_and_plot('F30', make_wrapper(composition.f30), 3000.0, 'Composition: F15;
```

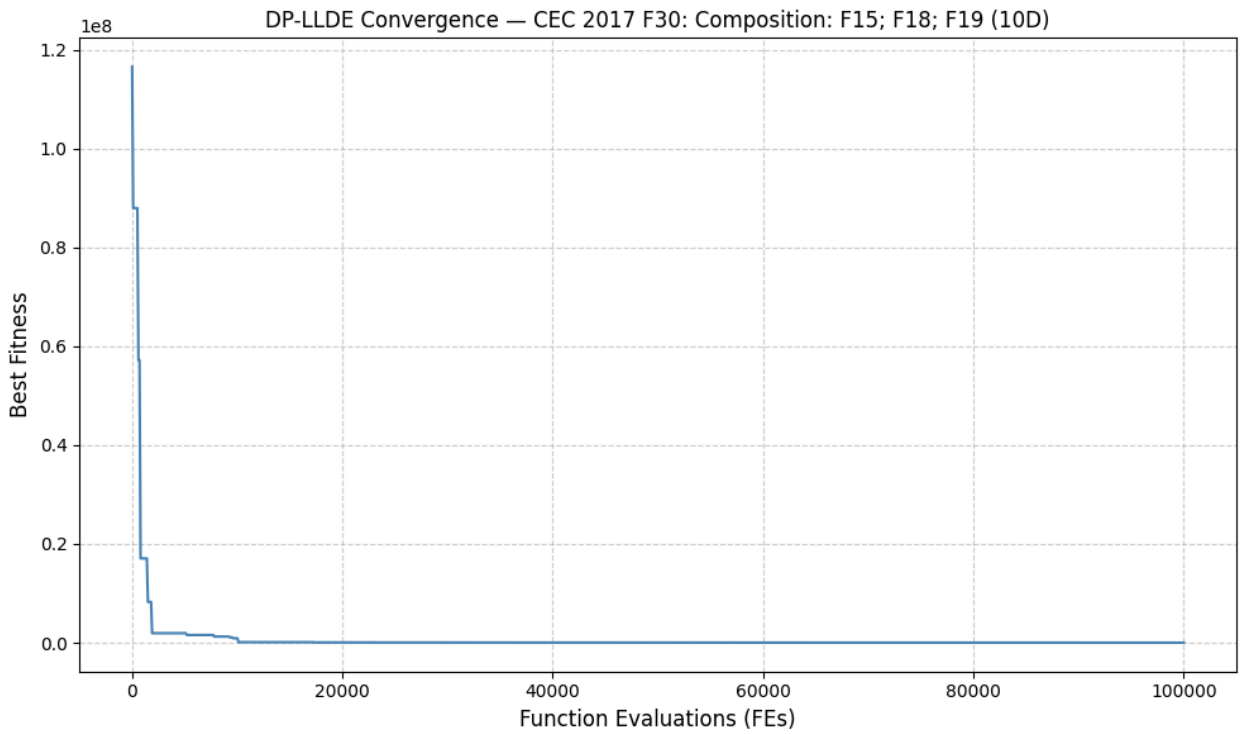
```
================================================================================
=
DP-LLDE on CEC 2017 F30 — Single Run
================================================================================
=
Function       : Composition: F15; F18; F19 (10D)
Bounds         : [-100, 100]
Global optimum : 3000.0
Initial D^(0) = 2.5607e+02  |  Diversity threshold = 2.5607e+01
Generation 0: Best Fitness = 1.165390e+08
Generation   10: Best = 1.705750e+07  FES =   1100  μ_F = 0.562  μ_CR = 0.339
Generation   20: Best = 1.931435e+06  FES =   2100  μ_F = 0.630  μ_CR = 0.350
Generation   30: Best = 1.931435e+06  FES =   3100  μ_F = 0.684  μ_CR = 0.323
Generation   40: Best = 1.931435e+06  FES =   4100  μ_F = 0.736  μ_CR = 0.282
Generation   50: Best = 1.931435e+06  FES =   5100  μ_F = 0.779  μ_CR = 0.251
Generation   60: Best = 1.558576e+06  FES =   6100  μ_F = 0.787  μ_CR = 0.200
Generation   70: Best = 1.558576e+06  FES =   7100  μ_F = 0.817  μ_CR = 0.165
Generation   80: Best = 1.241456e+06  FES =   8100  μ_F = 0.820  μ_CR = 0.153
Generation   90: Best = 1.241456e+06  FES =   9100  μ_F = 0.840  μ_CR = 0.141
Generation  100: Best = 9.037795e+05  FES =  10100  μ_F = 0.849  μ_CR = 0.121
Generation  110: Best = 1.137592e+05  FES =  11100  μ_F = 0.852  μ_CR = 0.111
Generation  120: Best = 1.014216e+05  FES =  12100  μ_F = 0.874  μ_CR = 0.105
Generation  130: Best = 1.014216e+05  FES =  13100  μ_F = 0.868  μ_CR = 0.097
Generation  140: Best = 1.014216e+05  FES =  14100  μ_F = 0.861  μ_CR = 0.098
Generation  150: Best = 1.014216e+05  FES =  15100  μ_F = 0.862  μ_CR = 0.093
Generation  160: Best = 1.014216e+05  FES =  16100  μ_F = 0.866  μ_CR = 0.088
Generation  170: Best = 1.014216e+05  FES =  17100  μ_F = 0.868  μ_CR = 0.093
Generation  180: Best = 6.099959e+04  FES =  18100  μ_F = 0.879  μ_CR = 0.089
Generation  190: Best = 5.701184e+04  FES =  19100  μ_F = 0.873  μ_CR = 0.095
Generation  200: Best = 5.701184e+04  FES =  20100  μ_F = 0.866  μ_CR = 0.077
Generation  210: Best = 5.701184e+04  FES =  21100  μ_F = 0.871  μ_CR = 0.087
Generation  220: Best = 5.701184e+04  FES =  22100  μ_F = 0.873  μ_CR = 0.082
Generation  230: Best = 5.701184e+04  FES =  23100  μ_F = 0.885  μ_CR = 0.082
Generation  240: Best = 5.701184e+04  FES =  24100  μ_F = 0.877  μ_CR = 0.077
Generation  250: Best = 5.701184e+04  FES =  25100  μ_F = 0.889  μ_CR = 0.061
Generation  260: Best = 5.701184e+04  FES =  26100  μ_F = 0.889  μ_CR = 0.068
Generation  270: Best = 5.701184e+04  FES =  27100  μ_F = 0.890  μ_CR = 0.072
Generation  280: Best = 5.701184e+04  FES =  28100  μ_F = 0.876  μ_CR = 0.078
Generation  290: Best = 5.701184e+04  FES =  29100  μ_F = 0.867  μ_CR = 0.078
Generation  300: Best = 5.701184e+04  FES =  30100  μ_F = 0.865  μ_CR = 0.076
Generation  310: Best = 5.701184e+04  FES =  31100  μ_F = 0.884  μ_CR = 0.067
Generation  320: Best = 4.468524e+04  FES =  32100  μ_F = 0.850  μ_CR = 0.070
Generation  330: Best = 4.078083e+04  FES =  33100  μ_F = 0.861  μ_CR = 0.078
Generation  340: Best = 3.359967e+04  FES =  34100  μ_F = 0.885  μ_CR = 0.092
Generation  350: Best = 3.117800e+04  FES =  35100  μ_F = 0.871  μ_CR = 0.098
Generation  360: Best = 3.117800e+04  FES =  36100  μ_F = 0.880  μ_CR = 0.085
Generation  370: Best = 3.117800e+04  FES =  37100  μ_F = 0.875  μ_CR = 0.070
Generation  380: Best = 3.117800e+04  FES =  38100  μ_F = 0.882  μ_CR = 0.064
Generation  390: Best = 3.117800e+04  FES =  39100  μ_F = 0.859  μ_CR = 0.064
Generation  400: Best = 3.117800e+04  FES =  40100  μ_F = 0.887  μ_CR = 0.063
Generation  410: Best = 3.117800e+04  FES =  41100  μ_F = 0.896  μ_CR = 0.070
Generation  420: Best = 3.117800e+04  FES =  42100  μ_F = 0.903  μ_CR = 0.080
Generation  430: Best = 3.117800e+04  FES =  43100  μ_F = 0.880  μ_CR = 0.070
Generation  440: Best = 3.117800e+04  FES =  44100  μ_F = 0.881  μ_CR = 0.076
```

```
Generation  450: Best = 3.117800e+04  FES =  45100  μ_F = 0.880  μ_CR = 0.087
Generation  460: Best = 3.117800e+04  FES =  46100  μ_F = 0.856  μ_CR = 0.074
Generation  470: Best = 3.117800e+04  FES =  47100  μ_F = 0.853  μ_CR = 0.077
Generation  480: Best = 3.117800e+04  FES =  48100  μ_F = 0.844  μ_CR = 0.072
Generation  490: Best = 3.117800e+04  FES =  49100  μ_F = 0.846  μ_CR = 0.061
Generation  500: Best = 3.117800e+04  FES =  50100  μ_F = 0.845  μ_CR = 0.065
Generation  510: Best = 3.117800e+04  FES =  51100  μ_F = 0.865  μ_CR = 0.068
Generation  520: Best = 3.117800e+04  FES =  52100  μ_F = 0.873  μ_CR = 0.077
Generation  530: Best = 3.117800e+04  FES =  53100  μ_F = 0.878  μ_CR = 0.067
Generation  540: Best = 3.117800e+04  FES =  54100  μ_F = 0.882  μ_CR = 0.064
Generation  550: Best = 2.230961e+04  FES =  55100  μ_F = 0.879  μ_CR = 0.067
Generation  560: Best = 2.230961e+04  FES =  56100  μ_F = 0.889  μ_CR = 0.071
Generation  570: Best = 2.230961e+04  FES =  57100  μ_F = 0.889  μ_CR = 0.071
Generation  580: Best = 2.230961e+04  FES =  58100  μ_F = 0.846  μ_CR = 0.070
Generation  590: Best = 2.230961e+04  FES =  59100  μ_F = 0.855  μ_CR = 0.084
Generation  600: Best = 2.230961e+04  FES =  60100  μ_F = 0.863  μ_CR = 0.076
Generation  610: Best = 2.230961e+04  FES =  61100  μ_F = 0.868  μ_CR = 0.086
Generation  620: Best = 2.230961e+04  FES =  62100  μ_F = 0.871  μ_CR = 0.078
Generation  630: Best = 2.230961e+04  FES =  63100  μ_F = 0.883  μ_CR = 0.074
Generation  640: Best = 2.230961e+04  FES =  64100  μ_F = 0.873  μ_CR = 0.073
Generation  650: Best = 2.230961e+04  FES =  65100  μ_F = 0.867  μ_CR = 0.078
Generation  660: Best = 2.230961e+04  FES =  66100  μ_F = 0.874  μ_CR = 0.076
Generation  670: Best = 2.230961e+04  FES =  67100  μ_F = 0.877  μ_CR = 0.077
Generation  680: Best = 2.230961e+04  FES =  68100  μ_F = 0.874  μ_CR = 0.073
Generation  690: Best = 2.230961e+04  FES =  69100  μ_F = 0.872  μ_CR = 0.074
Generation  700: Best = 2.230961e+04  FES =  70100  μ_F = 0.879  μ_CR = 0.081
Generation  710: Best = 2.230961e+04  FES =  71100  μ_F = 0.868  μ_CR = 0.089
Generation  720: Best = 6.485822e+03  FES =  72100  μ_F = 0.899  μ_CR = 0.071
Generation  730: Best = 6.485822e+03  FES =  73100  μ_F = 0.884  μ_CR = 0.080
Generation  740: Best = 6.485822e+03  FES =  74100  μ_F = 0.875  μ_CR = 0.072
Generation  750: Best = 6.485822e+03  FES =  75100  μ_F = 0.889  μ_CR = 0.076
Generation  760: Best = 6.485822e+03  FES =  76100  μ_F = 0.875  μ_CR = 0.069
Generation  770: Best = 6.485822e+03  FES =  77100  μ_F = 0.879  μ_CR = 0.073
Generation  780: Best = 6.485822e+03  FES =  78100  μ_F = 0.878  μ_CR = 0.081
Generation  790: Best = 6.485822e+03  FES =  79100  μ_F = 0.828  μ_CR = 0.071
Generation  800: Best = 6.485822e+03  FES =  80100  μ_F = 0.854  μ_CR = 0.064
Generation  810: Best = 6.485822e+03  FES =  81100  μ_F = 0.879  μ_CR = 0.069
Generation  820: Best = 6.485822e+03  FES =  82100  μ_F = 0.871  μ_CR = 0.076
Generation  830: Best = 6.485822e+03  FES =  83100  μ_F = 0.895  μ_CR = 0.068
Generation  840: Best = 6.485822e+03  FES =  84100  μ_F = 0.908  μ_CR = 0.078
Generation  850: Best = 6.485822e+03  FES =  85100  μ_F = 0.882  μ_CR = 0.078
Generation  860: Best = 6.485822e+03  FES =  86100  μ_F = 0.878  μ_CR = 0.088
Generation  870: Best = 6.485822e+03  FES =  87100  μ_F = 0.895  μ_CR = 0.088
Generation  880: Best = 6.485822e+03  FES =  88100  μ_F = 0.902  μ_CR = 0.083
Generation  890: Best = 6.485822e+03  FES =  89100  μ_F = 0.918  μ_CR = 0.080
Generation  900: Best = 6.485822e+03  FES =  90100  μ_F = 0.897  μ_CR = 0.073
Generation  910: Best = 6.485822e+03  FES =  91100  μ_F = 0.884  μ_CR = 0.083
Generation  920: Best = 6.485822e+03  FES =  92100  μ_F = 0.864  μ_CR = 0.073
Generation  930: Best = 6.485822e+03  FES =  93100  μ_F = 0.884  μ_CR = 0.071
Generation  940: Best = 6.485822e+03  FES =  94100  μ_F = 0.885  μ_CR = 0.069
Generation  950: Best = 6.479274e+03  FES =  95100  μ_F = 0.882  μ_CR = 0.075
Generation  960: Best = 6.479274e+03  FES =  96100  μ_F = 0.867  μ_CR = 0.067
Generation  970: Best = 6.479274e+03  FES =  97100  μ_F = 0.866  μ_CR = 0.067
Generation  980: Best = 6.479274e+03  FES =  98100  μ_F = 0.895  μ_CR = 0.069
```

```
Generation  990: Best = 6.479274e+03  FES =  99100  μ_F = 0.886  μ_CR = 0.090

Optimization complete.
Best Fitness             = 6.479274e+03
Total Function Evals     = 100000
Total Successful Updates = 7049

================================================================================
=
Results:
  Best Fitness           : 6.479274e+03
  Global Optimum         : 3.000000e+03
  Error (|best-optimum|) : 3.479274e+03
  Best Solution (5 dim)  : [ -84.07645117   82.01252836  100.          100.
-100.        ]
  D^(0) (init. diversity): 2.5607e+02
================================================================================
=
```




```
Graph saved as convergence_F30.png
```

Section 7. Compiled Results

Print out a summary of all results in a tabular format

In [41]:
```
print('=' * 80)
print('SUMMARY — DP-LLDE on CEC 2017 (10D), Single Run, Seed=42')
print('=' * 80)
```

```
print(f'{"Function":<10} {"Best Fitness":>18} {"Global Optimum":>18} {"Error |
print('-' * 70)
for fn in [f'F{i}' for i in range(1, 31)]:
    if fn not in all_results:
        continue
    r = all_results[fn]
    if r['best_fit'] is None:
        print(f'{fn:<10} {"N/A (deprecated)":>18} {r["f_global"]:>18.2f} {"N/A
    else:
        print(f'{fn:<10} {r["best_fit"]:>18.6e} {r["f_global"]:>18.2f} {r["err
print('=' * 80)
```

```
================================================================================
=
SUMMARY — DP-LLDE on CEC 2017 (10D), Single Run, Seed=42
================================================================================
=
Function         Best Fitness     Global Optimum   Error |best-opt|
----------------------------------------------------------------------
F1               1.137057e+02             100.00       1.370568e+01
F2               2.000000e+02             200.00       3.642452e-05
F3               3.118899e+02             300.00       1.188993e+01
F4               4.000000e+02             400.00       0.000000e+00
F5               5.077587e+02             500.00       7.758743e+00
F6               6.000000e+02             600.00       8.256476e-06
F7               7.156720e+02             700.00       1.567204e+01
F8               8.021592e+02             800.00       2.159158e+00
F9               9.000000e+02             900.00       0.000000e+00
F10              1.255191e+03            1000.00       2.551913e+02
F11              1.103074e+03            1100.00       3.073868e+00
F12              2.235439e+04            1200.00       2.115439e+04
F13              1.308351e+03            1300.00       8.350809e+00
F14              1.400232e+03            1400.00       2.320530e-01
F15              1.500884e+03            1500.00       8.843735e-01
F16              1.604330e+03            1600.00       4.329604e+00
F17              1.700472e+03            1700.00       4.721823e-01
F18              1.800463e+03            1800.00       4.630561e-01
F19              1.900012e+03            1900.00       1.161373e-02
F20              2.000011e+03            2000.00       1.148618e-02
F21              2.306755e+03            2100.00       2.067551e+02
F22              2.300180e+03            2200.00       1.001800e+02
F23              2.608867e+03            2300.00       3.088673e+02
F24              2.740196e+03            2400.00       3.401957e+02
F25              2.943510e+03            2500.00       4.435096e+02
F26              2.947149e+03            2600.00       3.471489e+02
F27              3.090047e+03            2700.00       3.900475e+02
F28              3.411822e+03            2800.00       6.118217e+02
F29              3.149627e+03            2900.00       2.496270e+02
F30              6.479274e+03            3000.00       3.479274e+03
================================================================================
=
```